4

Vector Algebra

Learning Objectives

After studying this chapter you should be able to:

- describe a vector by its magnitude and direction and by its components in $\mathbb{R}^2$ and $\mathbb{R}^3$, and compute the vector between two points, its length and its unit vector;
- add, subtract and scale vectors, both algebraically and geometrically;
- compute and interpret the dot product (angle, orthogonality, projection) and the cross product (perpendicular vector, area);
- use the scalar triple product to find volumes and to test whether three vectors are coplanar;
- find projections, reflections and direction cosines;
- decide whether vectors are linearly independent;
- apply vectors to forces in equilibrium, work and power, torque, rotation, the magnetic force and navigation.

Imagine a boat trying to cross a river. The boat's engine pushes it with a certain speed in one direction, but the river's current is also pushing it with its own speed in a different direction (Figure 4.1).

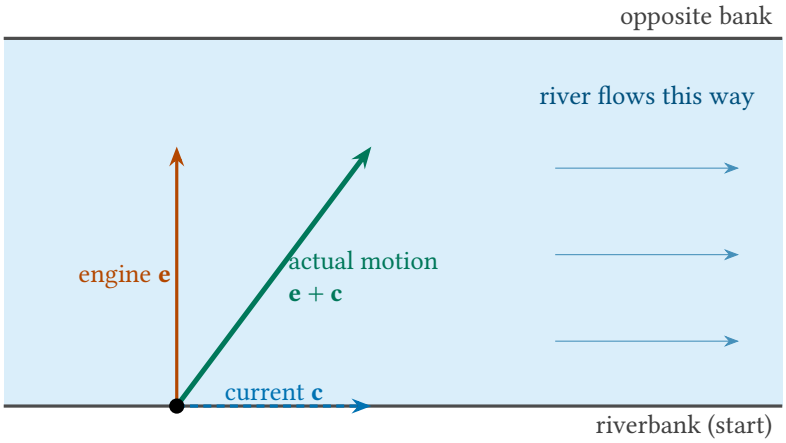


Figure 4.1: The engine pushes the boat straight across (**e**), the current pushes it downstream (**c**). The boat actually moves along the diagonal **e** + **c**.

The problem is: *What is the boat's actual speed and direction relative to the riverbank?* We cannot describe the boat's motion with just a single number (its speed), because its direction is equally important. The same is true for the river's current. Each of these quantities has both a magnitude (how fast it is moving) and a direction (where it is going). To solve this problem, we need to introduce a new mathematical object called a **vector** that can represent these quantities. We will then define rules for adding vectors and finding the magnitude of the resulting vector. We will come back to the boat at the end of Section 4.2 and answer the question with numbers (the *Boat on the River* example).

Intuition. Quantities such as time, mass, temperature and distance need only one number: they are *scalars*. Displacement, velocity, force and torque need a direction as well: they are *vectors*. If you walk 3 km east and then 4 km north, you have walked a *distance* of 7 km, but your *displacement* from the start is only 5 km, because the two legs point in different directions.

4.1 Vectors

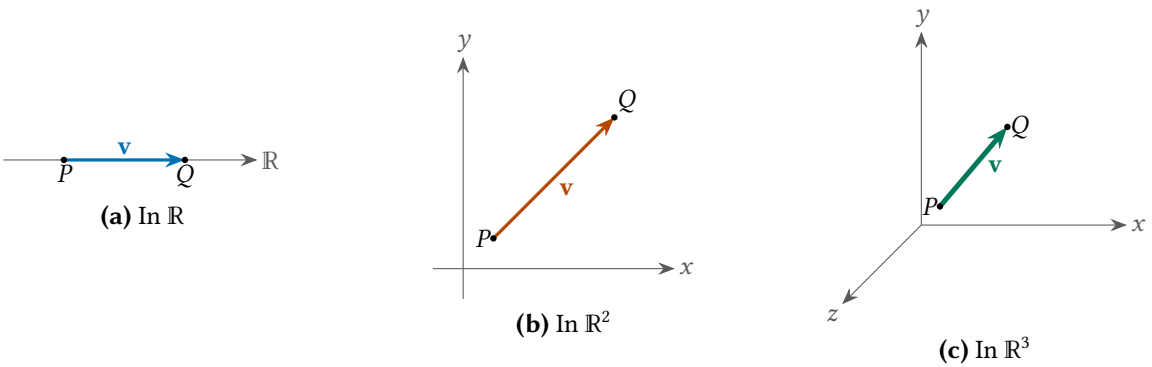
Definition

A **vector** is a directed line segment with both **magnitude** (length) and **direction**. Vectors are used to represent quantities such as displacement, velocity and force. Geometrically, a vector is drawn as an arrow from an **initial point** (tail) to a **terminal point** (tip). The direction of the vector is the orientation of the arrow, and its magnitude is the length of the arrow.
In Euclidean space $\mathbb{R}^n$, the vector $\mathbf{v}$ from the initial point $P(p_1, p_2, \dots, p_n)$ to the terminal point $Q(q_1, q_2, \dots, q_n)$ is the displacement from P to Q:

$$\mathbf{v} = \overrightarrow{PQ} = (q_1 - p_1,\ q_2 - p_2,\ \dots,\ q_n - p_n) = Q - P.$$

Notation. Vectors are written in boldface, $\mathbf{v}$, or with an arrow, $\vec{v}$; in handwriting the arrow is normally used. Points are written with capital letters, $P(x, y)$; a vector is never a point, although $Q - P$ is a vector.

Common mistake. Two *points* cannot be added: $P + Q$ has no meaning. What *is* meaningful is the vector $Q - P$ between two points, and a point plus a vector, $P + \mathbf{v}$, which is again a point. Likewise, "dividing by a vector" has no meaning: we never write $\mathbf{u}/\mathbf{v}$.



(a) In $\mathbb{R}$ **(b)** In $\mathbb{R}^2$ **(c)** In $\mathbb{R}^3$

Figure 4.2: A vector $\mathbf{v} = \overrightarrow{PQ}$ with initial point P and terminal point Q on the line, in the plane and in space.

Component form in $\mathbb{R}$, $\mathbb{R}^2$ and $\mathbb{R}^3$: the components of $\overrightarrow{PQ}$ are obtained by subtracting the coordinates of the initial point P from those of the terminal point Q:

- in $\mathbb{R}$: if $P = (p_1)$ and $Q = (q_1)$, then $\mathbf{v} = (v_1) = (q_1 - p_1)$;
- in $\mathbb{R}^2$: if $P = (p_1, p_2)$ and $Q = (q_1, q_2)$, then

$$\mathbf{v} = (v_1, v_2) = (q_1 - p_1,\ q_2 - p_2);$$

- in $\mathbb{R}^3$: if $P = (p_1, p_2, p_3)$ and $Q = (q_1, q_2, q_3)$, then

$$\mathbf{v} = (v_1, v_2, v_3) = (q_1 - p_1,\ q_2 - p_2,\ q_3 - p_3).$$

Free vectors

A **translation** of a vector $\mathbf{v}$ moves both its initial and terminal points by the same displacement. The new vector $\mathbf{v}'$ has a new initial point P' and a new terminal point Q' but the *same components*, hence the same length and the same direction. Therefore $\mathbf{v}' = \mathbf{v}$ (Figure 4.3).
This is why vectors are called **free vectors**: a vector is determined by its magnitude and direction only, not by where it is drawn.

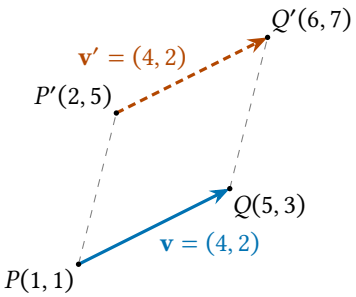
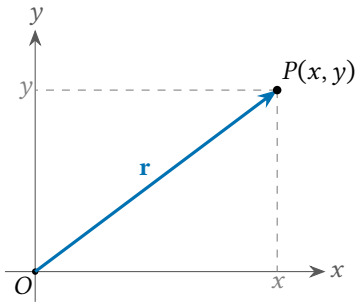


Figure 4.3: Translating a vector: both arrows have components $(4, 2)$, so they are the same free vector (the dashed grey lines show the shift by $(1, 4)$).

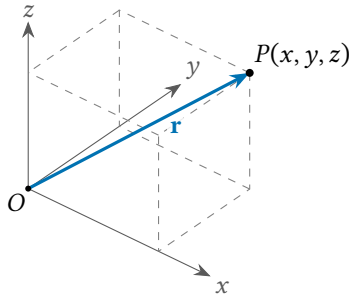
A **position vector** is a vector that describes the location of a point P relative to a fixed origin O; its initial point is always the origin. For a point $P(x, y)$ in the plane, or $P(x, y, z)$ in space,

$$\mathbf{r} = \overrightarrow{OP} = (x, y) \qquad \text{or} \qquad \mathbf{r} = \overrightarrow{OP} = (x, y, z).$$

So the position vector of a point has the same components as the coordinates of the point (Figure 4.4). Unlike a free vector, a position vector must *not* be translated: when we describe lines, curves or planes by vectors, the tail must stay at the origin so that the vector really identifies a point.



(a) In the plane



(b) In space

Figure 4.4: The position vector $\mathbf{r} = \overrightarrow{OP}$ of a point P; its components are the coordinates of P.

4.1.1 Magnitude and Unit Vectors

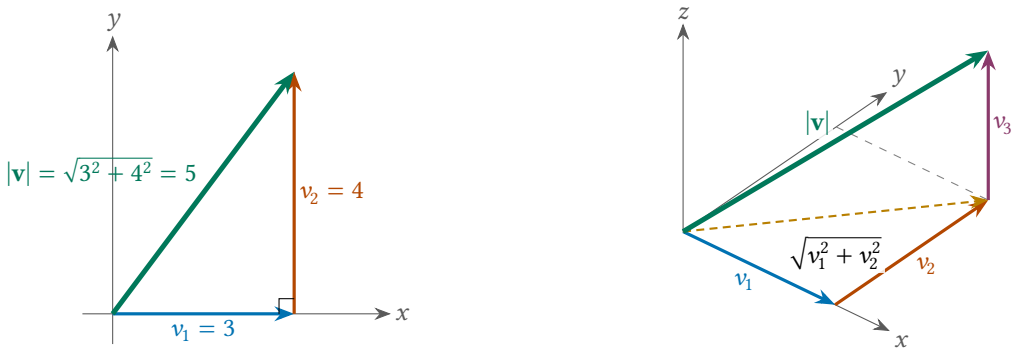
Definition

The **magnitude** (or **length**) of a vector $\mathbf{v} = (v_1, v_2, \dots, v_n)$ in $\mathbb{R}^n$, denoted $|\mathbf{v}|$, is

$$|\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$$

In particular, $|\mathbf{v}| = \sqrt{v_1^2 + v_2^2}$ in $\mathbb{R}^2$ and $|\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}$ in $\mathbb{R}^3$. A vector of length 1 is a **unit vector**. The only vector of length 0 is the **zero vector** $\mathbf{0} = (0, \dots, 0)$, which has no direction.

The formula is just the Pythagorean theorem: in the plane $|\mathbf{v}|$ is the hypotenuse of a right triangle with legs $|v_1|$ and $|v_2|$; in space, apply the theorem twice (Figure 4.5).



(a) In $\mathbb{R}^2$ **(b)** In $\mathbb{R}^3$: $|\mathbf{v}|^2 = (v_1^2 + v_2^2) + v_3^2$

Figure 4.5: The length of a vector from the Pythagorean theorem.

To obtain the **unit vector** in the direction of a nonzero vector $\mathbf{v}$, divide $\mathbf{v}$ by its magnitude:

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{|\mathbf{v}|}, \qquad |\hat{\mathbf{v}}| = 1.$$

Example: Length and Unit Vector

For $\mathbf{v} = (3, 4)$ we get $|\mathbf{v}| = \sqrt{9 + 16} = 5$ and

$$\hat{\mathbf{v}} = \left(\tfrac{3}{5}, \tfrac{4}{5}\right), \qquad |\hat{\mathbf{v}}| = \sqrt{\tfrac{9}{25} + \tfrac{16}{25}} = 1.$$

In $\mathbb{R}^3$, for $\mathbf{w} = (2, -1, 2)$: $|\mathbf{w}| = \sqrt{4 + 1 + 4} = 3$, so $\hat{\mathbf{w}} = \left(\tfrac{2}{3}, -\tfrac{1}{3}, \tfrac{2}{3}\right)$.

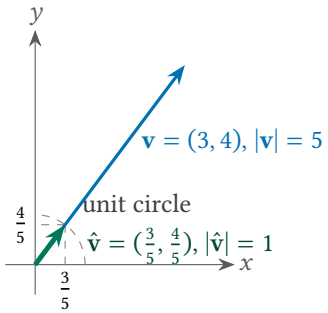


Figure 4.6: Normalising a vector: $\hat{\mathbf{v}} = \mathbf{v}/|\mathbf{v}|$ points the same way as $\mathbf{v} = (3, 4)$ but has length 1, so its tip lies on the unit circle. Its components $(\frac{3}{5}, \frac{4}{5})$ are the cosines of the angles with the axes (direction cosines, Section 4.2.9).

4.1.2 Representation of Vectors Using Standard Basis Vectors

The **standard basis vectors** form the natural coordinate system. In $\mathbb{R}^2$ they are

$$\mathbf{i} = (1, 0) \text{ (along the } x\text{-axis)}, \qquad \mathbf{j} = (0, 1) \text{ (along the } y\text{-axis)},$$

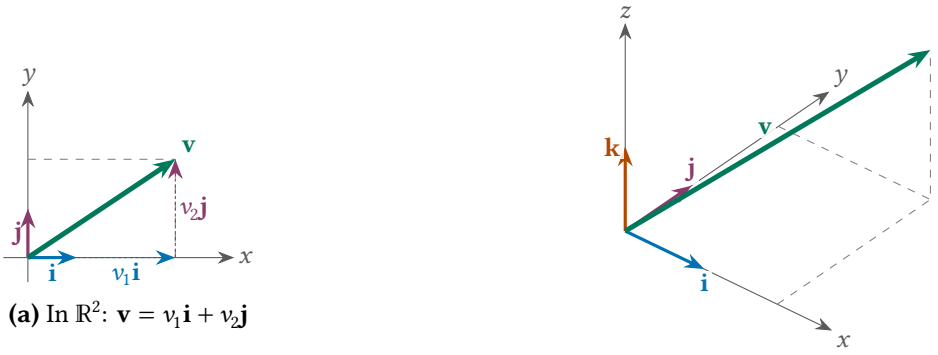
and in $\mathbb{R}^3$

$$\mathbf{i} = (1, 0, 0), \qquad \mathbf{j} = (0, 1, 0), \qquad \mathbf{k} = (0, 0, 1) \text{ (along the } z\text{-axis)}.$$

Every vector is a **linear combination** of the basis vectors:

$$\mathbf{v} = (v_1, v_2) = v_1\mathbf{i} + v_2\mathbf{j}, \qquad \mathbf{v} = (v_1, v_2, v_3) = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}.$$

The components v_1, v_2, v_3 are the coefficients of this expansion (Figure 4.7): to reach the tip of $\mathbf{v}$ we walk v_1 steps along $\mathbf{i}$, then v_2 steps along $\mathbf{j}$, then v_3 steps along $\mathbf{k}$.



(a) In $\mathbb{R}^2$: $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j}$

(b) In $\mathbb{R}^3$: $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$

Figure 4.7: Components of a vector with respect to the standard basis.

4.2 Operations on Vectors

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are **equal** if they have the same components, i.e. $u_i = v_i$ for all i. Geometrically, equal vectors have the same magnitude and direction, but they may have different initial points (Figure 4.8).

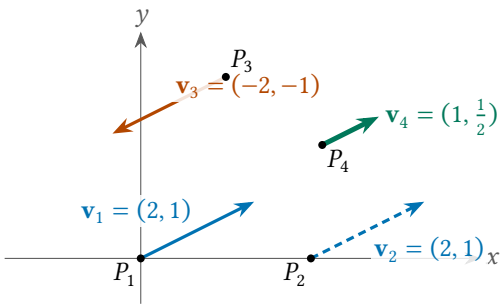


Figure 4.8: In the plane: $\mathbf{v}_1 = \mathbf{v}_2$ (same components, different initial points); $\mathbf{v}_3 = -\mathbf{v}_1$ has the opposite direction; $\mathbf{v}_4 = \frac{1}{2}\mathbf{v}_1$ has the same direction but half the length.

4.2.1 Multiplication by a Scalar

Definition

The **multiplication** of a vector $\mathbf{v}$ by a scalar $k \in \mathbb{R}$ in $\mathbb{R}^2$ (with $\mathbf{v} = (v_1, v_2)$) and $\mathbb{R}^3$ (with $\mathbf{v} = (v_1, v_2, v_3)$) is defined by

$$k\mathbf{v} = (kv_1, kv_2) \quad \text{and} \quad (kv_1, kv_2, kv_3), \quad \text{respectively.}$$

Geometric meaning: multiplication by k scales the magnitude by $|k|$:

$$|k\mathbf{v}| = |k|\,|\mathbf{v}|,$$

and affects the direction as follows:

- if $k > 0$, the direction is the same;
- if $k < 0$, the direction is reversed;
- if $k = 0$, the result is the zero vector $\mathbf{0}$.

Two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are **parallel** if and only if one is a scalar multiple of the other, i.e. $\mathbf{v} = k\mathbf{u}$ for some scalar $k \neq 0$. Parallel vectors have the same or opposite direction. (By convention the zero vector is parallel to every vector.)

Example: Scalar Multiplication in $\mathbb{R}^2$

Let $\mathbf{v} = (2, 1)$ (see Figure 4.9). Then $|\mathbf{v}| = \sqrt{2^2 + 1^2} = \sqrt{5} \approx 2.236$ and

k	$k\mathbf{v}$	$\lvert k\mathbf{v}\rvert$	direction
2	$(4, 2)$	$\sqrt{20} = 2\sqrt{5} \approx 4.472$	same as $\mathbf{v}$
-1	$(-2, -1)$	$\sqrt{5} \approx 2.236$	opposite to $\mathbf{v}$
0.5	$(1, 0.5)$	$\sqrt{1.25} \approx 1.118$	same as $\mathbf{v}$

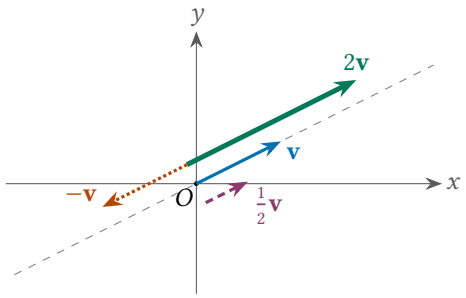


Figure 4.9: Multiples of $\mathbf{v} = (2, 1)$: $2\mathbf{v}$ (green, long), $\mathbf{v}$ (blue), $\frac{1}{2}\mathbf{v}$ (purple, dashed) and $-\mathbf{v}$ (vermilion, dotted). All are parallel; they are drawn slightly shifted only to keep them visible. The negative multiple points the other way.

4.2.2 Vector Addition and Subtraction

Definition

The **sum** of two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^2$ (with $\mathbf{u} = (u_1, u_2)$, $\mathbf{v} = (v_1, v_2)$) and $\mathbb{R}^3$ (with $\mathbf{u} = (u_1, u_2, u_3)$, $\mathbf{v} = (v_1, v_2, v_3)$) is defined by

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1,\ u_2 + v_2) \quad \text{and} \quad (u_1 + v_1,\ u_2 + v_2,\ u_3 + v_3), \quad \text{respectively.}$$

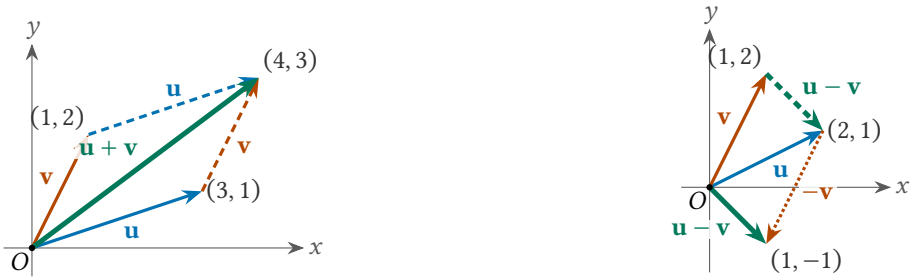
The **difference** $\mathbf{u} - \mathbf{v}$ is defined componentwise in the same manner, yielding

$$\mathbf{u} - \mathbf{v} = (u_1 - v_1,\ u_2 - v_2) \quad \text{and} \quad (u_1 - v_1,\ u_2 - v_2,\ u_3 - v_3), \quad \text{respectively.}$$

Geometric interpretation. Vector addition follows the **tail-to-tip rule** (equivalently, the **parallelogram rule**): place the tail of $\mathbf{v}$ at the tip of $\mathbf{u}$; the sum $\mathbf{u} + \mathbf{v}$ is the vector from the tail of $\mathbf{u}$ to the tip of $\mathbf{v}$ (Figure 4.10a). Subtraction $\mathbf{u} - \mathbf{v}$ is the same as adding $\mathbf{u}$ and $-\mathbf{v}$ (Figure 4.10b). Note also the useful fact that $\mathbf{u} - \mathbf{v}$ is the vector that goes *from the tip of* $\mathbf{v}$ *to the tip of* $\mathbf{u}$ when both start at the same point.

Properties of vector addition and scalar multiplication. For vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and scalars k, l:

1. **Commutative**: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
2. **Associative**: $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
3. **Identity**: the zero vector $\mathbf{0}$ satisfies $\mathbf{u} + \mathbf{0} = \mathbf{u}$.
4. **Inverse**: for every $\mathbf{u}$ there is $-\mathbf{u}$ with $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.
5. **Distributive**: $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$ and $(k + l)\mathbf{u} = k\mathbf{u} + l\mathbf{u}$.
6. **Subtraction**: $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$.



(a) Vector addition **(b)** Vector subtraction

Figure 4.10: Geometric representation of vector addition and subtraction in $\mathbb{R}^2$.

Common mistake. $|\mathbf{u}+\mathbf{v}|$ is *not* $|\mathbf{u}|+|\mathbf{v}|$. For perpendicular vectors of lengths 3 and 4 the sum has length 5, not 7. In general $|\mathbf{u}+\mathbf{v}|$ can be anything from $\big||\mathbf{u}|-|\mathbf{v}|\big|$ (opposite directions) to $|\mathbf{u}|+|\mathbf{v}|$ (same direction).

Triangle inequality

The vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{u}+\mathbf{v}$ form a triangle (Figure 4.11a), and in a triangle no side is longer than the sum of the other two:

$$|\mathbf{u}+\mathbf{v}| \le |\mathbf{u}|+|\mathbf{v}|,$$

with equality exactly when $\mathbf{u}$ and $\mathbf{v}$ point in the same direction (or one of them is $\mathbf{0}$). (For the vectors of Figure 4.11b: $|\mathbf{u}+\mathbf{v}| = 5$, while $|\mathbf{u}|+|\mathbf{v}| = \sqrt{10}+\sqrt{5} \approx 5.40$.)



(a) Different directions: $|\mathbf{u}+\mathbf{v}| < |\mathbf{u}|+|\mathbf{v}|$ **(b)** Same direction: $|\mathbf{u}+\mathbf{v}| = |\mathbf{u}|+|\mathbf{v}|$

Figure 4.11: The triangle inequality: the straight route $\mathbf{u}+\mathbf{v}$ is never longer than the detour along $\mathbf{u}$ and then $\mathbf{v}$. In (b) the sum is drawn slightly below the other two vectors so that all three stay visible.

Quick Check 1

1. If $|\mathbf{u}| = 3$ and $|\mathbf{v}| = 4$, what are the largest and the smallest possible values of $|\mathbf{u}+\mathbf{v}|$? When does each occur?

Example: Addition, Subtraction and Scalar Multiplication

Let $\mathbf{u} = (2, 3, -3)$ and $\mathbf{v} = (1, -1, 4)$. Then:

$$\begin{aligned}
\mathbf{u}+\mathbf{v} &= (2+1,\ 3-1,\ -3+4) = (3, 2, 1),\\
\mathbf{u}-\mathbf{v} &= (2-1,\ 3+1,\ -3-4) = (1, 4, -7),\\
2(\mathbf{u}+\mathbf{v}) &= 2(3, 2, 1) = (6, 4, 2) = 2\mathbf{u}+2\mathbf{v}.
\end{aligned}$$

Example: Boat on the River

For the boat of Figure 4.1 (river width 200 m, current 3 m/s, engine speed 4 m/s straight across), setting the x-axis downstream and y-axis across yields $\mathbf{e} = (0, 4)$ and $\mathbf{c} = (3, 0)$. The true velocity vector is:

$$\mathbf{e} + \mathbf{c} = (3, 4), \quad |\mathbf{e} + \mathbf{c}| = 5 \text{ m/s}, \quad \tan\varphi = \tfrac{3}{4} \implies \varphi \approx 36.9^\circ.$$

Crossing the 200 m river takes $200/4 = 50$ s, yielding a true speed of 5 m/s and a drift of $3(50) = 150$ m downstream.

4.2.3 Dot Product

Why we need it. *Standard multiplication only works for single numbers, but physical interactions often depend on alignment. We need the dot product to answer questions like:* How much of an engine's thrust actually pushes a vehicle forward? *or* How much work does gravity perform along a slope? *By multiplying components that point in the same direction, the dot product converts two vector inputs into a single measure of directional agreement, alignment, or work done.*

The dot product multiplies two vectors and produces a *scalar*. This operation can be defined for vectors of any finite dimension, provided that both vectors have the same number of components. We use it to find the angle between vectors, determine if they are orthogonal (perpendicular), and calculate projections—making it essential for computing work, force components, and directional alignment in physics and engineering.

Definition

The **dot product** (or **scalar product**) of vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^2$ (with $\mathbf{u} = (u_1, u_2)$, $\mathbf{v} = (v_1, v_2)$) and $\mathbb{R}^3$ (with $\mathbf{u} = (u_1, u_2, u_3)$, $\mathbf{v} = (v_1, v_2, v_3)$) yields a scalar, defined by

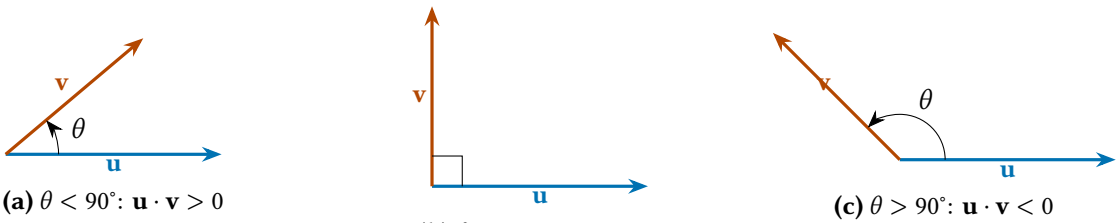
$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 \quad \text{and} \quad u_1 v_1 + u_2 v_2 + u_3 v_3, \quad \text{respectively.}$$

Geometric meaning. For nonzero vectors,

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}|\,|\mathbf{v}| \cos\theta, \tag{4.1}$$

where θ ($0 \le \theta \le \pi$) is the angle between the vectors when they are placed tail to tail. Hence the sign of the dot product tells us about the angle (Figure 4.12):

- $\theta < 90^\circ$ (acute angle): $\cos\theta > 0$, so $\mathbf{u} \cdot \mathbf{v} > 0$; the maximum $|\mathbf{u}||\mathbf{v}|$ occurs at $\theta = 0^\circ$ (parallel, same direction);
- $\theta = 90^\circ$: $\cos\theta = 0$, so $\mathbf{u} \cdot \mathbf{v} = 0$; the vectors are **perpendicular** (also called **orthogonal**);
- $\theta > 90^\circ$ (obtuse angle): $\mathbf{u} \cdot \mathbf{v} < 0$; the minimum $-|\mathbf{u}||\mathbf{v}|$ occurs at $\theta = 180^\circ$ (opposite directions).



(a) $\theta < 90^\circ$: $\mathbf{u} \cdot \mathbf{v} > 0$ **(b)** $\theta = 90^\circ$: $\mathbf{u} \cdot \mathbf{v} = 0$ **(c)** $\theta > 90^\circ$: $\mathbf{u} \cdot \mathbf{v} < 0$

Figure 4.12: The sign of the dot product is determined by the angle between the vectors.

Intuition. The dot product measures *how much two vectors agree in direction*, scaled by their lengths: it is large and positive when they point the same way, zero when they are independent of each other (perpendicular), and negative when they work against each other. Equivalently, $\mathbf{u}\cdot\mathbf{v}$ is the length of $\mathbf{v}$ times the length of the shadow of $\mathbf{u}$ on $\mathbf{v}$ (Section 4.2.7).

Why does formula (4.1) hold?

The vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{u}-\mathbf{v}$ form a triangle whose sides have lengths $|\mathbf{u}|$, $|\mathbf{v}|$, $|\mathbf{u}-\mathbf{v}|$, and the angle between the first two is θ. The law of cosines gives

$$|\mathbf{u}-\mathbf{v}|^2 = |\mathbf{u}|^2 + |\mathbf{v}|^2 - 2|\mathbf{u}||\mathbf{v}|\cos\theta.$$

On the other hand, expanding with components (or with the properties of the dot product below),

$$|\mathbf{u}-\mathbf{v}|^2 = (\mathbf{u}-\mathbf{v})\cdot(\mathbf{u}-\mathbf{v}) = |\mathbf{u}|^2 - 2\,\mathbf{u}\cdot\mathbf{v} + |\mathbf{v}|^2.$$

Comparing the two expressions gives $\mathbf{u}\cdot\mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos\theta$.

Solving (4.1) for the angle we get the two most important uses of the dot product:

$$\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{u}|\,|\mathbf{v}|}, \qquad \mathbf{u}\perp\mathbf{v} \iff \mathbf{u}\cdot\mathbf{v} = 0 \quad (\mathbf{u},\mathbf{v}\neq\mathbf{0}).$$

Example: Dot Product and Angle in $\mathbb{R}^2$

Let $\mathbf{u} = (2,1)$ and $\mathbf{v} = (1,3)$. Then $\mathbf{u}\cdot\mathbf{v} = 2(1)+1(3) = 5$, $|\mathbf{u}| = \sqrt{5}$, $|\mathbf{v}| = \sqrt{10}$, and

$$\cos\theta = \frac{5}{\sqrt{5}\sqrt{10}} = \frac{1}{\sqrt{2}} \quad\Longrightarrow\quad \theta = 45^\circ.$$

Example: Testing Orthogonality in $\mathbb{R}^3$

For $\mathbf{a} = (2,-1,3)$ and $\mathbf{b} = (1,5,1)$, since $\mathbf{a}\cdot\mathbf{b} = 2(1)+(-1)(5)+3(1) = 0$, we have $\mathbf{a}\perp\mathbf{b}$. In contrast, $\mathbf{a}\cdot(1,0,0) = 2 \neq 0$, so $\mathbf{a}$ is not orthogonal to the x-axis.

Common mistake. Three habits from ordinary numbers that fail for the dot product: (i) $\mathbf{u}\cdot\mathbf{v}=0$ does *not* mean $\mathbf{u}=\mathbf{0}$ or $\mathbf{v}=\mathbf{0}$ (perpendicular vectors also give 0); (ii) $\mathbf{u}\cdot\mathbf{v} = \mathbf{u}\cdot\mathbf{w}$ does *not* imply $\mathbf{v}=\mathbf{w}$ (it only says $\mathbf{u}\perp(\mathbf{v}-\mathbf{w})$); (iii) $\mathbf{u}\cdot\mathbf{v}\cdot\mathbf{w}$ is meaningless. These are the vector versions of the matrix traps of Chapter 1.

Properties of the dot product. For vectors $\mathbf{u},\mathbf{v},\mathbf{w}$ and a scalar k:

1. **Commutative**: $\mathbf{u}\cdot\mathbf{v} = \mathbf{v}\cdot\mathbf{u}$.
2. **Distributive**: $\mathbf{u}\cdot(\mathbf{v}+\mathbf{w}) = \mathbf{u}\cdot\mathbf{v}+\mathbf{u}\cdot\mathbf{w}$.
3. **Scalars**: $(k\mathbf{u})\cdot\mathbf{v} = k(\mathbf{u}\cdot\mathbf{v}) = \mathbf{u}\cdot(k\mathbf{v})$.
4. **Zero vector**: $\mathbf{u}\cdot\mathbf{0} = 0$.
5. **Length**: $\mathbf{u}\cdot\mathbf{u} = |\mathbf{u}|^2$, so $|\mathbf{u}| = \sqrt{\mathbf{u}\cdot\mathbf{u}}$.

There is no associative law: $(\mathbf{u}\cdot\mathbf{v})\cdot\mathbf{w}$ makes no sense, because $\mathbf{u}\cdot\mathbf{v}$ is a number.

Two inequalities

Since $|\cos\theta| \le 1$, formula (4.1) gives the **Cauchy–Schwarz inequality** $|\mathbf{u}\cdot\mathbf{v}| \le |\mathbf{u}||\mathbf{v}|$. It implies the triangle inequality of the previous section:

$$|\mathbf{u}+\mathbf{v}|^2 = |\mathbf{u}|^2 + 2\,\mathbf{u}\cdot\mathbf{v} + |\mathbf{v}|^2 \le |\mathbf{u}|^2 + 2|\mathbf{u}||\mathbf{v}| + |\mathbf{v}|^2 = (|\mathbf{u}| + |\mathbf{v}|)^2.$$

Quick Check 2

1. Are $(1,2,3)$ and $(2,-1,0)$ orthogonal?
2. If $\mathbf{u}\cdot\mathbf{v} = 6$, $|\mathbf{u}| = 2$ and $|\mathbf{v}| = 6$, what is the angle between $\mathbf{u}$ and $\mathbf{v}$?
3. Can $\mathbf{u}\cdot\mathbf{v} = 7$ if $|\mathbf{u}| = 2$ and $|\mathbf{v}| = 3$? Explain.

4.2.4 Cross Product in $\mathbb{R}^3$

Why we need it. *While the dot product measures alignment, rotational phenomena—like turning a wrench, turbine rotation, or aerodynamic torque—act* perpendicular *to the plane of motion. We need the cross product because it takes two vectors in a plane and constructs a third vector strictly perpendicular to both. Its magnitude measures rotational tendency or enclosed area, while its direction defines the precise axis of rotation using the right-hand rule.*

Definition

The **cross product** of $\mathbf{u} = (u_1,u_2,u_3) = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ and $\mathbf{v} = (v_1,v_2,v_3) = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$ is the vector

$$\mathbf{u}\times\mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = (u_2v_3 - u_3v_2)\,\mathbf{i} - (u_1v_3 - u_3v_1)\,\mathbf{j} + (u_1v_2 - u_2v_1)\,\mathbf{k}.$$

Geometric meaning. The cross product $\mathbf{u}\times\mathbf{v}$ is a vector that is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$, with magnitude

$$|\mathbf{u}\times\mathbf{v}| = |\mathbf{u}|\,|\mathbf{v}|\sin\theta,$$

where θ ($0 \le \theta \le \pi$) is the angle between $\mathbf{u}$ and $\mathbf{v}$. This number is exactly the area of the parallelogram spanned by $\mathbf{u}$ and $\mathbf{v}$ (base $|\mathbf{u}|$ times height $|\mathbf{v}|\sin\theta$). The direction is given by the **right-hand rule**: point the fingers of your right hand along $\mathbf{u}$, curl them towards $\mathbf{v}$ (through the smaller angle); the thumb points along $\mathbf{u}\times\mathbf{v}$ (Figure 4.13). Swapping the two factors reverses the direction, so $\mathbf{v}\times\mathbf{u} = -\mathbf{u}\times\mathbf{v}$.

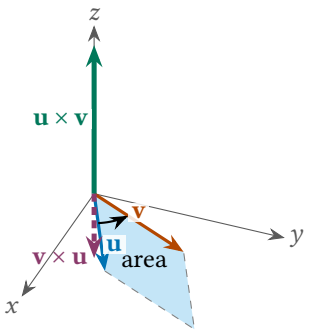


Figure 4.13: For $\mathbf{u} = (2,1,0)$ and $\mathbf{v} = (1,2,0)$ in the xy-plane the cross product $\mathbf{u}\times\mathbf{v} = (0,0,3)$ points up (right-hand rule: curl from $\mathbf{u}$ towards $\mathbf{v}$). Its length 3 equals the area of the shaded parallelogram. The dashed vector is $\mathbf{v}\times\mathbf{u} = -\mathbf{u}\times\mathbf{v}$ (drawn shorter).

Intuition. The dot product asks "how much do the vectors *agree*?", the cross product asks "how much do they *disagree*, and in which sense of rotation?". Its length is the area of the parallelogram (zero when the vectors are parallel, largest when they are perpendicular), and its direction is the axis about which $\mathbf{u}$ turns into $\mathbf{v}$. That is exactly the structure of a torque, a rotation axis or a magnetic force (Section 7.6).

Cross products of the basis vectors

Going around the cycle $\mathbf{i} \to \mathbf{j} \to \mathbf{k} \to \mathbf{i}$ gives a $+$ sign, going against it gives a $-$ sign:

$$\mathbf{i}\times\mathbf{j} = \mathbf{k}, \quad \mathbf{j}\times\mathbf{k} = \mathbf{i}, \quad \mathbf{k}\times\mathbf{i} = \mathbf{j}, \qquad \mathbf{j}\times\mathbf{i} = -\mathbf{k}, \quad \mathbf{k}\times\mathbf{j} = -\mathbf{i}, \quad \mathbf{i}\times\mathbf{k} = -\mathbf{j},$$

and $\mathbf{i}\times\mathbf{i} = \mathbf{j}\times\mathbf{j} = \mathbf{k}\times\mathbf{k} = \mathbf{0}$.

Example: Cross Product in $\mathbb{R}^3$

Compute $\mathbf{u}\times\mathbf{v}$ for $\mathbf{u} = (1,2,3)$ and $\mathbf{v} = (0,1,2)$.

$$\mathbf{u}\times\mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 3 \\ 0 & 1 & 2 \end{vmatrix} = \mathbf{i}(2\cdot 2 - 3\cdot 1) - \mathbf{j}(1\cdot 2 - 3\cdot 0) + \mathbf{k}(1\cdot 1 - 2\cdot 0) = \mathbf{i} - 2\mathbf{j} + \mathbf{k} = (1,-2,1).$$

Check (always worthwhile): the result must be perpendicular to both factors:

$$(1,-2,1)\cdot(1,2,3) = 1 - 4 + 3 = 0, \qquad (1,-2,1)\cdot(0,1,2) = 0 - 2 + 2 = 0.$$

Common mistake. The cross product is *anticommutative*, $\mathbf{u}\times\mathbf{v} = -\mathbf{v}\times\mathbf{u}$, so the order of the factors matters; it is *not* associative; it exists only in $\mathbb{R}^3$; and $\mathbf{u}\times\mathbf{v} = \mathbf{0}$ means the vectors are *parallel*, not that one of them is zero. A reliable check after every computation: dot the result with both factors and verify that you get 0.

4.2.5 Area of a Parallelogram and of a Triangle

The area of the parallelogram with adjacent sides $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^3$ is the magnitude of their cross product, and the area of the triangle with the same two sides is half of it:

$$A_{\text{parallelogram}} = |\mathbf{u}\times\mathbf{v}|, \qquad A_{\text{triangle}} = \frac{1}{2}|\mathbf{u}\times\mathbf{v}|.$$

Example: Area of a Triangle Using the Cross Product

Find the area of the triangle with vertices $A(1,0,0)$, $B(0,2,0)$ and $C(0,0,3)$ (Figure 4.14).

Solution. Take $\mathbf{u} = \overrightarrow{AB} = B - A = (-1,2,0)$ and $\mathbf{v} = \overrightarrow{AC} = C - A = (-1,0,3)$. Then

$$\mathbf{u}\times\mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 2 & 0 \\ -1 & 0 & 3 \end{vmatrix} = \mathbf{i}(2\cdot 3 - 0\cdot 0) - \mathbf{j}\big((-1)\cdot 3 - 0\cdot(-1)\big) + \mathbf{k}\big((-1)\cdot 0 - 2\cdot(-1)\big) = (6,3,2),$$

$$|\mathbf{u}\times\mathbf{v}| = \sqrt{6^2 + 3^2 + 2^2} = \sqrt{49} = 7, \qquad A_{\text{triangle}} = \frac{1}{2}\cdot 7 = 3.5.$$

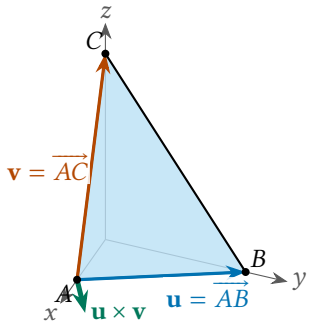


Figure 4.14: Triangle ABC with $A(1,0,0)$, $B(0,2,0)$, $C(0,0,3)$. The vector $\mathbf{u}\times\mathbf{v} = (6,3,2)$ is perpendicular to the plane of the triangle (drawn shortened).

Properties of the cross product. For vectors $\mathbf{u},\mathbf{v},\mathbf{w}\in\mathbb{R}^3$ and a scalar k:

1. **Anticommutative**: $\mathbf{u}\times\mathbf{v} = -\mathbf{v}\times\mathbf{u}$. In particular $\mathbf{u}\times\mathbf{u} = \mathbf{0}$.
2. **Distributive**: $\mathbf{u}\times(\mathbf{v}+\mathbf{w}) = \mathbf{u}\times\mathbf{v} + \mathbf{u}\times\mathbf{w}$.
3. **Scalars**: $(k\mathbf{u})\times\mathbf{v} = \mathbf{u}\times(k\mathbf{v}) = k(\mathbf{u}\times\mathbf{v})$.
4. **Zero vector**: $\mathbf{u}\times\mathbf{0} = \mathbf{0}\times\mathbf{u} = \mathbf{0}$.
5. **Not associative**: in general $\mathbf{u}\times(\mathbf{v}\times\mathbf{w}) \neq (\mathbf{u}\times\mathbf{v})\times\mathbf{w}$. (For example $\mathbf{i}\times(\mathbf{i}\times\mathbf{j}) = \mathbf{i}\times\mathbf{k} = -\mathbf{j}$, but $(\mathbf{i}\times\mathbf{i})\times\mathbf{j} = \mathbf{0}$.)
6. **Orthogonality**: $(\mathbf{u}\times\mathbf{v})\cdot\mathbf{u} = 0$ and $(\mathbf{u}\times\mathbf{v})\cdot\mathbf{v} = 0$.
7. **Magnitude**: $|\mathbf{u}\times\mathbf{v}| = |\mathbf{u}||\mathbf{v}|\sin\theta$.
8. **Parallel vectors**: $\mathbf{u}\times\mathbf{v} = \mathbf{0}$ if and only if $\mathbf{u}$ and $\mathbf{v}$ are parallel (this includes the case that one of them is the zero vector).

Dot and cross product together

Since $\cos^2\theta + \sin^2\theta = 1$, the two formulas for the dot and cross product combine to the **Lagrange identity**

$$|\mathbf{u}\times\mathbf{v}|^2 + (\mathbf{u}\cdot\mathbf{v})^2 = |\mathbf{u}|^2\,|\mathbf{v}|^2.$$

This is a good way to check a computation. For $\mathbf{u} = (1,2,3)$ and $\mathbf{v} = (0,1,2)$: $|\mathbf{u}\times\mathbf{v}|^2 = 6$, $(\mathbf{u}\cdot\mathbf{v})^2 = 8^2 = 64$ and $|\mathbf{u}|^2|\mathbf{v}|^2 = 14\cdot 5 = 70$, and indeed $6 + 64 = 70$.

4.2.6 Scalar Triple Product in $\mathbb{R}^3$

Why we need it. *Real-world engineering objects inhabit three dimensions, where volume and orientation determine load-bearing capabilities, fluid container capacities, and geometric stability. The scalar triple product combines area (via the cross product) with depth (via the dot product) to calculate 3D volumes directly from coordinate points. Crucially, if the calculated volume equals zero, it instantly alerts us that three structural beams or forces lie flat in the same plane.*

Definition

The **scalar triple product** of three vectors $\mathbf{a} = (a_1,a_2,a_3)$, $\mathbf{b} = (b_1,b_2,b_3)$ and $\mathbf{c} = (c_1,c_2,c_3)$ in $\mathbb{R}^3$ is the number

$$\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) = \det\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} = a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1).$$

Geometric meaning. The absolute value $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$ is the *volume of the parallelepiped* spanned by $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$. The reason is visible in Figure 4.15: the base spanned by $\mathbf{b}$ and $\mathbf{c}$ has area $|\mathbf{b} \times \mathbf{c}|$, and the height h is the length of the component of $\mathbf{a}$ along the perpendicular direction $\mathbf{b} \times \mathbf{c}$, so

$$V = (\text{area of base}) \cdot h = |\mathbf{b} \times \mathbf{c}| \cdot |\mathbf{a}|\,|\cos\varphi| = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|,$$

where φ is the angle between $\mathbf{a}$ and $\mathbf{b} \times \mathbf{c}$. The sign is positive if $\mathbf{a}, \mathbf{b}, \mathbf{c}$ form a right-handed system and negative otherwise. The volume of the *tetrahedron* (triangular pyramid) with the same three edges is one sixth of the parallelepiped volume.

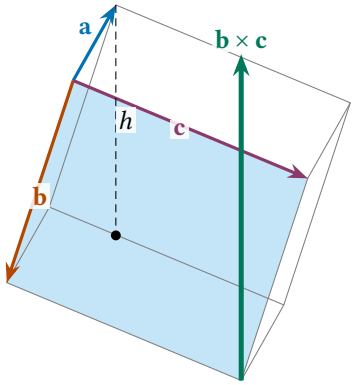


Figure 4.15: The parallelepiped spanned by $\mathbf{a}, \mathbf{b}, \mathbf{c}$. The base spanned by $\mathbf{b}, \mathbf{c}$ (shaded) has area $|\mathbf{b} \times \mathbf{c}|$ and the vector $\mathbf{b} \times \mathbf{c}$ (drawn at a corner of the base) is perpendicular to it; the dashed segment is the height h, so $V = |\mathbf{b} \times \mathbf{c}|\, h = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$.

Intuition. The scalar triple product is a *signed volume*: area of the base times height. The sign records the orientation (right-handed or left-handed order of $\mathbf{a}, \mathbf{b}, \mathbf{c}$), which is why swapping two vectors flips the sign. Zero volume means the box has collapsed into a plane, that is, the three vectors are linearly dependent.

Example: Are Three Vectors Coplanar?

Let $\mathbf{a} = (1, 2, 3)$, $\mathbf{b} = (4, 5, 6)$, $\mathbf{c} = (7, 8, 9)$. Then

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \det \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = 1(45 - 48) - 2(36 - 42) + 3(32 - 35) = -3 + 12 - 9 = 0.$$

The parallelepiped is flat (zero volume), so the vectors lie in one plane. Indeed $\mathbf{c} = 2\mathbf{b} - \mathbf{a}$.

Properties of the scalar triple product.

1. **Cyclic permutations** do not change the value: $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b})$. Swapping two vectors changes the sign: $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = -\,\mathbf{b} \cdot (\mathbf{a} \times \mathbf{c})$.
2. **Coplanarity test**: $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$ if and only if $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are coplanar (linearly dependent); this includes the case that one of them is the zero vector.
3. **Scalars**: $(k\mathbf{a}) \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{a} \cdot (k\mathbf{b} \times \mathbf{c}) = \mathbf{a} \cdot (\mathbf{b} \times k\mathbf{c}) = k\big(\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})\big)$.
4. **Distributive**: $\mathbf{a} \cdot ((\mathbf{b} + \mathbf{d}) \times \mathbf{c}) = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) + \mathbf{a} \cdot (\mathbf{d} \times \mathbf{c})$.
5. **Orientation**: The sign determines the chirality or handedness of the coordinate system: it is positive if $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ forms a right-handed triple and negative if it is left-handed.

Product	Formula	Geometric meaning	Zero when
Dot product $\mathbf{u}\cdot\mathbf{v}$ (a number)	$u_1v_1+u_2v_2+u_3v_3$ $=\lvert\mathbf{u}\rvert\lvert\mathbf{v}\rvert\cos\theta$	angle between vectors; projection; work $=\mathbf{F}\cdot\mathbf{d}$	$\mathbf{u}\perp\mathbf{v}$ (or one is $\mathbf{0}$)
Cross product $\mathbf{u}\times\mathbf{v}$ (a vector)	determinant with $\mathbf{i},\mathbf{j},\mathbf{k}$; $\lvert\mathbf{u}\times\mathbf{v}\rvert=\lvert\mathbf{u}\rvert\lvert\mathbf{v}\rvert\sin\theta$	perpendicular vector; area of parallelogram; torque $=\mathbf{r}\times\mathbf{F}$	$\mathbf{u}\parallel\mathbf{v}$ (or one is $\mathbf{0}$)
Triple product $\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$ (a number)	3×3 determinant of the components	signed volume of the parallelepiped	$\mathbf{a},\mathbf{b},\mathbf{c}$ coplanar

Table 4.1: The three products of vectors at a glance.

4.2.7 Projection of a Vector onto Another Vector

Why we need it. *Forces in nature rarely align perfectly with structural axes or coordinate systems. To analyze stress on a bridge truss or an aircraft wing, engineers must isolate the portion of a force acting* along *a beam from the portion pushing* across *it. Orthogonal projection breaks any arbitrary vector into two independent, perpendicular components, allowing us to analyze complex multidimensional forces one axis at a time.*

Definition

The **projection** of a vector $\mathbf{u}$ onto a nonzero vector $\mathbf{v}$ in $\mathbb{R}^2$ or $\mathbb{R}^3$ is the vector along $\mathbf{v}$

$$\operatorname{proj}_{\mathbf{v}}\mathbf{u}=\left(\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|^2}\right)\mathbf{v}.$$

The signed number $\operatorname{comp}_{\mathbf{v}}\mathbf{u}=\dfrac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|}=|\mathbf{u}|\cos\theta$ is the **scalar projection**; the length of the projection is $|\operatorname{proj}_{\mathbf{v}}\mathbf{u}|=|\mathbf{u}|\,|\cos\theta|$, where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$.

Geometric interpretation. $\operatorname{proj}_{\mathbf{v}}\mathbf{u}$ is the "shadow" of $\mathbf{u}$ on the line spanned by $\mathbf{v}$ when light falls perpendicular to $\mathbf{v}$ (Figure 4.16). It is obtained as follows:

1. place $\mathbf{u}$ and $\mathbf{v}$ tail to tail;
2. draw the line containing $\mathbf{v}$;
3. drop a perpendicular from the tip of $\mathbf{u}$ to this line;
4. $\operatorname{proj}_{\mathbf{v}}\mathbf{u}$ is the vector from the common tail to the foot of the perpendicular. It points along $\mathbf{v}$ if $\cos\theta>0$ and opposite to $\mathbf{v}$ if $\cos\theta<0$.

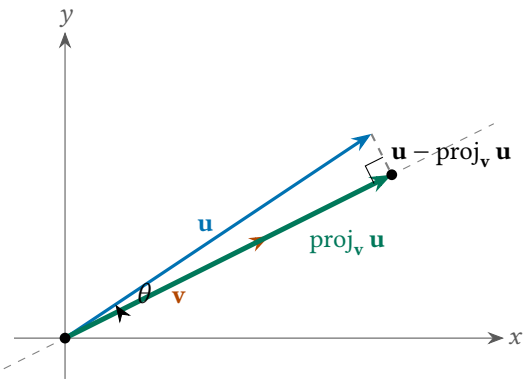


Figure 4.16: Orthogonal projection of $\mathbf{u}=(3,2)$ onto $\mathbf{v}=(2,1)$: $\operatorname{proj}_{\mathbf{v}}\mathbf{u}=\frac{8}{5}(2,1)=(3.2,1.6)$ (green, starting at the origin). The remainder $\mathbf{u}-\operatorname{proj}_{\mathbf{v}}\mathbf{u}$ is perpendicular to $\mathbf{v}$.

Example: Projection in $\mathbb{R}^3$

Let $\mathbf{u} = (3, 1, 2)$ and $\mathbf{v} = (1, 0, 1)$. Then $\mathbf{u} \cdot \mathbf{v} = 3 + 0 + 2 = 5$ and $|\mathbf{v}|^2 = 2$, so

$$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \frac{5}{2}\mathbf{v} = \left(\frac{5}{2}, 0, \frac{5}{2}\right), \qquad \operatorname{comp}_{\mathbf{v}} \mathbf{u} = \frac{5}{\sqrt{2}} \approx 3.54.$$

Check: $\mathbf{u} - \operatorname{proj}_{\mathbf{v}} \mathbf{u} = \left(\frac{1}{2}, 1, -\frac{1}{2}\right)$ and $\left(\frac{1}{2}, 1, -\frac{1}{2}\right) \cdot (1, 0, 1) = \frac{1}{2} - \frac{1}{2} = 0$, so the remainder is perpendicular to $\mathbf{v}$, as it must be.

Common mistake. $\operatorname{proj}_{\mathbf{v}} \mathbf{u}$ and $\operatorname{proj}_{\mathbf{u}} \mathbf{v}$ are different vectors: the first is parallel to $\mathbf{v}$, the second to $\mathbf{u}$. In the formula the *divisor* is $|\mathbf{v}|^2$, the squared length of the vector you project *onto*, and the result is a multiple of $\mathbf{v}$.

Orthogonal decomposition

Every vector $\mathbf{u}$ splits uniquely into a part parallel to $\mathbf{v}$ and a part perpendicular to $\mathbf{v}$:

$$\mathbf{u} = \underbrace{\operatorname{proj}_{\mathbf{v}} \mathbf{u}}_{\parallel \mathbf{v}} + \underbrace{(\mathbf{u} - \operatorname{proj}_{\mathbf{v}} \mathbf{u})}_{\perp \mathbf{v}}.$$

This is the idea behind splitting a force into components along and across a beam.

Properties of the projection. For a nonzero vector $\mathbf{v}$:

1. **Parallel to $\mathbf{v}$**: $\operatorname{proj}_{\mathbf{v}} \mathbf{u} = k\mathbf{v}$ with $k = \dfrac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2}$.
2. **Perpendicular remainder**: $(\mathbf{u} - \operatorname{proj}_{\mathbf{v}} \mathbf{u}) \cdot \mathbf{v} = 0$.
3. **Self-projection**: $\operatorname{proj}_{\mathbf{v}} \mathbf{v} = \mathbf{v}$.
4. **Zero projection**: if $\mathbf{u} \perp \mathbf{v}$, then $\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \mathbf{0}$.
5. **Linearity in $\mathbf{u}$**: $\operatorname{proj}_{\mathbf{v}}(k\mathbf{u}) = k \operatorname{proj}_{\mathbf{v}} \mathbf{u}$ and $\operatorname{proj}_{\mathbf{v}}(\mathbf{u} + \mathbf{w}) = \operatorname{proj}_{\mathbf{v}} \mathbf{u} + \operatorname{proj}_{\mathbf{v}} \mathbf{w}$.
6. **Size**: $|\operatorname{proj}_{\mathbf{v}} \mathbf{u}| \le |\mathbf{u}|$, with equality iff $\mathbf{u} \parallel \mathbf{v}$.

4.2.8 Reflection of a Vector in a Line

Reflecting $\mathbf{u}$ in the line through the origin spanned by a nonzero vector $\mathbf{v}$ (think of the line as a mirror) keeps the component along $\mathbf{v}$ and reverses the perpendicular component:

$$\operatorname{refl}_{\mathbf{v}} \mathbf{u} = \operatorname{proj}_{\mathbf{v}} \mathbf{u} - (\mathbf{u} - \operatorname{proj}_{\mathbf{v}} \mathbf{u}) = 2 \operatorname{proj}_{\mathbf{v}} \mathbf{u} - \mathbf{u}.$$

In the plane this is the mirror image of $\mathbf{u}$ (Figure 4.17). In $\mathbb{R}^3$ the same formula gives the vector obtained by rotating $\mathbf{u}$ by 180° about the line spanned by $\mathbf{v}$.

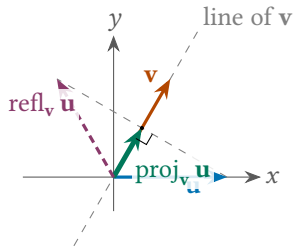


Figure 4.17: Reflection of $\mathbf{u} = (1, 0)$ in the line spanned by $\mathbf{v} = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$. The dashed grey segment from $\mathbf{u}$ to its mirror image $\operatorname{refl}_{\mathbf{v}} \mathbf{u} = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ is perpendicular to the line and is cut in half by it (at the foot of the projection).

Properties of the Reflection.

- **Involutive property:** $\text{refl}_{\mathbf{v}}(\text{refl}_{\mathbf{v}}\,\mathbf{u}) = \mathbf{u}$ (reflecting a vector twice returns it to its original position).
- **Linearity:** The reflection mapping is a linear transformation: $\text{refl}_{\mathbf{v}}(a\mathbf{u}_1 + b\mathbf{u}_2) = a\,\text{refl}_{\mathbf{v}}\,\mathbf{u}_1 + b\,\text{refl}_{\mathbf{v}}\,\mathbf{u}_2$.
- **Norm preservation (Isometry):** $\|\text{refl}_{\mathbf{v}}\,\mathbf{u}\| = \|\mathbf{u}\|$, meaning lengths and distances are preserved.
- **Action on orthogonal components:** Vectors parallel to $\mathbf{v}$ are fixed ($\text{refl}_{\mathbf{v}}\,\mathbf{u} = \mathbf{u}$), while vectors orthogonal to $\mathbf{v}$ are inverted ($\text{refl}_{\mathbf{v}}\,\mathbf{u} = -\mathbf{u}$).

Why we need it. *Refinement and boundary conditions in engineering—such as light ray trajectories, acoustic reflections in auditorium design, or elastic particle collisions against walls—depend on symmetry. Vector reflection provides an exact algebraic formula to compute incoming and outgoing trajectories off surfaces without drawing complex geometric angles.*

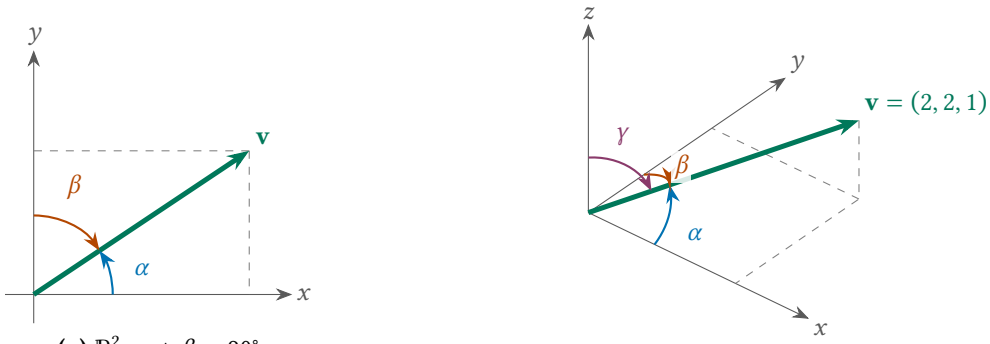
4.2.9 Direction Cosines in $\mathbb{R}^2$ and $\mathbb{R}^3$

Definition

The **direction cosines** of a nonzero vector $\mathbf{v}$ are the cosines of the angles that $\mathbf{v}$ makes with the positive coordinate axes in $\mathbb{R}^2$ (with angles α, β) and $\mathbb{R}^3$ (with angles α, β, γ), defined by

$$\cos\alpha = \frac{v_1}{|\mathbf{v}|}, \quad \cos\beta = \frac{v_2}{|\mathbf{v}|}, \quad \text{and} \quad \cos\gamma = \frac{v_3}{|\mathbf{v}|} \quad (\text{for } \mathbb{R}^3), \quad \text{respectively.}$$

The formulas come from the dot product: $\cos\alpha = \dfrac{\mathbf{v}\cdot\mathbf{i}}{|\mathbf{v}||\mathbf{i}|} = \dfrac{v_1}{|\mathbf{v}|}$, and similarly for the other angles. In other words, the direction cosines are the components of the unit vector $\hat{\mathbf{v}}$ (Figure 4.18).



(a) $\mathbb{R}^2$: $\alpha + \beta = 90^\circ$ **(b)** $\mathbb{R}^3$: $\cos\alpha = \cos\beta = \frac{2}{3}$, $\cos\gamma = \frac{1}{3}$

Figure 4.18: The angles α, β, γ between a vector and the coordinate axes. In (b), $\alpha = \beta \approx 48.2^\circ$ and $\gamma \approx 70.5^\circ$.

Properties of direction cosines.

1. **Sum of squares**: $\cos^2\alpha + \cos^2\beta = 1$ in $\mathbb{R}^2$ and $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$ in $\mathbb{R}^3$, because they are the components of the unit vector $\hat{\mathbf{v}}$, whose length is 1.
2. **Unit vector**: $\hat{\mathbf{v}} = (\cos\alpha, \cos\beta)$ in $\mathbb{R}^2$ and $\hat{\mathbf{v}} = (\cos\alpha, \cos\beta, \cos\gamma)$ in $\mathbb{R}^3$. So $\mathbf{v} = |\mathbf{v}|\,(\cos\alpha, \cos\beta, \cos\gamma)$.
3. **Perpendicular to an axis**: $\cos\alpha = 0$ means $\mathbf{v} \perp x$-axis, and similarly for β, γ.
4. Since $\text{proj}_{\mathbf{i}}\,\mathbf{v} = v_1\mathbf{i}$, $\text{proj}_{\mathbf{j}}\,\mathbf{v} = v_2\mathbf{j}$ and $\text{proj}_{\mathbf{k}}\,\mathbf{v} = v_3\mathbf{k}$, the direction cosines are the scalar projections of $\mathbf{v}$ on the axes divided by the length: $\cos\alpha = \text{comp}_{\mathbf{i}}\,\mathbf{v}/|\mathbf{v}|$, etc.
5. **Zero vector**: the direction cosines of $\mathbf{0}$ are undefined.
6. **Bounds**: Because they are trigonometric cosines, every direction cosine must lie within the closed interval $[-1, 1]$.
7. **Direction numbers**: Any set of numbers proportional to the direction cosines (such as the raw vector components v_1, v_2, v_3) is known as a set of *direction numbers* for the vector or line.

Example: Direction Cosines

For $\mathbf{v} = (2, 2, 1)$ we have $|\mathbf{v}| = \sqrt{4+4+1} = 3$, so

$$\cos\alpha = \frac{2}{3}, \quad \cos\beta = \frac{2}{3}, \quad \cos\gamma = \frac{1}{3}, \qquad \frac{4}{9} + \frac{4}{9} + \frac{1}{9} = 1$$

and $\alpha = \beta = \arccos\frac{2}{3} \approx 48.2^\circ$, $\gamma = \arccos\frac{1}{3} \approx 70.5^\circ$.

4.3 Linear Independence

A set of vectors is linearly independent if no vector in the set can be expressed as a linear combination of the others—meaning every vector introduces a genuinely new, non-redundant direction.

We need them because they form the foundational building blocks of bases and coordinate systems. They guarantee that we can describe any point, force, or movement in a space uniquely and efficiently, without redundant coordinates or wasted dimensions that carry no new information (which is essential when defining independent axes for physical systems like aircraft motion and structural loads).

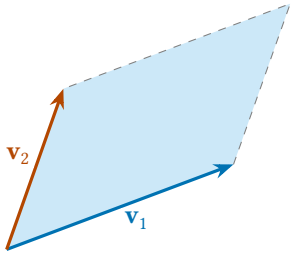
Definition

Vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ in $\mathbb{R}^n$ are **linearly independent** if the only solution of

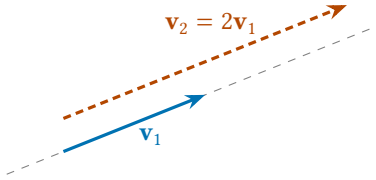
$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$$

is $c_1 = c_2 = \cdots = c_k = 0$ (the trivial solution). If there is a nontrivial solution (not all $c_i = 0$), they are **linearly dependent**.

Equivalently: the vectors are dependent exactly when at least one of them is a linear combination of the others. Geometrically (Figures 4.19 and 4.20): two vectors in the plane are dependent if they are parallel; three vectors in space are dependent if they are coplanar.



(a) independent: not parallel; combinations $c_1\mathbf{v}_1 + c_2\mathbf{v}_2$ fill the whole plane

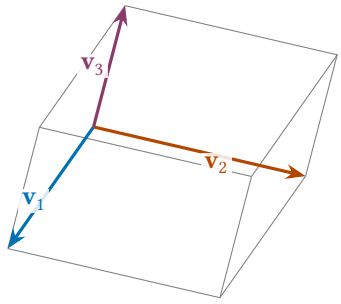


(b) dependent: parallel; $2\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0}$, combinations only fill a line

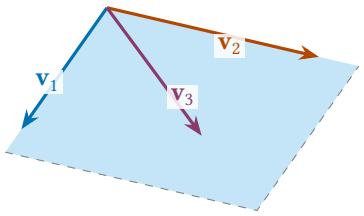
Figure 4.19: Two vectors in the plane. (In (b) the vector $\mathbf{v}_2$ is drawn slightly shifted so that both are visible.)

How to decide. Form the matrix M whose *columns* are the vectors. The equation $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$ is the homogeneous system $M\mathbf{c} = \mathbf{0}$ (Chapter 3), so:

- if $\operatorname{rank}(M) = k$ (the number of vectors), the vectors are linearly independent;
- if $\operatorname{rank}(M) < k$, they are linearly dependent;
- if $k = n$, M is square and this is the same as $\det M \neq 0$ (independent) or $\det M = 0$ (dependent);
- if $k > n$, the vectors are always dependent (there are more unknowns c_i than equations).



(a) independent: the vectors span a box of volume $\neq 0$



(b) dependent: all three lie in one plane (here $\mathbf{v}_3 = \frac{3}{4}\mathbf{v}_1 + \frac{3}{4}\mathbf{v}_2$)

Figure 4.20: Three vectors in space.

Linear Independence in $\mathbb{R}^2$ and $\mathbb{R}^3$

Vectors $\mathbf{v}_1, \mathbf{v}_2$ in $\mathbb{R}^2$ and $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ in $\mathbb{R}^3$ are linearly independent if and only if their determinant is non-zero:

$$\det \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \neq 0 \quad \text{and} \quad \det \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \neq 0, \quad \text{respectively,}$$

meaning they are not parallel in $\mathbb{R}^2$, or not coplanar with a non-zero scalar triple product $\mathbf{v}_1 \cdot (\mathbf{v}_2 \times \mathbf{v}_3) \neq 0$ in $\mathbb{R}^3$.

Intuition. Linearly independent vectors contain *no redundancy*: each one adds a genuinely new direction. Two independent vectors in the plane, or three in space, can serve as a coordinate system (a *basis*): every vector has exactly one "address" (c_1, c_2) or (c_1, c_2, c_3) in terms of them (Figure 4.21).

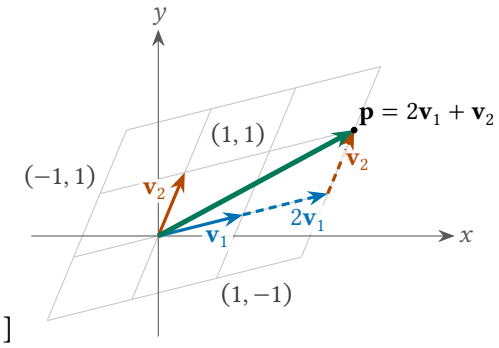


Figure 4.21: Two linearly independent vectors form a (skew) coordinate system. The grey lines are the points with constant c_1 or constant c_2 in $c_1\mathbf{v}_1 + c_2\mathbf{v}_2$. The point $\mathbf{p}$ is reached in exactly one way, with coordinates $(2, 1)$. For dependent vectors the grid would collapse onto a single line.

Examples: Linear Independence in $\mathbb{R}^2$ and $\mathbb{R}^3$

- **In $\mathbb{R}^2$:** For $\mathbf{v}_1 = (1, 2)$ and $\mathbf{v}_2 = (3, 4)$, $\det = -2 \neq 0$ (independent); for $\mathbf{w}_1 = (1, 2)$ and $\mathbf{w}_2 = (2, 4)$, $\det = 0$ ($\mathbf{w}_2 = 2\mathbf{w}_1$, dependent).
- **In $\mathbb{R}^3$:** For $\mathbf{v}_1 = (1, 2, 3)$, $\mathbf{v}_2 = (4, 5, 6)$, and $\mathbf{v}_3 = (7, 8, 9)$, $\det = 0$ (dependent) with relation $1\mathbf{v}_1 - 2\mathbf{v}_2 + 1\mathbf{v}_3 = \mathbf{0}$, showing $\mathbf{v}_3 = 2\mathbf{v}_2 - \mathbf{v}_1$ lies in the span of $\mathbf{v}_1$ and $\mathbf{v}_2$.

4.4 Applications in Physics and Engineering

Vectors are the working language of mechanics, electromagnetism and navigation. The *dot product* answers questions of the type "how much of one vector acts along another?" (work, power, flux); the *cross product* answers questions about *turning* (torque, rotation, magnetic force). In every application below we do the same three things: (1) state the physical law as a vector formula, (2) draw the situation, (3) compute.

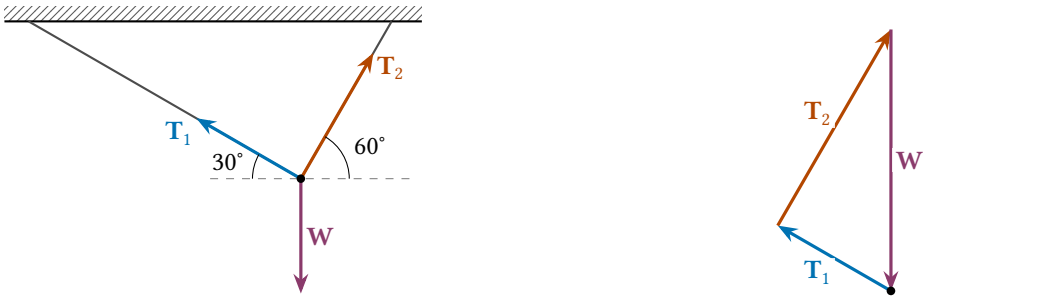
4.4.1 Statics: Forces in Equilibrium

Physics & Engineering: Equilibrium of a Point

Physical law (Newton's first law). A body that stays at rest has no net force acting on it:

$$\sum_i \mathbf{F}_i = \mathbf{0}.$$

In $\mathbb{R}^2$ this is two scalar equations, in $\mathbb{R}^3$ three. Geometrically, the forces placed tail to tip close up into a polygon (Figure 4.22b). Each force is written as magnitude times direction, $\mathbf{F}_i = |\mathbf{F}_i|\,\hat{\mathbf{u}}_i$, and the unknown magnitudes are found from the component equations.



(a) Two cables hold a sign **(b)** Force polygon: $\mathbf{T}_1 + \mathbf{T}_2 + \mathbf{W} = \mathbf{0}$

Figure 4.22: A sign of weight $W = 100$ N hangs from two cables at 30° and 60° to the horizontal. At the knot the forces $\mathbf{T}_1$, $\mathbf{T}_2$ (cable tensions) and $\mathbf{W}$ (weight) must add up to $\mathbf{0}$; placed tail to tip they form a closed triangle.

Example: Tensions in Two Cables

With the angles of Figure 4.22a the unit vectors along the cables (away from the knot) are

$$\hat{\mathbf{u}}_1 = (-\cos 30^\circ,\ \sin 30^\circ), \quad \hat{\mathbf{u}}_2 = (\cos 60^\circ,\ \sin 60^\circ), \quad \mathbf{W} = (0, -100)N.$$

Equilibrium $T_1\hat{\mathbf{u}}_1 + T_2\hat{\mathbf{u}}_2 + \mathbf{W} = \mathbf{0}$ reads

$$-\tfrac{\sqrt{3}}{2}\,T_1 + \tfrac{1}{2}\,T_2 = 0, \qquad \tfrac{1}{2}\,T_1 + \tfrac{\sqrt{3}}{2}\,T_2 = 100.$$

The first equation gives $T_2 = \sqrt{3}\,T_1$; the second becomes $\tfrac{1}{2}T_1 + \tfrac{3}{2}T_1 = 2T_1 = 100$. Hence

$$T_1 = 50 \text{ N}, \qquad T_2 = 50\sqrt{3} \approx 86.6 \text{ N}.$$

Check: $T_1\hat{\mathbf{u}}_1 + T_2\hat{\mathbf{u}}_2 = (-43.3 + 43.3,\ 25 + 75) = (0, 100)$, which cancels the weight. ☒

Intuition. A shortcut with the dot product: here

$$\hat{\mathbf{u}}_1 \cdot \hat{\mathbf{u}}_2 = -\cos 30^\circ \cos 60^\circ + \sin 30^\circ \sin 60^\circ = 0,$$

so the cables are perpendicular. Dotting the equilibrium equation with $\hat{\mathbf{u}}_1$ kills the unknown T_2 and leaves $T_1 + \mathbf{W} \cdot \hat{\mathbf{u}}_1 = 0$, so $T_1 = -\mathbf{W} \cdot \hat{\mathbf{u}}_1 = 50$ N. Dotting with a direction that is perpendicular to an unknown force eliminates that unknown.

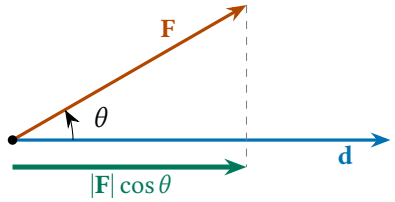
4.4.2 Work and Power

Physics & Engineering: Work and Power

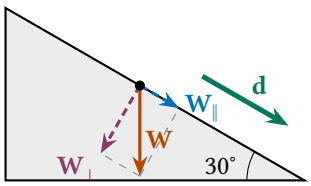
Physical law. A constant force $\mathbf{F}$ acting on a body that undergoes the displacement $\mathbf{d}$ does the work

$$W = \mathbf{F} \cdot \mathbf{d} = |\mathbf{F}|\,|\mathbf{d}| \cos\theta = |\mathbf{d}|\ \mathrm{comp}_{\mathbf{d}}\,\mathbf{F}$$

(joules for newtons and metres): only the component of $\mathbf{F}$ *along the motion* does work (Figure 4.23a). If the body moves with velocity $\mathbf{v}$, the **power** (rate of doing work) is $P = \mathbf{F} \cdot \mathbf{v}$ (watts).



(a) Only the part of $\mathbf{F}$ along $\mathbf{d}$ works



(b) A body sliding down a 30° incline

Figure 4.23: (a) Work $W = \mathbf{F} \cdot \mathbf{d} = |\mathbf{d}| \cdot |\mathbf{F}| \cos\theta$. (b) The weight $\mathbf{W}$ splits (orthogonal decomposition, Section 4.2.7) into a part $\mathbf{W}_\parallel$ along the slope and a part $\mathbf{W}_\perp$ pressing into it; only $\mathbf{W}_\parallel$ does work along the displacement $\mathbf{d}$.

Example: Pulling a Crate and Sliding on an Incline

(a) A rope pulls a crate with a force of 100 N at 30° above the horizontal over a distance of 20 m. With $\mathbf{F} = (100\cos 30^\circ,\ 100\sin 30^\circ) = (86.6,\ 50)$ N and $\mathbf{d} = (20, 0)$ m,

$$W = \mathbf{F} \cdot \mathbf{d} = 86.6 \cdot 20 + 50 \cdot 0 = 1732 \text{ J}.$$

The vertical part of the force does no work, because it is perpendicular to the motion.
(b) A body of mass 10 kg ($g = 9.8$ m/s^2) slides 5 m down a 30° incline (Figure 4.23b). Its weight is $\mathbf{W} = (0, -98)$ N and $\mathbf{d} = 5(\cos 30^\circ,\ -\sin 30^\circ) = (4.33,\ -2.5)$ m, so

$$W_{\text{gravity}} = \mathbf{W} \cdot \mathbf{d} = 0 \cdot 4.33 + (-98)(-2.5) = 245 \text{ J}.$$

The same result follows from the components: $|\mathbf{W}_\parallel| = 98\sin 30^\circ = 49$ N along the slope, so $W = 49 \cdot 5 = 245$ J; the normal part $|\mathbf{W}_\perp| = 98\cos 30^\circ \approx 84.9$ N is perpendicular to $\mathbf{d}$ and does no work. (It also equals mgh with the height drop $h = 2.5$ m.)
(c) A force $\mathbf{F} = (3, 4)$ N acting on a particle moving with velocity $\mathbf{v} = (2, -1)$ m/s delivers the power $P = \mathbf{F} \cdot \mathbf{v} = 6 - 4 = 2$ W.

Intuition. The *sign* of the work tells the story: $W > 0$ when the force helps the motion ($\theta < 90^\circ$), $W < 0$ when it opposes it (friction, $\theta = 180^\circ$), and $W = 0$ when it is perpendicular to the motion. That is why the normal force of a table, the centripetal force on a satellite in a circular orbit, and the magnetic force on a charge (Section 4.4.5) never change the speed: $P = \mathbf{F} \cdot \mathbf{v} = 0$.

4.4.3 Navigation: Wind and Current

Physics & Engineering: Relative Velocity

Physical law. The velocity relative to the ground is the velocity relative to the moving medium plus the velocity of the medium $\mathbf{g} = \mathbf{a} + \mathbf{w}$ (aircraft in wind, boat in a current, swimmer in a river). The pilot controls only $\mathbf{a}$ (its length is the airspeed, its direction the heading) and must choose the heading so that $\mathbf{g}$ points along the desired track.

Example: Flying Due North in a Crosswind

The airspeed is 200 km/h and the wind blows from the west at 50 km/h, so $\mathbf{w} = (50, 0)$. Let the heading be the angle δ west of north: $\mathbf{a} = 200(-\sin\delta,\ \cos\delta)$. The ground velocity $\mathbf{g} = \mathbf{a} + \mathbf{w}$ must have zero x-component:

$$-200\sin\delta + 50 = 0 \implies \sin\delta = \tfrac{1}{4}, \qquad \delta = \arcsin\tfrac{1}{4} \approx 14.5^\circ.$$

The ground speed is $g_y = 200\cos\delta = \sqrt{200^2 - 50^2} = 50\sqrt{15} \approx 193.6$ km/h, so 500 km take $500/193.6 \approx 2.58$ h. Compare Homework Problem 9: heading *straight* north is faster along the path (206 km/h) but the aircraft drifts 14.0° off the track (Figure 4.24).

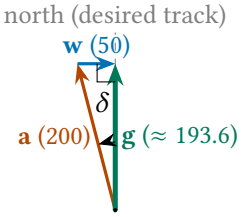


Figure 4.24: The wind triangle $\mathbf{g} = \mathbf{a} + \mathbf{w}$ (speeds in km/h). To fly due north in a wind from the west, the aircraft must head by the angle δ *into* the wind; then $\mathbf{g} \perp \mathbf{w}$ and the ground speed is $\sqrt{200^2 - 50^2}$.

4.4.4 Torque and Moments

Physics & Engineering: Moment of a Force

Physical law. The **torque** (moment) of a force $\mathbf{F}$ about a point O is

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}, \qquad |\boldsymbol{\tau}| = |\mathbf{r}|\,|\mathbf{F}|\sin\theta = |\mathbf{F}|\,d_\perp,$$

where $\mathbf{r}$ points from O to the point where the force acts and $d_\perp = |\mathbf{r}|\sin\theta$ is the *lever arm* (the perpendicular distance from O to the line of action of $\mathbf{F}$).

The vector $\boldsymbol{\tau}$ points along the axis of the turning tendency (right-hand rule). A body is in **rotational equilibrium** if $\sum \boldsymbol{\tau}_i = \mathbf{0}$ (together with $\sum \mathbf{F}_i = \mathbf{0}$). For forces in the xy-plane, $\mathbf{r} = (x, y, 0)$ and $\mathbf{F} = (F_x, F_y, 0)$, the torque has only a z-component, $\tau_z = xF_y - yF_x$: positive for an anticlockwise turn.

Examples: Torque and Rotational Equilibrium

1. Wrench (Figure 4.25a): A wrench of length 0.4 m is pushed with a force of 200 N at a 60° angle to the handle. Using the position vector $\mathbf{r} = (0.4, 0, 0)$ m and force vector $\mathbf{F} = 200(\cos 60^\circ, \sin 60^\circ, 0) = (100, 173.2, 0)$ N, the cross-product gives:

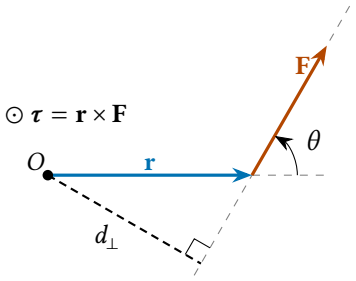
$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = (0, 0, 0.4 \cdot 173.2) = (0, 0, 69.3)\ \text{N} \cdot \text{m}.$$

The optimal push is perpendicular to the handle ($200 \cdot 0.4 = 80$ N·m), while a push directed along the line of the handle yields zero torque ($\theta = 0^\circ$).

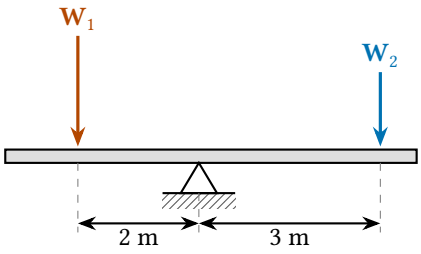
2. Balanced Lever (Figure 4.25b): With the pivot set at the origin, weight $\mathbf{W}_1 = (0, -300)$ N acts at position $\mathbf{r}_1 = (-2, 0)$ m and unknown weight $\mathbf{W}_2 = (0, -W_2)$ acts at $\mathbf{r}_2 = (3, 0)$ m. The torques about the pivot are:

$$\tau_1 = x_1F_{1y} - y_1F_{1x} = (-2)(-300) = +600\ \text{N} \cdot \text{m}, \qquad \tau_2 = x_2F_{2y} - y_2F_{2x} = (3)(-W_2) = -3W_2\ \text{N} \cdot \text{m}.$$

Applying the condition for rotational equilibrium ($\tau_1 + \tau_2 = 0$) yields $600 - 3W_2 = 0$, which gives $W_2 = 200$ N. The pivot supports the combined total weight of $300 + 200 = 500$ N, but because its reaction force acts directly at the origin ($\mathbf{r} = \mathbf{0}$), it contributes zero torque.



(a) Lever arm $d_\perp = |\mathbf{r}| \sin\theta$



(b) Balancing a lever about the pivot

Figure 4.25: (a) The torque of $\mathbf{F}$ about O points out of the page (⊙, anticlockwise turn); its size is the force times the lever arm. A force along the line through O ($\theta = 0^\circ$) has no lever arm and no torque. (b) A beam balanced on a pivot: the two weights turn it in opposite senses, and the torques must cancel (arrow lengths are proportional to the forces).

4.4.5 Rotation and Magnetic Force

Physics & Engineering: Velocity of a Point on a Rotating Body

Physical law. A rigid body rotates about an axis through the origin with **angular velocity vector** $\boldsymbol{\omega}$: its direction is the axis (right-hand rule) and $|\boldsymbol{\omega}|$ is the rate of rotation in rad/s. The point at position $\mathbf{r}$ moves with

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}, \qquad |\mathbf{v}| = |\boldsymbol{\omega}|\,|\mathbf{r}| \sin\theta = \omega R,$$

where R is the distance of the point from the axis. The velocity is perpendicular to the axis and to $\mathbf{r}$, and the (centripetal) acceleration is $\mathbf{a} = \boldsymbol{\omega} \times \mathbf{v}$, which points towards the axis.

Physics & Engineering: Magnetic Force and Flux

Physical law (Lorentz force). A charge q moving with velocity $\mathbf{v}$ in a magnetic field $\mathbf{B}$ (in tesla) experiences

$$\mathbf{F} = q\,\mathbf{v} \times \mathbf{B}, \qquad |\mathbf{F}| = |q|\,|\mathbf{v}|\,|\mathbf{B}| \sin\theta.$$

For $q > 0$ the direction follows the right-hand rule (curl from $\mathbf{v}$ towards $\mathbf{B}$). The force is perpendicular to $\mathbf{v}$, so $P = \mathbf{F} \cdot \mathbf{v} = 0$: it changes the *direction* of the motion but not the speed. The **magnetic flux** through a flat surface is $\Phi = \mathbf{B} \cdot \mathbf{A}$ (weber), where the area vector $\mathbf{A}$ is perpendicular to the surface and its length is the area; for the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$ we have $\mathbf{A} = \mathbf{a} \times \mathbf{b}$, so $\Phi = \mathbf{B} \cdot (\mathbf{a} \times \mathbf{b})$ is a scalar triple product.

Example: A Rotating Wheel, a Proton and a Coil

(a) Rotation. (Figure 4.26a) Let $\boldsymbol{\omega} = (0, 0, 2)$ rad/s and $\mathbf{r} = (3, 4, 5)$ m. Then

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} = (0 \cdot 5 - 2 \cdot 4,\ 2 \cdot 3 - 0 \cdot 5,\ 0) = (-8,\ 6,\ 0) \text{ m/s}, \qquad |\mathbf{v}| = 10 \text{ m/s} = \omega R \ \ (R = 5 \text{ m}).$$

(b)Lorentz force. (Figure 4.26b) A proton ($q = 1.6 \cdot 10^{-19}$ C) moves with $\mathbf{v} = (2 \cdot 10^6, 0, 0)$ m/s in $\mathbf{B} = (0, 0.5, 0)$ T. Then $\mathbf{v} \times \mathbf{B} = (0, 0, 10^6)$ and

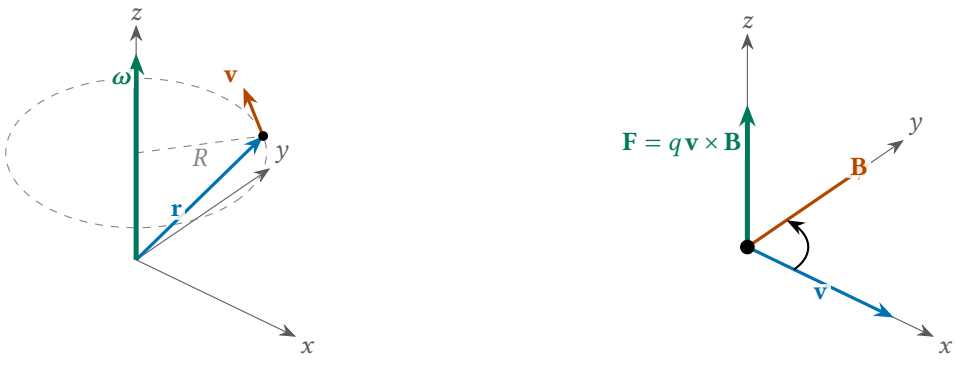
$$\mathbf{F} = q\,\mathbf{v} \times \mathbf{B} = (0,\ 0,\ 1.6 \cdot 10^{-13}) \text{ N}, \qquad \mathbf{F} \cdot \mathbf{v} = 0.$$

The force does no work and has constant size qvB, so it acts as a centripetal force: $qvB = mv^2/r$ gives the radius $r = \dfrac{mv}{qB} \approx \dfrac{1.67 \cdot 10^{-27} \cdot 2 \cdot 10^6}{1.6 \cdot 10^{-19} \cdot 0.5} \approx 0.042$ m, a circle of about 4.2 cm.

(c) Flux. A flat coil is spanned by $\mathbf{a} = (0.1, 0, 0)$ m and $\mathbf{b} = (0, 0.1, 0.1)$ m in the field $\mathbf{B} = (0, 0, 0.3)$ T. Its area vector is $\mathbf{a} \times \mathbf{b} = (0, -0.01, 0.01)$ m^2 (area 0.0141 m^2), so

$$\Phi = \mathbf{B} \cdot (\mathbf{a} \times \mathbf{b}) = 0.3 \cdot 0.01 = 3 \cdot 10^{-3} \text{ Wb}.$$

Check: the coil's normal makes 45° with $\mathbf{B}$, and $0.3 \cdot 0.0141 \cdot \cos 45^\circ = 3 \cdot 10^{-3}$. ☒



(a) $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$ **(b) $\mathbf{F} = q\,\mathbf{v} \times \mathbf{B}$ $(q > 0)$**

Figure 4.26: (a) A point at $\mathbf{r}$ on a body rotating about the z-axis moves on a circle of radius R with velocity $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$, tangent to the circle. (b) The magnetic force on a positive charge is perpendicular to both $\mathbf{v}$ and $\mathbf{B}$ (right-hand rule: curl the fingers from $\mathbf{v}$ towards $\mathbf{B}$, the thumb gives $\mathbf{F}$).

Key Formulas

- Vector from P to Q: $\overrightarrow{PQ} = Q - P$; length $|\mathbf{v}| = \sqrt{v_1^2 + \cdots + v_n^2}$; unit vector $\hat{\mathbf{v}} = \mathbf{v}/|\mathbf{v}|$.
- Dot product: $\mathbf{u} \cdot \mathbf{v} = \sum u_i v_i = |\mathbf{u}||\mathbf{v}| \cos\theta$; $\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0$.
- Cross product (in $\mathbb{R}^3$): $\mathbf{u} \times \mathbf{v} \perp \mathbf{u}, \mathbf{v}$; $|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}| \sin\theta$ = area of the parallelogram; $\mathbf{v} \times \mathbf{u} = -\mathbf{u} \times \mathbf{v}$.
- Triangle area $\frac{1}{2}|\mathbf{u} \times \mathbf{v}|$; parallelepiped volume $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$; tetrahedron volume $\frac{1}{6}|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$.
- Projection: $\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \dfrac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2}\,\mathbf{v}$; reflection in the line of $\mathbf{v}$: $2\operatorname{proj}_{\mathbf{v}} \mathbf{u} - \mathbf{u}$.
- Direction cosines: $(\cos\alpha, \cos\beta, \cos\gamma) = \hat{\mathbf{v}}$, so $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$.
- $\mathbf{v}_1, \ldots, \mathbf{v}_k$ are independent $\iff$ the matrix with these columns has rank k; for three vectors in $\mathbb{R}^3$: $\iff \mathbf{v}_1 \cdot (\mathbf{v}_2 \times \mathbf{v}_3) \neq 0$.
- Physics: equilibrium $\sum \mathbf{F}_i = \mathbf{0}$; work $W = \mathbf{F} \cdot \mathbf{d}$, power $P = \mathbf{F} \cdot \mathbf{v}$; torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$; rotation $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$; Lorentz force $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$; flux $\Phi = \mathbf{B} \cdot \mathbf{A}$; relative velocity $\mathbf{g} = \mathbf{a} + \mathbf{w}$.

4.5 Worked Problems

Problem: Vector Between Two Points, Length and Direction Cosines in $\mathbb{R}^3$

Find the vector from $P(1, 2, 3)$ to $Q(4, 0, 1)$, its length and its direction cosines.

Solution.

1. **The vector:** $\mathbf{v} = \overrightarrow{PQ} = Q - P = (4-1,\ 0-2,\ 1-3) = (3, -2, -2)$.
2. **The length:** $|\mathbf{v}| = \sqrt{3^2 + (-2)^2 + (-2)^2} = \sqrt{17}$.
3. **The direction cosines:**

$$\cos\alpha = \frac{3}{\sqrt{17}} \approx 0.728, \quad \cos\beta = \frac{-2}{\sqrt{17}} \approx -0.485, \quad \cos\gamma = \frac{-2}{\sqrt{17}} \approx -0.485,$$

so $\alpha \approx 43.3^\circ$ and $\beta = \gamma \approx 119.0^\circ$ (obtuse, because the components are negative).

Problem: Cross Product, Parallelogram Area and Parallelepiped Volume in $\mathbb{R}^3$

Let $\mathbf{u} = (1, 2, 0)$, $\mathbf{v} = (0, 1, 3)$ and $\mathbf{w} = (2, 0, 1)$. Find
(1) a vector perpendicular to both $\mathbf{u}$ and $\mathbf{v}$;
(2) the area of the parallelogram spanned by $\mathbf{u}$ and $\mathbf{v}$;
(3) the volume of the parallelepiped spanned by $\mathbf{u}, \mathbf{v}, \mathbf{w}$.

Solution.

1. $\mathbf{u} \times \mathbf{v}$ is perpendicular to both:

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 0 \\ 0 & 1 & 3 \end{vmatrix} = \mathbf{i}(2 \cdot 3 - 0 \cdot 1) - \mathbf{j}(1 \cdot 3 - 0 \cdot 0) + \mathbf{k}(1 \cdot 1 - 2 \cdot 0) = (6, -3, 1).$$

2. $A = |\mathbf{u} \times \mathbf{v}| = \sqrt{6^2 + (-3)^2 + 1^2} = \sqrt{46} \approx 6.78$.
3. $V = |\mathbf{w} \cdot (\mathbf{u} \times \mathbf{v})| = |2 \cdot 6 + 0 \cdot (-3) + 1 \cdot 1| = 13$. (Equivalently, the determinant with rows $\mathbf{u}, \mathbf{v}, \mathbf{w}$ equals $1(1-0) - 2(0-6) + 0 = 13$; the order $\mathbf{w}, \mathbf{u}, \mathbf{v}$ is a cyclic permutation of $\mathbf{u}, \mathbf{v}, \mathbf{w}$, so the sign is the same.)

Problem 1: Vector Equation with Exactly One Solution in $\mathbb{R}^3$

Let $\mathbf{u} = (1, 2, 0)$ and $\mathbf{v} = (0, 1, 1)$. Find scalars a and b such that $a\mathbf{u} + b\mathbf{v} = (2, 7, 3)$.

Solution. Compute the linear combination:

$$a(1, 2, 0) + b(0, 1, 1) = (a,\ 2a + b,\ b).$$

Setting this equal to $(2, 7, 3)$ gives three equations in two unknowns:

$$a = 2, \qquad 2a + b = 7, \qquad b = 3.$$

The first and third equations give $a = 2$, $b = 3$. **Check the remaining equation:** $2a + b = 4 + 3 = 7$. ☒ Since the system is overdetermined, this check is essential; it holds here, so the solution $a = 2$, $b = 3$ exists and is unique (the vectors $\mathbf{u}, \mathbf{v}$ are not parallel).

Problem 2: Dot Product, Angle, Projection and Reflection in $\mathbb{R}^3$

Let $\mathbf{u} = (1, 0, 0)$ and $\mathbf{v} = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right)$ (Figure 4.17). Compute
(1) $\mathbf{u} \cdot \mathbf{v}$, (2) the angle θ between them,
(3) $\operatorname{proj}_{\mathbf{v}} \mathbf{u}$, (4) the reflection of $\mathbf{u}$ in the line spanned by $\mathbf{v}$.

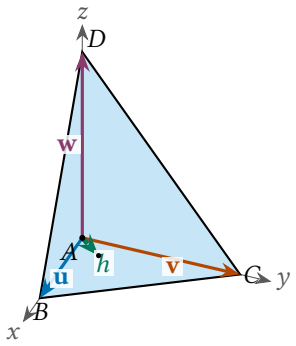


Figure 4.27: The tetrahedron $ABCD$. The dashed green segment is the height h from A, perpendicular to the face BCD (shaded).

Solution.

1. $\mathbf{u} \cdot \mathbf{v} = 1 \cdot \frac{1}{2} + 0 \cdot \frac{\sqrt{3}}{2} + 0 \cdot 0 = \frac{1}{2}$.
2. $|\mathbf{u}| = 1$ and $|\mathbf{v}| = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1$, so $\cos\theta = \dfrac{1/2}{1 \cdot 1} = \dfrac{1}{2}$ and $\theta = 60^\circ$.
3. Since $|\mathbf{v}|^2 = 1$,
$$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2}\,\mathbf{v} = \frac{1}{2}\left(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right) = \left(\frac{1}{4}, \frac{\sqrt{3}}{4}, 0\right).$$
4. Use $\operatorname{refl}_{\mathbf{v}} \mathbf{u} = 2\operatorname{proj}_{\mathbf{v}} \mathbf{u} - \mathbf{u}$ (Figure 4.17):
$$\operatorname{refl}_{\mathbf{v}} \mathbf{u} = 2\left(\frac{1}{4}, \frac{\sqrt{3}}{4}, 0\right) - (1, 0, 0) = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right).$$

Check: the mirror image has the same length, $|\operatorname{refl}_{\mathbf{v}} \mathbf{u}| = 1$, and makes the angle 60° with $\mathbf{v}$ on the other side (it points in the direction 120°, while $\mathbf{u}$ points in the direction 0° and the mirror line in the direction 60°).

Problem 3: Volume and Height of a Tetrahedron

The tetrahedron has vertices $A(0,0,0)$, $B(2,0,0)$, $C(0,3,0)$ and $D(0,0,4)$. Find its volume with the scalar triple product and the height of the tetrahedron drawn from A (the distance from A to the face BCD).

Solution.

1. Take the edges from A (Figure 4.27): $\mathbf{u} = \overrightarrow{AB} = (2,0,0)$, $\mathbf{v} = \overrightarrow{AC} = (0,3,0)$, $\mathbf{w} = \overrightarrow{AD} = (0,0,4)$.
2. The scalar triple product is

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \det \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix} = 2 \cdot 3 \cdot 4 = 24.$$

3. A tetrahedron has one sixth of the volume of the parallelepiped: $V = \dfrac{1}{6} \cdot 24 = 4$.
4. The volume is also $V = \dfrac{1}{3}(\text{area of face } BCD) \cdot h$. To get the area of the face BCD, take $\overrightarrow{BC} = C - B = (-2,3,0)$ and $\overrightarrow{BD} = D - B = (-2,0,4)$:

$$\overrightarrow{BC} \times \overrightarrow{BD} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 3 & 0 \\ -2 & 0 & 4 \end{vmatrix} = \mathbf{i}(12-0) - \mathbf{j}(-8-0) + \mathbf{k}(0+6) = (12,8,6),$$

$$A_{BCD} = \tfrac{1}{2}\sqrt{12^2 + 8^2 + 6^2} = \tfrac{1}{2}\sqrt{244} = \sqrt{61}.$$

5. Therefore $h = \dfrac{3V}{A_{BCD}} = \dfrac{12}{\sqrt{61}} = \dfrac{12\sqrt{61}}{61} \approx 1.54$.

Problem 4: Linear Independence and Related Quantities in $\mathbb{R}^3$

Let $\mathbf{u} = (1,2,0)$, $\mathbf{v} = (0,1,2)$, $\mathbf{w} = (2,0,1)$. (1) Are they linearly independent? (2) Find the volume of the parallelepiped they span. (3) Express $\mathbf{z} = (3,5,4)$ as $a\mathbf{u} + b\mathbf{v} + c\mathbf{w}$, if possible.

Solution.

1. Put the vectors into the columns of a matrix and compute its determinant:

$$\det \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} = 1 \cdot (1 \cdot 1 - 0 \cdot 2) - 0 \cdot (2 \cdot 1 - 0 \cdot 0) + 2 \cdot (2 \cdot 2 - 1 \cdot 0) = 1 + 8 = 9 \neq 0.$$

   So the vectors are linearly independent (the matrix has rank 3).
2. $V = |\det| = 9$. Directly: $\mathbf{v} \times \mathbf{w} = (1 \cdot 1 - 2 \cdot 0,\ 2 \cdot 2 - 0 \cdot 1,\ 0 \cdot 0 - 1 \cdot 2) = (1,4,-2)$ and $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = 1 + 8 + 0 = 9$.
3. We need $a\mathbf{u} + b\mathbf{v} + c\mathbf{w} = (a + 2c,\ 2a + b,\ 2b + c) = (3,5,4)$, i.e.

$$a + 2c = 3, \qquad 2a + b = 5, \qquad 2b + c = 4.$$

   From the third equation $c = 4 - 2b$; the first gives $a = 4b - 5$; the second becomes $2(4b-5) + b = 5$, so $9b = 15$ and $b = \frac{5}{3}$, $c = \frac{2}{3}$, $a = \frac{5}{3}$. **Check:** $\frac{5}{3} + \frac{4}{3} = 3$, $\frac{10}{3} + \frac{5}{3} = 5$, $\frac{10}{3} + \frac{2}{3} = 4$. ☒ Because the vectors are independent, this representation is unique.

Problem 5: Direction Cosines in a Tetrahedron

The tetrahedron $ABCD$ has vertices $A(0,0,0)$, $B(2,0,0)$, $C(0,2,0)$, $D(1,1,1)$ (Figure 4.28). A vector $\mathbf{v}$ with initial point A has length $k > 0$ and direction cosines $\cos\alpha = \cos\beta = \cos\gamma = \frac{1}{\sqrt{3}}$. Compute
(1) $\mathbf{v} \cdot \overrightarrow{AB}$;
(2) $\text{proj}_{\overrightarrow{AB}} \mathbf{v}$;
(3) the volume of the tetrahedron.

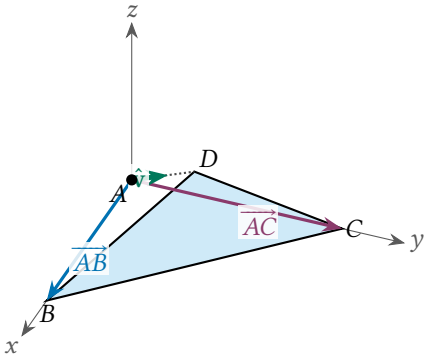


Figure 4.28: The tetrahedron $ABCD$ with $D(1,1,1)$ and the unit vector $\hat{\mathbf{v}} = (\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$ from A; it lies along the (dotted) edge AD.

Solution.

1. The direction cosines are the components of the unit vector, so $\mathbf{v} = k\,\hat{\mathbf{v}} = k\big(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\big)$. (Note that $\overrightarrow{AD} = (1,1,1)$ has the same direction, so $\mathbf{v}$ points along the edge AD; see Figure 4.28.) Then
$$\mathbf{v} \cdot \overrightarrow{AB} = \frac{k}{\sqrt{3}}\,(1,1,1) \cdot (2,0,0) = \frac{2k}{\sqrt{3}}.$$

2. With $|\overrightarrow{AB}|^2 = 4$:
$$\text{proj}_{\overrightarrow{AB}} \mathbf{v} = \frac{2k/\sqrt{3}}{4}\,(2,0,0) = \Big(\frac{k}{\sqrt{3}}, 0, 0\Big).$$

3. From A: $\overrightarrow{AB} = (2,0,0)$, $\overrightarrow{AC} = (0,2,0)$, $\overrightarrow{AD} = (1,1,1)$. Then
$$\overrightarrow{AC} \times \overrightarrow{AD} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 2 & 0 \\ 1 & 1 & 1 \end{vmatrix} = (2,0,-2), \qquad \overrightarrow{AB} \cdot (2,0,-2) = 4,$$
so $V = \frac{1}{6} \cdot 4 = \frac{2}{3}$.

Problem 6: Resultant, Independence and Work of Three Forces

Three forces act on a body that we model as a particle at the origin $O(0,0,0)$: $\mathbf{F}_1 = (2,0,0)$ kN (along the x-axis), $\mathbf{F}_2 = (0,3,0)$ kN (along the y-axis) and $\mathbf{F}_3 = (1,1,2)$ kN. The body moves by $\mathbf{d} = (1,1,1)$ m. Find (a) the resultant $\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3$ and its magnitude; (b) whether $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3$ are linearly independent; (c) the work done by $\mathbf{R}$ along $\mathbf{d}$ (Figure 4.29).

Solution.

1. $\mathbf{R} = (2,0,0) + (0,3,0) + (1,1,2) = (3,4,2)$ kN and $|\mathbf{R}| = \sqrt{9+16+4} = \sqrt{29} \approx 5.39$ kN.

2. With the forces as columns, expanding along the first row,

$$\det\begin{bmatrix}2&0&1\\0&3&1\\0&0&2\end{bmatrix} = 2\begin{vmatrix}3&1\\0&2\end{vmatrix} - 0\begin{vmatrix}0&1\\0&2\end{vmatrix} + 1\begin{vmatrix}0&3\\0&0\end{vmatrix} = 2\cdot 6 - 0 + 0 = 12 \neq 0.$$

   (The matrix is upper triangular, so the determinant is the product of the diagonal entries $2{\cdot}3{\cdot}2 = 12$.) The forces are linearly independent: they are not coplanar, so no force can be replaced by a combination of the other two.
3. $W = \mathbf{R}\cdot\mathbf{d} = 3\cdot 1 + 4\cdot 1 + 2\cdot 1 = 9$ kN·m $= 9$ kJ.

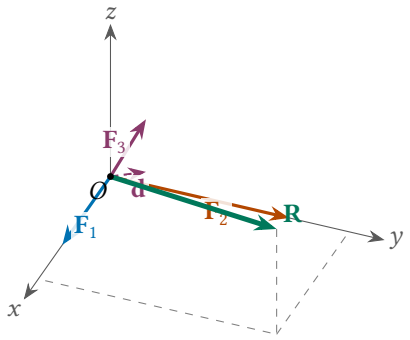


Figure 4.29: The forces $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3$ (kN), the displacement $\mathbf{d}$ (m, dashed) and the resultant $\mathbf{R} = (3,4,2)$ (green) on the body at O.

Problem 7: Torque of a Force

A wrench is attached at the point given by the position vector $\mathbf{r} = (0.5, 0, 0)$ m, and a force $\mathbf{F} = (0, 20, 10)$ N acts there. The torque about the origin is $\boldsymbol{\tau} = \mathbf{r}\times\mathbf{F}$. Find $\boldsymbol{\tau}$ and $|\boldsymbol{\tau}|$.

Solution.

$$\boldsymbol{\tau} = \begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\0.5&0&0\\0&20&10\end{vmatrix} = \mathbf{i}(0-0) - \mathbf{j}(0.5\cdot 10 - 0) + \mathbf{k}(0.5\cdot 20 - 0) = (0,\,-5,\,10)\ \text{N}\cdot\text{m},$$

$$|\boldsymbol{\tau}| = \sqrt{25 + 100} = 5\sqrt{5} \approx 11.2\ \text{N}\cdot\text{m}.$$

The torque is perpendicular to both $\mathbf{r}$ and $\mathbf{F}$ (check: $\boldsymbol{\tau}\cdot\mathbf{r} = 0$ and $\boldsymbol{\tau}\cdot\mathbf{F} = -100 + 100 = 0$), and its axis is the axis about which the wrench tends to turn. Only the component of $\mathbf{F}$ perpendicular to $\mathbf{r}$ contributes: here $|\mathbf{r}|\,|\mathbf{F}_\perp| = 0.5\cdot\sqrt{500} = 5\sqrt{5}$, where $\mathbf{F}_\perp = (0,20,10)$ since $\mathbf{F}\perp\mathbf{r}$.

4.6 Conceptual & Computational Exercises

4.6.1 Conceptual Problems

Try these *before* computing: most can be answered by thinking about types, geometry or physical meaning. Answers and hints follow the homework problems.

1. **Type checking.** For $\mathbf{u},\mathbf{v},\mathbf{w} \in \mathbb{R}^3$ decide whether each expression is a scalar, a vector, or meaningless: (a) $\mathbf{u}\cdot\mathbf{v} + k$ with $k\in\mathbb{R}$; (b) $(\mathbf{u}\cdot\mathbf{v})\,\mathbf{w}$; (c) $(\mathbf{u}\cdot\mathbf{v})\cdot\mathbf{w}$; (d) $\mathbf{u}\times(\mathbf{v}\cdot\mathbf{w})$; (e) $\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})$; (f) $(\mathbf{u}\times\mathbf{v})\times\mathbf{w}$; (g) $|\mathbf{u}|\,\mathbf{v}$; (h) $\mathbf{u}/\mathbf{v}$.
2. **Loci.** Let $\mathbf{u}\neq\mathbf{0}$. Describe geometrically all $\mathbf{v}$ with $\mathbf{u}\cdot\mathbf{v} = 0$, (i) in $\mathbb{R}^2$ and (ii) in $\mathbb{R}^3$. Describe all $\mathbf{v}\in\mathbb{R}^3$ with $\mathbf{u}\times\mathbf{v} = \mathbf{0}$.
3. **No cancellation.**

   (a) For $\mathbf{u} = (1, 0)$, $\mathbf{v} = (2, 3)$, $\mathbf{w} = (2, -5)$ show that $\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot \mathbf{w}$ although $\mathbf{v} \neq \mathbf{w}$. How is $\mathbf{v} - \mathbf{w}$ related to $\mathbf{u}$?
   (b) For $\mathbf{u} = (0, 0, 1)$ find $\mathbf{v} \neq \mathbf{w}$ with $\mathbf{u} \times \mathbf{v} = \mathbf{u} \times \mathbf{w}$. How is $\mathbf{v} - \mathbf{w}$ related to $\mathbf{u}$ now?
   (c) Prove: if $\mathbf{u} \neq \mathbf{0}$, $\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot \mathbf{w}$ *and* $\mathbf{u} \times \mathbf{v} = \mathbf{u} \times \mathbf{w}$, then $\mathbf{v} = \mathbf{w}$.

4. **True or false?** Justify each answer by a proof or a counterexample. (a) $|\mathbf{u} + \mathbf{v}| = |\mathbf{u}| + |\mathbf{v}|$ for all $\mathbf{u}, \mathbf{v}$. (b) If $\mathbf{u} \cdot \mathbf{v} = 0$, then $\mathbf{u} = \mathbf{0}$ or $\mathbf{v} = \mathbf{0}$. (c) $|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}||\mathbf{v}|$. (d) $\mathbf{u} \times \mathbf{v} = \mathbf{v} \times \mathbf{u}$ only if $\mathbf{u} \parallel \mathbf{v}$. (e) $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} = 0$ for all $\mathbf{u}, \mathbf{v}$. (f) Three nonzero, pairwise orthogonal vectors are linearly independent. (g) Any three vectors in $\mathbb{R}^2$ are linearly dependent. (h) The zero vector is a unit vector.
5. **Extremes.** (a) For unit vectors $\mathbf{u}, \mathbf{v}$, which values can $\mathbf{u} \cdot \mathbf{v}$ take, and when are the extreme values attained? (b) For fixed lengths $|\mathbf{u}|, |\mathbf{v}|$, for which angle is $|\mathbf{u} \times \mathbf{v}|$ largest?
6. **Shears do not change areas or volumes.** (a) Show that $\mathbf{u} \times (\mathbf{v} + k\mathbf{u}) = \mathbf{u} \times \mathbf{v}$. What does this say about the parallelograms spanned by $\mathbf{u}, \mathbf{v}$ and by $\mathbf{u}, \mathbf{v} + k\mathbf{u}$? (b) Show that $\mathbf{a} \cdot \big(\mathbf{b} \times (\mathbf{c} + p\mathbf{a} + q\mathbf{b})\big) = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ and interpret it.
7. **Scaling.** Every vector is multiplied by 3. By what factor do $|\mathbf{u}|$, $\mathbf{u} \cdot \mathbf{v}$, $|\mathbf{u} \times \mathbf{v}|$, $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$ and $\operatorname{proj}_{\mathbf{v}} \mathbf{u}$ change?
8. **Reading the physics.**
   (a) Why does a force perpendicular to the velocity never change the speed of a body?
   (b) Why can you not loosen a bolt by pulling a wrench *along* its handle?
   (c) A ball is thrown upwards and caught at the same height. What is the total work of gravity, and why?
   (d) Two symmetric cables, each at the angle θ above the horizontal, carry a weight W. Show that the tension is $T = W/(2\sin\theta)$. What happens as $\theta \to 0$, and what does this say about stretching a cable perfectly straight?
9. **Independence.** Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$ be linearly independent. Are (a) $\{\mathbf{u}, \mathbf{v}, \mathbf{u} + \mathbf{v}\}$ and (b) $\{\mathbf{u}, \mathbf{v}, \mathbf{u} \times \mathbf{v}\}$ linearly independent? (c) What changes in (b) if $\mathbf{u} \parallel \mathbf{v}$?
10. **Direction cosines.** (a) A vector makes the angles 60° with the x-axis and with the y-axis. What can its angle with the z-axis be? (b) Can a vector make 30° with all three axes? (c) Which angles does $(1, 1, 1)$ make with the axes?
11. **Geometry with dot products.** (a) Prove that the diagonals of a rhombus are perpendicular. (b) Prove the Pythagorean theorem: if $\mathbf{u} \perp \mathbf{v}$, then $|\mathbf{u} + \mathbf{v}|^2 = |\mathbf{u}|^2 + |\mathbf{v}|^2$.
12. **Reading the triple product.** (a) What does a *negative* value of $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ mean? (b) What happens to it if you swap $\mathbf{b}$ and $\mathbf{c}$? (c) If $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})| = |\mathbf{a}||\mathbf{b}||\mathbf{c}|$ for nonzero vectors, what can you say about them?

4.6.2 Computational Problems

Solve the following problems. Show all work and give exact answers unless otherwise specified.

1. Let $P(1, 1, 0)$, $Q(3, 0, 2)$ and $R(0, 2, 1)$.
   (a) Find the vector $\overrightarrow{PQ}$ and its magnitude.
   (b) Compute the direction cosines of $\overrightarrow{PQ}$.
   (c) Find a vector $\mathbf{w}$ of magnitude 5 with the same direction as $\overrightarrow{PQ}$.
   (d) Find the area of the triangle PQR.
2. Let $\mathbf{u} = (2, 1, -1)$ and $\mathbf{v} = (-1, 2, 2)$.
   (a) Compute the dot product $\mathbf{u} \cdot \mathbf{v}$.
   (b) Find the angle θ between $\mathbf{u}$ and $\mathbf{v}$ (to two decimal places, in degrees). Is it acute or obtuse?
   (c) Compute the projection of $\mathbf{u}$ onto $\mathbf{v}$.
   (d) Compute the reflection of $\mathbf{u}$ in the line spanned by $\mathbf{v}$.

3. Let $\mathbf{a} = (1, 0, 2)$ and $\mathbf{b} = (0, 3, 1)$.
   (a) Find a vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.
   (b) Find the area of the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$.
   (c) Find the angle between $\mathbf{a}$ and $\mathbf{b}$ (to two decimal places) and check that $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin\theta$.
4. A tetrahedron has vertices $A(0,0,0)$, $B(4,0,0)$, $C(0,2,0)$ and $D(1,1,2)$. Find its volume using the scalar triple product and the height drawn from A.
5. Let $\mathbf{u} = (1,1,0)$, $\mathbf{v} = (0,1,1)$ and $\mathbf{w} = (2,0,1)$.
   (a) Are $\mathbf{u}, \mathbf{v}, \mathbf{w}$ linearly independent?
   (b) If possible, find a, b, c with $a\mathbf{u} + b\mathbf{v} + c\mathbf{w} = (4, 3, 2)$.
6. **Physics.** Three forces act on a particle at the origin: $\mathbf{F}_1 = (3,0,0)$ N, $\mathbf{F}_2 = (0,2,1)$ N and $\mathbf{F}_3 = (1,-1,2)$ N. The particle is displaced by $\mathbf{d} = (2,1,1)$ m.
   (a) Find the resultant force $\mathbf{R}$ and its magnitude.
   (b) Find the work done by $\mathbf{R}$ along $\mathbf{d}$.
7. **A parameter.** For which value of k are the vectors $\mathbf{u} = (1,1,1)$, $\mathbf{v} = (1,2,3)$, $\mathbf{w} = (2,3,k)$ linearly dependent? For this k write $\mathbf{w}$ as a combination of $\mathbf{u}$ and $\mathbf{v}$, and explain the geometric meaning. What is the volume of the parallelepiped spanned by them for $k = 7$?
8. **Another parameter.** Find t such that the vectors $\mathbf{u} = (1, t, 2)$ and $\mathbf{v} = (t, 4, -1)$ are orthogonal. Then, with this t, compute $\cos\theta$ for $\mathbf{u}$ and $\mathbf{w} = (1,0,1)$.
9. **Aviation.** An aircraft flies with an airspeed of 200 km/h heading due north. A wind blows from the west (towards the east) at 50 km/h. Take the x-axis pointing east and the y-axis pointing north.
   (a) Write the velocity of the aircraft relative to the air and the velocity of the wind as vectors.
   (b) Find the ground velocity, the ground speed and the angle by which the aircraft drifts away from north.
   (c) How far east is the aircraft from its starting point after 2 hours?
10. **Statics.** A traffic light of weight 200 N hangs at the junction of two cables; the left cable makes 30° and the right cable 45° with the horizontal. Find both tensions (write the equilibrium condition in components) and draw the force polygon.
11. **Work.** A crate is dragged 8 m along a level floor by a rope with tension 150 N at 40° above the horizontal, while friction exerts 100 N opposite to the motion. Find the work done by the rope, by friction, and the net work. Which of the forces (rope, friction, weight, normal force) do no work, and why?
12. **Torque.** A seesaw of length 4 m is pivoted at its centre; take the x-axis along the seesaw (pivot at the origin) and the y-axis upwards. A child of weight 400 N sits 1.2 m to the left of the pivot. (a) Compute the torque vector of this weight about the pivot as $\mathbf{r} \times \mathbf{F}$. (b) Where must a child of weight 300 N sit to balance it? (c) What is the force exerted by the pivot?
13. **Rotation.** A wheel rotates about the z-axis with $\boldsymbol{\omega} = (0,0,-3)$ rad/s. Find the velocity $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$ and the acceleration $\mathbf{a} = \boldsymbol{\omega} \times \mathbf{v}$ of the point with position vector $\mathbf{r} = (2,0,1)$ m. Check that $\mathbf{v} \perp \mathbf{r}$ and that $|\mathbf{v}| = \omega R$.
14. **Electromagnetism.** A particle of charge $q = 3\ \mu$C moves with $\mathbf{v} = (0,4,0)$ m/s in the field $\mathbf{B} = (2,0,1)$ T. (a) Find the magnetic force $\mathbf{F} = q\,\mathbf{v} \times \mathbf{B}$ and verify that it is perpendicular to both $\mathbf{v}$ and $\mathbf{B}$. (b) Find the flux of $\mathbf{B}$ through the rectangle spanned by $\mathbf{a} = (0.2, 0, 0)$ m and $\mathbf{b} = (0, 0.1, 0)$ m.
15. **Navigation.** A boat has speed 6 km/h relative to the water in a river of width 1.5 km that flows east at 2 km/h. (a) At what angle to north must it head to travel due north? (b) How long does the crossing take? (c) If it heads straight north instead, how far downstream does it land?

5

Straight Lines in the Plane

Learning Objectives

After studying this chapter you should be able to:

- pass between the general, point-slope, slope-intercept, intercept, canonical, parametric and normal forms of a line;
- find a line from a point and a normal or direction vector, from a slope, or from two points;
- find where two lines meet, and decide from the coefficients whether they are parallel, perpendicular or coincident;
- compute the distance from a point to a line and the distance between two parallel lines, and use the sign of $Ax + By + C$ to tell on which side of a line a point lies;
- find the angle between two lines using normals, direction vectors, or slopes;
- reflect a point, or a whole line, across a mirror line, and find equidistant points and angle bisectors;
- use straight lines to model calibration curves, motion, light rays, grades and clearances in physics and engineering.

Think of a line in the plane in two complementary ways. You can describe it by a vector that is **tangent** to it, showing which way it runs: this is the **direction vector**. Or you can describe it by a vector that is **perpendicular** to it, showing which way is "across" it: this is the **normal vector**. Every equation of a line in this chapter comes from one of these two pictures (Figure 5.1).

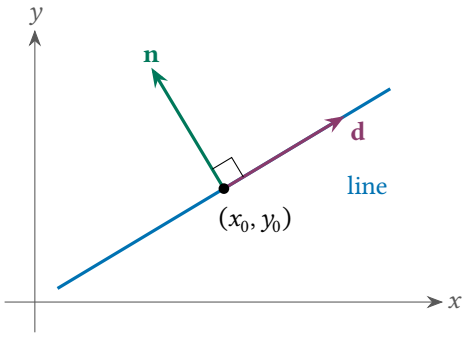


Figure 5.1: Two ways to pin down a line through (x_0, y_0): a direction vector $\mathbf{d} = (1.6, 0.96)$ that runs along it, or a normal vector $\mathbf{n} = (-0.96, 1.6)$ that is perpendicular to it. Indeed $\mathbf{n} \cdot \mathbf{d} = -0.96 \cdot 1.6 + 1.6 \cdot 0.96 = 0$.

Intuition. Geometrically, a line is an one-dimensional, possessing only length and zero width or depth, which makes it fundamentally invisible and purely conceptual. What we draw on a board or screen can never be a true line; instead, it is a physical object, a trail of graphite, ink, or glowing pixels, that actually has length, width, and depth. Because these physical dimensions are so tiny as to be entirely negligible at our scale, our drawing acts as a practical approximation of an 1D line.

5.1 Forms of the Equation of a Line

5.1.1 General Form and the Normal Vector

Definition

The **general form** of the equation of a line in $\mathbb{R}^2$ is

$$Ax + By + C = 0,$$

where A, B, C are real constants and A, B are not both zero. The vector $\mathbf{n} = (A, B)$ is called the **normal vector** of the line: it is perpendicular to the line.

Given a point (x_0, y_0) and a normal vector $\mathbf{n} = (A, B)$, the line through (x_0, y_0) with normal $\mathbf{n}$ is

$$A(x - x_0) + B(y - y_0) = 0,$$

which expands to the general form with $C = -Ax_0 - By_0$.

Why is $\mathbf{n} = (A, B)$ perpendicular to the line? If (x_1, y_1) and (x_2, y_2) both satisfy $Ax + By + C = 0$, subtracting the two equations gives

$$A(x_2 - x_1) + B(y_2 - y_1) = 0, \qquad \text{i.e.} \qquad \mathbf{n} \cdot (x_2 - x_1,\ y_2 - y_1) = 0.$$

The vector $(x_2 - x_1, y_2 - y_1)$ points along the line, so $\mathbf{n}$ is orthogonal to every direction along the line.

> **Common mistake. (A, B) is the normal, not the direction.** For $2x + 3y - 6 = 0$ the normal is $\mathbf{n} = (2, 3)$, a direction along the line is $\mathbf{d} = (3, -2)$ (or $(-3, 2)$), and the slope is $-\frac{2}{3}$. It is a very common slip to read $(2, 3)$ as a direction (slope $\frac{3}{2}$) or to forget the minus sign in $m = -\frac{A}{B}$. Always test: $\mathbf{n} \cdot \mathbf{d} = 2 \cdot 3 + 3 \cdot (-2) = 0$. Also remember that $Ax + By + C = 0$ and $kAx + kBy + kC = 0$ ($k \neq 0$) are the *same* line: the general form is not unique.

5.1.2 Direction Vector, Canonical and Parametric Forms

Definition

A **direction vector** $\mathbf{d} = (d_1, d_2)$ of a line is a nonzero vector parallel to the line. If (x_1, y_1) and (x_2, y_2) lie on the line, $(x_2 - x_1, y_2 - y_1)$ is a scalar multiple of $\mathbf{d}$, and $\mathbf{d}$ is perpendicular to the normal vector, $\mathbf{n} \cdot \mathbf{d} = 0$.

Given a point (x_0, y_0) and direction vector $\mathbf{d} = (d_1, d_2)$, the line has **parametric equations**

$$x = x_0 + d_1 t, \qquad y = y_0 + d_2 t, \qquad t \in \mathbb{R},$$

and, if $d_1, d_2 \neq 0$, the equivalent **canonical (symmetric) form**

$$\frac{x - x_0}{d_1} = \frac{y - y_0}{d_2}.$$

If $d_1 = 0$ the line is vertical, $x = x_0$; if $d_2 = 0$ it is horizontal, $y = y_0$. Eliminating t from the parametric form, or cross-multiplying the canonical form, gives the general form with normal vector $\mathbf{n} = (d_2, -d_1)$ (this is $\mathbf{d}$ rotated 90°, and indeed $\mathbf{n} \cdot \mathbf{d} = d_2 d_1 - d_1 d_2 = 0$):

$$d_2(x - x_0) - d_1(y - y_0) = 0.$$

Intuition. The parametric form is a moving particle. Think of t as time and $\mathbf{d}$ as a constant velocity. At $t = 0$ the particle is at (x_0, y_0); at $t = 1$ it has moved by $\mathbf{d}$; at $t = \frac{1}{2}$ it is halfway; for $t < 0$ it was there before. The line is the whole path. This picture explains why one line has infinitely many parametrisations (start at another point, or move at another speed) and why two particles on different lines need *different* time symbols t and s: they need not be at the intersection at the same moment.

Example: Lines Defined by Normal or Direction Vectors

(a) Using a Normal Vector. Find the line through $(3, 2)$ with normal vector $\mathbf{n} = (1, 2)$ (Figure 5.2a).

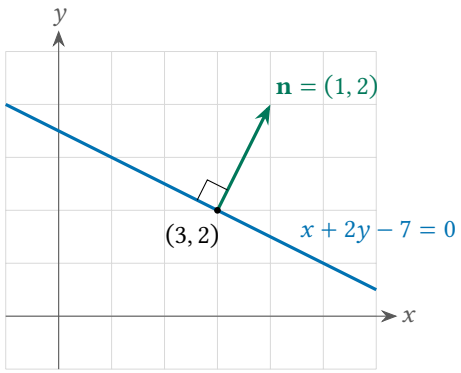
$$A(x - x_0) + B(y - y_0) = 0 \implies 1(x - 3) + 2(y - 2) = 0 \implies x + 2y - 7 = 0.$$

(b) Using a Direction Vector. A line passes through $(2, -3)$ with direction vector $\mathbf{d} = (-2, 5)$ (Figure 5.2b). Its canonical and parametric forms are

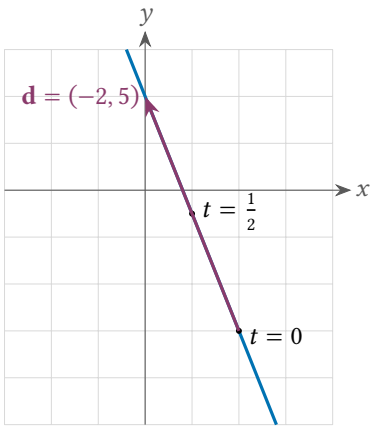
$$\text{Canonical:} \quad \frac{x - 2}{-2} = \frac{y + 3}{5}, \qquad \text{Parametric:} \quad x = 2 - 2t, \;\; y = -3 + 5t, \;\; t \in \mathbb{R}.$$

A normal vector is $\mathbf{n} = (d_2, -d_1) = (5, 2)$, giving the general form:

$$5(x - 2) + 2(y + 3) = 0 \implies 5x + 2y - 4 = 0.$$



(a) Normal vector $(1, 2)$ is perpendicular to the line



(b) Direction vector $(-2, 5)$ from $(2, -3)$

Figure 5.2: (a) The line $x + 2y - 7 = 0$ with its normal vector. (b) The line through $(2, -3)$ with direction vector $\mathbf{d} = (-2, 5)$; the points $t = 0$ and $t = \frac{1}{2}$ of the parametrisation are marked, and $t = 1$ is the tip of $\mathbf{d}$.

A line has infinitely many direction vectors and normal vectors (any non-zero scalar multiple $k\mathbf{d}$ or $k\mathbf{n}$ works). However, once a point on the line is fixed, using any of these proportional vectors yields the exact same equation, because multiplying an entire linear equation by a non-zero constant k leaves its solution set unchanged.

Common mistake. Canonical form and zero components. The fraction $\frac{x-x_0}{d_1} = \frac{y-y_0}{d_2}$ makes no sense if $d_1 = 0$ or $d_2 = 0$ (division by zero). Use the cross-multiplied form $d_2(x-x_0)-d_1(y-y_0) = 0$ instead, which always works: for $\mathbf{d} = (0, 3)$ it gives $3(x - x_0) = 0$, the vertical line $x = x_0$.

5.2 Slope, Intercepts and Related Forms

Definition

The **slope** of a line, denoted m, measures its steepness: for two distinct points (x_1, y_1), (x_2, y_2) on the line with $x_1 \neq x_2$,

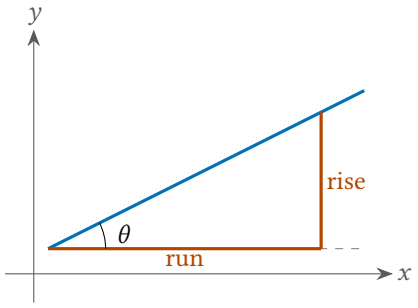
$$m = \frac{y_2 - y_1}{x_2 - x_1} \text{ (rise over run).}$$

For a line in general form $Ax + By + C = 0$ with $B \neq 0$, $m = -\frac{A}{B}$; if $B = 0$ the line is vertical and the slope is undefined. Geometrically, $m = \tan\theta$, where $\theta \in [0^\circ, 180^\circ)$ is the angle the line makes with the positive x-axis (Figure 5.3). The **point-slope form** through (x_0, y_0) with slope m is

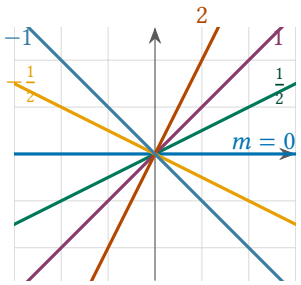
$$y - y_0 = m(x - x_0),$$

which is the general form with $A = m$, $B = -1$, $C = y_0 - mx_0$.

Intuition. **Slope is a "rate".** The slope m says: for every 1 unit you move to the right, the line rises by m units (or falls, if $m < 0$). Since $\tan\theta = \frac{\text{rise}}{\text{run}}$ in the right triangle formed by the line, the slope *is* the tangent of its angle of inclination. A horizontal line has $\theta = 0$ and $m = 0$; as the line tips up towards the vertical, $\theta \to 90^\circ$ and $m \to \infty$, which is why a vertical line has *no* slope at all.



(a) $m = \frac{\text{rise}}{\text{run}} = \tan\theta$



(b) lines through the origin with various slopes

Figure 5.3: (a) The slope is the tangent of the angle θ between the line and the positive x-axis. (b) Larger positive slope means a steeper rise; negative slope means the line falls to the right; slope 0 is horizontal.

Common mistake. **Vertical lines and the slope formula.** A vertical line $x = x_0$ has *no* slope, cannot be written as $y = mx + b$, and cannot be fed into formulas that use m (the slope-intercept form, the perpendicularity test $m_1 m_2 = -1$, the angle formula with $\tan\alpha$). Use the general form, a direction vector, or a normal vector instead. Also note that *negative* slopes correspond to inclination angles between 90° and 180°: $m = -1$ means $\theta = 135^\circ$, not -45°.

Example: Line Using Point-Slope Form

The equation of a line passes through $(3, -2)$ with slope $m = \frac{1}{2}$ is

$$y + 2 = \tfrac{1}{2}(x - 3) \implies x - 2y - 7 = 0.$$

The ***x*-intercept** of a line is the point $(a, 0)$ where it crosses the x-axis (set $y = 0$); the ***y*-intercept** is $(0, b)$ (set $x = 0$). For general form $Ax + By + C = 0$: x-intercept $\left(-\frac{C}{A}, 0\right)$ if $A \neq 0$, y-intercept $\left(0, -\frac{C}{B}\right)$

if $B \neq 0$. The **slope-intercept form** is $y = mx + b$ (m the slope, b the y-intercept); this is the general form with $A = m$, $B = -1$, $C = b$. The **intercept form** (line not vertical or horizontal, both intercepts nonzero) is

$$\frac{x}{a} + \frac{y}{b} = 1,$$

where a is the x-intercept and b the y-intercept; general form with $A = \frac{1}{a}$, $B = \frac{1}{b}$, $C = -1$.

Common mistake. **When the intercept form fails.** A line through the origin has $a = b = 0$, and a line parallel to an axis has only one intercept, so $\frac{x}{a} + \frac{y}{b} = 1$ cannot describe them. Likewise $y = mx + b$ cannot describe vertical lines. The general form $Ax + By + C = 0$ describes *every* line, which is why it is the safest starting point.

To find the line through two points (x_1, y_1), (x_2, y_2) with $x_1 \neq x_2$: compute $m = \frac{y_2 - y_1}{x_2 - x_1}$ and use point-slope with either point. (If $x_1 = x_2$ the line is vertical, $x = x_1$.) A direction vector is $\mathbf{d} = (x_2 - x_1, y_2 - y_1)$ and a normal vector is $\mathbf{n} = (y_2 - y_1,\ -(x_2 - x_1))$.

Example: Line Through Two Points

To find the equation of the line through $(1, 2)$ and $(4, -1)$, first compute the slope: $m = \dfrac{-1 - 2}{4 - 1} = -1$. Using the point-slope form with $(1, 2)$, we get $y - 2 = -(x - 1) \implies y = -x + 3$.

Definition

The **normal (Hesse) form** of a line writes its normal vector as a *unit* vector $\hat{\mathbf{n}} = (\cos\alpha, \sin\alpha)$, where α is the angle of $\hat{\mathbf{n}}$ measured from the positive x-axis. Dividing $Ax + By + C = 0$ by $\pm\sqrt{A^2 + B^2}$ gives

$$x\cos\alpha + y\sin\alpha = p, \qquad \cos\alpha = \frac{\pm A}{\sqrt{A^2 + B^2}}, \quad \sin\alpha = \frac{\pm B}{\sqrt{A^2 + B^2}},$$

where $p \geq 0$ is the perpendicular distance from the origin to the line (the sign is chosen so that $p \geq 0$). For a line through (x_0, y_0) with unit normal $\hat{\mathbf{n}}$, $p = x_0\cos\alpha + y_0\sin\alpha$.

Intuition. **Why p is the distance from the origin.** The expression $x\cos\alpha + y\sin\alpha = (x, y) \cdot \hat{\mathbf{n}}$ is the length of the shadow (projection) of the point (x, y) onto the unit normal direction. Every point of the line has the *same* shadow length p along $\hat{\mathbf{n}}$ (that is what makes the line perpendicular to $\hat{\mathbf{n}}$), so p is exactly how far the line lies from the origin, measured along its normal.

Example: Normal Equation of a Line

A line passes through $(2, 3)$, and its unit normal vector makes an angle of $\alpha = \frac{\pi}{4}$ with the positive x-axis. Since $\cos\frac{\pi}{4} = \sin\frac{\pi}{4} = \frac{\sqrt{2}}{2}$, the distance from the origin is

$$p = 2 \cdot \frac{\sqrt{2}}{2} + 3 \cdot \frac{\sqrt{2}}{2} = \frac{5\sqrt{2}}{2}.$$

Thus, the normal equation is $\frac{\sqrt{2}}{2}x + \frac{\sqrt{2}}{2}y = \frac{5\sqrt{2}}{2}$, which simplifies to $x + y = 5$.

5.2.1 One Line, Seven Descriptions

The line $3x+4y-12=0$ meets the axes at $(4,0)$ and $(0,3)$, has slope $-\frac{3}{4}$, normal $\mathbf{n}=(3,4)$ and direction $\mathbf{d}=(4,-3)$. Figure 5.4 shows all of these on one picture, and Table 5.1 lists the equation in every form of this section.

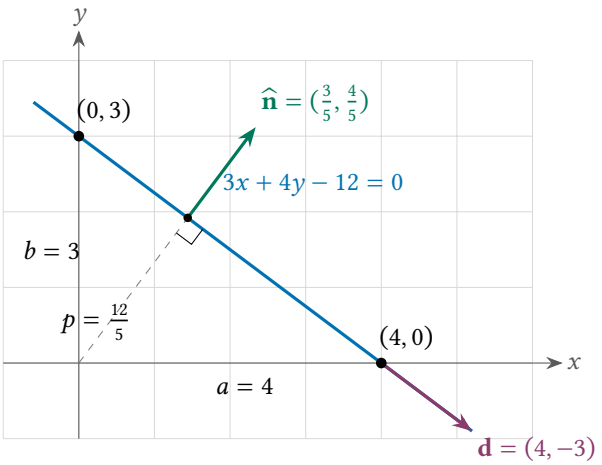


Figure 5.4: The line $3x+4y-12=0$. Its intercepts are $a=4$, $b=3$ and its slope is $-\frac{b}{a}=-\frac{3}{4}$. The unit normal $(\frac{3}{5},\frac{4}{5})$ drawn at the foot of the perpendicular from the origin shows the distance $p=\frac{12}{5}$. The direction vector $(4,-3)$ runs along the line; $\mathbf{n}\cdot\mathbf{d}=12-12=0$.

Form	Equation of $3x+4y-12=0$	Fails or needs care for
General	$3x+4y-12=0$	nothing; not unique (multiply by k)
Slope-intercept	$y=-\frac{3}{4}x+3$	vertical lines
Point-slope (through $(4,0)$)	$y-0=-\frac{3}{4}(x-4)$	vertical lines
Intercept	$\frac{x}{4}+\frac{y}{3}=1$	lines through O, axis-parallel lines
Two-point	through $(4,0)$ and $(0,3)$: $m=\frac{3-0}{0-4}=-\frac{3}{4}$	$x_1=x_2$
Canonical	$\frac{x-4}{4}=\frac{y}{-3}$	$d_1=0$ or $d_2=0$
Parametric	$x=4+4t,\ y=-3t$	not unique (choice of point, speed)
Normal (Hesse)	$\frac{3}{5}x+\frac{4}{5}y=\frac{12}{5}$	sign of $\hat{\mathbf{n}}$ fixed by $p\geq 0$

Table 5.1: The same line in every form of this chapter, with the cases where each form breaks down.

5.3 Intersection of Two Lines

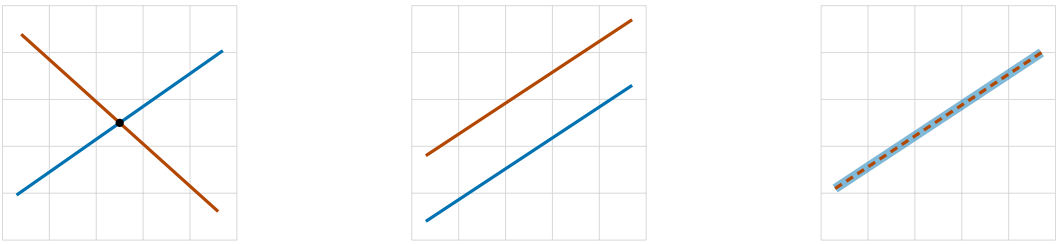
Two lines $A_1x+B_1y+C_1=0$ and $A_2x+B_2y+C_2=0$ meet where both equations hold, i.e. at the solution of a 2×2 linear system. With

$$D=\begin{vmatrix}A_1 & B_1\\ A_2 & B_2\end{vmatrix}=A_1B_2-A_2B_1$$

(the determinant of the two normal vectors), therefore: If $D\neq 0$ the lines meet in exactly one point

$$x=\frac{B_1C_2-B_2C_1}{D},\qquad y=\frac{C_1A_2-C_2A_1}{D}.$$

If $D = 0$ the normals are parallel, so the lines are **parallel**: either **coincident** (all three of A, B, C proportional: $A_1 : B_1 : C_1 = A_2 : B_2 : C_2$) or **distinct and disjoint** (Figure 5.5).



$D \neq 0$: one point $D = 0$: parallel, no point $D = 0$: coincident

Figure 5.5: The three possibilities for two lines in the plane. Which one occurs is decided by the single number $D = A_1B_2 - A_2B_1$.

Example: Intersection of Two Lines

To find the point of intersection of the lines $x - 2y - 7 = 0$ and $2x + y - 3 = 0$, we compute $D = 1 \cdot 1 - 2 \cdot (-2) = 5 \neq 0$, so there is one point:

$$x = \frac{(-2)(-3) - (1)(-7)}{5} = \frac{13}{5}, \qquad y = \frac{(-7)(2) - (-3)(1)}{5} = -\frac{11}{5}.$$

Intuition. D **measures how "non-parallel" the lines are.** In fact $D = |\mathbf{n}_1||\mathbf{n}_2| \sin\varphi$, where φ is the angle between the normals. When the lines are almost parallel D is tiny, and a tiny shift of one line moves the intersection point a long way along the other (think of two nearly parallel train tracks crossing very far away). This is the geometric meaning of an ill-conditioned 2×2 system.

Common mistake. **Intersecting parametric lines.** For two lines $\mathbf{r} = \mathbf{p} + t\mathbf{d}_1$ and $\mathbf{r} = \mathbf{q} + s\mathbf{d}_2$ use *different* parameters t and s. Using the same letter for both silently asks that the two "particles" arrive at the intersection at the same time, which is not required, and gives a wrong answer or no answer. Always check $D \neq 0$ first: if $D = 0$ the system has either no solution or infinitely many.

5.4 Distance From a Point to a Line

Mathematically, a distance (or *metric*) d is an objective referee for measuring space that obeys four common-sense rules: (1) distances are never negative; (2) the distance between two points is zero if and only if they are the exact same point; (3) distance is symmetric (from A to B equals from B to A); and (4) the triangle inequality ($d(x, z) \leq d(x, y) + d(y, z)$), meaning a straight line is always the shortest path.

Distance in $\mathbb{R}^n$. Extending the Pythagorean theorem to n dimensions, the distance between two points $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$ is the root of the sum of squared coordinate differences:

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}.$$

Point-to-Line Distance (The Shortest Path). Among all paths from a point to a line, the perpendicular drop is always the shortest. Any other path forms a right triangle where your chosen path is the hypotenuse, making it strictly longer than the straight perpendicular line.

The **distance** from a point (x_0, y_0) to a line $Ax + By + C = 0$ is the length of the perpendicular segment from the point to the line:

$$d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}.$$

Without the absolute value, $\dfrac{Ax_0 + By_0 + C}{\sqrt{A^2 + B^2}}$ is the **signed distance**: it is positive on the side of the line towards which $\mathbf{n} = (A, B)$ points and negative on the other side, and 0 exactly on the line.

> **Intuition. Why the denominator $\sqrt{A^2 + B^2}$?** The number $Ax_0 + By_0 + C$ is the "height" of the tilted plane of the introduction, but the plane may be steep or gentle, depending on $|\mathbf{n}|$. Doubling the equation ($2Ax + 2By + 2C = 0$) describes the *same* line yet doubles the height, so the raw height cannot be a distance. Dividing by $|\mathbf{n}| = \sqrt{A^2 + B^2}$ converts "height" into "length" (like converting units), and the result no longer depends on how the equation was scaled.

Where does this formula come from? Write $\mathbf{n} = (A, B)$ and pick any point $Q = (x_1, y_1)$ on the line, so $Ax_1 + By_1 + C = 0$. The distance from $P = (x_0, y_0)$ to the line is the length of the projection of $\overrightarrow{QP}$ onto the unit normal $\hat{\mathbf{n}} = \mathbf{n}/|\mathbf{n}|$:

$$d = \left|\overrightarrow{QP} \cdot \hat{\mathbf{n}}\right| = \frac{|A(x_0 - x_1) + B(y_0 - y_1)|}{\sqrt{A^2 + B^2}} = \frac{|Ax_0 + By_0 - (Ax_1 + By_1)|}{\sqrt{A^2 + B^2}} = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}},$$

using $Ax_1 + By_1 = -C$ in the last step.

> **Common mistake.** Write the line as $Ax + By + C = 0$ first. For $y = 2x + 1$ rewrite $2x - y + 1 = 0$, so $A = 2$, $B = -1$, $C = 1$ (not $C = -1$, and not $B = 1$).

Distance between parallel lines. Two lines $Ax + By + C_1 = 0$ and $Ax + By + C_2 = 0$ share the normal vector $\mathbf{n} = (A, B)$, so they are parallel, and the (perpendicular) distance between them is

$$d = \frac{|C_2 - C_1|}{\sqrt{A^2 + B^2}}.$$

(Pick any point on the first line and apply the point-to-line formula to the second; the $Ax_0 + By_0$ part cancels with $-C_1$, leaving $|C_2 - C_1|$.)

> **Common mistake. Parallel lines: the coefficients A, B must be identical.** The formula $\frac{|C_2 - C_1|}{\sqrt{A^2 + B^2}}$ is valid only after both equations are scaled so that they have the *same* (A, B). For $2x - 3y + 4 = 0$ and $4x - 6y - 12 = 0$ first divide the second by 2: $2x - 3y - 6 = 0$, and then $d = \frac{|-6-4|}{\sqrt{13}} = \frac{10}{\sqrt{13}}$. Using $C_1 = 4$, $C_2 = -12$ directly would give a wrong answer.

Example: Distance from a Point to a Line

The distance from $(2, 1)$ to the line $3x - 4y + 5 = 0$ (Figure 5.6).is

$$d = \frac{|3 \cdot 2 + (-4) \cdot 1 + 5|}{\sqrt{3^2 + (-4)^2}} = \frac{|6 - 4 + 5|}{\sqrt{25}} = \frac{7}{5}.$$

The foot of the perpendicular is $F = P - \dfrac{3 \cdot 2 - 4 \cdot 1 + 5}{3^2 + (-4)^2}(3, -4) = (2, 1) - \frac{7}{25}(3, -4) = \left(\frac{29}{25}, \frac{53}{25}\right)$.

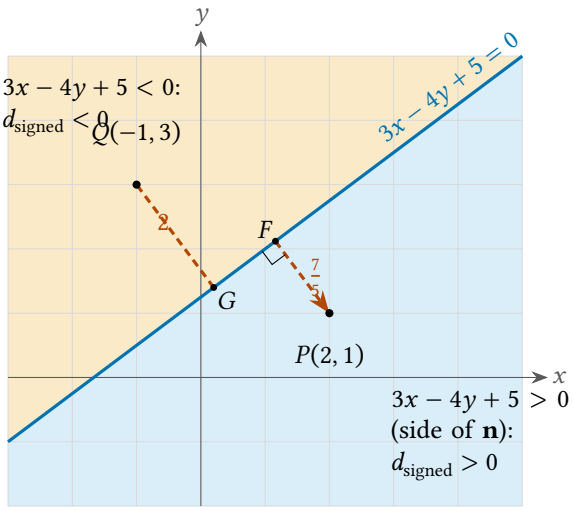


Figure 5.6: Distance and side. For $P(2,1)$: $3\cdot 2 - 4\cdot 1 + 5 = 7 > 0$, so $d = \frac{7}{5}$ on the blue side (the side the normal $\mathbf{n} = (3,-4)$ points to). For $Q(-1,3)$: $-3 - 12 + 5 = -10 < 0$, so Q lies on the other side, at distance $\frac{10}{5} = 2$. F and G are the feet of the perpendiculars (dashed segments are parallel to $\mathbf{n}$).

5.5 The Angle Between Two Lines

The **angle between two lines** is the acute angle α formed where they cross (and $\alpha = 90^\circ$ if they are perpendicular, $\alpha = 0$ if they are parallel). For lines $A_1x + B_1y + C_1 = 0$ and $A_2x + B_2y + C_2 = 0$, there are three equivalent ways to find it.

A) Using normal vectors. $\mathbf{n}_1 = (A_1, B_1)$ and $\mathbf{n}_2 = (A_2, B_2)$ are perpendicular to the two lines, so rotating both lines by the same angle rotates both normals by that angle: the angle α between the lines equals the angle between $\mathbf{n}_1, \mathbf{n}_2$, or its supplement. Either way $\cos\alpha \geq 0$, so we take the absolute value:

$$\cos\alpha = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1||\mathbf{n}_2|} = \frac{|A_1A_2 + B_1B_2|}{\sqrt{A_1^2 + B_1^2}\sqrt{A_2^2 + B_2^2}}.$$

B) Using direction vectors. Direction vectors are $\mathbf{d}_1 = (-B_1, A_1)$, $\mathbf{d}_2 = (-B_2, A_2)$ (each is $\mathbf{n}$ rotated 90°, so it is automatically perpendicular to $\mathbf{n}$). The same reasoning gives

$$\cos\alpha = \frac{|\mathbf{d}_1 \cdot \mathbf{d}_2|}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{|B_1B_2 + A_1A_2|}{\sqrt{A_1^2 + B_1^2}\sqrt{A_2^2 + B_2^2}},$$

exactly the same formula as method A, unsurprisingly, since $\mathbf{d}_i$ is just $\mathbf{n}_i$ rotated.

C) Using slopes. If neither line is vertical, with slopes $m_1 = -A_1/B_1$, $m_2 = -A_2/B_2$,

$$\tan\alpha = \left|\frac{m_2 - m_1}{1 + m_1m_2}\right|.$$

Intuition. Where the slope formula comes from. The lines have inclination angles θ_1, θ_2 with $\tan\theta_i = m_i$. The angle between them is $\theta_2 - \theta_1$ (up to sign), and the tangent subtraction identity gives

$$\tan(\theta_2 - \theta_1) = \frac{\tan\theta_2 - \tan\theta_1}{1 + \tan\theta_1\tan\theta_2} = \frac{m_2 - m_1}{1 + m_1m_2}.$$

The absolute value selects the acute angle. When $1 + m_1m_2 = 0$ the denominator vanishes and $\tan\alpha$ is infinite: $\alpha = 90^\circ$, which is exactly the perpendicularity condition $m_1m_2 = -1$.

Example: Angle Between Two Lines, Using Normals

To find the acute angle between $L_1 : 3x - 4y + 1 = 0$ and $L_2 : 2x + 5y - 3 = 0$ (Figure 5.7), we use their normal vectors $\mathbf{n}_1 = (3, -4)$ and $\mathbf{n}_2 = (2, 5)$. Since $\mathbf{n}_1 \cdot \mathbf{n}_2 = 6 - 20 = -14$, $|\mathbf{n}_1| = 5$, and $|\mathbf{n}_2| = \sqrt{29}$, we have

$$\cos\alpha = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1||\mathbf{n}_2|} = \frac{|-14|}{5\sqrt{29}} \approx 0.520 \implies \alpha \approx 58.7^\circ.$$

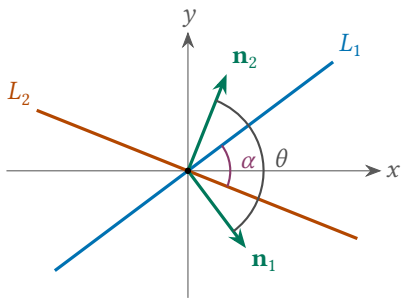
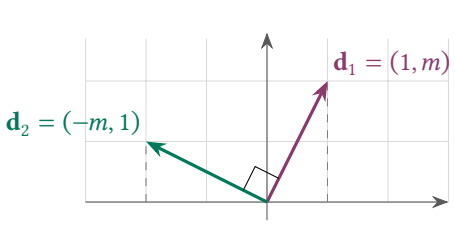
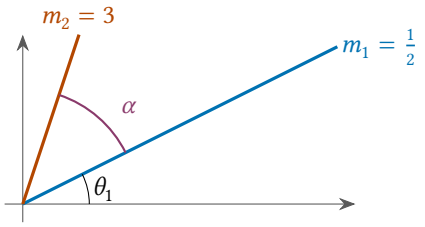


Figure 5.7: Lines $L_1 : 3x - 4y = 0$ and $L_2 : 2x + 5y = 0$ (shifted to pass through the origin; the angle does not depend on where they cross) and their normals $\mathbf{n}_1 = (3, -4)$, $\mathbf{n}_2 = (2, 5)$. The acute angle between the lines is $\alpha \approx 58.7^\circ$; the angle between the normals is $\theta \approx 121.3^\circ = 180^\circ - \alpha$. Using $|\cos\theta|$ always produces the acute angle.

Intuition. Why do perpendicular slopes multiply to -1? Take a line with direction $(1, m)$. Rotating a vector (a, b) by 90° counter-clockwise gives $(-b, a)$, so the perpendicular direction is $(-m, 1)$, with slope $\frac{1}{-m} = -\frac{1}{m}$. The product of the two slopes is $m \cdot (-\frac{1}{m}) = -1$. The rotation swaps the roles of rise and run and reverses one sign (Figure 5.8a).



(a) Rotating $(1, m)$ by 90° gives $(-m, 1)$



(b) Angle between lines with $m_1 = 0.5$ and $m_2 = 3$

Figure 5.8: Geometric interpretation of perpendicular slopes and angles between lines.

Common mistake. Parallel, perpendicular, and when slopes are not enough.

- *Parallel:* $m_1 = m_2$ (equivalently $A_1B_2 = A_2B_1$, i.e. $D = 0$). *Perpendicular:* $m_1m_2 = -1$ (equivalently $A_1A_2 + B_1B_2 = 0$). In the general form the perpendicularity test needs *no* division and works for vertical lines too.
- A horizontal line is perpendicular to a vertical line, but $m_1m_2 = -1$ cannot test this because the vertical line has no slope.
- Always take $|\cdot|$ in the angle formulas. A negative value of $\tan\alpha$ or $\cos\alpha$ just means you found the obtuse angle $180^\circ - \alpha$ between the *vectors*; the angle between the *lines* is the acute one.
- $m_1m_2 = 1$ does *not* mean perpendicular (the lines $y = 3x$ and $y = \frac{1}{3}x$ are mirror images in $y = x$, at the angle $\arctan\frac{3-1/3}{1+1} = \arctan\frac{4}{3} \approx 53.1^\circ$).

5.6 Reflecting a Point and a Line

A **reflection** across a line ℓ (called the *mirror*) is a geometric transformation that maps every point in the plane to a corresponding image on the opposite side of ℓ. Think of it as placing an ideal flat mirror along the line: every point is flipped directly across the axis to maintain an exact symmetric counterpart.

Definition

The **image** P' of a point P reflected across a line ℓ (the mirror) is the point such that ℓ is the perpendicular bisector of PP'. If $\ell : Ax+By+C=0$ has normal $\mathbf{n}=(A,B)$ and $d = Ax_0+By_0+C$ for $P=(x_0,y_0)$, then treating P and P' as position vectors $\mathbf{p}=(x_0,y_0)$ and $\mathbf{p}'$,

$$\mathbf{p}' = \mathbf{p} - \frac{2d}{A^2+B^2}\,\mathbf{n}.$$

Intuition. **"Walk to the mirror, then walk the same distance again."** The foot of the perpendicular from P to ℓ is $F = \mathbf{p} - \dfrac{d}{A^2+B^2}\mathbf{n}$ (this is the distance formula *without* the absolute value: it moves P along $\mathbf{n}$ towards the line). Since F is the midpoint of PP', we go from P to F and then the *same* step again: $\mathbf{p}' = 2F - \mathbf{p} = \mathbf{p} - \dfrac{2d}{A^2+B^2}\mathbf{n}$. That is the whole origin of the factor 2.

Problem: Image of a Point Across a Line

Reflect $P(3,1)$ across the line $2x-y+4=0$, Figure 7.22.

Solution. $\mathbf{n}=(2,-1)$, $A^2+B^2=5$, $d = 2\cdot 3 - 1\cdot 1 + 4 = 9$:

$$P' = (3,1) - \frac{2\cdot 9}{5}(2,-1) = (3,1) - \left(\tfrac{36}{5},-\tfrac{18}{5}\right) = \left(-\tfrac{21}{5},\tfrac{23}{5}\right).$$

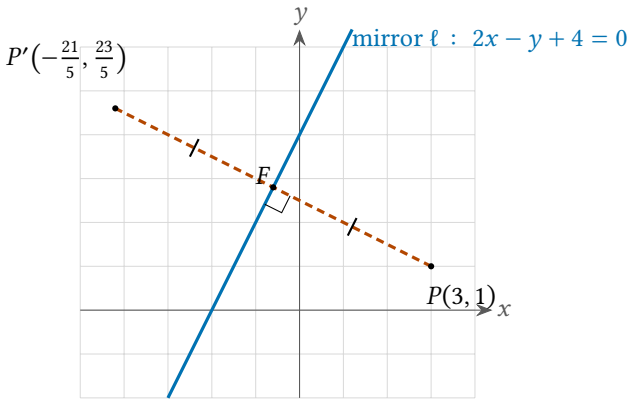


Figure 5.9: Reflecting $P(3,1)$ across the mirror $2x-y+4=0$: the foot $F=(-\frac{3}{5},\frac{14}{5})$ is the midpoint of PP' (the two tick marks show equal lengths), and PP' is perpendicular to the mirror. This is the worked example below.

Common mistake. **Keep the sign of d.** In the reflection formula $d = Ax_0+By_0+C$ is the *signed* value; putting an absolute value there sends P' to the wrong side (to P's own side, twice as far from the mirror). Also, a reflection is not a rotation: it reverses orientation, and it fixes every point of the mirror (for those points $d=0$).

5.6.1 Reflecting a Whole Line

A reflection preserves distances and angles and maps lines to lines. Let L be a line and ℓ the mirror.

- If L and ℓ **intersect** at a point X, then X is on the mirror, so X is fixed, and the image L' passes through X and makes the *same* angle with ℓ on the other side. To find L' reflect *one other* point of L.
- If $L \perp \ell$, then $L' = L$: the line is mapped onto itself (each point of L goes to the other side of ℓ along L). This holds for **every** mirror, not only for special ones.
- If $L \parallel \ell$, then L' is the parallel line on the other side of ℓ, at the same distance.

In all cases the practical method is: *reflect two points of L*, then draw the line through the two images.

Problem: Image of a Line

Reflect the line $L:\ 3x + y - 5 = 0$ across the mirror $\ell:\ x - y + 1 = 0$ (Figure 5.10).

Solution. Two points of L: $(0, 5)$ and $(1, 2)$. For $(1, 2)$: $d = 1 - 2 + 1 = 0$, so it lies on the mirror (it is $L \cap \ell$) and is fixed. For $(0, 5)$: $d = 0 - 5 + 1 = -4$, $A^2 + B^2 = 2$, so

$$(0, 5) - \frac{2(-4)}{2}(1, -1) = (0, 5) + 4(1, -1) = (4, 1).$$

The image passes through $(1, 2)$ and $(4, 1)$, slope $\frac{1-2}{4-1} = -\frac{1}{3}$: $y - 2 = -\frac{1}{3}(x - 1)$, i.e. $\boxed{x + 3y - 7 = 0}$.

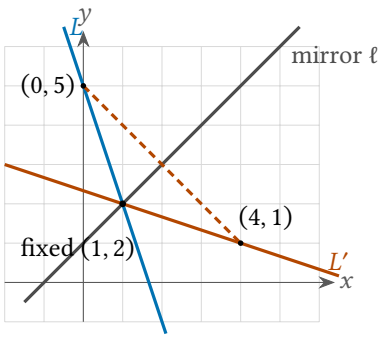


Figure 5.10: The line $L:\ 3x + y - 5 = 0$ and its image L' in the mirror $x - y + 1 = 0$. They meet on the mirror at the fixed point $(1, 2)$; the dashed segment joins $(0, 5)$ to its image $(4, 1)$ and is perpendicular to the mirror.

5.7 Equidistant Points: Perpendicular and Angle Bisectors

Equidistant from two points. The set of points equally far from A and B is the **perpendicular bisector** of AB: the line through the midpoint of AB with normal vector $\overrightarrow{AB}$.

Equidistant from two lines. Points (x, y) at equal distance from L_1 and L_2 satisfy

$$\frac{|A_1x + B_1y + C_1|}{\sqrt{A_1^2 + B_1^2}} = \frac{|A_2x + B_2y + C_2|}{\sqrt{A_2^2 + B_2^2}}.$$

Removing the absolute values gives *two* lines, the **angle bisectors**:

$$\frac{A_1x + B_1y + C_1}{\sqrt{A_1^2 + B_1^2}} = \pm\frac{A_2x + B_2y + C_2}{\sqrt{A_2^2 + B_2^2}}.$$

The two bisectors are perpendicular and pass through $L_1 \cap L_2$ (one bisects the acute angle, the other the obtuse). If $L_1 \parallel L_2$ and both equations are written with the same normal vector, the "+" case has no solution and the "−" case is the *midline* between them.

Problem: Angle Bisectors

Find the bisectors of $L_1 : 3x - 4y + 5 = 0$ and $L_2 : 4x - 3y - 5 = 0$ (Figure 5.11).

Solution. Both have $\sqrt{A^2 + B^2} = 5$, so the factors cancel:

$$3x - 4y + 5 = +(4x - 3y - 5) \implies x + y - 10 = 0,$$
$$3x - 4y + 5 = -(4x - 3y - 5) \implies 7x - 7y = 0 \iff x - y = 0.$$

The lines meet at $(5, 5)$, which lies on both bisectors; the bisectors are perpendicular (normals $(1, 1)$ and $(1, -1)$). As a check, $T(8, 8)$ lies on $x - y = 0$ and is at distance $\frac{|24-32+5|}{5} = \frac{3}{5}$ from L_1 and $\frac{|32-24-5|}{5} = \frac{3}{5}$ from L_2.

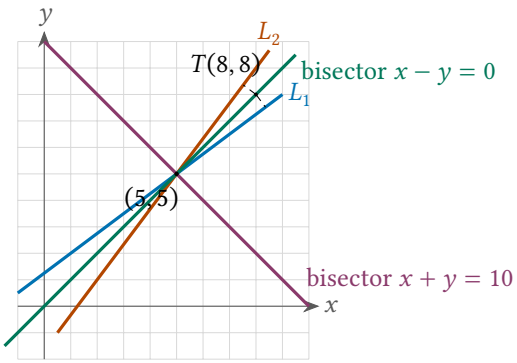


Figure 5.11: The two lines $3x - 4y + 5 = 0$ (blue) and $4x - 3y - 5 = 0$ (red) and their angle bisectors. Every point of a bisector is equally far from both lines; the short dashed segments from $T(8, 8)$ are the two equal perpendicular distances $\frac{3}{5}$.

Common mistake. Which sign gives which bisector? The "+" and "−" cases are *not* labelled "acute" and "obtuse" in advance: it depends on how the two equations are written (multiplying an equation by -1 swaps the roles). Use the sign of $\mathbf{n}_1 \cdot \mathbf{n}_2$: if $\mathbf{n}_1 \cdot \mathbf{n}_2 > 0$, the "−" equation is the bisector of the *acute* angle between the lines; if $\mathbf{n}_1 \cdot \mathbf{n}_2 < 0$, it is the "+" equation. In the example $\mathbf{n}_1 \cdot \mathbf{n}_2 = (3, -4) \cdot (4, -3) = 24 > 0$, so the acute bisector is $x - y = 0$, which indeed has slope 1, between the slopes $\frac{3}{4}$ and $\frac{4}{3}$.

5.8 Straight Lines in Physics and Engineering

A straight line is the simplest model of how one quantity depends on another. In physics and engineering it describes calibration curves, motion at constant velocity, the path of a light ray, the grade of a road, and the safety margin between a robot and an obstacle. Each application below turns a feature of the line (its slope, its intercept, its normal vector, its distance from a point) into a physical statement.

5.8.1 Linear Models and Calibration

If an output y depends linearly on an input x, then $y = mx + b$ and the two numbers have physical meaning: the **slope** m is the *sensitivity* (change of output per unit input) and the **intercept** b is the *offset* (output when the input is zero). Many laws are of this form: Hooke's law $F = kx$ (slope k, through the origin), Ohm's law $V = RI$ (slope R), and constant-velocity motion $s = s_0 + vt$ (slope v, intercept s_0).

Problem: Calibrating a Temperature Sensor

A sensor is known to respond linearly (Figure 5.12). Two calibration measurements give 1.05 V at 20°C and 1.85 V at 100°C. Find the calibration line $V(T)$, and the temperature when the sensor reads 1.37 V.

Solution. Slope: $m = \dfrac{1.85 - 1.05}{100 - 20} = 0.01$ V/°C (a sensitivity of 10 mV per degree). Using $(20, 1.05)$: $V - 1.05 = 0.01(T - 20)$, so

$$V = 0.85 + 0.01\,T.$$

The intercept 0.85 V is the offset at 0°C. Inverting the line, $T = \dfrac{V - 0.85}{0.01}$, so for $V = 1.37$ V: $T = \dfrac{0.52}{0.01} = 52$°C

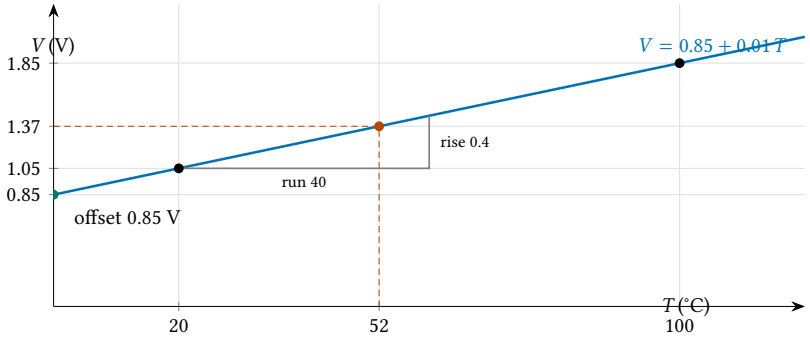


Figure 5.12: The calibration line of the sensor. The slope triangle (40 units run, 0.4 V rise) gives the sensitivity 0.01 V/°C; the green dot is the offset (the y-intercept); the red dashed path converts a reading of 1.37 V into 52°C by reading the line "backwards".

Common mistake. Slope has units, and a line is only trusted inside the calibrated range. The slope above is 0.01 *volts per degree*, not just "0.01". If you convert the temperature to kelvin or the voltage to millivolts the numbers change (10 mV/°C). Also, a line fitted between 20°C and 100°C says nothing reliable at 500°C: real sensors saturate or become non-linear, and extrapolating the line far beyond the data is a classic source of error.

5.8.2 Motion at Constant Velocity: Meeting and Overtaking

On a graph of position x against time t, motion at constant velocity is a straight line $x = x_0 + vt$: the **slope is the velocity**, the intercept is the starting position. Two objects meet exactly where their lines *intersect* (Section 5.3); parallel lines mean equal velocities and no meeting.

Example: Two Vehicles Approaching Each Other

Car A leaves town 0 at 60 km/h; car B leaves town 150 km away at the same moment, driving towards A at 40 km/h (Figure 5.13). Then $x_A = 60t$ and $x_B = 150 - 40t$. They meet when $60t = 150 - 40t$, i.e. at $t = 1.5$ h, at the position $x = 90$ km.

Overtaking. A truck at 50 km/h passes town 0 at $t = 0$; half an hour later a car leaves town 0 at 80 km/h. With $x_T = 50t$ and $x_C = 80(t - 0.5)$, the car catches the truck when $50t = 80t - 40$, i.e. $t = \frac{4}{3}$ h, at $x = \frac{200}{3} \approx 66.7$ km.

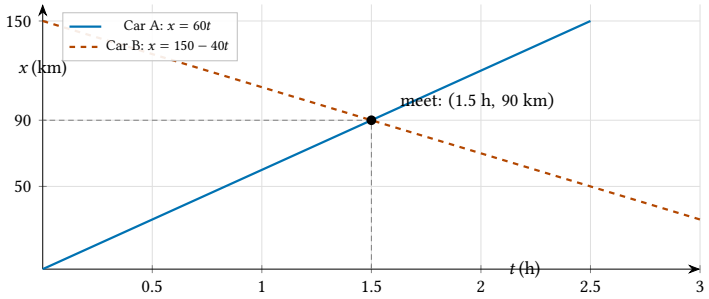


Figure 5.13: Position–time graph of two cars. The slopes are the velocities (+60 and −40 km/h); the intersection point is the meeting event. A steeper line means a faster vehicle, and a falling line means motion towards the origin.

5.8.3 Light and Mirrors: the Law of Reflection

A ray of light reflecting from a flat mirror leaves at the same angle at which it arrived. Equivalently, the reflected ray looks as if it came straight from the *mirror image* of the source. This turns a reflection problem into a straight-line problem: to send light from A to B via a mirror, reflect B to B' across the mirror and aim at B'; the ray strikes the mirror where the segment AB' crosses it. Among all paths from A to B touching the mirror this is also the *shortest*.

Example: Laser Bounce

A laser at $A(1,4)$ must hit a sensor at $B(2,5)$ after reflecting off the mirror $y = x$ (both points lie above it, Figure 5.14) . Reflecting B across $y = x$ gives $B'(5,2)$. The line AB' has slope $\frac{2-4}{5-1} = -\frac{1}{2}$:

$$y - 4 = -\frac{1}{2}(x - 1).$$

It meets the mirror $y = x$ at

$$x - 4 = -\frac{1}{2}x + \frac{1}{2},$$

i.e. $x = 3$: the reflection point is $R(3,3)$. The total path length is $|AB'| = \sqrt{16 + 4} = 2\sqrt{5} \approx 4.47$ which equals $|AR| + |RB| = \sqrt{4 + 1} + \sqrt{1 + 4} = 2\sqrt{5}$, so the reflected path is exactly as long as the straight segment to the image.

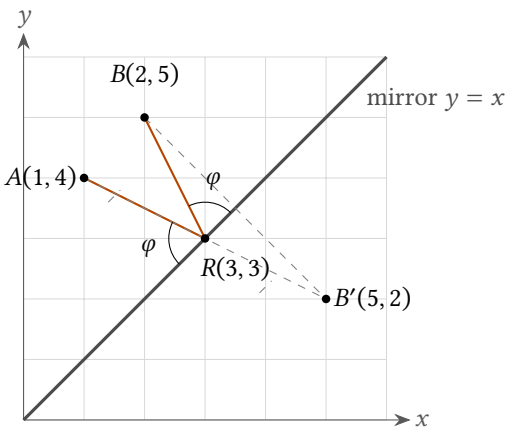


Figure 5.14: The image method. Reflect B to B' in the mirror; the straight segment AB' crosses the mirror at the reflection point R. At R the incoming and outgoing rays make equal angles $\varphi \approx 71.6^\circ$ with the mirror, and $|AR| + |RB| = |AB'|$ is the shortest path.

Intuition. Retroreflectors. Reflect a direction in two *perpendicular* mirrors and it comes out reversed: the composition of two reflections in perpendicular lines is a rotation by 180° (in coordinates: $(x, y) \mapsto (y, x) \mapsto (-x, -y)$ for the mirrors $y = x$ and $y = -x$). That is why a "corner" of two perpendicular mirrors sends light straight back to its source, the principle of bicycle and road-sign reflectors, and of the corner-cube reflectors left on the Moon for laser ranging.

5.8.4 Grades and Slopes in Civil Engineering

Roads, ramps and roofs are specified by their **grade**, the slope written as a percentage (Figure 5.15): grade $= 100\,m$ %. A road that rises 45 m over 600 m of horizontal distance has $m = 0.075$, a 7.5% grade, and an inclination angle $\theta = \arctan 0.075 \approx 4.3^\circ$. The sloping length is $\sqrt{600^2 + 45^2} \approx 601.7$ m, hardly longer than the horizontal distance.

Grade (%)	Slope m	Angle $\theta = \arctan m$
5	0.05	2.86°
8.33 (1 : 12 ramp)	0.0833	4.76°
10	0.10	5.71°
15	0.15	8.53°
100	1	45°

Table 5.2: Grade versus angle. A wheelchair ramp of ratio 1 : 12 has a grade of 8.33%; a 100% grade is a slope of 1, i.e. 45°, *not* a vertical wall.

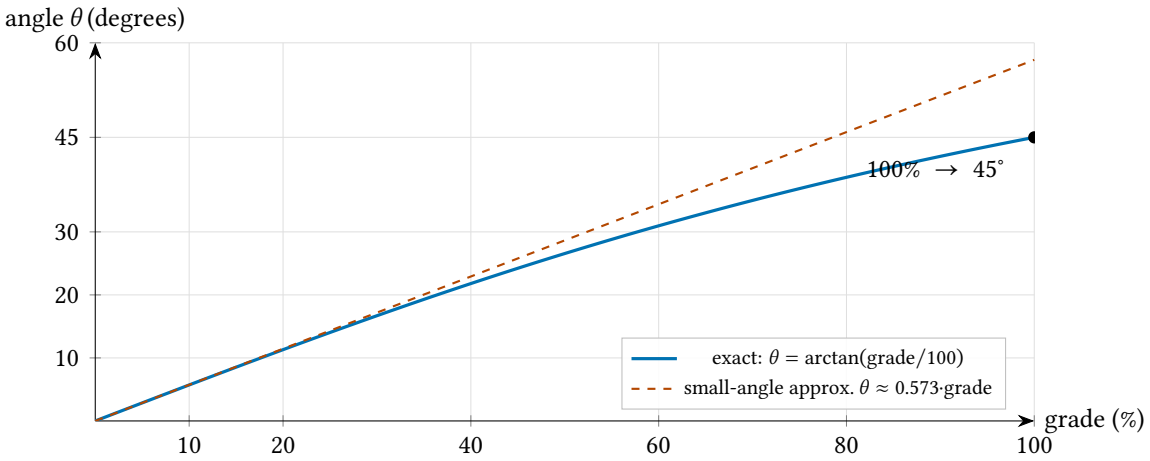


Figure 5.15: Angle of inclination as a function of grade. The relation is almost linear for small grades (up to about 10% the dashed line is within 1% of the truth) but bends over: doubling the grade does not double the angle.

5.8.5 Clearances: Distance to a Planned Path

Navigation, surveying and robotics constantly ask: *how far is this obstacle from my planned straight path, and on which side?* Both answers come from the signed-distance formula $\dfrac{Ax_0 + By_0 + C}{\sqrt{A^2 + B^2}}$: its size is the clearance and its sign is the side.

Example: Robot Path and Obstacles

A robot of radius 0.5 m follows the straight path $3x - 4y + 5 = 0$ (metres). Obstacles of radius 1.0 m are centred at $P(2, 1)$ and $Q(-1, 3)$. A collision is possible if the centre-to-path distance is less than $0.5 + 1.0 = 1.5$ m.

P: $\dfrac{3 \cdot 2 - 4 \cdot 1 + 5}{5} = +1.4$ m: it is on the positive side and $1.4 < 1.5$, so **collision** (the robot must be re-planned).

Q: $\dfrac{-3 - 12 + 5}{5} = -2$ m: it is on the other side and $2 > 1.5$, so it is **clear** (see Figure 5.16).

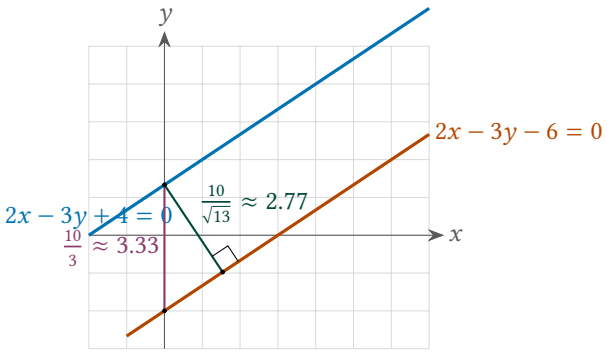


Figure 5.16: Two parallel pipes (or rails). The true spacing is the perpendicular distance $\frac{10}{\sqrt{13}} \approx 2.77$ (green); measuring straight down gives the larger vertical gap $\frac{|C_2 - C_1|}{|B|} = \frac{10}{3} \approx 3.33$ (purple), which overstates the spacing whenever the lines are tilted.

5.8.6 Fitting a Line to Data

When measured points (x_i, y_i) nearly lie on a line, the *least-squares* line chooses m, b to minimise the sum of the squared *vertical* gaps $y_i - (mx_i + b)$. If both coordinates are noisy it is more natural to minimise the squared *perpendicular* distances from the points to the line, which are given by exactly the point-to-line formula of this chapter ("orthogonal" or total least squares). The two lines differ more and more as the data become steeper.

5.9 Facts and Curiosities

1. Three numbers, two degrees of freedom. The general form uses three constants (A, B, C), but a line has only two degrees of freedom (a slope and a position), because (A, B, C) and (kA, kB, kC) describe the same line. The distance formula is built to be unchanged under this rescaling.

2. A 100% grade is 45°. Slope measures rise per run, not angle. The slopes $\frac{1}{2}, 1, 2$ correspond to 26.6°, 45° and 63.4°, so equal steps in slope are *not* equal steps in angle.

3. The normal vector is a gradient. For the function $f(x, y) = Ax + By + C$ the vector of partial derivatives is $\nabla f = (A, B) = \mathbf{n}$. In multivariable calculus you will see that the gradient is always perpendicular to the level curves of f, and the lines $Ax + By + C = c$ are exactly the level curves of the plane f.

4. Parallel lines meet "at infinity". In the intersection formulas the denominator is $D = A_1B_2 - A_2B_1$. As two lines approach parallelism $D \to 0$ and the intersection point runs off to infinity. Projective geometry makes this precise by adding a "point at infinity" for each direction, so that any two distinct lines meet in exactly one point.

5. Reflection in $y = x$ swaps slopes with their reciprocals. Reflecting the line $y = mx + b$ in $y = x$ swaps x and y, giving $x = my + b$, i.e. $y = \frac{x-b}{m}$ with slope $\frac{1}{m}$. This is why the graph of an inverse function is the mirror image of the original, and why the derivative of an inverse is the reciprocal of the original derivative (Chapter **??**).

6. Collinearity is an area test. Three points are collinear exactly when the triangle they span has zero area, i.e. when $\begin{vmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{vmatrix} = 0$. For $(0,1), (2,5), (3,7)$ this determinant is 0: all lie on $y = 2x + 1$.

7. The shortest path off a mirror. The image trick for reflections goes back to Heron of Alexandria (1st century). Nearly two millennia later Fermat showed that it is a special case of the principle that light takes the path of least time.

8. A name. The normal form $x\cos\alpha + y\sin\alpha = p$ is named after the German mathematician Otto Hesse (1811–1874).

Lines in Practice: Linear Approximation and Regression. In real-world engineering and physics, experimental data rarely lies on a strictly straight line due to measurement noise and physical imperfections. However, we can model the underlying trend by finding the best-fit line using linear regression (the method of least squares).

Given a set of data points (x_i, y_i), the optimal line $y = mx + b$ is chosen to minimize the sum of the squared vertical distances (residuals) between the data points and the line:

$$S(m, b) = \sum_{i=1}^{n} (mx_i + b - y_i)^2 .$$

This transforms abstract geometry into an essential tool for data analysis, sensor calibration, and empirical modeling across all engineering disciplines.

5.10 Worked Problems

Problem 1: Slopes and Equations of Lines

(1) Find the slope of the line through $(1, 4)$ and $(3, -2)$, and its equation in slope-intercept and general form. (2) For a line through $(2, 1)$ with $\tan\theta = -\frac{1}{3}$ (angle to the positive x-axis), find its equation in both forms. (3) Are the two lines parallel or perpendicular?

Solution.

1. $m_1 = \dfrac{-2-4}{3-1} = -3$. Using $(1,4)$: $y - 4 = -3(x-1) \implies y = -3x + 7$, i.e. $3x + y - 7 = 0$.
2. $m_2 = \tan\theta = -\frac{1}{3}$. Using $(2,1)$: $y - 1 = -\frac{1}{3}(x-2) \implies y = -\frac{1}{3}x + \frac{5}{3}$, i.e. $x + 3y - 5 = 0$.
3. Parallel? $m_1 \neq m_2$: no. Perpendicular? $m_1 m_2 = (-3)(-\frac{1}{3}) = 1 \neq -1$: no.

Problem 2: Intercept Form and Distance Between Parallel Lines

(1) Find the line through $(2, 3)$ and $(4, 1)$ in intercept form and general form. (2) Find the distance between $2x - 3y + 4 = 0$ and $2x - 3y - 6 = 0$.

Solution.

1. $m = \dfrac{1-3}{4-2} = -1$, so $y - 3 = -(x-2) \implies y = -x + 5$. Intercepts: $x = 5$ when $y = 0$, $y = 5$ when $x = 0$, so $a = b = 5$:
$$\frac{x}{5} + \frac{y}{5} = 1 \iff x + y - 5 = 0.$$
2. $A = 2, B = -3, C_1 = 4, C_2 = -6$: $d = \dfrac{|-6-4|}{\sqrt{4+9}} = \dfrac{10}{\sqrt{13}}$.

Problem 3: Slopes and Angle Between Lines

For $5x - 12y + 13 = 0$ and $12x + 5y - 10 = 0$: (1) find both slopes;
(2) find the acute angle using slopes; (3) verify with direction vectors; (4) are the lines parallel or perpendicular?

Solution.

1. $5x - 12y + 13 = 0 \Rightarrow y = \frac{5}{12}x + \frac{13}{12}$, so $m_1 = \frac{5}{12}$. $12x + 5y - 10 = 0 \Rightarrow y = -\frac{12}{5}x + 2$, so $m_2 = -\frac{12}{5}$.
2. $1 + m_1 m_2 = 1 + \frac{5}{12} \cdot \left(-\frac{12}{5}\right) = 1 - 1 = 0$: the formula's denominator vanishes, so $\tan\alpha \to \infty$ and $\alpha = 90^\circ$.
3. $\mathbf{d}_1 = (12, 5)$, $\mathbf{d}_2 = (-5, 12)$: $\mathbf{d}_1 \cdot \mathbf{d}_2 = -60 + 60 = 0$, confirming $\alpha = 90^\circ$.
4. $m_1 m_2 = \frac{5}{12} \cdot \left(-\frac{12}{5}\right) = -1$: the lines are perpendicular.

Problem 4: Parametric Equations and Point of Intersection

Line 1 passes through $(1, -2)$ with $\mathbf{d}_1 = (2, 3)$; Line 2 passes through $(-1, 4)$ with $\mathbf{d}_2 = (-1, 2)$. Find the parametric equations of both lines and their point of intersection.

Solution. $x = 1 + 2t,\ y = -2 + 3t$ and $x = -1 - s,\ y = 4 + 2s$ (different parameters t, s!). Setting equal:

$$1 + 2t = -1 - s \implies s = -2 - 2t; \qquad -2 + 3t = 4 + 2s = 4 - 4 - 4t \implies 7t = 2 \implies t = \frac{2}{7}.$$

Then $s = -\frac{18}{7}$, and the intersection is $x = 1 + \frac{4}{7} = \frac{11}{7}$, $y = -2 + \frac{6}{7} = -\frac{8}{7}$, i.e. $\left(\frac{11}{7}, -\frac{8}{7}\right)$ (check on Line 2:

$$x = -1 + \frac{18}{7} = \frac{11}{7}, \quad y = 4 - \frac{36}{7} = -\frac{8}{7}$$

☒).

Cross-check with the formula: the general forms are $3x - 2y - 7 = 0$ and $2x + y - 2 = 0$; $D = 3 \cdot 1 - 2 \cdot (-2) = 7$, $x = \frac{(-2)(-2)-(1)(-7)}{7} = \frac{11}{7}$, $y = \frac{(-7)(2)-(-2)(3)}{7} = -\frac{8}{7}$ ☒.

Problem 5: Height and Area of a Triangle

For $A(1, 2)$, $B(4, -1)$, $C(-2, 3)$, find the length of the altitude from A to line BC, and the area of the triangle.

Solution. $m_{BC} = \dfrac{3-(-1)}{-2-4} = -\frac{2}{3}$; using $B(4, -1)$: $y + 1 = -\frac{2}{3}(x - 4) \implies 2x + 3y - 5 = 0$. Then

$$h = \frac{|2 \cdot 1 + 3 \cdot 2 - 5|}{\sqrt{2^2 + 3^2}} = \frac{3}{\sqrt{13}}.$$

$|BC| = \sqrt{6^2 + 4^2} = 2\sqrt{13}$, so the area is $\frac{1}{2} \cdot 2\sqrt{13} \cdot \frac{3}{\sqrt{13}} = 3$.

(Check with the determinant: $\frac{1}{2}\begin{Vmatrix} 3 & -3 \\ -3 & 1 \end{Vmatrix} = \frac{1}{2}|3 - 9| = 3$ ☒, with $\overrightarrow{AB} = (3, -3)$, $\overrightarrow{AC} = (-3, 1)$.)

Problem 6: Line from a Normal Vector and Distance

A line has normal vector $\mathbf{n} = (3, -4)$ and distance $p = 2$ from the origin. Find it in normal and general form.

Solution. $|\mathbf{n}| = 5$, so $\cos\alpha = \frac{3}{5}$, $\sin\alpha = -\frac{4}{5}$:

$$\tfrac{3}{5}x - \tfrac{4}{5}y = 2 \iff 3x - 4y - 10 = 0.$$

Problem 7: Image of a Point Across a Line

Reflect $Q(-2, 5)$ across the mirror $x + y - 1 = 0$.

Solution. $\mathbf{n} = (1, 1)$, $A^2 + B^2 = 2$, $d = -2 + 5 - 1 = 2$:

$$Q' = (-2, 5) - \frac{2 \cdot 2}{2}(1, 1) = (-4, 3).$$

Check: the midpoint $(-3, 4)$ satisfies $-3 + 4 - 1 = 0$ ☒, and $QQ' = (-2, -2)$ is parallel to $\mathbf{n}$ ☒.

Problem 8: Line Equidistant from Two Points

Find the line equidistant from $A(2, 3)$ and $B(-1, -1)$.

Solution. This is the perpendicular bisector of AB. Midpoint $M = \left(\frac{1}{2}, 1\right)$; $m_{AB} = \dfrac{-1-3}{-1-2} = \frac{4}{3}$, so the bisector has slope $-\frac{3}{4}$:

$$y - 1 = -\tfrac{3}{4}\left(x - \tfrac{1}{2}\right) \implies y = -\tfrac{3}{4}x + \tfrac{11}{8} \implies 6x + 8y - 11 = 0.$$

(Check: the normal vector $\overrightarrow{AB} = (-3, -4)$ gives $6x + 8y = \text{const}$ up to scaling ☒.)

Problem 9: Locus with a Constant Sum of Distances to Two Parallel Lines

Find the locus of points whose distances to $3x - 4y + 5 = 0$ and $3x - 4y - 5 = 0$ sum to 4.

Solution. Write $u = 3x - 4y$; both lines have $\sqrt{A^2 + B^2} = 5$, so $d_1 = \frac{|u+5|}{5}$, $d_2 = \frac{|u-5|}{5}$, and $d_1 + d_2 = 4 \iff |u + 5| + |u - 5| = 20$. Splitting into cases (only the two below are consistent with their own assumption):

- If $u \geq 5$: $(u + 5) + (u - 5) = 2u = 20 \implies u = 10$ (≥ 5 ☒).
- If $u < -5$: $-(u + 5) - (u - 5) = -2u = 20 \implies u = -10$ (< -5 ☒).

(For $-5 \leq u \leq 5$ the sum is the constant 10, never 20.) The locus is the pair of lines $3x - 4y - 10 = 0$ and $3x - 4y + 10 = 0$.

Intuition. A shortcut for Problem 9. The two given lines are $\frac{|5-(-5)|}{5} = 2$ apart. For any point *between* them $d_1 + d_2 = 2$. A point outside the strip, at distance s from the nearer line, has $d_1 + d_2 = 2 + 2s$ (it is s farther from *both*). So a sum of 4 means $s = 1$: the locus is the pair of lines parallel to the given ones, one unit outside each. One unit of distance is $|\mathbf{n}| = 5$ units of $u = 3x - 4y$, so these lines are at $u = \pm(5 + 5) = \pm 10$, matching the case analysis.

Problem 10: Reflecting a Line Across Another Line

Find the image of the line $x + y - 2 = 0$ reflected across the mirror $x - y = 0$.

Solution. Reflection across $y = x$ swaps coordinates: $(x, y) \mapsto (y, x)$. A point (x_0, y_0) on $x+y-2 = 0$ satisfies $x_0 + y_0 = 2$; its image (y_0, x_0) satisfies $y_0 + x_0 = 2$ as well, the *same* equation. So the line is mapped onto itself:

$$x + y - 2 = 0 \quad \text{(the image is the same line).}$$

Intuition. Why does this line map to itself? The line $x + y = 2$ has slope -1 and the mirror $y = x$ has slope 1: the line is *perpendicular* to the mirror. A reflection moves every point along the perpendicular to the mirror through that point, i.e. here along the line $x + y = 2$ itself. Hence a line perpendicular to the mirror is always carried onto itself, for *any* mirror (not only for $y = x$). For a line that merely crosses the mirror at another angle the image is a *different* line through the same intersection point, as in the example of Figure 5.10; and for a line parallel to the mirror the image is the parallel line on the other side.

Problem 11: Cable Alignment (Structural Engineering)

A cable is anchored at $A(2, 3)$ and runs parallel to a wall $3x - 4y + 12 = 0$. It meets a boundary line $x + y - 6 = 0$ at B. Find the cable's equation and the coordinates of B.

Solution. The wall has slope $\frac{3}{4}$, so the cable does too: $y - 3 = \frac{3}{4}(x - 2) \implies 3x - 4y + 6 = 0$. Solving with $x + y = 6$ (so $y = 6 - x$):

$$3x - 4(6 - x) + 6 = 0 \implies 7x = 18 \implies x = \tfrac{18}{7}, \quad y = \tfrac{24}{7}, \qquad B = \left(\tfrac{18}{7}, \tfrac{24}{7}\right).$$

Problem 12: Family of Lines Through a Fixed Point

Show that every line of the form $(2x + y - 3) + k(x - y + 1) = 0$, as k ranges over $\mathbb{R}$, passes through the intersection of $2x + y - 3 = 0$ and $x - y + 1 = 0$, and find that common point.

Solution. Solving $2x + y = 3$ and $x - y = -1$: adding gives $3x = 2$, so $x = \frac{2}{3}$, $y = x + 1 = \frac{5}{3}$ (the formula with $D = -3$ gives the same). At this point both $2x + y - 3$ and $x - y + 1$ vanish, so *every* member of the family vanishes there too, for any k: every line of the family passes through $\left(\frac{2}{3}, \frac{5}{3}\right)$. (The family is the *pencil* of all lines through the intersection point, except the second line itself, which is reached only as $k \to \infty$.)

Problem 13: A Line Equidistant from Two Points, Through a Given Point

Find the lines through $P(1, 2)$ that are equally far from $A(0, 0)$ and $B(4, 0)$.

Solution. A line is equally far from A and B in exactly two ways: it is *parallel* to AB, or it *passes through the midpoint* $M(2,0)$ of AB. (i) Parallel to AB (the x-axis) through P: $y = 2$, which is at distance 2 from both. (ii) Through $P(1,2)$ and $M(2,0)$: slope -2, $2x + y - 4 = 0$; distances $\frac{|0+0-4|}{\sqrt{5}} = \frac{4}{\sqrt{5}}$ and $\frac{|8+0-4|}{\sqrt{5}} = \frac{4}{\sqrt{5}}$ ☒.

Problem 14: Collision Check

A robot of radius 0.5 m follows $3x - 4y + 5 = 0$. Which of the circular obstacles centred at $P(2,1)$ and $Q(-1,3)$, each of radius 1.0 m, can it hit?

Solution. The signed distances are $+\frac{7}{5} = 1.4$ m for P and $-\frac{10}{5} = -2$ m for Q. Collision requires a distance below $0.5 + 1.0 = 1.5$ m: P is hit ($1.4 < 1.5$), Q is not ($2 > 1.5$). The signs show that the obstacles are on opposite sides of the path.

5.11 Conceptual & Computational Exercises

5.11.1 Conceptual Questions

Answer by *thinking*, with little or no calculation; give a reason or a counterexample. Answers are in the answer key.

1. **True or false?**
   (a) Every line in the plane has a slope.
   (b) $2x - 4y + 6 = 0$ and $x - 2y + 3 = 0$ describe the same line.
   (c) The (unsigned) distance from a point to a line can be negative.
   (d) Two distinct lines in the plane always meet in exactly one point.
   (e) A line is determined by its slope alone.
   (f) Reflecting a line in a mirror always gives a line parallel to the original.
   (g) If $m_1 m_2 = -1$, the lines are perpendicular.
2. A line has direction vector $(3, -2)$. Give a normal vector and the slope.
3. Why does the distance formula contain $\sqrt{A^2 + B^2}$ in the denominator? What would go wrong without it?
4. What does the *sign* of $Ax_0 + By_0 + C$ tell you about the point (x_0, y_0)?
5. Two lines have slopes 2 and $\frac{1}{2}$. Are they perpendicular? What is the relation between them?
6. For which values of k are the lines $kx + 2y - 1 = 0$ and $2x + ky + 3 = 0$ (a) parallel, (b) perpendicular?
7. Why does the factor 2 appear in the reflection formula $P' = P - \frac{2d}{A^2+B^2}\mathbf{n}$?
8. Why can the intercept form $\frac{x}{a} + \frac{y}{b} = 1$ not describe a line through the origin?
9. Why must the equations have the same (A, B) before using $\frac{|C_2 - C_1|}{\sqrt{A^2+B^2}}$ for the distance between parallel lines?
10. What is the image of a line L under reflection in a mirror ℓ if $L \perp \ell$? If $L \parallel \ell$?
11. What happens to $\tan\alpha = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right|$ if one of the lines is vertical? How can you still find the angle?
12. Why are the two angle bisectors of two intersecting lines perpendicular to each other?
13. Find a line through $P(1,2)$ that is equally far from $A(0,0)$ and $B(4,0)$. (There are two.) Explain geometrically.

5.11.2 Computational Problems

Solve the following problems. Show all work and give exact answers unless otherwise specified.

1. Given $A(-2,3)$ and $B(4,-1)$, find the slope of line AB and its equation in slope-intercept and general form.
2. A line passes through $P(0,5)$ with $\tan\theta = 2$ (angle to the positive x-axis). Find its equation in slope-intercept and general form, and check whether it is parallel or perpendicular to $y = -0.5x + 3$.
3. Find the acute angle between $x + y - 1 = 0$ and $2x - y + 3 = 0$ using slopes, and verify using direction vectors.
4. Write the parametric equations of the line through $P(2,-4)$ with direction vector $\mathbf{d} = (3,-2)$, and find its intersection with the line $x = -1 + 2t$, $y = 3 - t$, if it exists.
5. For a triangle with vertices $A(1,3)$, $B(-3,0)$, $C(2,-2)$, find the length of the altitude from B to line AC, and the area of the triangle.
6. A line has normal vector $\mathbf{n} = (1,-5)$ and distance $p = 4$ from the origin. Find its equation in normal and general form.
7. Find the image of $P(-1,2)$ reflected across the line $2x + 3y - 6 = 0$.
8. A support cable passes through $P(5,2)$ and is parallel to a girder $4x - y + 8 = 0$. Find the cable's equation in general form and its intersection with the boundary line $x - 2y + 3 = 0$.
9. Find the locus of points whose distances to $2x - y + 4 = 0$ and $2x - y - 4 = 0$ sum to 6.
10. Find the image of the line $2x + y - 4 = 0$ reflected across the mirror $x - y - 1 = 0$.
11. Find the intersection of $3x - 2y + 4 = 0$ and $5x + y - 3 = 0$ using the determinant formula, and decide whether the lines are perpendicular.
12. Find the equations of the two angle bisectors of $3x + 4y - 7 = 0$ and $4x - 3y + 1 = 0$.
13. Show that $2x - y - 3 = 0$, $x + y - 6 = 0$ and $3x - 2y - 3 = 0$ are concurrent (use a determinant) and find the common point.
14. A sensor responds linearly: 0.50 V at 0°C and 2.00 V at 50°C. Find $V(T)$, its sensitivity in mV/°C, and the temperature when $V = 1.43$ V.
15. A truck passes town O at $t = 0$ at 50 km/h. A car leaves O at $t = 0.5$ h at 80 km/h in the same direction. When and where does the car catch the truck?
16. A laser at $A(1,5)$ must reach $B(6,2)$ after reflecting off the x-axis. Find the reflection point and the total path length.
17. A road rises 45 m over a horizontal distance of 600 m. Find its grade, its angle of inclination and its length. Is it steeper than a 7° slope?
18. A robot of radius 0.6 m follows the path $5x - 12y + 13 = 0$ (metres). An obstacle of radius 0.5 m is centred at $(3,-2)$. On which side of the path is it, how far is its centre from the path, and is there a collision?

6

Straight Lines in Space

Learning Objectives

After studying this chapter you should be able to:

- write the equation of a line in space in canonical (symmetric), parametric and vector form, from a point and a direction vector, or from two points;
- find the distance from a point to a line, and the foot of the perpendicular;
- classify a pair of lines as coincident, parallel, intersecting or skew, and find the angle between two lines;
- find the distance between two parallel lines, and between two skew lines;
- use lines to model uniform motion and closest approach, pipe clearance, triangulation, the field of a straight wire, surveying and tool paths.

A line in the plane could be pinned down by a single normal vector, because in $\mathbb{R}^2$ there is only one perpendicular direction at each point (up to sign). In $\mathbb{R}^3$ that is no longer true: infinitely many directions are perpendicular to a given line, they sweep out a whole plane. So a normal vector alone cannot describe a line in space – but a **direction vector**, the vector the line runs along, still works exactly as before. Every equation of a line in this chapter comes from this one idea: a point on the line, plus a vector along it.

6.1 Equations of a Line in Space

Definition: Direction Vector and Canonical Form

A **direction vector** $\mathbf{d} = (a, b, c)$ of a line in space is a nonzero vector parallel to the line. The **canonical (symmetric) form** of the line through (x_0, y_0, z_0) with direction vector $\mathbf{d} = (a, b, c)$, where a, b, c are all nonzero, is

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

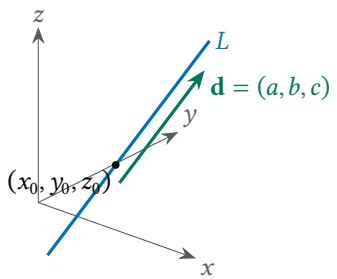


Figure 6.1: The line L through (x_0, y_0, z_0) with direction vector $\mathbf{d}$. Any nonzero scalar multiple of $\mathbf{d}$ describes the same line.

Example: Canonical Form

The line through $(2, -1, 3)$ with direction vector $\mathbf{d} = (1, -2, 4)$:

$$\frac{x-2}{1} = \frac{y+1}{-2} = \frac{z-3}{4}.$$

Common mistake. The canonical form $\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$ is really *two* equations (the line is the intersection of two planes); a single equation $Ax + By + Cz = D$ describes a plane, never a line. Watch the signs: in $\frac{x-2}{1} = \frac{y+1}{-2} = \frac{z-3}{4}$ the line passes through $(2,\ -1,\ 3)$ (the numerator $y+1$ means $y_0 = -1$), while the denominators $(1, -2, 4)$ are the direction vector.

Definition: Parametric Form

The **parametric form** of the line through (x_0, y_0, z_0) with direction vector $\mathbf{d} = (a, b, c)$ is

$$x = x_0 + at, \qquad y = y_0 + bt, \qquad z = z_0 + ct, \qquad t \in \mathbb{R}.$$

Unlike the canonical form, this allows any of a, b, c to be zero. In vector form, writing $\mathbf{r}_0 = (x_0, y_0, z_0)$, this is $\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{d}$.

Intuition. Think of $\mathbf{r}(t) = \mathbf{r}_0 + t\,\mathbf{d}$ as a recipe for a trip: start at the point $\mathbf{r}_0$ and keep walking in the direction $\mathbf{d}$. The number t plays the role of *time*: $t = 0$ is the start, $t = 1$ is one "step" $\mathbf{d}$ later, and negative t is where you came from. A straight line is the path of a body moving with constant velocity, which is why this chapter also describes uniform motion (Section 6.4).

Example: A Zero Component

The line through $(1, 0, 2)$ with $\mathbf{d} = (0, 3, -1)$: since $a = 0$, use

$$x = 1, \qquad \frac{y}{3} = \frac{z-2}{-1},$$

or, equivalently, the parametric form $x = 1,\ y = 3t,\ z = 2 - t$ (Figure 6.2).

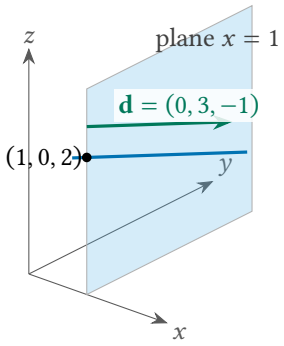


Figure 6.2: The line $x = 1,\ y = 3t,\ z = 2 - t$ of the example has $\mathbf{d} = (0, 3, -1)$. It never changes its x-coordinate, so it lies entirely in the plane $x = 1$ (shaded), which is parallel to the yz-plane. The fraction $\frac{x-x_0}{a}$ would divide by 0, so the equation $x = 1$ is stated on its own. (The arrow $\mathbf{d}$ is drawn slightly above the line so that both stay visible.)

A zero component

If, say, $a = 0$ in $\mathbf{d} = (a, b, c)$, the canonical form's first fraction $\frac{x-x_0}{a}$ is undefined – the line does not change its x-coordinate at all, so write $x = x_0$ as a separate equation and keep the canonical form only for the remaining two coordinates:

$$x = x_0, \qquad \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

Line Through Two Points: The line through (x_1, y_1, z_1) and (x_2, y_2, z_2) has direction vector $\mathbf{d} = (x_2 - x_1,\ y_2 - y_1,\ z_2 - z_1)$, giving the parametric form

$$x = x_1 + (x_2 - x_1)t, \quad y = y_1 + (y_2 - y_1)t, \quad z = z_1 + (z_2 - z_1)t,$$

and, if no component of $\mathbf{d}$ is zero, the canonical form

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}.$$

In vector form, $\mathbf{r}(t) = \mathbf{r}_1 + t(\mathbf{r}_2 - \mathbf{r}_1)$, where $\mathbf{r}_1, \mathbf{r}_2$ are the position vectors of the two points. (At $t = 0$ this gives the first point, at $t = 1$ the second, and $0 \le t \le 1$ traces the segment between them.)

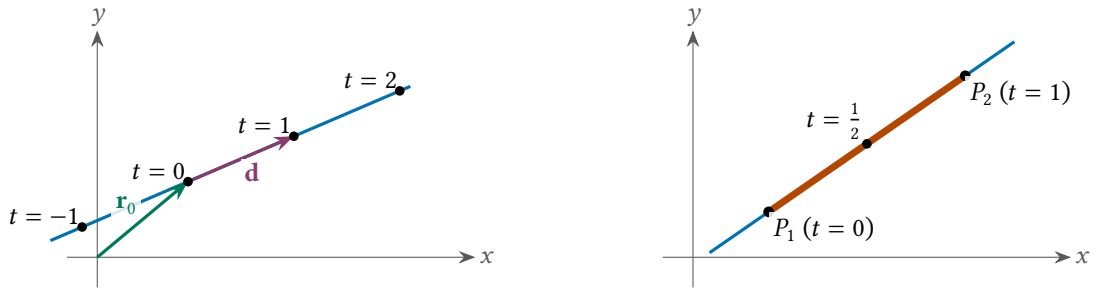
Problem: Line Through Two Points

Find the line through $(1, 2, -3)$ and $(4, -1, 2)$.

Solution. $\mathbf{d} = (4 - 1,\ -1 - 2,\ 2 - (-3)) = (3, -3, 5)$. Parametric form:

$$x = 1 + 3t, \qquad y = 2 - 3t, \qquad z = -3 + 5t.$$

Canonical form: $\dfrac{x - 1}{3} = \dfrac{y - 2}{-3} = \dfrac{z + 3}{5}$ (Figure 6.3).



(a) parametric form **(b)** line through two points

Figure 6.3: (a) The points $\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{d}$ for $t = -1, 0, 1, 2$ are equally spaced along the line, one step $\mathbf{d}$ apart. (b) For the line through P_1 and P_2 the thick orange piece is the segment $0 \le t \le 1$; $t = \frac{1}{2}$ is its midpoint, and $t = 2$ would be the point $2P_2 - P_1$, the reflection of P_1 in P_2. (Drawn in a plane for clarity; the same picture holds in space.)

Quick Check 1

1. Why can the canonical form not be written for $\mathbf{d} = (0, 3, -1)$, and what do you write instead?
2. Write the line through $(1, 2, 3)$ and $(3, 2, 5)$ in parametric form. Which coordinate stays constant?

6.2 Distance From a Point to a Line

Definition: Distance from a Point to a Line

The distance from a point P to the line through Q with direction vector $\mathbf{d}$ is

$$d = \frac{|\overrightarrow{QP} \times \mathbf{d}|}{|\mathbf{d}|}.$$

Why does this work?

The vectors $\overrightarrow{QP}$ and $\mathbf{d}$ span a parallelogram (Figure 6.4) with base $|\mathbf{d}|$ and the same perpendicular height d that we want – the distance from P down to the line. Since the parallelogram's area is both $|\overrightarrow{QP} \times \mathbf{d}|$ (Chapter 4) and (base) × (height) = $|\mathbf{d}|\, d$, solving for d gives the formula.

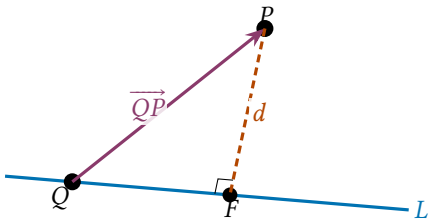


Figure 6.4: The distance d from P to line L is the height of the parallelogram spanned by $\overrightarrow{QP}$ and $\mathbf{d}$; F is the foot of the perpendicular.

Example: Distance Calculation

Find the distance from $P(1, 2, 3)$ to the line through $Q(0, 1, 2)$ with $\mathbf{d} = (1, -1, 2)$.

Solution. $\overrightarrow{QP} = (1, 1, 1)$, and

$$\overrightarrow{QP} \times \mathbf{d} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 1 & -1 & 2 \end{vmatrix} = \mathbf{i}(2+1) - \mathbf{j}(2-1) + \mathbf{k}(-1-1) = (3, -1, -2).$$

So $|\overrightarrow{QP} \times \mathbf{d}| = \sqrt{9+1+4} = \sqrt{14}$, $|\mathbf{d}| = \sqrt{1+1+4} = \sqrt{6}$, and $d = \frac{\sqrt{14}}{\sqrt{6}} = \frac{\sqrt{21}}{3}$.

Common mistake. The division by $|\mathbf{d}|$ in $d = |\overrightarrow{QP} \times \mathbf{d}| \,/\, |\mathbf{d}|$ is essential: the cross product alone is the *area* of the parallelogram, not its height. Two quick sanity checks for every answer: replacing $\mathbf{d}$ by $2\mathbf{d}$ must not change the result (numerator and denominator both double), and the distance can never exceed $|\overrightarrow{QP}|$.

Finding the foot of the perpendicular

Sometimes you need the actual point F where the perpendicular meets L, not just its distance. Write $F = Q + t\mathbf{d}$ and use $\overrightarrow{FP} \cdot \mathbf{d} = 0$ (since $\overrightarrow{FP}$ is perpendicular to L) to solve for t; then $|FP|$ is the distance, giving a check on the cross-product formula. Worked Problem 3 below does this in full.

A calculus view of the foot of the perpendicular

Walking along the line, $\mathbf{r}(t) = Q + t\mathbf{d}$, the squared distance to P is the quadratic function $f(t) = |P - \mathbf{r}(t)|^2$. It is smallest where $f'(t) = -2\,(P - \mathbf{r}(t)) \cdot \mathbf{d} = 0$, that is, where $\overrightarrow{FP} \cdot \mathbf{d} = 0$: the perpendicularity condition is exactly the "derivative equals zero" condition (Figure 6.5). This confirms that the shortest distance from a point to a line is the perpendicular one, and that the minimum value of f is d^2.

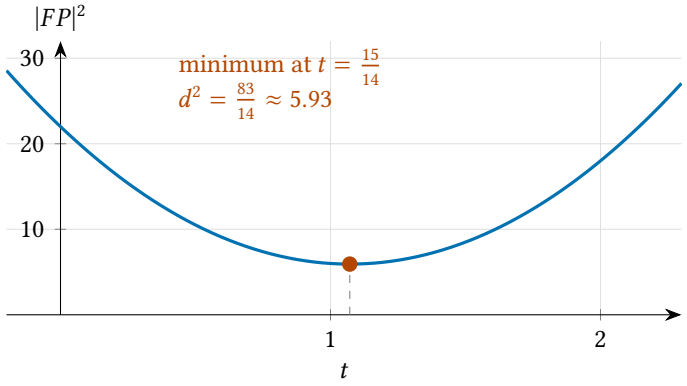


Figure 6.5: The squared distance from $P(4, -1, 2)$ to the point $Q + t\mathbf{d}$ of the line in Worked Problem 3 is $f(t) = 14t^2 - 30t + 22$. Its minimum is at $t = \frac{15}{14}$ (the foot F) and equals $\frac{83}{14}$, so $d = \sqrt{83/14} \approx 2.435$.

6.3 Positional Relationship of Straight Lines in Space

Definition: Angle Between Two Lines

The angle between two lines with direction vectors $\mathbf{d}_1 = (a_1, b_1, c_1)$ and $\mathbf{d}_2 = (a_2, b_2, c_2)$ (Figure 6.6) is the acute angle θ ($0^\circ \le \theta \le 90^\circ$) with

$$\cos\theta = \frac{|\mathbf{d}_1 \cdot \mathbf{d}_2|}{|\mathbf{d}_1|\,|\mathbf{d}_2|}.$$

The absolute value guarantees θ is acute, since the two lines' direction vectors could just as well have been taken pointing the other way. If $\cos\theta = 0$ the lines are perpendicular; if $\cos\theta = 1$ they are parallel.

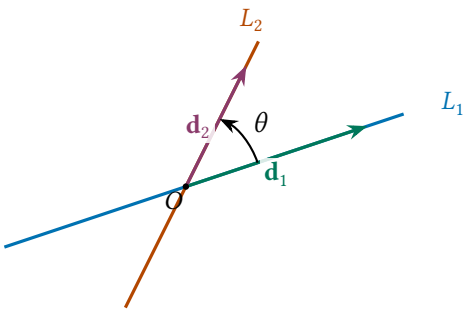


Figure 6.6: The angle θ between two lines equals the acute angle between any pair of their direction vectors.

Problem: Angle Between Two Lines

Find the angle between $L_1 : \mathbf{r}(t) = (1,0,2) + t(2,-1,2)$ and $L_2 : \mathbf{r}(s) = (0,1,3) + s(1,2,-1)$.

Solution. $\mathbf{d}_1 = (2,-1,2)$, $\mathbf{d}_2 = (1,2,-1)$, $\mathbf{d}_1 \cdot \mathbf{d}_2 = 2 - 2 - 2 = -2$, $|\mathbf{d}_1| = 3$, $|\mathbf{d}_2| = \sqrt{6}$:

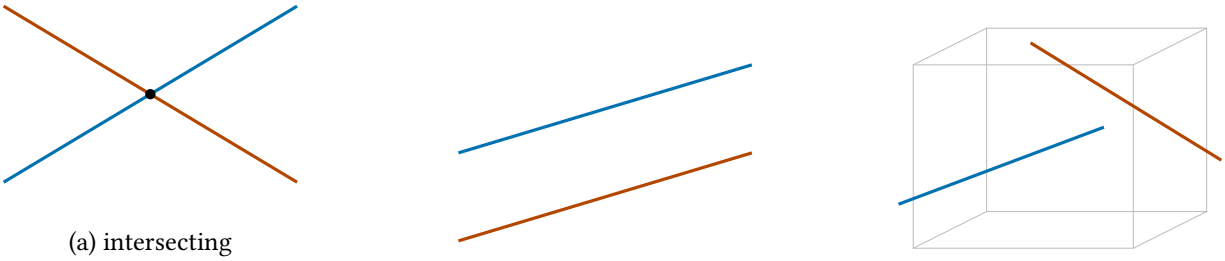
$$\cos\theta = \frac{|-2|}{3\sqrt{6}} = \frac{2}{3\sqrt{6}} = \frac{\sqrt{6}}{9} \approx 0.272 \implies \theta \approx 74.2^\circ.$$

Intuition. Two skew lines never meet, so there is no vertex at which to measure an angle. The definition therefore works with the direction vectors: slide one line parallel to itself until it meets the other, and measure the angle there. This is why θ depends only on $\mathbf{d}_1$ and $\mathbf{d}_2$, not on where the lines are.

Definition: Positional Relationships of Lines

Let Line 1 pass through P_1 with direction vector $\mathbf{d}_1$, and Line 2 through P_2 with direction vector $\mathbf{d}_2$. There are four possibilities (Figure 6.7):

- **Coincident**: the lines are identical (every point of one lies on the other).
- **Parallel**: $\mathbf{d}_1 \parallel \mathbf{d}_2$, but the lines are distinct and never meet.
- **Intersecting**: the lines are not parallel and share exactly one point.
- **Skew**: the lines are not parallel and share no point. This has no analogue in the plane – it is possible only because space gives lines room to miss each other without being parallel.



(a) intersecting (b) parallel (c) skew

Figure 6.7: The three ways two *non-coincident* lines can sit in space: (a) intersecting at one point, (b) parallel and never meeting, (c) skew – not parallel, yet still never meeting, because they pass at different "depths" (the faint box is only there to suggest the third dimension).

How to classify two lines (Figure 6.8)

1. **Test for parallelism.** The lines are parallel exactly when $\mathbf{d}_1 \times \mathbf{d}_2 = \mathbf{0}$.
2. **If parallel, test for coincidence.** Pick a point P_1 on Line 1; the lines are coincident if $\overrightarrow{P_1P_2} \times \mathbf{d}_1 = \mathbf{0}$ (i.e. P_2 also lies on Line 1), and merely parallel otherwise.
3. **If not parallel, test for intersection.** Set the two parametric forms equal and solve for t, s. Since this is *three* equations in only *two* unknowns, a solution is not guaranteed: if one exists, the lines intersect there; if the system is inconsistent, the lines are skew.

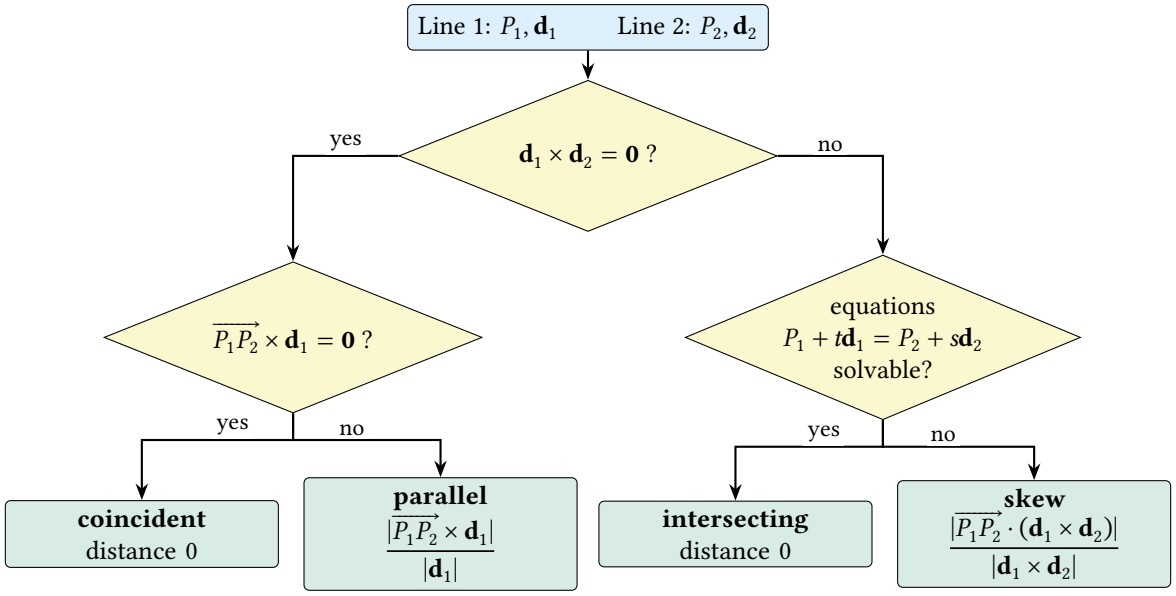


Figure 6.8: Classifying two lines. Each box at the bottom also gives the distance between the lines. The first question separates the parallel family (left) from the non-parallel family (right).

Example: Relative Positions of Two Lines in Space

Determine the relative positions of each pair of lines:

(a) Parallel Lines. $L_1 : \mathbf{r}_1(t) = (1,2,3) + t(2,4,6)$ and $L_2 : \mathbf{r}_2(s) = (1,1,1) + s(2,4,6)$.
Solution. The direction vectors are identical ($\mathbf{d}_1 = \mathbf{d}_2 = (2,4,6)$), but the point $(1,2,3)$ does not satisfy L_2's equation for any s. Thus, the lines have the same direction but different base points, making them parallel and distinct.

(b) Intersecting Lines. $L_1 : \mathbf{r}_1(t) = t(1,1,0)$ and $L_2 : \mathbf{r}_2(s) = (2,0,0) + s(-1,1,0)$.
Solution. Equating the coordinates yields $(t,t,0) = (2-s,s,0)$. From the y-component: $t = s$. Substituting into the x-component gives $t = 2 - t \implies t = 1$, which also results in $s = 1$. Since the equations are consistent, the lines intersect at $\mathbf{r}_1(1) = (1,1,0)$.

(c) Skew Lines. $L_1 : \mathbf{r}_1(t) = (1,0,0) + t(1,1,0)$ and $L_2 : \mathbf{r}_2(s) = (0,0,1) + s(1,1,1)$.
Solution. Equating the coordinates gives $(1+t,\ t,\ 0) = (s,\ s,\ 1+s)$. From the y-component: $t = s$. Substituting this into the x-component yields $1 + t = t$, or $1 = 0$, which is a contradiction. (The z-component $0 = 1 + s$ independently forces $s = -1$, confirming inconsistency.) Since the system has no solution and the direction vectors $\mathbf{d}_1 = (1,1,0)$ and $\mathbf{d}_2 = (1,1,1)$ are not proportional, the lines are skew.

6.3.1 Distance Between Lines

The distance between two lines depends on their geometric relationship to one another:

- **Parallel lines**: pick any points P_1, P_2, one on each line, and any direction vector $\mathbf{d}$ of either line;

$$\text{Distance} = \frac{|(P_2 - P_1) \times \mathbf{d}|}{|\mathbf{d}|}$$

  (this is just the point-to-line formula, applied to P_2 and the line through P_1 (Figure 6.9)).
- **Intersecting lines**: the distance is 0.

- **Skew lines**: the shortest segment joining them is perpendicular to *both*, so it lies along $\mathbf{d}_1 \times \mathbf{d}_2$. Its length is the projection of $\overrightarrow{P_1P_2}$ onto that common perpendicular direction:

$$\text{Distance} = \frac{|(P_2 - P_1) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|}{|\mathbf{d}_1 \times \mathbf{d}_2|}$$

(the numerator is a scalar triple product: geometrically, the volume of the parallelepiped spanned by $\overrightarrow{P_1P_2}$, $\mathbf{d}_1$, $\mathbf{d}_2$, divided by the area of its base – exactly the height of that parallelepiped, which is the common perpendicular distance).

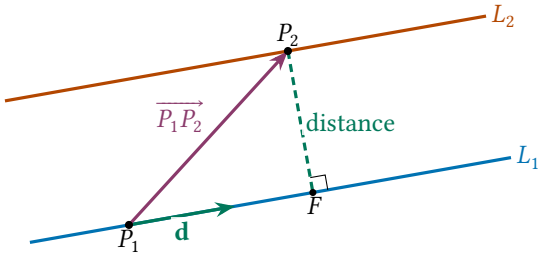


Figure 6.9: Distance between two parallel lines: it is the distance from any point P_2 of L_2 to L_1, the height of the parallelogram spanned by $\overrightarrow{P_1P_2}$ and $\mathbf{d}$.

Example: Distances Between Lines in Space

Classify each pair of lines and find the distance between them:

(a) Parallel Lines. $L_1 : \mathbf{r}_1(t) = (1,0,0) + t(1,2,3)$ and $L_2 : \mathbf{r}_2(s) = (0,1,1) + s(2,4,6)$.
Solution. Since $\mathbf{d}_2 = (2,4,6) = 2\mathbf{d}_1$ and the point $(0,1,1)$ does not lie on L_1, the lines are parallel. Choosing base points $P_1 = (1,0,0)$ on L_1 and $P_2 = (0,1,1)$ on L_2, we have $P_2 - P_1 = (-1,1,1)$. The cross product with the direction vector $\mathbf{d} = (1,2,3)$ is

$$(P_2 - P_1) \times \mathbf{d} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 1 & 1 \\ 1 & 2 & 3 \end{vmatrix} = (1,4,-3).$$

The distance is given by the ratio of magnitudes:

$$\text{Distance} = \frac{|(P_2 - P_1) \times \mathbf{d}|}{|\mathbf{d}|} = \frac{\sqrt{1+16+9}}{\sqrt{1+4+9}} = \frac{\sqrt{26}}{\sqrt{14}} = \frac{\sqrt{91}}{7}.$$

(b) Skew Lines. $L_1 : \mathbf{r}_1(t) = t(1,1,0)$ and $L_2 : \mathbf{r}_2(s) = (1,0,1) + s(0,1,1)$.
Solution. The direction vectors $\mathbf{d}_1 = (1,1,0)$ and $\mathbf{d}_2 = (0,1,1)$ are not parallel, and checking the system reveals no intersection, so the lines are skew. Choosing base points $P_1 = (0,0,0)$ and $P_2 = (1,0,1)$, we find their difference $P_2 - P_1 = (1,0,1)$. The cross product of the direction vectors is

$$\mathbf{d}_1 \times \mathbf{d}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = (1,-1,1).$$

The distance between skew lines is the magnitude of the scalar projection of $P_2 - P_1$ onto this normal vector:

$$\text{Distance} = \frac{|(P_2 - P_1) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|}{|\mathbf{d}_1 \times \mathbf{d}_2|} = \frac{|(1,0,1) \cdot (1,-1,1)|}{\sqrt{1+1+1}} = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}.$$

Where are the closest points?

For skew lines the shortest segment AB joins $A = P_1 + t\mathbf{d}_1$ on Line 1 to $B = P_2 + s\mathbf{d}_2$ on Line 2 with $\overrightarrow{AB} \perp \mathbf{d}_1$ and $\overrightarrow{AB} \perp \mathbf{d}_2$ (Figure 6.10). These are two *linear* equations for t and s,

$$\big(P_2 + s\mathbf{d}_2 - P_1 - t\mathbf{d}_1\big)\cdot \mathbf{d}_1 = 0, \qquad \big(P_2 + s\mathbf{d}_2 - P_1 - t\mathbf{d}_1\big)\cdot \mathbf{d}_2 = 0,$$

and they have exactly one solution whenever the lines are not parallel (the determinant has absolute value $|\mathbf{d}_1|^2|\mathbf{d}_2|^2 - (\mathbf{d}_1\cdot\mathbf{d}_2)^2 = |\mathbf{d}_1\times\mathbf{d}_2|^2 \neq 0$). For the skew lines of the last example ($P_1 = (0,0,0)$, $\mathbf{d}_1 = (1,1,0)$, $P_2 = (1,0,1)$, $\mathbf{d}_2 = (0,1,1)$) the equations read $2t - s - 1 = 0$ and $t - 2s - 1 = 0$. Hence $t = \frac{1}{3}$, $s = -\frac{1}{3}$, $A = (\frac{1}{3}, \frac{1}{3}, 0)$, $B = (1, -\frac{1}{3}, \frac{2}{3})$ and $\overrightarrow{AB} = \frac{2}{3}(1,-1,1)$, whose length $\frac{2}{\sqrt{3}}$ agrees with the distance found above.

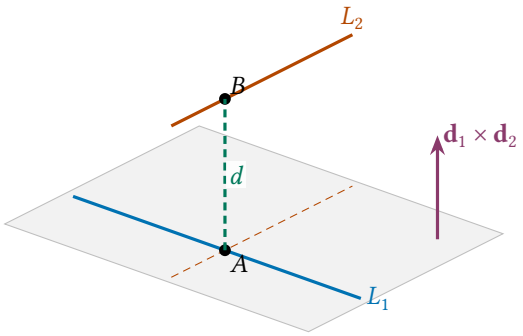


Figure 6.10: Two skew lines and their common perpendicular AB, which points along $\mathbf{d}_1 \times \mathbf{d}_2$. The shaded plane contains L_1 and is parallel to L_2; the dashed orange line is the shadow of L_2 in this plane, and it crosses L_1 exactly at the foot A. The distance $d = |AB|$ is the height at which L_2 floats above that plane.

Quick Check 3

1. If $\cos\theta = 0$ for the angle between two lines, must the lines intersect?
2. Which formula do you use for the distance between two lines if $\mathbf{d}_1 \times \mathbf{d}_2 = \mathbf{0}$? Why does the skew-line formula fail then?

6.4 Applications in Physics and Engineering

Straight lines model the path of a body that is not being pushed, a laser or radar beam, a pipe, a wire, a road centreline or a machine tool. Two ideas return in every application: the parameter t is often *time*, and the distance formulas of Sections 6.2 and 6.3 answer the practical question "how close do these two things come?".

6.4.1 Uniform Motion and Closest Approach

Physical law (Newton's first law). A body on which no net force acts moves with constant velocity $\mathbf{v}$, so its position is

$$\mathbf{r}(t) = \mathbf{r}_0 + t\,\mathbf{v}\ :$$

a straight line whose direction vector is the velocity and whose parameter is *time*. Two bodies A and B are closest when their relative position

$$\Delta(t) = \mathbf{r}_A(t) - \mathbf{r}_B(t) = \Delta_0 + t\,\mathbf{v}_{\text{rel}}, \qquad \mathbf{v}_{\text{rel}} = \mathbf{v}_A - \mathbf{v}_B,$$

is shortest. But $\boldsymbol{\Delta}(t)$ is itself a line (the path of A as seen from B), so the *minimum separation is the distance from the origin to this line*:

$$d_{\min} = \frac{|\boldsymbol{\Delta}_0 \times \mathbf{v}_{\mathrm{rel}}|}{|\mathbf{v}_{\mathrm{rel}}|}, \qquad t^* = -\frac{\boldsymbol{\Delta}_0 \cdot \mathbf{v}_{\mathrm{rel}}}{|\mathbf{v}_{\mathrm{rel}}|^2}$$

(the time t^* is the parameter of the foot of the perpendicular; if $t^* < 0$ the closest approach lies in the past).

Intuition. "The paths intersect" and "the bodies collide" are different statements. A line knows nothing about time, but a trajectory does: a collision needs the *same* value of t in $\mathbf{r}_A(t) = \mathbf{r}_B(t)$, which is three equations for *one* unknown and therefore almost never solvable. This is why air-traffic and collision-avoidance systems compute the minimum separation $d_{\min}$ and the time t^*, not just whether the routes cross.

6.4.2 Surveying and Machining: Offset, Station and Tool Paths

Physics & Engineering: Surveying and Machining

Surveying. A point is located relative to a straight centreline (road, pipeline, rail) by two numbers: its **station**, the distance along the line from the starting point Q to the foot F of the perpendicular, and its **offset**, the perpendicular distance. For the line $\mathbf{r}(t) = Q + t\mathbf{d}$ and a point P,

$$\text{offset} = \frac{|\overrightarrow{QP} \times \mathbf{d}|}{|\mathbf{d}|}, \qquad \text{station} = t^*|\mathbf{d}| = \frac{\overrightarrow{QP} \cdot \mathbf{d}}{|\mathbf{d}|} \quad \text{(signed)}.$$

Machining. A straight CNC move ("G01") from P_1 to P_2 at the feed rate f is the line through two points traversed at constant speed:

$$\mathbf{r}(t) = P_1 + \frac{ft}{L}(P_2 - P_1), \qquad 0 \le t \le \frac{L}{f}, \qquad L = |P_2 - P_1|,$$

with velocity $f\dfrac{P_2 - P_1}{L}$ and duration L/f.

Example: Station and Offset; a G01 Move

(a) Survey. For the centreline of Worked Problem 3, $Q(1, 2, 0)$ and $\mathbf{d} = (2, -1, 3)$ (metres), and a building corner $P(4, -1, 2)$ we found the offset $\sqrt{83/14} \approx 2.44$ m and $t^* = \frac{15}{14}$. The station is

$$t^*|\mathbf{d}| = \frac{15}{14}\sqrt{14} = \frac{15}{\sqrt{14}} \approx 4.01 \text{ m},$$

so the corner lies 4.01 m along the line from Q and 2.44 m to the side.
(b) Machining. A tool moves from $P_1 = (10, 20, 5)$ mm to $P_2 = (70, 100, 5)$ mm at $f = 25$ mm/s. Then $P_2 - P_1 = (60, 80, 0)$, $L = 100$ mm, the duration is 4 s and the velocity is $25 \cdot (0.6, 0.8, 0) = (15, 20, 0)$ mm/s. At $t = 2.5$ s the tool is at $(10, 20, 5) + 2.5 \cdot (15, 20, 0) = (47.5,\ 70,\ 5)$ mm, which is 62.5 % of the way along the segment.

6.4.3 Skew Lines in Practice: Pipe Clearance and Triangulation

Physics & Engineering: Clearance and Triangulation

Pipe clearance. In a plant model each straight pipe is a centre line plus a radius. Two pipes with radii r_1, r_2 and skew centre lines have the *clearance* $D - r_1 - r_2$, where D is the distance between the centre lines (Section 6.3); design rules require the clearance to exceed a minimum value. Software that checks this for thousands of pipes ("clash detection") evaluates exactly this formula.
Triangulation. Two stations S_1, S_2 measure the direction towards a target (radio direction finding, cameras, sonar). Each measurement defines a ray; because of noise the two rays are skew and do not meet. The best estimate of the target is the *midpoint M of the common perpendicular AB* (Figure 6.11), and the length $|AB|$, the *miss distance*, measures the quality of the data: a large miss distance warns that the measurements are unreliable.

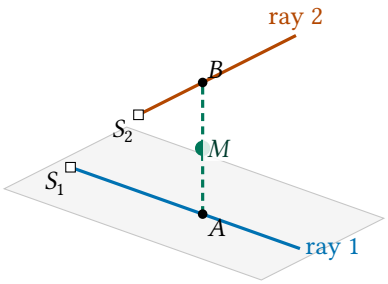


Figure 6.11: Triangulation with two noisy bearings (schematic). Ray 1 starts at station S_1 and ray 2 at station S_2; they do not intersect. AB is the common perpendicular (the closest pair of points), its midpoint M is the position estimate, and $|AB|$ is the miss distance.

Example: Pipes and Bearings

(a) Pipe clearance. Two pipes have the centre lines of the skew-line example, $L_1 : t(1,1,0)$ and $L_2 : (1,0,1) + s(0,1,1)$ (metres), and outer radii 0.30 m and 0.40 m. We found $D = \frac{2}{\sqrt{3}} \approx 1.155$ m, so the clearance is

$$1.155 - 0.30 - 0.40 \approx 0.455 \text{ m}.$$

The angle between the pipes is given by $\cos\theta = \dfrac{|\mathbf{d}_1 \cdot \mathbf{d}_2|}{|\mathbf{d}_1||\mathbf{d}_2|} = \dfrac{1}{\sqrt{2}\sqrt{2}} = \dfrac{1}{2}$, so $\theta = 60^\circ$.
(b) Triangulation. (Distances in km; the miss distance is exaggerated to keep the numbers simple.) Station $S_1 = (0,0,0)$ has the bearing $\mathbf{d}_1 = (1,1,0)$ and station $S_2 = (2,0,1)$ the bearing $\mathbf{d}_2 = (-1,1,1)$: $\mathbf{r}_1(t) = t(1,1,0)$, $\mathbf{r}_2(s) = (2,0,1) + s(-1,1,1)$. Then $\mathbf{r}_2 - \mathbf{r}_1 = (2 - s - t,\ s - t,\ 1 + s)$ and the two perpendicularity conditions are

$$(\mathbf{r}_2 - \mathbf{r}_1)\cdot\mathbf{d}_1 = 2 - 2t = 0, \qquad (\mathbf{r}_2 - \mathbf{r}_1)\cdot\mathbf{d}_2 = -1 + 3s = 0,$$

so $t = 1$, $s = \frac{1}{3}$, $A = (1,1,0)$, $B = (\frac{5}{3}, \frac{1}{3}, \frac{4}{3})$ and

$$M = \tfrac{1}{2}(A + B) = \left(\tfrac{4}{3}, \tfrac{2}{3}, \tfrac{2}{3}\right), \qquad |AB| = \left|\tfrac{2}{3}(1,-1,2)\right| = \frac{2\sqrt{6}}{3} \approx 1.63 \text{ km}.$$

Check: $\mathbf{d}_1 \times \mathbf{d}_2 = (1,-1,2)$ and $\dfrac{|(S_2 - S_1)\cdot(\mathbf{d}_1 \times \mathbf{d}_2)|}{|\mathbf{d}_1 \times \mathbf{d}_2|} = \dfrac{4}{\sqrt{6}} = \dfrac{2\sqrt{6}}{3}$. ☒

6.4.4 The Magnetic Field of a Straight Wire

Physics & Engineering: Field of a Long Straight Current

Physical law (Biot–Savart / Ampère). A long straight wire along the line L, carrying the current I in the direction of $\mathbf{d}$, produces at the point P a magnetic field of magnitude

$$|\mathbf{B}| = \frac{\mu_0 I}{2\pi d}, \qquad \mu_0 = 4\pi \cdot 10^{-7}\ \mathrm{T\cdot m/A},$$

where d is the *distance from P to L*, the quantity of Section 6.2. The field is tangent to circles around the wire (right-hand rule: thumb along the current, fingers curl along $\mathbf{B}$) and points along $\mathbf{d}\times\overrightarrow{FP}$, where F is the foot of the perpendicular. Since $\mathbf{d}\perp\overrightarrow{FP}$, we have $|\mathbf{d}\times\overrightarrow{FP}| = |\mathbf{d}|\,d$ (Figure 6.12).

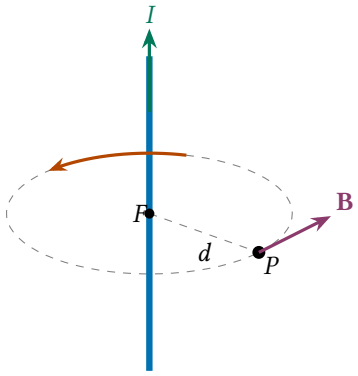


Figure 6.12: The field of a long straight wire. In coordinates in which the wire is the vertical line, the field $\mathbf{B}$ at P (distance d from the wire, foot F) is tangent to the dashed circle of radius d around the wire; the orange arrow shows the sense of circulation given by the right-hand rule (thumb along the current I). Its size depends on d only.

Example: Field Near a Wire

A wire along the line through $Q(0,1,2)$ with $\mathbf{d} = (1,-1,2)$ (metres) carries $I = 10$ A in the direction of $\mathbf{d}$. Find $\mathbf{B}$ at $P(1,2,3)$.

Solution. From the example of Section 6.2, $d = \frac{\sqrt{21}}{3} \approx 1.53$ m, so

$$|\mathbf{B}| = \frac{4\pi\cdot 10^{-7}\cdot 10}{2\pi\cdot 1.5275} = \frac{2\cdot 10^{-7}\cdot 10}{1.5275} \approx 1.31\cdot 10^{-6}\ \mathrm{T}.$$

For the direction, the foot is $F = Q + t\mathbf{d}$ with $t = \frac{\overrightarrow{QP}\cdot\mathbf{d}}{|\mathbf{d}|^2} = \frac{2}{6} = \frac{1}{3}$, so $F = (\frac{1}{3},\frac{2}{3},\frac{8}{3})$ and $\overrightarrow{FP} = (\frac{2}{3},\frac{4}{3},\frac{1}{3})$. Then

$$\mathbf{d}\times\overrightarrow{FP} = (-3,\ 1,\ 2), \qquad \mathbf{B} \approx 1.31\cdot 10^{-6}\cdot\frac{(-3,1,2)}{\sqrt{14}} \approx (-1.05,\ 0.35,\ 0.70)\cdot 10^{-6}\ \mathrm{T}.$$

Check: $(-3,1,2)\cdot\mathbf{d} = -3-1+4 = 0$ and $(-3,1,2)\cdot\overrightarrow{FP} = -2+\frac{4}{3}+\frac{2}{3} = 0$: $\mathbf{B}$ is perpendicular to both the wire and the radius. ☒

Key Formulas of This Chapter

- Line through $\mathbf{r}_0$ with direction $\mathbf{d} = (a, b, c)$: $\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{d}$; canonical form $\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$ (if $a, b, c \neq 0$). Through two points: $\mathbf{d} = \mathbf{r}_2 - \mathbf{r}_1$, segment $0 \le t \le 1$.
- Distance from P to the line through Q with direction $\mathbf{d}$: $\dfrac{|\overrightarrow{QP} \times \mathbf{d}|}{|\mathbf{d}|}$; foot: $F = Q + t\mathbf{d}$ with $\overrightarrow{FP} \cdot \mathbf{d} = 0$, $t = \dfrac{\overrightarrow{QP} \cdot \mathbf{d}}{|\mathbf{d}|^2}$; symmetric point $Q' = 2F - P$.
- Angle between lines: $\cos\theta = \dfrac{|\mathbf{d}_1 \cdot \mathbf{d}_2|}{|\mathbf{d}_1||\mathbf{d}_2|}$, $0^\circ \le \theta \le 90^\circ$.
- Classification: $\mathbf{d}_1 \times \mathbf{d}_2 = \mathbf{0} \Rightarrow$ parallel or coincident (point test); otherwise solve $\mathbf{r}_1(t) = \mathbf{r}_2(s)$: solvable $\Rightarrow$ intersecting, not solvable $\Rightarrow$ skew.
- Distance between lines: parallel $\dfrac{|\overrightarrow{P_1P_2} \times \mathbf{d}|}{|\mathbf{d}|}$; skew $\dfrac{|\overrightarrow{P_1P_2} \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|}{|\mathbf{d}_1 \times \mathbf{d}_2|}$; intersecting or coincident 0.
- Physics: uniform motion $\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}$; closest approach $d_{\min} = \dfrac{|\Delta_0 \times \mathbf{v}_{\text{rel}}|}{|\mathbf{v}_{\text{rel}}|}$ at $t^* = -\dfrac{\Delta_0 \cdot \mathbf{v}_{\text{rel}}}{|\mathbf{v}_{\text{rel}}|^2}$; wire field $|\mathbf{B}| = \dfrac{\mu_0 I}{2\pi d}$; triangulation estimate = midpoint of the common perpendicular.

6.5 Worked Problems

Problem 1: Line Perpendicular to Two Vectors

Find the line through $P(1, -2, 3)$ perpendicular to both $\mathbf{v}_1 = (2, 1, 0)$ and $\mathbf{v}_2 = (0, 1, 2)$.

Solution. A vector perpendicular to both is their cross product:

$$\mathbf{v}_1 \times \mathbf{v}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{vmatrix} = (2, -4, 2).$$

Any nonzero multiple works as a direction vector; taking half of it, $\mathbf{d} = (1, -2, 1)$, gives

$$x = 1 + t, \quad y = -2 - 2t, \quad z = 3 + t, \qquad \frac{x-1}{1} = \frac{y+2}{-2} = \frac{z-3}{1}.$$

Problem 2: Classify Two Lines and Find Their Distance

Line 1: $x = 2 + 3t$, $y = 1 - t$, $z = 4 + 2t$. Line 2: $\dfrac{x-3}{1} = \dfrac{y+2}{-2} = \dfrac{z-1}{1}$. Classify their relationship and find the distance between them.

Solution. $\mathbf{d}_1 = (3, -1, 2)$, $\mathbf{d}_2 = (1, -2, 1)$: no scalar multiple relates them, so the lines are not parallel. Setting the parametric forms equal,

$$2 + 3t = 3 + s, \qquad 1 - t = -2 - 2s, \qquad 4 + 2t = 1 + s,$$

the first two equations give $t = -\frac{1}{5}$, $s = -\frac{8}{5}$; substituting into the third, the left side is $4 + 2(-\frac{1}{5}) = \frac{18}{5}$ while the right side is $1 + (-\frac{8}{5}) = -\frac{3}{5}$ – these disagree, so the system is inconsistent and the lines do *not* intersect. Being neither parallel nor intersecting, the lines are **skew**.

For the distance, use $P_1 = (2,1,4)$, $P_2 = (3,-2,1)$:

$$\mathbf{d}_1 \times \mathbf{d}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -1 & 2 \\ 1 & -2 & 1 \end{vmatrix} = (-1+4,\ -(3-2),\ -6+1) = (3,-1,-5),$$

$$P_2 - P_1 = (1,-3,-3), \qquad (P_2-P_1)\cdot(\mathbf{d}_1\times\mathbf{d}_2) = 3+3+15 = 21, \qquad |\mathbf{d}_1\times\mathbf{d}_2| = \sqrt{35},$$

$$d = \frac{|21|}{\sqrt{35}} = \frac{3\sqrt{35}}{5}.$$

Problem 3: Distance from a Point to a Line, and the Foot of the Perpendicular

$P(4,-1,2)$; line L through $Q(1,2,0)$ with $\mathbf{d} = (2,-1,3)$. Find (1) the distance from P to L; (2) the foot of the perpendicular.

Solution. $\overrightarrow{QP} = (3,-3,2)$, so

$$\overrightarrow{QP}\times\mathbf{d} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -3 & 2 \\ 2 & -1 & 3 \end{vmatrix} = \big(-9-(-2),\ -(9-4),\ -3-(-6)\big) = (-7,-5,3),$$

$$|\overrightarrow{QP}\times\mathbf{d}| = \sqrt{49+25+9} = \sqrt{83}, \qquad |\mathbf{d}| = \sqrt{14}, \qquad d = \frac{\sqrt{83}}{\sqrt{14}} = \frac{\sqrt{1162}}{14} \approx 2.435.$$

For the foot, let $F = Q + t\mathbf{d} = (1+2t,\ 2-t,\ 3t)$. Then $\overrightarrow{FP} = P - F = (3-2t,\ -3+t,\ 2-3t)$, and this must be perpendicular to $\mathbf{d}$:

$$\overrightarrow{FP}\cdot\mathbf{d} = 2(3-2t) - (-3+t) + 3(2-3t) = 6-4t+3-t+6-9t = 15-14t = 0 \implies t = \frac{15}{14}.$$

So

$$F = \Big(1+\tfrac{15}{7},\ 2-\tfrac{15}{14},\ \tfrac{45}{14}\Big) = \Big(\frac{22}{7},\ \frac{13}{14},\ \frac{45}{14}\Big),$$

and as a check, $|FP| = \sqrt{83}/\sqrt{14} = \sqrt{1162}/14$, matching part (1). ☒

Problem 4: Perpendicularity Check with Mixed Forms

Line 1: $\dfrac{x-2}{1} = \dfrac{y+1}{-2} = \dfrac{z-3}{1}$. Line 2: $x = 2t,\ y = 1+4t,\ z = 2-t$. Are they perpendicular?

Solution. $\mathbf{d}_1 = (1,-2,1)$, $\mathbf{d}_2 = (2,4,-1)$:

$$\mathbf{d}_1\cdot\mathbf{d}_2 = 2-8-1 = -7 \neq 0,$$

so the lines are *not* perpendicular.

Problem 5: Symmetric Point About a Line (Figure 6.13)

Line: $x = 1+t$, $y = 2-2t$, $z = 3+t$. Find the point Q symmetric to $P(0,4,1)$ about this line.

Solution. $\mathbf{d} = (1,-2,1)$. Let $F = (1+t,\ 2-2t,\ 3+t)$; we need $\overrightarrow{PF} \perp \mathbf{d}$, where $\overrightarrow{PF} = F - P$:

$$(1+t,\ -2-2t,\ 2+t)\cdot(1,-2,1) = (1+t)+(4+4t)+(2+t) = 7+6t = 0 \implies t = -\frac{7}{6}.$$

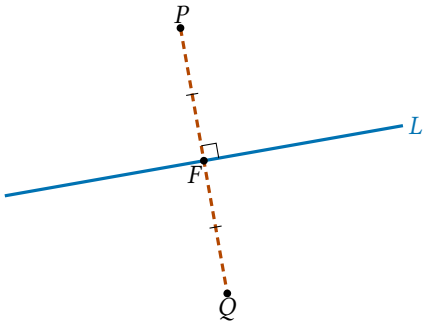


Figure 6.13: The symmetric point Q of P about the line L. F is the foot of the perpendicular from P; Q lies on the other side with $|FQ| = |FP|$ (equal tick marks), so F is the midpoint of PQ and $Q = 2F - P$.

$$F = \left(1 - \tfrac{7}{6},\ 2 + \tfrac{7}{3},\ 3 - \tfrac{7}{6}\right) = \left(-\tfrac{1}{6},\ \tfrac{13}{3},\ \tfrac{11}{6}\right).$$

Since F is the midpoint of PQ: $Q = 2F - P = \left(-\tfrac{1}{3},\ \tfrac{14}{3},\ \tfrac{8}{3}\right)$.

Problem 6: Engineering Application – Cable Alignment

A cable follows the line through $(0, 0, 10)$ with $\mathbf{d} = (1, 1, 0)$. A second cable through $(5, -2, 8)$, with direction $\mathbf{v} = (a, b, c)$, must be perpendicular to the first and cross it. Find $\mathbf{v}$ and the intersection point.

Solution. Perpendicularity: $\mathbf{d} \cdot \mathbf{v} = a + b = 0 \implies b = -a$, so $\mathbf{v} = (a, -a, c)$ for some a, c. For intersection, set

$$(t, t, 10) = (5 + sa,\ -2 - sa,\ 8 + sc).$$

From the first two: $t = 5 + sa$ and $t = -2 - sa$, so $5 + sa = -2 - sa \implies sa = -\tfrac{7}{2} \implies t = 5 + sa = \tfrac{3}{2}$. From the third: $10 = 8 + sc \implies sc = 2 \implies c = 2/s = 2/(-\tfrac{7}{2a}) = -\tfrac{4a}{7}$. So

$$\mathbf{v} = a\left(1,\ -1,\ -\tfrac{4}{7}\right), \quad a \neq 0, \qquad \text{intersection at } \left(\tfrac{3}{2},\ \tfrac{3}{2},\ 10\right).$$

(This makes sense: the answer is a whole family of direction vectors, all parallel to $\left(1, -1, -\tfrac{4}{7}\right)$, since scaling $\mathbf{v}$ does not change which line it describes.)

6.6 Conceptual & Computational Exercises

6.6.1 Conceptual Problems

Try these *before* computing: most can be answered by geometry, by checking types, or by thinking about the physical meaning. Answers and hints follow the homework problems.

1. **True or false?** Justify each answer. (a) Two lines with the same direction vector are parallel. (b) Two lines with no common point are parallel. (c) Three lines in space can be pairwise skew. (d) If $\mathbf{d}_1 \cdot \mathbf{d}_2 = 0$, the lines intersect. (e) The angle between two lines is always at most 90°. (f) The distance between two intersecting lines is 0. (g) One linear equation $Ax + By + Cz = D$ describes a line in space. (h) A line is determined by a point and a direction vector, and also by two of its points.
2. **Why a direction vector?** Describe all lines through a point P that are perpendicular to a given line with direction vector $\mathbf{d}$. Why can no single normal vector describe a line in $\mathbb{R}^3$?
3. **Same line, different equations.** Show that $(1, 2, 3)+t(2, -2, 4)$ and $(3, 0, 7)+s(-1, 1, -2)$ describe the same line. Which two things must you check?

4. **Special lines.** Write (a) the z-axis and (b) the line through $(1,2,0)$ parallel to the y-axis in parametric form. Which coordinates stay constant, and why does the canonical form not apply?
5. **Two points.** Why can two distinct lines have at most one common point?
6. **A test for "flat".** Show that two lines are coplanar (parallel, coincident or intersecting) exactly when $(P_2-P_1)\cdot(\mathbf{d}_1\times\mathbf{d}_2)=0$. Use it to confirm that $L_1:(1,0,0)+t(1,1,0)$ and $L_2:(0,0,1)+s(1,1,1)$ are skew.
7. **Invariance.** Show that $\dfrac{|\overrightarrow{QP}\times\mathbf{d}|}{|\mathbf{d}|}$ does not change if (i) $\mathbf{d}$ is replaced by $k\mathbf{d}$ ($k\neq 0$), or (ii) Q is replaced by another point $Q+s\mathbf{d}$ of the line.
8. **Pythagoras.** Using the Lagrange identity of Chapter 4, show that $d^2=|\overrightarrow{QP}|^2-\dfrac{(\overrightarrow{QP}\cdot\mathbf{d})^2}{|\mathbf{d}|^2}$, and explain this with a right triangle.
9. **Paths versus bodies.** Body A moves with $\mathbf{r}_A(t)=(t,0,0)$ and body B with $\mathbf{r}_B(t)=(0,t-3,0)$. Do the paths intersect? Do the bodies collide? How close do they come, and when?
10. **Calculus.** For $\mathbf{r}(t)=\Delta_0+t\mathbf{v}$ let $f(t)=|\mathbf{r}(t)|^2$. Solve $f'(t)=0$ and show that $f(t^*)=\dfrac{|\Delta_0\times\mathbf{v}|^2}{|\mathbf{v}|^2}$. Why is t^* a minimum?
11. **Triangulation.** Why is the midpoint M of the common perpendicular a natural position estimate for two skew rays? How far is M from each of the two lines?
12. **Symmetric point.** Why does $Q=2F-P$ give the point symmetric to P about the line? Show that $\overrightarrow{PQ}\perp L$.

6.6.2 Computational Problems

Solve the following problems. Show all work and give exact answers unless otherwise specified.

1. A point $P(2,1,-1)$ and direction vector $\mathbf{d}=(3,-2,1)$ are given. Write the line through P in parametric and canonical form.
2. Find the distance from $Q(0,-3,2)$ to the line $x=1+2t$, $y=4-t$, $z=3+3t$.
3. Line 1: $x=t$, $y=2+2t$, $z=1-t$. Line 2: $\dfrac{x-1}{1}=\dfrac{y-4}{2}=\dfrac{z+1}{-1}$. Determine their positional relationship, and find the intersection point if they intersect.
4. Find the angle between the line through $(1,0,3)$ with $\mathbf{d}_1=(1,-1,0)$ and the line through $(2,1,-2)$ with $\mathbf{d}_2=(2,2,1)$.
5. Line 1: $x=2+3t$, $y=-1+t$, $z=0$. Line 2: $x=5+3s$, $y=2+s$, $z=2$. Verify the lines are parallel and find the perpendicular distance between them.
6. A line passes through $(1,2,3)$ and $(4,0,5)$. Write its equation in parametric form, and find the point symmetric to $(0,1,2)$ about this line.
7. $P(3,-1,2)$; line L: $x=1+t$, $y=2t$, $z=-1+t$. Find the distance from P to L.
8. Find the angle between $L_1:\mathbf{r}_1(t)=(0,1,2)+t(1,-1,2)$ and $L_2:\mathbf{r}_2(s)=(1,0,0)+s(2,1,-1)$.
9. Are $L_1:\dfrac{x-1}{2}=\dfrac{y+1}{-1}=\dfrac{z-3}{1}$ and $L_2:x=t,\ y=2+3t,\ z=1-t$ perpendicular?
10. $L_1:\mathbf{r}_1(t)=(0,0,1)+t(1,2,-1)$, $L_2:\mathbf{r}_2(s)=(1,-1,0)+s(2,-1,1)$. Check whether they intersect; if not, find the distance between them.
11. A line is given by $x=1+2t$, $y=-t$, $z=2+3t$, and a point $P(4,2,1)$. Find the point Q symmetric to P about the line.
12. **Pipes.** Two pipes have the centre lines $L_1:\ t(2,1,0)$ and $L_2:\ (0,2,1)+s(0,1,1)$ (metres) and outer radii 0.15 m and 0.20 m. (a) Show that the lines are skew and find the distance between them. (b) Find the clearance between the pipes. (c) Find the angle between the pipes.

13. **Wire.** A long straight wire through $Q(0,0,0)$ with direction $\mathbf{d} = (0,3,4)$ carries 20 A in the direction of $\mathbf{d}$. At $P(2,0,0)$ (metres), find (a) the distance from P to the wire, (b) $|\mathbf{B}|$, (c) the vector $\mathbf{B}$ (use $\mu_0 = 4\pi \cdot 10^{-7}$ T·m/A).
14. **CNC.** A tool moves along a straight line from $A(10,10,0)$ mm to $B(58,10,14)$ mm at 20 mm/s. (a) Find the length of the move, its duration and the velocity vector. (b) A clamp is at $C(34,14,7)$ mm. Find the distance from C to the tool path and the time at which the tool is closest to the clamp.
15. **Triangulation.** Station $S_1(0,0,0)$ measures the bearing $(1,0,0)$ and station $S_2(4,-2,1)$ the bearing $(-1,1,0)$. Find the closest points A, B on the two rays, the miss distance, and the triangulated position M.

7

Planes in Space

Learning Objectives

After working through this chapter you will be able to:

- describe a plane by a point and a *normal vector*, and explain why this is enough to fix it;
- write the equation of a plane from a point and a normal, from its intercepts, from three points, from a point and two directions, or from a point and a line;
- classify two planes (or a line and a plane) as parallel, perpendicular, coincident or intersecting, and measure the angle between them;
- find the line in which two planes meet and the point in which a line pierces a plane;
- compute the signed distance from a point to a plane, the foot of the perpendicular, and the mirror image of a point;
- use planes to model a banked aircraft, a glide path, an obstacle-clearance surface, a ramp, a flat mirror and a flatness check.

A runway, the floor of a hangar, a wing panel, the glass of a cockpit window: each is, to a good approximation, a piece of a **plane**, a perfectly flat sheet that extends without limit in every direction (Figure 7.1). In Chapter 6 a line in space was pinned down by one *direction* running along it. A plane has two independent directions *inside* it, so it is pinned down the other way round: by the single direction that stands perpendicular to the whole sheet, the **normal vector** (Figure 7.2). That is why a plane is described by *one* linear equation, whereas a line in space needs two.

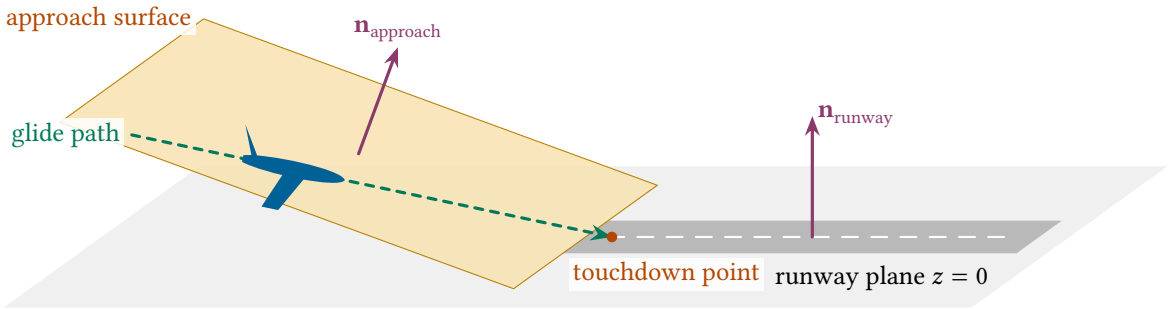


Figure 7.1: Three planes that every pilot meets: the runway (the plane $z = 0$), the invisible *approach surface* that obstacles must stay below, and the glide path, a line that crosses the runway plane at the touchdown point. The purple arrows are normal vectors; they fix the orientation of the two planes. Slopes are exaggerated. Sections 7.6.2 and 7.6.3 model this situation.

To see how this works algebraically, let $P_0(x_0, y_0, z_0)$ be a known point lying on the plane, and let $\mathbf{n} = (a, b, c)$ be a vector perpendicular to the plane (the normal vector). If $P(x, y, z)$ is any arbitrary point on the plane, the vector $\overrightarrow{P_0P}$ lies entirely within the plane. Because the normal vector is perpendicular to the entire surface, it must be orthogonal to every vector contained inside it (Figure 7.2):

$$\mathbf{n} \cdot \overrightarrow{P_0P} = 0.$$

Expanding this dot product gives the standard linear equation of a plane.

Intuition. Geometrically, a plane is a two-dimensional surface, possessing infinite length and width but absolutely zero thickness or depth, making it perfectly flat and purely conceptual. What we draw on a chalkboard, print on a page, or render on a screen can never be a true plane; instead, it is a physical object—a thin sheet of paper, a whiteboard coating, or a bounded panel—that actually possesses a tiny, non-zero depth. Because this physical thickness is microscopic relative to our scale, our surface acts as a practical, bounded approximation of an ideal 2D plane.

7.1 Equations of a Plane

7.1.1 A Point and a Normal Vector

Take a point P_0 and a direction $\mathbf{n}$. Think of P_0 as a nail in the wall and $\mathbf{n}$ as a rod stuck into the nail at a right angle. Lay a sheet of cardboard flat against the nail so that it is perpendicular to the rod: there is exactly one such sheet. Every point P of the sheet is reached from P_0 by an arrow $\overrightarrow{P_0P}$ lying *in* the sheet, and every such arrow makes a right angle with the rod (Figure 7.2). Turning this sentence into algebra gives the equation of the plane.

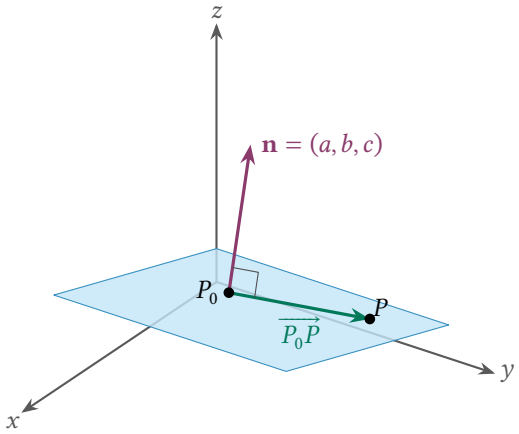


Figure 7.2: A plane through P_0 with normal vector $\mathbf{n}$. The vector $\overrightarrow{P_0P}$ to any other point P of the plane lies in the plane and is therefore perpendicular to $\mathbf{n}$ (small square: right angle).

Theorem 7.1: Point–normal equation of a plane

Let $P_0(x_0, y_0, z_0)$ be a point and $\mathbf{n} = (a, b, c) \neq \mathbf{0}$ a vector. A point $P(x, y, z)$ lies on the plane through P_0 perpendicular to $\mathbf{n}$ if and only if

$$\mathbf{n} \cdot \overrightarrow{P_0P} = 0 \qquad \Longleftrightarrow \qquad a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

Proof. P is on the plane exactly when $\overrightarrow{P_0P}$ lies in the plane, that is, exactly when $\overrightarrow{P_0P} \perp \mathbf{n}$, that is, when $\mathbf{n} \cdot \overrightarrow{P_0P} = 0$. Since $\overrightarrow{P_0P} = (x - x_0,\ y - y_0,\ z - z_0)$, writing out the dot product gives the stated equation. (The case $P = P_0$ gives $\overrightarrow{P_0P} = \mathbf{0}$, which is perpendicular to everything, so P_0 is on the plane, as it should be.) ∎

Multiplying out and collecting the constants gives the form used most often. The **general equation** of a plane is

$$ax + by + cz + d = 0,$$

where $\mathbf{n} = (a, b, c) \neq \mathbf{0}$ is a **normal vector** (perpendicular to the plane) and $d = -(ax_0 + by_0 + cz_0)$ for any point $P_0(x_0, y_0, z_0)$ of the plane. The number d fixes the *position* of the plane; the vector $\mathbf{n}$ fixes its *orientation*.

Problem: Finding a Plane's Equation

Find the plane through $P_0(1, 0, -2)$ with normal vector $\mathbf{n} = (2, -1, 3)$.

Solution. Insert into the point–normal equation:

$$2(x-1) - (y-0) + 3(z+2) = 0 \implies 2x - y + 3z + 4 = 0.$$

Check. The coordinates of P_0 must satisfy the result: $2(1) - 0 + 3(-2) + 4 = 0$. ✓

Common mistake. The numbers (a, b, c) in front of x, y, z are the components of the *normal vector*, not a point on the plane. The plane $x + 2y + 3z = 6$ has normal $(1, 2, 3)$; the point $(1, 2, 3)$ is not on it, because $1 + 4 + 9 = 14 \neq 6$. Likewise, d is *not* part of the normal vector.

Intuition. Many equations, one plane. Multiplying an equation by a nonzero number k does not move the plane: $2x - y + 3z + 4 = 0$ and $-4x + 2y - 6z - 8 = 0$ describe the same plane, with normals $\mathbf{n}$ and $-2\mathbf{n}$ pointing along the same line (one way or the other). Only the *direction* of $\mathbf{n}$ matters, never its length or sign. This freedom is useful: you can rescale an equation to remove fractions, or to make $\mathbf{n}$ a unit vector (Section 7.1.4).

The two sides of a plane

A plane cuts space into two half-spaces. The function $f(x, y, z) = ax + by + cz + d$ tells you on which side a point lies: $f = 0$ on the plane, $f > 0$ on the side into which $\mathbf{n}$ points, $f < 0$ on the other side (Figure 7.3). To see why, note that $f(P) = \mathbf{n} \cdot \overrightarrow{P_0P}$, which is positive when $\overrightarrow{P_0P}$ has a component along $\mathbf{n}$ and negative when it has a component against $\mathbf{n}$. This simple sign test is used in computer graphics and in aviation, for example to decide whether an obstacle sticks up through a safety surface (Section 7.6).

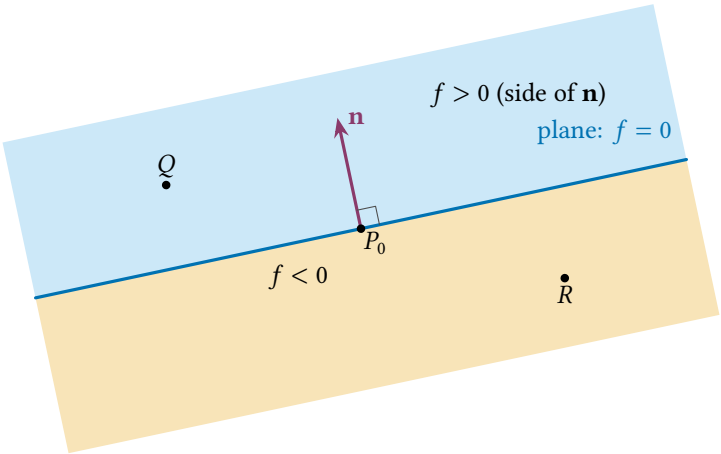


Figure 7.3: A plane seen edge-on. With $f(x, y, z) = ax + by + cz + d$, the point Q on the side of $\mathbf{n}$ has $f(Q) > 0$, the point P_0 on the plane has $f = 0$, and R on the far side has $f(R) < 0$.

Quick Check 1

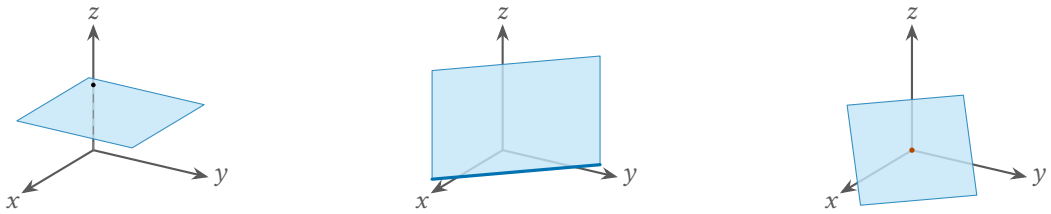
1. Is the point $(1, 1, 1)$ on the plane $2x - y + 3z + 4 = 0$? On which side of it does it lie?
2. Which plane is described by $z = 0$? Give a normal vector.
3. The planes $x + 2y - z = 3$ and $2x + 4y - 2z = 6$: are they the same plane?

7.1.2 Planes in Special Positions

Because $\mathbf{n} = (a, b, c)$ is perpendicular to the plane, a zero coefficient says that the plane does not "see" that axis direction: if $c = 0$ then $\mathbf{n} \perp \mathbf{k}$, so $\mathbf{k}$ points *along* the plane, which is therefore parallel to the z-axis. A zero constant term $d = 0$ means $(0, 0, 0)$ satisfies the equation, so the plane passes through the origin (see Table 7.1 and Figure 7.4).

Equation	Position of the plane
$z = c \quad (a = b = 0)$	parallel to the xy-plane (a "horizontal floor"); normal $\parallel \mathbf{k}$
$ax + by + d = 0 \quad (c = 0)$	parallel to the z-axis (a "vertical wall"); the picture is the same line $ax + by + d = 0$ at every height
$ax + by + cz = 0 \quad (d = 0)$	passes through the origin

Table 7.1: Reading the position of a plane from its equation. The same pattern holds for x and y in place of z.



(a) $z = c$ **(b)** $ax + by + d = 0$ **(c)** $ax + by + cz = 0$

Figure 7.4: Three special positions. (a) A plane parallel to the coordinate plane xy. (b) A plane with no z term is a vertical wall over the line $ax + by + d = 0$ of the xy-plane. (c) With $d = 0$ the plane passes through the origin (dot).

7.1.3 Intercept Form

If a plane meets the three axes at $(a, 0, 0)$, $(0, b, 0)$ and $(0, 0, c)$, these **intercepts** give a very readable equation, and a quick sketch: draw the three points and join them (Figure 7.5). A plane with x-, y- and z-intercepts a, b, c (all *nonzero*) has equation

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1.$$

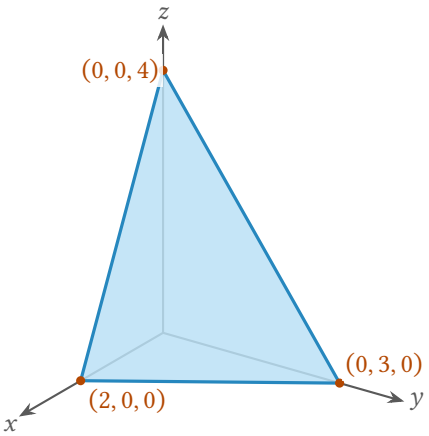


Figure 7.5: The plane $\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = 1$ cuts the axes at 2, 3 and 4. The triangle is the part of the plane in the first octant.

When the intercept form does not apply

Intercept form needs three *nonzero* intercepts. Two things can go wrong.

- **A plane parallel to an axis has no intercept on that axis.** For example, a plane parallel to the x-axis never meets it, and its equation has no x term: $\frac{y}{b}+\frac{z}{c}=1$.
- **A plane through the origin has all its intercepts equal to** 0, and the fractions $\frac{x}{0}$ are meaningless. Use the general form $ax+by+cz=0$ instead.

7.1.4 Normalised Form and Direction Cosines

Dividing the general equation $ax+by+cz+d'=0$ by the length $|\mathbf{n}|=\sqrt{a^2+b^2+c^2}$ turns $\mathbf{n}$ into a *unit* vector $\hat{\mathbf{n}}=(\cos\alpha,\cos\beta,\cos\gamma)$. Its components are the **direction cosines** (the cosines of the angles that $\hat{\mathbf{n}}$ makes with the three axes), and they satisfy $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$. The result is the **normalised (Hesse) equation**

$$\cos\alpha\,x+\cos\beta\,y+\cos\gamma\,z+d=0, \qquad d=\frac{d'}{\sqrt{a^2+b^2+c^2}}.$$

Theorem 7.2: Distance from the origin

In the normalised equation $\hat{\mathbf{n}}\cdot\mathbf{r}+d=0$, the perpendicular distance from the origin to the plane is $|d|$.

Proof. Let F be the point of the plane closest to the origin. The segment OF is perpendicular to the plane, so it is parallel to $\hat{\mathbf{n}}$: $\overrightarrow{OF}=s\hat{\mathbf{n}}$. Substituting into the equation gives $s+d=0$, so $s=-d$ and $|OF|=|s|\,|\hat{\mathbf{n}}|=|d|$. ∎

This is exactly the Hesse normal form of a line in Chapter 5, one dimension up (Figure 7.6).

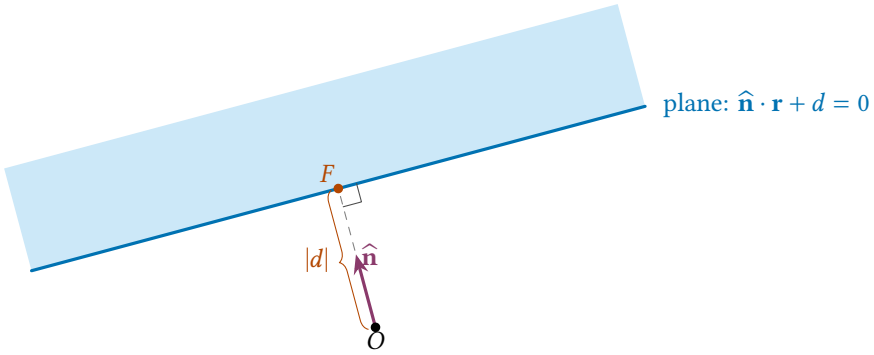


Figure 7.6: Normalised equation: the distance from the origin O to the plane is $|d|$, measured along the unit normal $\hat{\mathbf{n}}$ to the foot F of the perpendicular.

Problem: Normalising an Equation

Normalise $2x-3y+6z-14=0$ and find its distance from the origin.

Solution. $|\mathbf{n}|=\sqrt{4+9+36}=7$, so

$$\tfrac{2}{7}x-\tfrac{3}{7}y+\tfrac{6}{7}z-2=0, \qquad (\cos\alpha,\cos\beta,\cos\gamma)=\left(\tfrac{2}{7},-\tfrac{3}{7},\tfrac{6}{7}\right).$$

The distance from the origin is $|d|=2$. The angles with the axes are $\alpha\approx 73.4^\circ$, $\beta\approx 115.4^\circ$ and $\gamma\approx 31.0^\circ$, and indeed $\frac{4+9+36}{49}=1$.

7.2 Two Planes: Angle and Position

7.2.1 The Angle Between Two Planes

Definition: Angle Between Two Planes

The angle between two planes with normal vectors $\mathbf{n}_1 = (a_1, b_1, c_1)$ and $\mathbf{n}_2 = (a_2, b_2, c_2)$ is the *acute* angle θ between the normals:

$$\cos\theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1|\,|\mathbf{n}_2|}, \qquad 0^\circ \le \theta \le 90^\circ.$$

To see why the angle between the planes equals the angle between the normals, look at the two planes *edge-on*, along their common line (Figure 7.7). Each normal is perpendicular to its plane. Turning plane 1 into plane 2 means rotating by θ, and the normals, which are glued at right angles to their planes, rotate by the same θ.

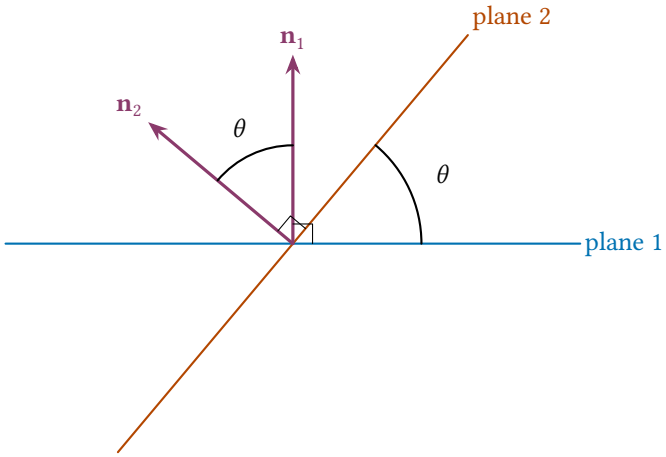


Figure 7.7: Two planes seen edge-on (the view along their line of intersection, which shrinks to the point O). The angle θ between the planes is the same as the angle between their normal vectors.

Intuition. Every plane has two normals, $\mathbf{n}$ and $-\mathbf{n}$, which point to opposite sides. Depending on which one you pick, the angle between the normals is θ or $180^\circ - \theta$. The absolute value in the formula removes this ambiguity: two intersecting planes make a pair of equal acute angles and a pair of equal obtuse angles, and by convention we report the acute one.

Example: Inclination of a Plane to the Horizontal

The angle between a plane and the horizontal xy-plane (normal $\mathbf{k} = (0, 0, 1)$) is called its **inclination**. For $x + y + z = 1$, $\mathbf{n} = (1, 1, 1)$:

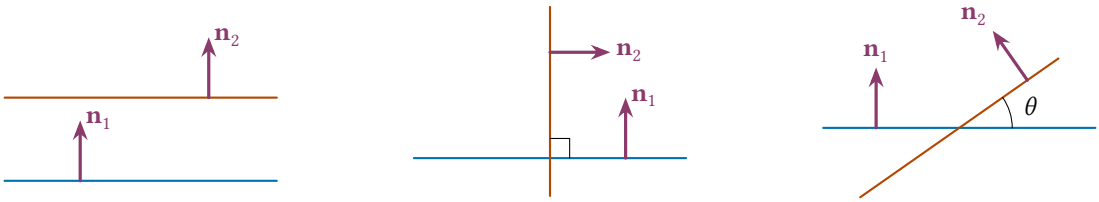
$$\cos\theta = \frac{|1\cdot 0 + 1\cdot 0 + 1\cdot 1|}{\sqrt{3}\cdot 1} = \frac{1}{\sqrt{3}} \approx 0.577 \implies \theta \approx 54.7^\circ.$$

In general the inclination of $ax + by + cz + d = 0$ satisfies $\cos\theta = |c|/|\mathbf{n}|$. For a ramp, a roof or a banked wing, this is the slope angle (see Section 7.6).

Common mistake. Forgetting the absolute value. If $\mathbf{n}_1 \cdot \mathbf{n}_2 < 0$ the plain formula gives an obtuse angle, which is the angle between the *normals you happened to choose*, not the angle between the planes. Always take $|\mathbf{n}_1 \cdot \mathbf{n}_2|$.

7.2.2 Parallel, Perpendicular, Coincident

Two planes are exactly in one of three situations: they never meet (**parallel**), they are the same plane (**coincident**), or they meet along a line (**intersecting**); perpendicular planes are the special intersecting case with $\theta = 90^\circ$ (Figure 7.8).



(a) parallel: $\mathbf{n}_1 \parallel \mathbf{n}_2$ **(b)** perpendicular: $\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$ **(c)** intersecting: $0^\circ < \theta < 90^\circ$

Figure 7.8: The three relative positions of two planes, seen edge-on. Parallel planes have parallel normals; perpendicular planes have perpendicular normals.

Position	Test on the normals	Additional check
parallel (distinct)	$\mathbf{n}_1 = k\,\mathbf{n}_2$, i.e. $\mathbf{n}_1 \times \mathbf{n}_2 = \mathbf{0}$	a point of one plane is *not* on the other
coincident	$\mathbf{n}_1 = k\,\mathbf{n}_2$	a point of one plane *is* on the other
perpendicular	$\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$	none
intersecting	$\mathbf{n}_1 \times \mathbf{n}_2 \neq \mathbf{0}$	they meet along a line of direction $\mathbf{n}_1 \times \mathbf{n}_2$

Example: Relative Positions of Planes

- **Testing for Perpendicularity ($\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$):**
  - $2x + 3y - z + 1 = 0$ and $x + y + 5z - 4 = 0$: $\mathbf{n}_1 \cdot \mathbf{n}_2 = 2 + 3 - 5 = 0$. The planes are *perpendicular*.
  - $2x + 3y - z + 1 = 0$ and $-6x + y + 4z - 5 = 0$: $\mathbf{n}_1 \cdot \mathbf{n}_2 = -12 + 3 - 4 = -13 \neq 0$. These planes are *not* perpendicular.
- **Parallel Versus Coincident ($\mathbf{n}_2 = k\mathbf{n}_1$):**
  - $2x - 4y + 6z - 3 = 0$ and $4x - 8y + 12z + 1 = 0$: here $\mathbf{n}_2 = 2\mathbf{n}_1$, so the normals are parallel. However, the constant does not scale by the same factor: $2 \cdot (-3) = -6 \neq 1$. So they are *distinct* parallel planes.
  - $3x - y + 2z = 5$ and $-6x + 2y - 4z = -10$: $\mathbf{n}_2 = -2\mathbf{n}_1$ and also $-10 = -2 \cdot 5$. Both normals and constants scale equally, so the planes are *coincident*.

7.2.3 Intersecting Planes and the Line of Intersection

If $\mathbf{n}_1 \times \mathbf{n}_2 \neq \mathbf{0}$, the planes are not parallel, and two non-parallel planes always meet along a line (Figure 7.9). The direction of that line is forced on us: the line lies in both planes, so it is perpendicular to both normals, and a vector perpendicular to two vectors is their cross product.

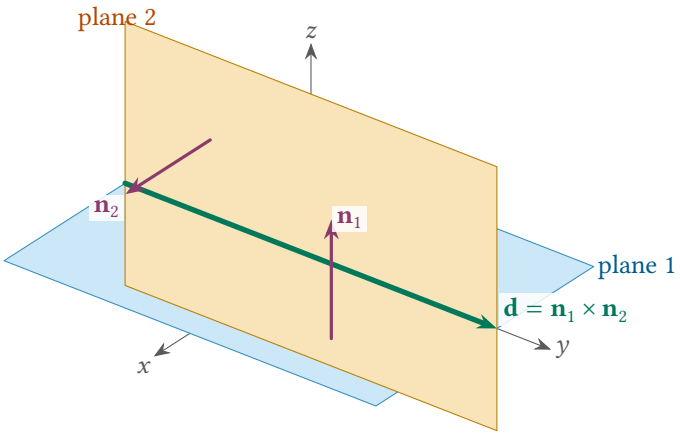


Figure 7.9: Two non-parallel planes meet along a line. The direction $\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2$ of this line is perpendicular to both normals. Here plane 1 is $z = 0$ (normal $\mathbf{k}$), plane 2 is $x = 0$ (normal $\mathbf{i}$), and $\mathbf{k} \times \mathbf{i} = \mathbf{j}$ points along the y-axis.

Finding the line of intersection

1. **Direction**: $\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2$.
2. **A point**: set one variable to a convenient value (say $z = 0$) and solve the resulting 2×2 system for the other two. If this has no solution, the line never crosses $z = 0$ (its direction has $d_z = 0$, so it is parallel to the xy-plane); fix another variable instead.
3. **The line**: through that point, with direction $\mathbf{d}$, in parametric or canonical form.
4. **Check**: $\mathbf{d} \cdot \mathbf{n}_1 = \mathbf{d} \cdot \mathbf{n}_2 = 0$, and the point satisfies both equations.

Problem: Line of Intersection

Find the line where $2x - y + 3z - 5 = 0$ and $x + 2y - z + 1 = 0$ meet.

Solution. $\mathbf{n}_1 = (2, -1, 3)$, $\mathbf{n}_2 = (1, 2, -1)$:

$$\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 3 \\ 1 & 2 & -1 \end{vmatrix} = \mathbf{i}(1 - 6) - \mathbf{j}(-2 - 3) + \mathbf{k}(4 + 1) = (-5, 5, 5).$$

Dividing by 5 gives $\mathbf{d} = (-1, 1, 1)$. Setting $z = 0$: $2x - y - 5 = 0$ and $x + 2y + 1 = 0$. From the second, $x = -2y - 1$; substituting, $2(-2y - 1) - y - 5 = 0 \implies -5y - 7 = 0 \implies y = -\frac{7}{5}$, $x = \frac{9}{5}$. So $P\left(\frac{9}{5}, -\frac{7}{5}, 0\right)$ lies on both planes, and

$$\mathbf{r}(t) = \left(\tfrac{9}{5}, -\tfrac{7}{5}, 0\right) + t(-1, 1, 1), \qquad \frac{x - \frac{9}{5}}{-1} = \frac{y + \frac{7}{5}}{1} = \frac{z}{1}.$$

Check. $\mathbf{d} \cdot \mathbf{n}_1 = -2 - 1 + 3 = 0$ and $\mathbf{d} \cdot \mathbf{n}_2 = -1 + 2 - 1 = 0$. Also $2 \cdot \frac{9}{5} + \frac{7}{5} - 5 = 0$ and $\frac{9}{5} - \frac{14}{5} + 1 = 0$. ✓

Intuition. Two equations, three unknowns. Each plane equation is a linear equation in x, y, z, so finding the intersection of two planes is solving a linear system of two equations with three unknowns. There is always at least one free variable. This is why the answer is a line (one free parameter, t) rather than a point.

While this dimension-counting argument explains the typical case when two planes intersect at an angle, the nature of the linear system changes entirely if their normal vectors are parallel. Depending

on whether the constant terms match proportionally, the two planes will either never meet or collapse into a single surface. These distinct geometric outcomes are summarized below:

Normals	Equations	Solution set
not parallel	always consistent	a line (one free parameter)
parallel	contradictory (0 = nonzero)	no solution (distinct parallel planes)
parallel	one equation is a multiple of the other	a whole plane (two free parameters)

7.3 A Line and a Plane

7.3.1 The Angle Between a Line and a Plane

Definition: Angle Between a Line and a Plane

For a line with direction vector $\mathbf{d}$ and a plane with normal $\mathbf{n}$, the angle θ between the *line and the plane itself* satisfies

$$\sin\theta = \frac{|\mathbf{d}\cdot\mathbf{n}|}{|\mathbf{d}|\,|\mathbf{n}|}, \qquad 0^\circ \le \theta \le 90^\circ.$$

Geometrically, θ is the angle between the line and its shadow (orthogonal projection) on the plane. It is the complement of the angle α between $\mathbf{d}$ and $\mathbf{n}$ (Figure 7.10): $\theta = 90^\circ - \alpha$.

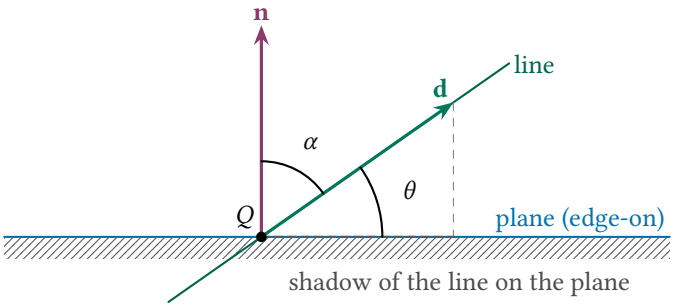


Figure 7.10: A line crossing a plane at the point Q, seen from the side with the plane edge-on. θ is the angle between the line and the plane; $\alpha = 90^\circ - \theta$ is the angle between the line's direction $\mathbf{d}$ and the normal $\mathbf{n}$.

Intuition. **Why sine here, but cosine for two planes?** The dot product $\mathbf{d}\cdot\mathbf{n}$ always measures the angle between two *vectors*, here $\mathbf{d}$ and $\mathbf{n}$, giving $\cos\alpha$. But the plane itself is not the vector $\mathbf{n}$; it is the sheet *perpendicular* to $\mathbf{n}$. So the angle to the plane is $90^\circ - \alpha$, and $\cos\alpha = \sin(90^\circ - \alpha) = \sin\theta$. A line along the normal ($\alpha = 0^\circ$) is perpendicular to the plane ($\theta = 90^\circ$); a line perpendicular to the normal ($\alpha = 90^\circ$) lies flat along the plane ($\theta = 0^\circ$).

Problem: Angle Between a Line and a Plane

Find the angle θ between the line $\mathbf{r}(t) = (1,0,2) + t(1,2,3)$ and the plane $x - y + z = 5$.

Solution. The direction vector of the line is $\mathbf{d} = (1,2,3)$, and the normal vector of the plane is $\mathbf{n} = (1,-1,1)$. Compute dot product and magnitudes:

$$\mathbf{d}\cdot\mathbf{n} = (1)(1) + (2)(-1) + (3)(1) = 2, \quad |\mathbf{d}| = \sqrt{14}, \quad |\mathbf{n}| = \sqrt{3}.$$

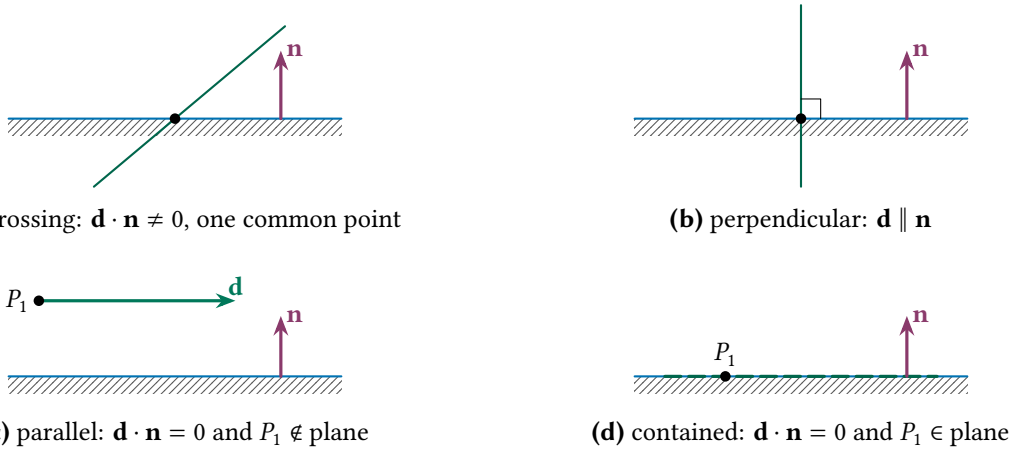
Since the angle θ between a line and a plane is complementary to the angle between its direction and the normal vector, we use sine:

$$\sin\theta = \frac{|\mathbf{d}\cdot\mathbf{n}|}{|\mathbf{d}||\mathbf{n}|} = \frac{2}{\sqrt{42}} \approx 0.309 \implies \theta \approx 18.0^\circ.$$

Common mistake. Using $\cos\theta$ instead of $\sin\theta$ in the line–plane formula gives the angle between the line and the *normal*, which is 90° minus the angle you want. A quick sanity check: a line almost parallel to the plane must give a *small* θ.

7.3.2 Four Positions of a Line Relative to a Plane

A line and a plane are in exactly one of four positions (Figure 7.11). Everything is decided by $\mathbf{d}\cdot\mathbf{n}$ and, in one case, by a single extra test.



(a) crossing: $\mathbf{d}\cdot\mathbf{n} \neq 0$, one common point

(b) perpendicular: $\mathbf{d} \parallel \mathbf{n}$

(c) parallel: $\mathbf{d}\cdot\mathbf{n} = 0$ and $P_1 \notin$ plane

(d) contained: $\mathbf{d}\cdot\mathbf{n} = 0$ and $P_1 \in$ plane

Figure 7.11: A line and a plane, with the plane edge-on. In (c) and (d) the line direction is perpendicular to the normal; the two cases differ only in whether a point P_1 of the line lies on the plane.

Position	Test	Common points
crossing	$\mathbf{d}\cdot\mathbf{n} \neq 0$	exactly one
perpendicular	$\mathbf{d}\times\mathbf{n} = \mathbf{0}$ (so $\mathbf{d} \parallel \mathbf{n}$)	exactly one
parallel	$\mathbf{d}\cdot\mathbf{n} = 0$, and a point of the line is *not* on the plane	none
contained	$\mathbf{d}\cdot\mathbf{n} = 0$, and a point of the line *is* on the plane	the whole line

Common mistake. $\mathbf{d}\cdot\mathbf{n} = 0$ does *not* by itself mean "no intersection". It only says the line points along the plane's directions; the line may be floating beside the plane (parallel) or lying in it (contained). Always test one point of the line.

What the equation for t tells you

Substituting a line into a plane always gives an equation of the form $\alpha t = \beta$, where $\alpha = \mathbf{d}\cdot\mathbf{n}$. If $\alpha \neq 0$ there is exactly one solution (one crossing point). If $\alpha = 0$ and $\beta \neq 0$ the equation is impossible (parallel). If $\alpha = 0$ and $\beta = 0$ it holds for every t (contained).

Example: Relative Positions of a Line and a Plane

Determine the intersection or relative position for each of the following line-plane pairs:

- **Perpendicular Case:** Line $\mathbf{r}(t) = (1, 2, 3) + t(2, 4, 6)$, plane $2x + 4y + 6z = 1$.
  - *Solution:* The direction vector $\mathbf{d} = (2, 4, 6)$ is parallel to the normal vector $\mathbf{n} = (2, 4, 6)$ (so $\mathbf{d} \times \mathbf{n} = \mathbf{0}$). Therefore, the line is *perpendicular* to the plane.
- **Parallel (Disjoint) Case:** Line $\mathbf{r}(t) = (1, 2, 3) + t(1, 0, 1)$, plane $x + 2y - z = 4$.
  - *Solution:* $\mathbf{d} \cdot \mathbf{n} = (1, 0, 1) \cdot (1, 2, -1) = 1 + 0 - 1 = 0$, meaning the line is parallel to the plane. Testing the base point $(1, 2, 3)$ gives $1 + 4 - 3 = 2 \neq 4$, so the point does not lie on the plane. Thus, the line is strictly *parallel* and disjoint.
- **Line Contained in the Plane:** Line $\mathbf{r}(t) = (1, 0, 0) + t(1, -1, 0)$, plane $x + y + z = 1$.
  - *Solution:* Here $\mathbf{d} \cdot \mathbf{n} = (1, -1, 0) \cdot (1, 1, 1) = 0$ (parallel), and the base point $(1, 0, 0)$ satisfies the plane equation $(1 + 0 + 0 = 1)$. Because a parallel line shares at least one point with the plane, the entire line lies *in* the plane (substituting $x = 1+t$, $y = -t$, $z = 0$ yields the identity $1 = 1$).
- **Single Intersection Point:** Line $\mathbf{r}(t) = (1, 2, 3) + t(1, -1, 2)$, plane $3x + 2y - z = 1$.
  - *Solution:* $\mathbf{d} \cdot \mathbf{n} = (1, -1, 2) \cdot (3, 2, -1) = 3 - 2 - 2 = -1 \neq 0$, so they intersect at a unique point. Substituting parametric expressions into the plane equation:

$$3(1 + t) + 2(2 - t) - (3 + 2t) = 1 \implies 4 - t = 1 \implies t = 3.$$

  Evaluating at $t = 3$ gives $(4, -1, 9)$. **Check:** $3(4) + 2(-1) - 9 = 1$. ✓

Quick Check 2

1. Two planes have normals $(1, 2, 2)$ and $(2, 4, 4)$. What can you say about the planes? What more do you need to know?
2. The line $\mathbf{r}(t) = (0, 0, 1) + t(1, 1, 0)$ and the plane $z = 1$: parallel, contained, or crossing?
3. Can two planes in space be perpendicular *and* parallel? Can a line be perpendicular to a plane and parallel to it?

7.4 Finding the Equation of a Plane

To write down a plane's equation we always need two ingredients: *a point on the plane* and *a normal vector*. The data in a problem differ, and so does the way the normal vector is obtained. The table gives the road map; the cases that follow explain each row.

What you are given	Point to use	Normal vector
a point and a normal vector	the point	the given $\mathbf{n}$
a point and two non-collinear vectors parallel to the plane (Case 1)	the point	$\mathbf{u} \times \mathbf{v}$
two points P_1, P_2 and a vector $\mathbf{v}$ parallel to the plane (Case 2)	P_1	$\overrightarrow{P_1P_2} \times \mathbf{v}$
three non-collinear points (Case 3)	P_1	$\overrightarrow{P_1P_2} \times \overrightarrow{P_1P_3}$
a line and a point not on it (Case 4)	the given point	$\mathbf{d} \times \overrightarrow{AP}$, with A on the line

Intuition. Why the cross product? A plane has two independent directions inside it, but a plane equation needs the *one* direction that sticks out of it. The cross product is exactly the tool that turns two in-plane directions into the direction perpendicular to both. That is why every case below ends with "$\mathbf{n} = \mathbf{u} \times \mathbf{v}$".

Definition: Collinear Vectors

Two vectors $\mathbf{v}, \mathbf{w}$ are **collinear** if $\mathbf{v} = k\mathbf{w}$ for some scalar k: they point along the same line (Figure 7.12), in the same or in opposite directions. **Collinear points** lie on a single line. Two *non*-collinear vectors that lie in a plane (or are parallel to it) determine its orientation, because their cross product is a normal to the plane. If the vectors *are* collinear, then $\mathbf{u} \times \mathbf{v} = \mathbf{0}$ and no plane is determined.

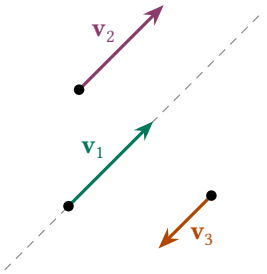


Figure 7.12: Collinear vectors: $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ all point along the same direction (the dashed guide line), $\mathbf{v}_3$ the opposite way. Their cross products with one another are all $\mathbf{0}$.

Case 1: a point and two non-collinear vectors parallel to the plane. Given $P_0(x_0, y_0, z_0)$ and non-collinear $\mathbf{u}, \mathbf{v}$ parallel to the plane, $\mathbf{n} = \mathbf{u} \times \mathbf{v}$ is a normal, and the plane is $A(x-x_0)+B(y-y_0)+C(z-z_0) = 0$ with $\mathbf{n} = (A, B, C)$ ((Figure 7.14a).

Case 2: two points and a vector parallel to the plane. Given P_1, P_2 on the plane and $\mathbf{v}$ parallel to it, form $\mathbf{u} = \overrightarrow{P_1P_2}$. Since $\mathbf{u}, \mathbf{v}$ both lie in (or parallel to) the plane, $\mathbf{n} = \mathbf{u} \times \mathbf{v}$ is a normal. (This needs $\mathbf{v}$ not parallel to $\overrightarrow{P_1P_2}$.) (Figure 7.14b).

Case 3: three non-collinear points. Given P_1, P_2, P_3, form $\mathbf{u} = \overrightarrow{P_1P_2}$, $\mathbf{v} = \overrightarrow{P_1P_3}$ and take $\mathbf{n} = \mathbf{u} \times \mathbf{v}$ (Figure 7.13). Equivalently, the plane is

$$\det \begin{bmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{bmatrix} = 0.$$

Expanding this determinant along the first row reproduces exactly $A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$ with $(A, B, C) = \mathbf{u} \times \mathbf{v}$. In words: the point P lies in the plane when the three vectors $\overrightarrow{P_1P}$, $\mathbf{u}$, $\mathbf{v}$ are coplanar, that is, when their scalar triple product is 0.

Case 4: a line and a point not on it. Let the line be $\mathbf{r}(t) = \mathbf{r}_A + t\,\mathbf{d}$ through the point A, and let P be a point not on it. The vectors $\mathbf{d}$ and $\overrightarrow{AP}$ both lie in the required plane and are not collinear (otherwise P would be on the line), so $\mathbf{n} = \mathbf{d} \times \overrightarrow{AP}$ (Figure 7.14c).

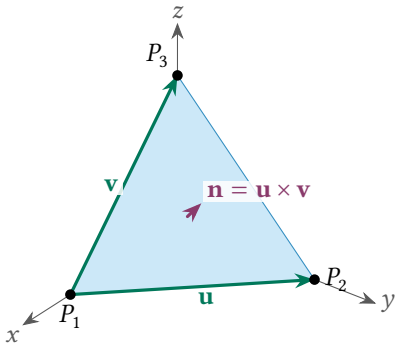
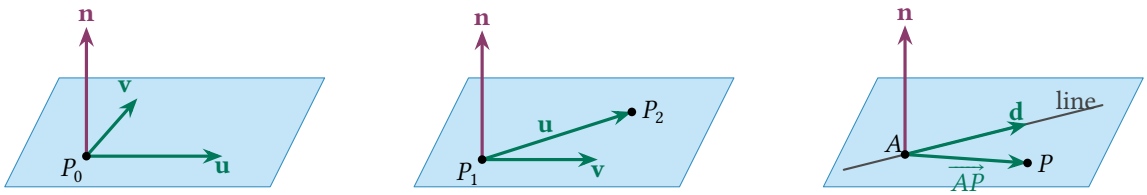


Figure 7.13: Case 3 for $P_1(1,0,0)$, $P_2(0,1,0)$, $P_3(0,0,1)$. The vectors $\mathbf{u} = \overrightarrow{P_1P_2}$ and $\mathbf{v} = \overrightarrow{P_1P_3}$ lie in the plane; $\mathbf{n} = \mathbf{u} \times \mathbf{v} = (1,1,1)$ is perpendicular to both.



(a) Case 1: $\mathbf{n} = \mathbf{u} \times \mathbf{v}$ **(b)** Case 2: $\mathbf{u} = \overrightarrow{P_1P_2}$, $\mathbf{n} = \mathbf{u} \times \mathbf{v}$ **(c)** Case 4: $\mathbf{n} = \mathbf{d} \times \overrightarrow{AP}$

Figure 7.14: Constructing the normal vector in different scenarios: (a) given two vectors at a point, (b) given two points and a parallel vector, and (c) given a line and an external point. Case 3 is shown in Figure 7.13.

Example: Constructing Equations of Planes

Find the equation of the plane for each of the following geometric configurations:

- **Case 1 (A point and two vectors):** Plane through $P_0(1,2,3)$ parallel to $\mathbf{u} = (1,0,-1)$ and $\mathbf{v} = (0,2,1)$.
  - *Solution:*

$$\mathbf{n} = \mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & -1 \\ 0 & 2 & 1 \end{vmatrix} = (2,-1,2).$$

    Using $P_0(1,2,3)$: $2(x-1) - (y-2) + 2(z-3) = 0 \implies 2x - y + 2z - 6 = 0$.
- **Case 2 (Two points and a parallel vector):** Plane through $P_1(1,0,0)$ and $P_2(0,1,0)$, parallel to $\mathbf{v} = (0,0,1)$.
  - *Solution:* $\mathbf{u} = \overrightarrow{P_1P_2} = (-1,1,0)$, so:

$$\mathbf{n} = \mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = (1,1,0).$$

    Using P_1: $(x-1) + y = 0 \implies x + y - 1 = 0$. (This is a vertical wall parallel to the z-axis, as shown in Figure 7.4(b).)
- **Case 3 (Three points):** Plane through $P_1(1,0,0)$, $P_2(0,1,0)$, and $P_3(0,0,1)$.
  - *Solution:* Form two vectors $\mathbf{u} = (-1,1,0)$ and $\mathbf{v} = (-1,0,1)$:

$$\mathbf{n} = \mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = (1,1,1).$$

    Equation: $(x-1) + y + z = 0 \implies x + y + z - 1 = 0$ (the standard intercept form).

Example: Constructing Equations of Planes

Case 4 (A line and an external point): Plane containing the line $\mathbf{r}(t) = (1,0,0) + t(1,1,1)$ and the point $P(0,0,1)$.

- *Solution:* Take base point $A = (1,0,0)$ on the line, direction $\mathbf{d} = (1,1,1)$, and vector $\overrightarrow{AP} = (-1,0,1)$:

$$\mathbf{n} = \mathbf{d} \times \overrightarrow{AP} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ -1 & 0 & 1 \end{vmatrix} = (1,-2,1).$$

  Equation: $(x-1) - 2y + z = 0 \implies x - 2y + z - 1 = 0$.
- **Check:** Point P: $0 - 0 + 1 - 1 = 0$. Line point $\mathbf{r}(1) = (2,1,1)$: $2 - 2 + 1 - 1 = 0$. ✓

Common mistake. Three points do not always determine a plane. If the points are collinear, then $\mathbf{u} \times \mathbf{v} = \mathbf{0}$ and you get $\mathbf{n} = \mathbf{0}$, which is not a normal vector. For example $(1,1,1)$, $(2,2,2)$, $(3,3,3)$ lie on one line, and infinitely many planes contain that line. Always check that $\mathbf{u} \times \mathbf{v} \neq \mathbf{0}$. Also, after finding the equation, *substitute the given points back in*: this catches almost every arithmetic slip.

7.5 Distances and Mirror Images

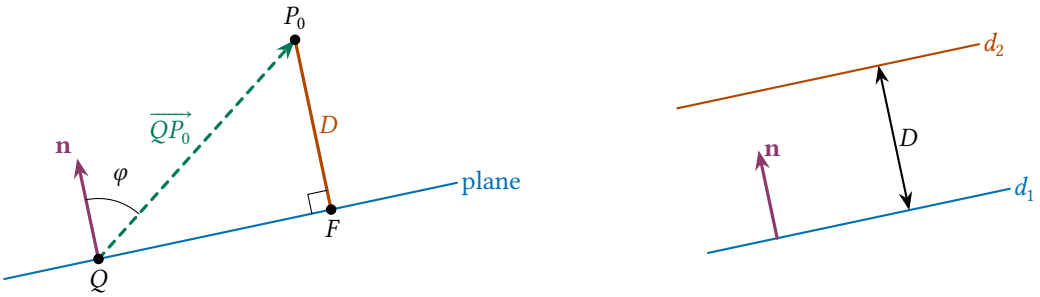
7.5.1 Distance From a Point to a Plane

Theorem 7.3: Distance from a point to a plane

The distance from $P_0(x_0, y_0, z_0)$ to the plane $ax + by + cz + d = 0$ is

$$D = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}} = \frac{|f(P_0)|}{|\mathbf{n}|}.$$

Proof. Pick any point Q on the plane. The shortest path from P_0 to the plane is the perpendicular one, i.e. the component of $\overrightarrow{QP_0}$ along the normal (Figure 7.15). Therefore $D = \left|\text{proj}_{\mathbf{n}} \overrightarrow{QP_0}\right| = \dfrac{|\mathbf{n} \cdot \overrightarrow{QP_0}|}{|\mathbf{n}|}$. Since Q is on the plane, $\mathbf{n} \cdot Q = -d$, so $\mathbf{n} \cdot \overrightarrow{QP_0} = \mathbf{n} \cdot P_0 + d = ax_0 + by_0 + cz_0 + d$. ■



(a) point and plane **(b)** two parallel planes

Figure 7.15: (a) The distance D from P_0 to the plane is the length of the component of $\overrightarrow{QP_0}$ along the normal: $D = |QP_0| \cos\varphi$. (b) For two parallel planes $ax + by + cz + d_1 = 0$ and $ax + by + cz + d_2 = 0$, the distance is the same everywhere: $D = |d_1 - d_2|/|\mathbf{n}|$.

Intuition. The value $f(P_0)$ is a distance in disguise. Substituting a point into the plane equation measures how far it is from satisfying the equation. The division by $|\mathbf{n}|$ converts that "gap" into an honest length. This is why multiplying the plane equation by 10 does not change the distance: both the numerator and $|\mathbf{n}|$ are multiplied by 10. The sign of $f(P_0)$ tells you the side of the plane (Figure 7.3), so $f(P_0)/|\mathbf{n}|$ is a *signed distance*.

Same idea, one dimension up

This is identical in spirit to the point-to-line distance in Chapter 5 and the point-to-line-in-space distance of Chapter 6: in each case, substituting the point into the (normalised) equation measures how far it is from satisfying it, and that gap equals the perpendicular distance.

7.5.2 Distance Between Parallel Planes

If two planes are parallel, every point of one is at the same distance from the other. So take any point of the first plane and apply the theorem. When both equations have *the same* left-hand side, $ax + by + cz + d_1 = 0$ and $ax + by + cz + d_2 = 0$, the result simplifies to

$$D = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}.$$

Common mistake. The formula $|d_1 - d_2|/|\mathbf{n}|$ is valid only when the two equations have *identical* coefficients (a, b, c). If one equation is, say, $2x - 4y + 6z - 3 = 0$ and the other $4x - 8y + 12z + 1 = 0$, first divide the second by 2 to get $2x - 4y + 6z + \frac{1}{2} = 0$, and only then compare the constants.

Example: Distance Between Parallel Planes

- $2x + y - 2z - 6 = 0$ and $2x + y - 2z + 3 = 0$: $|\mathbf{n}| = \sqrt{4 + 1 + 4} = 3$, so $D = \dfrac{|-6 - 3|}{3} = 3$.
- The two planes from the earlier example, $2x - 4y + 6z - 3 = 0$ and $2x - 4y + 6z + \frac{1}{2} = 0$: $|\mathbf{n}| = \sqrt{56} = 2\sqrt{14}$, so $D = \dfrac{|-3 - \frac{1}{2}|}{2\sqrt{14}} = \dfrac{7}{4\sqrt{14}} = \dfrac{\sqrt{14}}{8} \approx 0.47$.

7.5.3 Foot of the Perpendicular and Mirror Image

The nearest point of the plane to P is the **foot** F of the perpendicular from P. The **mirror image** P' of P lies on the other side, at the same distance from the plane, on the same perpendicular (Figure 7.16). Both can be written down at once, because the line from P to F is parallel to $\mathbf{n}$.

Theorem 7.4: Foot and mirror image

For the plane $f(x, y, z) = ax + by + cz + d = 0$ with $\mathbf{n} = (a, b, c)$ and a point P,

$$F = P - \frac{f(P)}{|\mathbf{n}|^2}\,\mathbf{n}, \qquad P' = P - \frac{2f(P)}{|\mathbf{n}|^2}\,\mathbf{n}.$$

Proof. Since $\overrightarrow{PF} \parallel \mathbf{n}$, write $F = P + t\mathbf{n}$. Then F is on the plane when $f(P + t\mathbf{n}) = f(P) + t|\mathbf{n}|^2 = 0$, so $t = -f(P)/|\mathbf{n}|^2$. The mirror image is the same step taken twice: $P' = P + 2t\mathbf{n}$. ∎

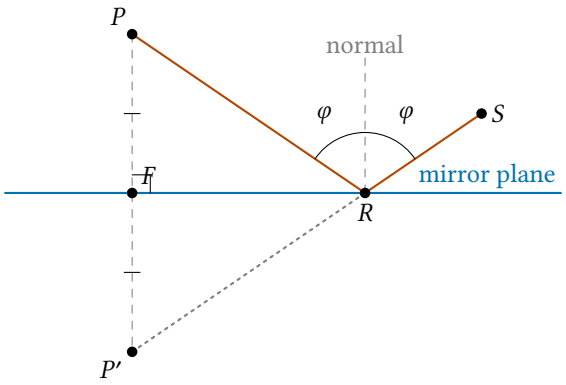


Figure 7.16: The mirror image P' of P in a plane: PP' is perpendicular to the plane and is bisected by it at F. A light ray from P that reflects at R and reaches S appears to come from P'; the angles of incidence and reflection (both φ) are equal.

Example: Foot and Mirror Image

For $P(1,-2,3)$ and the plane $2x-3y+4z-5=0$ we found $f(P)=15$ and $|\mathbf{n}|^2=29$. Hence

$$F = (1,-2,3) - \tfrac{15}{29}(2,-3,4) = \left(-\tfrac{1}{29}, -\tfrac{13}{29}, \tfrac{27}{29}\right),$$
$$P' = (1,-2,3) - \tfrac{30}{29}(2,-3,4) = \left(-\tfrac{31}{29}, \tfrac{32}{29}, -\tfrac{33}{29}\right).$$

Check. $f(F) = \frac{-2+39+108}{29} - 5 = 5 - 5 = 0$, so F is on the plane. The midpoint of PP' is F, and $|PF| = \frac{15}{\sqrt{29}}$ agrees with the distance computed above. ✓

Key Formulas

Point–normal form	$a(x-x_0)+b(y-y_0)+c(z-z_0)=0, \quad \mathbf{n}=(a,b,c)$
General and intercept form	$ax+by+cz+d=0; \quad \frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$
Normal from two vectors	$\mathbf{n}=\mathbf{u}\times\mathbf{v}$
Angle between two planes	$\cos\theta = \dfrac{\lvert\mathbf{n}_1\cdot\mathbf{n}_2\rvert}{\lvert\mathbf{n}_1\rvert\lvert\mathbf{n}_2\rvert}$
Line of intersection of two planes	direction $\mathbf{d}=\mathbf{n}_1\times\mathbf{n}_2$
Angle between line and plane	$\sin\theta = \dfrac{\lvert\mathbf{d}\cdot\mathbf{n}\rvert}{\lvert\mathbf{d}\rvert\lvert\mathbf{n}\rvert}$
Distance from point to plane	$D = \dfrac{\lvert ax_0+by_0+cz_0+d\rvert}{\sqrt{a^2+b^2+c^2}}$
Distance between parallel planes	$D = \dfrac{\lvert d_1-d_2\rvert}{\lvert\mathbf{n}\rvert}$
Foot and mirror image	$F = P - \dfrac{f(P)}{\lvert\mathbf{n}\rvert^2}\mathbf{n}, \quad P' = P - \dfrac{2f(P)}{\lvert\mathbf{n}\rvert^2}\mathbf{n}$

Quick Check 3

1. Why does the point–normal equation not change if you replace $\mathbf{n}$ by $5\mathbf{n}$?
2. You compute $\mathbf{u} \times \mathbf{v} = \mathbf{0}$ while looking for a plane through three points. What does this say about the points?
3. The signed values $f(P)$ and $f(Q)$ have opposite signs. What can you say about the segment PQ and the distances of P and Q from the plane?

7.6 Applications in Physics and Engineering

Planes are the standard model of every surface that is flat enough to treat as flat: a runway, a wing panel, a ramp, a mirror, a measuring table. In each application below the work splits into the same three steps.

1. **Describe the surface.** Choose axes, then find a point on the plane and a normal vector.
2. **Translate the question into a tool of this chapter.** "How steep?" is an angle between planes or between a line and a plane. "Does it clear?" is the sign of f. "How far?" is the distance formula. "Where does it hit?" is a line–plane intersection. "Where does the ray go?" is a mirror image.
3. **Compute, then interpret.** A number without a unit and a sign is not an answer. Always ask whether it is physically reasonable.

The first three applications come from aviation; the other three are from mechanics, optics and measurement. In all of them the normal vector is the hero.

7.6.1 Aviation: The Banked Turn

An aeroplane turns by tilting its wings. The reason is geometric: aerodynamic lift acts *perpendicular to the wing surface*, so when the wing plane tilts, the lift force tilts with it, and its sideways part bends the flight path into a curve.

Physics & Engineering: Lift in a Coordinated Turn

Physical law. The lift L acts along the normal of the wing plane; the weight $W = mg$ acts downwards. In a *steady, level, coordinated turn* the aircraft neither climbs nor descends, so the vertical part of the lift balances the weight, while the horizontal part is the centripetal force that keeps the aircraft on a circle of radius R at speed v.

Geometry. Take x forward, y to the left and z upwards. With the wings level, the wing plane is $z = 0$ with normal $\mathbf{k}$. Bank to the right by an angle φ: the wing plane still contains the flight direction $\mathbf{e}_x = (1, 0, 0)$, but the span direction tilts to $\mathbf{s} = (0, \cos\varphi, \sin\varphi)$ (left wing up). A normal vector of the wing plane is

$$\mathbf{n} = \mathbf{e}_x \times \mathbf{s} = (0, -\sin\varphi, \cos\varphi), \qquad |\mathbf{n}| = 1,$$

and the wing plane has the equation $-(\sin\varphi)\, y + (\cos\varphi)\, z = 0$. The angle θ between the wing plane and the horizontal plane $z = 0$ is the angle between their normals,

$$\cos\theta = |\mathbf{n} \cdot \mathbf{k}| = \cos\varphi \quad \Longrightarrow \quad \theta = \varphi.$$

So the bank angle *is* the angle between the wing plane and the horizontal plane (Figure 7.17).

Balance of forces.

$$L\cos\varphi = W, \qquad L\sin\varphi = \frac{mv^2}{R} \quad \Longrightarrow \quad n_L = \frac{L}{W} = \frac{1}{\cos\varphi}, \qquad \tan\varphi = \frac{v^2}{gR}.$$

The ratio $n_L = L/W$ is the **load factor**: the wings must carry n_L times the aircraft's weight.

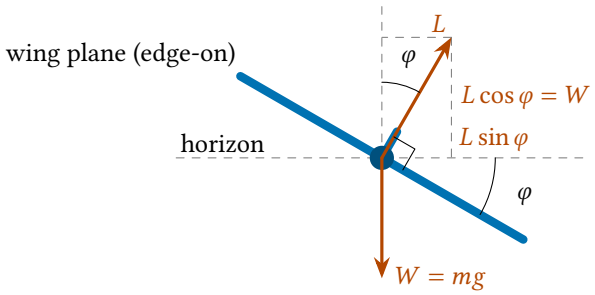


Figure 7.17: Rear view of an aircraft banked to the right by φ. The wing plane is seen edge-on; the lift L is perpendicular to it. The angle between the wing plane and the horizontal plane equals the angle between their normals, φ. The vertical part of L balances the weight; the horizontal part $L\sin\varphi$ points toward the centre of the turn.

Example: How Hard Does a 60° Bank Pull?

For $\varphi = 60^\circ$ the load factor is $n_L = 1/\cos 60^\circ = 2$. The wings carry twice the weight of the aircraft, and everyone on board feels twice their normal weight ("2 g"). To produce the extra lift the aircraft must fly faster or at a larger angle of attack, and its stall speed rises by the factor $\sqrt{n_L} = \sqrt{2} \approx 1.41$. The radius of the turn follows from $\tan\varphi = v^2/(gR)$. At $v = 70$ m/s (about 136 knots):

$$R = \frac{v^2}{g\tan\varphi} = \begin{cases} \dfrac{4900}{9.81 \cdot 0.577} \approx 865 \text{ m}, & \varphi = 30^\circ, \\[2ex] \dfrac{4900}{9.81 \cdot 1.732} \approx 288 \text{ m}, & \varphi = 60^\circ. \end{cases}$$

A steeper bank tightens the turn, but it costs load factor: 1.15 at 30°, 2.00 at 60°, 3.86 at 75° (Figure 7.18).

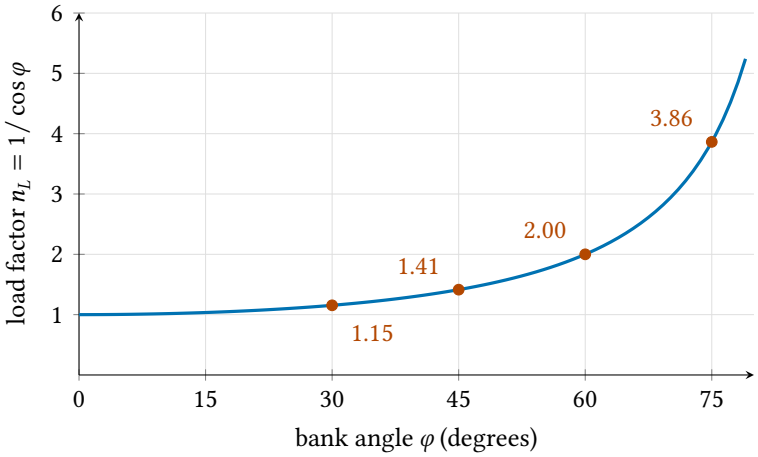


Figure 7.18: Load factor in a level coordinated turn, $n_L = 1/\cos\varphi$. The curve rises without bound as the bank angle approaches 90°, because the vertical part of the lift, $L\cos\varphi$, then tends to zero and cannot support the weight.

Common mistake. In a level coordinated turn the load factor depends *only* on the bank angle: $n_L = 1/\cos\varphi$. It does not depend on the mass or on the speed. A faster aircraft at the same bank angle flies a *larger* circle ($R \propto v^2$), but its wings carry the same multiple of its weight. Note also that the formula blows up as $\varphi \to 90^\circ$: no aircraft can hold a *level* turn with its wings vertical. (Compare the flat truss of Chapter 1, where the forces blow up as the matrix becomes singular.)

Example: A Solar-Powered Drone in a Turn

A flat solar panel of area $A = 1.5\ \text{m}^2$ lies on top of the wing of a drone. Sunlight of irradiance $I = 1000\ \text{W/m}^2$ (measured perpendicular to the rays) delivers the power $P = IA\cos\theta$, where θ is the angle between the panel's normal $\mathbf{n}$ and the direction to the sun. This is the same dot product as before: $\cos\theta = \mathbf{n}\cdot\hat{\mathbf{s}}$.
Sun overhead means $\hat{\mathbf{s}} = \mathbf{k}$. Level flight: $\mathbf{n} = \mathbf{k}$, $\cos\theta = 1$, $P = 1500$ W. In a 30° bank the panel's normal is $\mathbf{n} = (0, -\sin 30^\circ, \cos 30^\circ)$, so $\cos\theta = \mathbf{n}\cdot\mathbf{k} = \cos 30^\circ = 0.866$ and

$$P = 1500 \cdot 0.866 \approx 1299\ \text{W},$$

a loss of about 13 % caused by the tilt alone.

7.6.2 Aviation: Glide Path and Touchdown Point

A precision approach guides the aircraft along a straight descending path, the *glide path*, which is typically inclined at about 3° to the horizontal. The runway is a plane, the glide path is a line, and the questions are exactly those of Section 7.3: where does the line meet the plane, and at what angle?

Physics & Engineering: Where Does the Aircraft Touch Down?

Model. Put the origin at the runway threshold, the x-axis along the runway (in the landing direction), y to the left and z up. The runway plane is $z = 0$ with normal $\mathbf{k}$ (Figure 7.19). An aircraft flying at constant velocity $\mathbf{v}$ from the point A follows the line $\mathbf{r}(t) = A + t\,\mathbf{v}$. It touches the plane when $z(t) = 0$, and the angle of descent θ is the angle between this line and the plane:

$$t^* = -\frac{z_A}{v_z}, \qquad \sin\theta = \frac{|\mathbf{v}\cdot\mathbf{k}|}{|\mathbf{v}|} = \frac{|v_z|}{|\mathbf{v}|}.$$

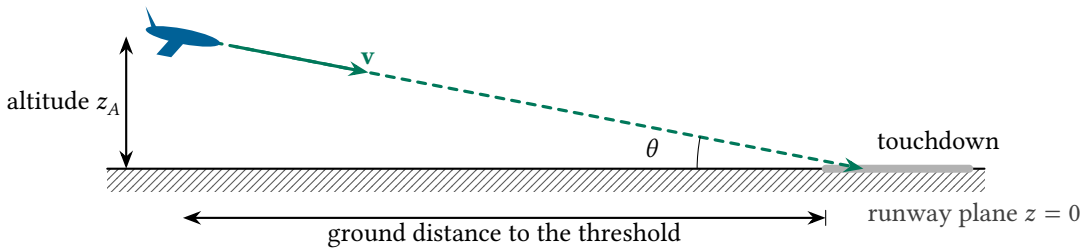


Figure 7.19: Side view of the approach. The angle θ between the glide path and the runway plane satisfies $\sin\theta = |v_z|/|\mathbf{v}|$. The angle is exaggerated: a real 3° path is almost flat, and the touchdown point lies a short distance beyond the threshold.

Example: Touchdown Point and Glide Angle

An aircraft is at $A(-9000, 450, 480)$ metres and moves with velocity $\mathbf{v} = (70, -3.5, -3.7)$ m/s. Where does it touch the runway if nothing changes, and what is the glide angle?

Solution. $z(t) = 480 - 3.7\,t = 0$ gives $t^* = 480/3.7 \approx 129.7$ s. Then

$$x = -9000 + 70 \cdot 129.7 \approx 81\ \text{m}, \qquad y = 450 - 3.5 \cdot 129.7 \approx -4\ \text{m}.$$

The aircraft touches down about 81 m beyond the threshold and 4 m to the right of the centreline ($y < 0$ is to the right). The glide angle follows from $|\mathbf{v}| = \sqrt{70^2 + 3.5^2 + 3.7^2} \approx 70.2$ m/s:

$$\sin\theta = \frac{3.7}{70.2} \approx 0.0527 \quad \Longrightarrow \quad \theta \approx 3.0^\circ.$$

Check. The descent rate 3.7 m/s is about 730 feet per minute, at a ground speed of about 136 knots: typical values for a 3° approach. ✓

Intuition. Why 3° is easy to remember. $\tan 3^\circ \approx 0.052$, so a 3° path loses about 52 metres of height per kilometre of ground distance, which is about 320 feet per nautical mile (pilots round this to "300 feet per mile"). From a height of 480 m the aircraft is therefore about 9 km from the threshold, as in the example.

Common mistake. The glide angle is the angle between the path and the runway *plane*, so the formula needs $\sin\theta = |\mathbf{v}\cdot\mathbf{k}|/|\mathbf{v}|$. With cos you would obtain 87°, which is the angle between the path and the vertical.

7.6.3 Aviation: Obstacle-Clearance Surface

Around every runway, aviation authorities define imaginary surfaces that no obstacle (a mast, a building, a tree) may pierce. The *approach surface* is a plane that rises away from the runway threshold. Checking an obstacle then means asking on which side of a plane a point lies, and by how much: this is the signed distance of Section 7.5.

Physics & Engineering: Does the Mast Clear the Surface?

Model. Use the coordinates of the previous example; the approach side is $x < 0$. Take an approach surface that rises with slope 1 : 50 (2 %) away from the threshold. This is an illustrative value; the actual slopes depend on the type of runway and on national regulations. The height of the surface above the ground at position x is $z = -x/50$, so the plane is

$$x + 50z = 0, \qquad \mathbf{n} = (1, 0, 50), \qquad |\mathbf{n}| = \sqrt{2501} \approx 50.01.$$

Put $f(P) = x + 50z$. Then $f(P) < 0$ means that P lies *below* the surface (allowed), $f(P) > 0$ means that it lies *above* (the obstacle pierces the surface), and the signed distance is $D = f(P)/|\mathbf{n}|$ (Figure 7.20).

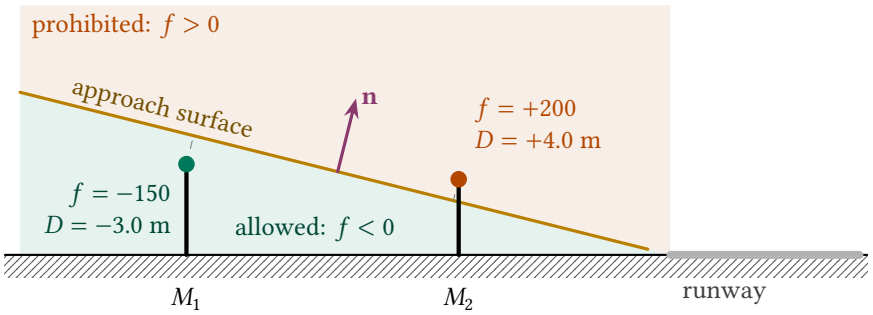


Figure 7.20: Side view of the approach surface $x + 50z = 0$ (the slope is greatly exaggerated; the real surface rises only 1 m for every 50 m). The perpendicular distance from a point to the surface is $|f|/|\mathbf{n}|$. The top of mast M_1 lies below the surface (clear), the top of M_2 lies above it (a penetration). The numbers belong to the example below.

Example: Two Masts

Mast M_1 has its top at $(-2400,\ 30,\ 45)$ and mast M_2 at $(-2400,\ -10,\ 52)$ (metres). At $x = -2400$ the surface is 48 m above the ground.

- M_1: $f = -2400 + 50 \cdot 45 = -150 < 0$, so $D = -150/50.01 \approx -3.0$ m. The top is 3.0 m *below* the surface: clear.
- M_2: $f = -2400 + 50 \cdot 52 = +200 > 0$, so $D = +200/50.01 \approx +4.0$ m. The top is 4.0 m *above* the surface: it pierces the surface and must be lowered, marked or removed.

The perpendicular distance and the vertical clearance almost agree ($52 - 48 = 4$ m), because the surface is so gently inclined: $\cos(\arctan \frac{1}{50}) \approx 0.9998$.

Limits of the model

A real obstacle-limitation surface is not an infinite plane. It has a finite, trapezoidal footprint and usually consists of several sections with different slopes. The plane test of this section therefore has to be combined with a check that the horizontal position (x, y) of the obstacle lies inside the footprint. The sign test itself, however, is exactly what a program does for every obstacle in a survey database.

Common mistake. Do not read the sign of f without knowing which way $\mathbf{n}$ points. Here $f = x + 50z > 0$ means "above" only because the coefficient of z is positive. If the same plane is written as $-x - 50z = 0$, every sign flips. Always calibrate with a point whose position you know: the point $(0, 0, 10)$ is clearly above the surface, and $f = 500 > 0$.

7.6.4 Mechanics: A Block on a Ramp

A smooth surface pushes on a body with the *normal force*: perpendicular to the surface, hence along the normal vector of the plane. Everything else the weight does, it does *along* the plane. Splitting the weight into these two parts is a projection onto $\mathbf{n}$ (Chapter 3).

Physics & Engineering: Forces on a Ramp (Figure 7.21)

Setting. A ramp is the plane $3x - 4z = 0$, rising in the direction of increasing x. A normal vector pointing out of the ramp (upwards) is $\mathbf{n} = (-3, 0, 4)$, $|\mathbf{n}| = 5$, so $\hat{\mathbf{n}} = (-0.6,\ 0,\ 0.8)$. The inclination φ of the ramp is the angle between the ramp plane and the horizontal plane: $\cos\varphi = |\mathbf{n}\cdot\mathbf{k}|/|\mathbf{n}| = \frac{4}{5}$, so $\varphi \approx 36.9^\circ$.

Forces. A block of mass $m = 10$ kg has weight $\mathbf{W} = (0, 0, -98.1)$ N. Its component along the normal is

$$\mathbf{W} \cdot \hat{\mathbf{n}} = -78.48 \text{ N}$$

(negative: the weight presses *into* the ramp). The ramp answers with the normal force $\mathbf{N} = 78.48\,\hat{\mathbf{n}} = (-47.09,\ 0,\ 62.78)$ N, of magnitude $mg\cos\varphi$. What is left of the weight is the part along the ramp:

$$\mathbf{W}_{\parallel} = \mathbf{W} + \mathbf{N} = (-47.09,\ 0,\ -35.32) \text{ N}, \qquad |\mathbf{W}_{\parallel}| = 58.86 \text{ N} = mg\sin\varphi,$$

directed down the slope. For the block to stay at rest, static friction must supply 58.86 N, which requires a coefficient of static friction $\mu_s \geq 58.86/78.48 = \tan\varphi = 0.75$.

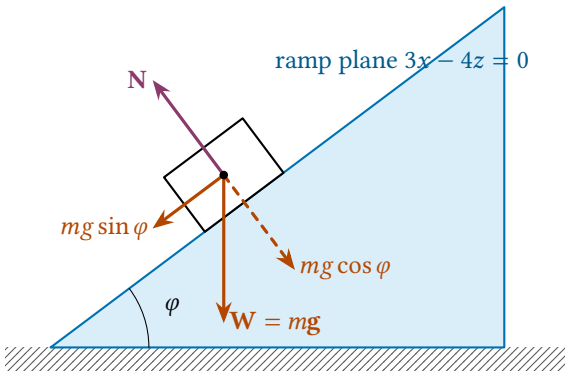


Figure 7.21: Forces on a block on a ramp. The normal force $\mathbf{N}$ points along the normal of the ramp plane. The weight $\mathbf{W}$ splits into a part $mg\cos\varphi$ into the ramp (dashed), which $\mathbf{N}$ cancels, and a part $mg\sin\varphi$ down the slope. Arrow lengths are proportional to the forces for $m = 10$ kg.

Intuition. Why does a steeper ramp make things slide? The normal force is the part of the weight that points into the ramp: $mg\cos\varphi$. As φ grows, $\cos\varphi$ shrinks and the ramp has less and less to push against, while the part along the slope, $mg\sin\varphi$, grows. For a vertical wall ($\varphi = 90^\circ$) the normal force is zero. The ratio of the two parts, $\tan\varphi$, is the friction coefficient needed to keep the block in place.

7.6.5 Optics and Radar: Reflection in a Plane

A flat mirror reflects a ray according to the law of reflection: the incident ray, the normal and the reflected ray lie in one plane, and the angle of incidence equals the angle of reflection. In vector form this is a short formula.

Physics & Engineering: Law of Reflection

Let $\mathbf{d}$ be the direction of an incoming ray and $\hat{\mathbf{n}}$ a unit normal of the mirror plane. The reflected direction is

$$\mathbf{r} = \mathbf{d} - 2(\mathbf{d}\cdot\hat{\mathbf{n}})\,\hat{\mathbf{n}}.$$

The component of $\mathbf{d}$ *along* the normal reverses its sign, and the component *within* the plane is unchanged. This is the mirror-image formula of Theorem 7.4 applied to a vector, i.e. with no constant term. The reflected direction is again a unit vector if $\mathbf{d}$ is: $|\mathbf{r}| = |\mathbf{d}|$.

Example: A Radar Wave Hits a Flat Facet

A radar wave travels in the direction $\mathbf{d} = (1,0,0)$ and hits a flat facet of an aircraft skin with normal $\mathbf{n} = (1,1,1)$, so $\hat{\mathbf{n}} = \frac{1}{\sqrt{3}}(1,1,1)$. Then $\mathbf{d}\cdot\hat{\mathbf{n}} = \frac{1}{\sqrt{3}}$ and

$$\mathbf{r} = (1,0,0) - \frac{2}{\sqrt{3}}\cdot\frac{1}{\sqrt{3}}(1,1,1) = \left(\tfrac{1}{3}, -\tfrac{2}{3}, -\tfrac{2}{3}\right).$$

Check. $|\mathbf{r}| = \sqrt{\frac{1}{9}+\frac{4}{9}+\frac{4}{9}} = 1$. ✓
The direction back toward the transmitter is $-\mathbf{d}$. The angle γ between $\mathbf{r}$ and $-\mathbf{d}$ satisfies $\cos\gamma = \mathbf{r}\cdot(-\mathbf{d}) = -\frac{1}{3}$, so $\gamma \approx 109.5^\circ$: the wave is sent away from the radar. Orienting flat facets so that reflections avoid the direction of the transmitter is one of the shaping principles of low-observable ("stealth") aircraft.

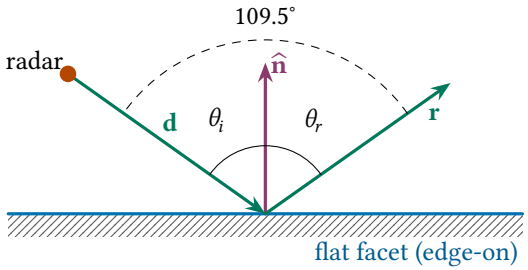


Figure 7.22: Reflection at a flat facet, drawn in the plane that contains $\mathbf{d}$ and $\hat{\mathbf{n}}$. For the example, $\theta_i = \theta_r = 54.7^\circ$. The reflected ray $\mathbf{r}$ makes an angle of 109.5° with the line back to the transmitter, so little energy returns to the radar.

7.6.6 Engineering Measurement: Datum Plane and Flatness

Is a machined wing-skin panel flat enough? Quality control measures the height of several points on the surface with a coordinate measuring machine and compares them with a reference plane, the *datum plane*. Three points fix a plane (Section 7.4); the distance of every further point from it is its deviation (Section 7.5).

Physics & Engineering: A Three-Point Datum Plane

Data. Coordinates in millimetres, with z the height above the measuring table: $P_1(0,0,0)$, $P_2(400,0,0.8)$, $P_3(0,300,-0.3)$ and a fourth point $P_4(400,300,1.0)$.

Datum plane. Case 3 of Section 7.4: $\mathbf{u} = \overrightarrow{P_1P_2} = (400,0,0.8)$ and $\mathbf{v} = \overrightarrow{P_1P_3} = (0,300,-0.3)$ give

$$\mathbf{n} = \mathbf{u} \times \mathbf{v} = (-240,\ 120,\ 120000) = 120\,(-2,\ 1,\ 1000).$$

So the datum plane is $-2x + y + 1000z = 0$, that is, $z = (2x - y)/1000$. (Check: at P_2, $z = 0.8$; at P_3, $z = -0.3$.)

Deviation of P_4. With $f(P_4) = -800 + 300 + 1000 = 500$ and $|\mathbf{n}| = \sqrt{4 + 1 + 10^6} \approx 1000.0$:

$$D = \frac{f(P_4)}{|\mathbf{n}|} \approx \frac{500}{1000.0} = 0.50 \text{ mm}$$

above the datum plane. If the drawing allows a deviation of at most 0.3 mm (an illustrative tolerance), the panel is out of tolerance.

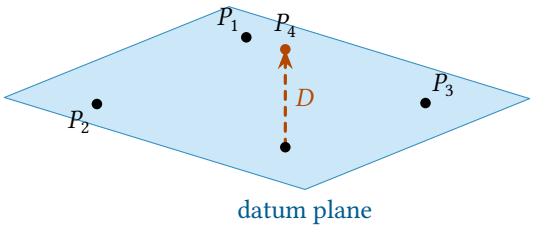


Figure 7.23: The datum plane is fixed by three measured points P_1, P_2, P_3; the deviation of a further point P_4 is its distance D from that plane. The picture is schematic and not to scale: in the example the deviations are fractions of a millimetre.

Real flatness checks

A real flatness specification uses *all* measured points. One standard version asks for two parallel planes, a certain distance apart, that enclose every point; the smallest such distance is the flatness error. The calculation behind it is the same: the distances from many points to a plane, and the distance between two parallel planes (Section 7.5).

7.6.7 Summary: Which Tool for Which Question?

Situation	Question	Tool	Section
Banked wing	How steep is the wing plane?	angle between planes, via the normals	7.2
Glide path	Where does the path hit the runway, and at what angle?	line–plane intersection and angle	7.3
Obstacle surface	Does the mast clear it, and by how much?	sign and size of $f(P)/\lvert\mathbf{n}\rvert$	7.5
Ramp	How much of the weight presses into the surface?	projection onto $\mathbf{n}$	7.6
Mirror, radar	Where does the ray go?	mirror image, $\mathbf{r} = \mathbf{d} - 2(\mathbf{d}\cdot\hat{\mathbf{n}})\hat{\mathbf{n}}$	7.5
Flatness	How far is a point from the reference plane?	plane through three points, then distance	7.4, 7.5

Table 7.2: Choosing the right tool of this chapter for an application.

Planes appear in many more places: in computer graphics (clipping planes decide which part of a scene is drawn), in CAD (a section through a solid is its intersection with a plane), in surveying (a levelled surface), and in robotics (a work table that a robot arm must not cross). Each time the steps are the same: find a point and a normal, then use the dot and cross products.

Quick Check 4

1. At which bank angle is the load factor in a level coordinated turn equal to 2? Does the answer change if the aircraft flies twice as fast?
2. For the approach surface $x + 50z = 0$, a surveyor reports $f = 0$ for the top of a mast. What does this mean, and is the mast allowed?
3. A ray with direction $\mathbf{d}$ meets a mirror perpendicularly ($\mathbf{d} \parallel \hat{\mathbf{n}}$). What is the reflected direction $\mathbf{r}$?

7.7 Worked Problems

Every solution below follows the same routine: *name what is given* (point, direction, normal), *choose the tool* from the summary table of Section 7.6, *compute*, and finally *check* the answer by substituting it back. The check is part of the solution, not an extra.

Problem 1: Line of Intersection

Find the line in which the planes $x + y + z = 3$ and $2x - y + z = 2$ meet.

Solution. The normals are $\mathbf{n}_1 = (1, 1, 1)$ and $\mathbf{n}_2 = (2, -1, 1)$. The line lies in both planes, so its direction is perpendicular to both normals:

$$\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{vmatrix} = \mathbf{i}(1+1) - \mathbf{j}(1-2) + \mathbf{k}(-1-2) = (2, 1, -3).$$

For a point on the line put $z = 0$: then $x + y = 3$ and $2x - y = 2$. Adding the two equations gives $3x = 5$, so $x = \frac{5}{3}$ and $y = \frac{4}{3}$. Hence

$$\mathbf{r}(t) = \left(\tfrac{5}{3}, \tfrac{4}{3}, 0\right) + t(2, 1, -3), \qquad x = \tfrac{5}{3} + 2t, \quad y = \tfrac{4}{3} + t, \quad z = -3t.$$

Check. $\mathbf{d} \cdot \mathbf{n}_1 = 2 + 1 - 3 = 0$ and $\mathbf{d} \cdot \mathbf{n}_2 = 4 - 1 - 3 = 0$, so $\mathbf{d}$ lies in both planes; and $\left(\frac{5}{3}, \frac{4}{3}, 0\right)$ satisfies both equations. ✓

Problem 2: A Line Meets a Plane

Find the point in which $\mathbf{r}(t) = (1, 2, 3) + t(2, -1, 1)$ meets the plane $x - y + 2z = 4$.

Solution. Here $\mathbf{d} \cdot \mathbf{n} = (2, -1, 1) \cdot (1, -1, 2) = 2 + 1 + 2 = 5 \neq 0$, so the line crosses the plane in exactly one point. Substitute $x = 1 + 2t$, $y = 2 - t$, $z = 3 + t$ into the equation of the plane:

$$(1 + 2t) - (2 - t) + 2(3 + t) = 4 \implies 5t + 5 = 4 \implies t = -\tfrac{1}{5}.$$

The point is $\left(1 - \frac{2}{5},\ 2 + \frac{1}{5},\ 3 - \frac{1}{5}\right) = \left(\frac{3}{5}, \frac{11}{5}, \frac{14}{5}\right)$.

Check. $\frac{3}{5} - \frac{11}{5} + \frac{28}{5} = \frac{20}{5} = 4$. ✓

Problem 3: Angle Between Two Planes

Find the angle between $2x - y + 2z = 5$ and $x + y - z = 3$ and classify the pair.

Solution. $\mathbf{n}_1 = (2, -1, 2)$ and $\mathbf{n}_2 = (1, 1, -1)$, so $\mathbf{n}_1 \cdot \mathbf{n}_2 = 2 - 1 - 2 = -1$, $|\mathbf{n}_1| = 3$, $|\mathbf{n}_2| = \sqrt{3}$. With the absolute value,

$$\cos\theta = \frac{|-1|}{3\sqrt{3}} = \frac{\sqrt{3}}{9} \approx 0.192 \implies \theta \approx 78.9^\circ.$$

The angle is neither 0° nor 90°, so the planes are neither parallel nor perpendicular: they intersect obliquely.

Problem 4: Angle Between a Line and a Plane

Find the angle between $\mathbf{r}(t) = (1, 0, 2) + t(2, -1, 3)$ and the plane $x - 2y + z = 4$ and classify the pair.

Solution. $\mathbf{d} = (2, -1, 3)$ and $\mathbf{n} = (1, -2, 1)$. Then $\mathbf{d} \cdot \mathbf{n} = 2 + 2 + 3 = 7$, $|\mathbf{d}| = \sqrt{14}$, $|\mathbf{n}| = \sqrt{6}$. For a line and a plane the formula has a *sine*:

$$\sin\theta = \frac{7}{\sqrt{14}\sqrt{6}} = \frac{7}{2\sqrt{21}} = \frac{\sqrt{21}}{6} \approx 0.764 \implies \theta \approx 49.8^\circ.$$

Not parallel ($\mathbf{d}\cdot\mathbf{n}\neq 0$) and not perpendicular ($\mathbf{d}\times\mathbf{n}=(5,1,-3)\neq\mathbf{0}$): the line crosses the plane obliquely.

Problem 5: A Point and Two Directions (Case 1)

Find the plane through $P_0(1,2,-1)$ parallel to $\mathbf{u}=(2,-1,1)$ and $\mathbf{v}=(1,1,2)$.

Solution.

$$\mathbf{n}=\mathbf{u}\times\mathbf{v}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&-1&1\\1&1&2\end{vmatrix}=(-3,-3,3).$$

Then $-3(x-1)-3(y-2)+3(z+1)=0$, that is, $-3x-3y+3z+12=0$. Dividing by -3:

$$x+y-z-4=0.$$

Check. $\mathbf{n}\cdot\mathbf{u}=-6+3+3=0$ and $\mathbf{n}\cdot\mathbf{v}=-3-3+6=0$; P_0: $1+2+1-4=0$. ✓

Problem 6: Two Points and a Direction (Case 2)

Find the plane through $P_1(1,0,2)$ and $P_2(3,-1,1)$ parallel to $\mathbf{v}=(2,1,1)$.

Solution. $\mathbf{u}=\overrightarrow{P_1P_2}=(2,-1,-1)$ and

$$\mathbf{n}=\mathbf{u}\times\mathbf{v}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&-1&-1\\2&1&1\end{vmatrix}$$
$$=\big((-1)(1)-(-1)(1),\ -[(2)(1)-(-1)(2)],\ (2)(1)-(-1)(2)\big)=(0,-4,4).$$

Using P_1: $0\cdot(x-1)-4(y-0)+4(z-2)=0$, so $-4y+4z-8=0$, that is

$$y-z+2=0.$$

Check. $\mathbf{v}\cdot\mathbf{n}=0-4+4=0$, so $\mathbf{v}$ is parallel to the plane; and $P_2(3,-1,1)$: $-1-1+2=0$. ✓

Problem 7: Three Points (Case 3)

Find the plane through $P_1(0,1,2)$, $P_2(1,0,1)$ and $P_3(2,1,0)$.

Solution. $\mathbf{u}=\overrightarrow{P_1P_2}=(1,-1,-1)$, $\mathbf{v}=\overrightarrow{P_1P_3}=(2,0,-2)$:

$$\mathbf{n}=\mathbf{u}\times\mathbf{v}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&-1&-1\\2&0&-2\end{vmatrix}=(2,0,2).$$

Then $2(x-0)+0\cdot(y-1)+2(z-2)=0$, so $2x+2z-4=0$, that is

$$x+z-2=0.$$

Check. P_1: $0+2-2=0$; P_2: $1+1-2=0$; P_3: $2+0-2=0$. ✓ Since $\mathbf{n}$ has no y-component, this plane is parallel to the y-axis.

Problem 8: Distance Between Parallel Planes

Find the distance between $6x - 3y + 2z - 14 = 0$ and $12x - 6y + 4z + 7 = 0$.

Solution. The normals $(6, -3, 2)$ and $(12, -6, 4)$ are parallel, so the planes are parallel. The formula $|d_1 - d_2|/|\mathbf{n}|$ needs *identical* coefficients, so first divide the second equation by 2: $6x - 3y + 2z + \frac{7}{2} = 0$. Now $|\mathbf{n}| = \sqrt{36 + 9 + 4} = 7$ and

$$D = \frac{\left|-14 - \frac{7}{2}\right|}{7} = \frac{35/2}{7} = \frac{5}{2} = 2.5.$$

(Without the division one would get $|-14 - 7|/\sqrt{144 + 36 + 16} = 21/14 = 1.5$: wrong, because $12x - 6y + 4z$ is twice $6x - 3y + 2z$, so the constants are not comparable. This is the mistake warned about in Section 7.5.)

Problem 9: Planes at a Given Distance

Find all planes parallel to $x + 2y - 2z - 1 = 0$ whose distance from it is 4.

Solution. A parallel plane has the same normal direction, so it has the form $x + 2y - 2z + d = 0$. Here $|\mathbf{n}| = \sqrt{1 + 4 + 4} = 3$, and the distance to the given plane (constant -1) is $|d - (-1)|/3 = |d + 1|/3$. Setting this equal to 4:

$$|d + 1| = 12 \implies d = 11 \text{ or } d = -13.$$

There are *two* such planes, one on each side of the given plane (Figure 7.24):

$$x + 2y - 2z + 11 = 0 \qquad \text{and} \qquad x + 2y - 2z - 13 = 0.$$

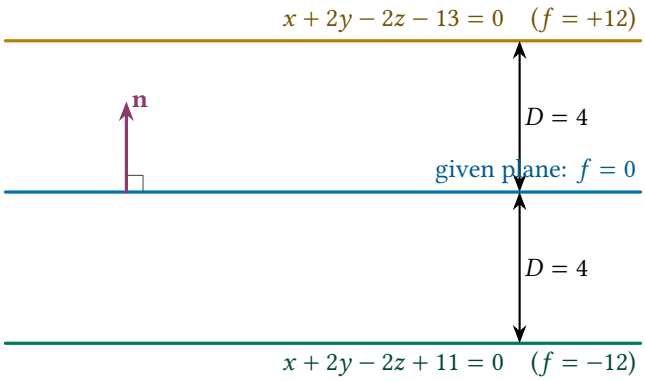


Figure 7.24: Problem 9. With $f = x + 2y - 2z - 1$ and $|\mathbf{n}| = 3$, the planes where $f = \pm 12$ are at distance $12/3 = 4$ from the given plane. The normal $\mathbf{n}$ points toward the plane with $f = +12$.

Check. A point of the upper plane is $(13, 0, 0)$; there $f = 13 - 1 = 12$ and $12/3 = 4$. ✓

Problem 10: Perpendicular to Two Planes

Find the plane through $P_0(1, 2, 3)$ that is perpendicular to both $x + y + z = 1$ and $2x - y + z = 3$.

Solution. A plane is perpendicular to another plane when their normals are perpendicular. So the unknown normal $\mathbf{n}$ must be perpendicular to both $\mathbf{n}_1 = (1, 1, 1)$ and $\mathbf{n}_2 = (2, -1, 1)$, which means

$$\mathbf{n} = \mathbf{n}_1 \times \mathbf{n}_2 = (2, 1, -3)$$

(computed in Problem 1). Geometrically, $\mathbf{n}$ is the direction of the line in which the two given planes meet: the required plane is the one that cuts that line at a right angle. Through P_0:

$$2(x-1)+(y-2)-3(z-3)=0 \implies 2x+y-3z+5=0.$$

Check. $\mathbf{n}\cdot\mathbf{n}_1 = 2+1-3=0$, $\mathbf{n}\cdot\mathbf{n}_2 = 4-1-3=0$; P_0: $2+2-9+5=0$. ✓

7.8 Conceptual & Computational Exercises

7.8.1 Conceptual Problems

Try these *before* touching a calculator: most can be settled by thinking about directions, signs, symmetry or a quick sketch. Hints and answers are collected at the end of the chapter in the edition of the notes that includes the answer key.

1. **One equation or two?** A line in space needs two linear equations, but a plane needs only one. Explain why, by counting free parameters.
2. **True or false?** Justify each answer with a short argument or a counterexample.
   (a) A plane has exactly one normal vector.
   (b) Two planes whose normals are parallel are parallel or coincident.
   (c) Two distinct planes in space always meet in a line.
   (d) Three points always determine exactly one plane.
   (e) If $\mathbf{d}\cdot\mathbf{n}=0$, the line has no point in common with the plane.
   (f) A line perpendicular to a plane is perpendicular to every line that lies in the plane.
   (g) If $\mathbf{n}_1\cdot\mathbf{n}_2=0$, the two planes are perpendicular.
3. **A family of planes.** Fix $(a,b,c)\neq\mathbf{0}$ and let d vary in $ax+by+cz+d=0$. What do all these planes have in common? How far apart are the planes with $d=0$ and $d=5$? How are the planes with d and $-d$ related?
4. **Distance without a formula.** The plane $x+y+2z=6$ passes through $(6,0,0)$. Using only the projection of the position vector $(6,0,0)$ onto a *unit* normal, find the distance of the plane from the origin.
5. **Inclination.** Without a calculator, find the angle between the plane $x+z=5$ and the horizontal plane. Does the plane $x-z=3$ make the same angle? What is different about the two planes?
6. **Symmetry.** Consider the plane through $(a,0,0)$, $(0,a,0)$ and $(0,0,a)$ with $a\neq 0$. Without computing a cross product, say which direction its normal must have, and write its equation. Then confirm with the intercept form.
7. **Reading equations.** Describe, in one sentence each, how the following planes lie relative to the coordinate axes: (a) $y=3$; (b) $x+z=1$; (c) $x+y+z=0$; (d) $2x-3y=6$.
8. **Spot the error.** A student claims: "The line in which two planes meet has the direction $\mathbf{n}_1+\mathbf{n}_2$." Test this claim on the planes $x=0$ and $y=0$, and state the correct rule with its reason.
9. **Perpendicular to two planes.** A plane is perpendicular to both $x+y+z=1$ and $2x-y+z=3$. What can you say about its normal? What does this mean for the line in which the two given planes meet?
10. **Which side?** Take the approach surface $x+50z=0$ of Section 7.6.3. For the points $A(-100,0,3)$ and $B(-100,0,1)$ decide on which side of the surface each lies. Can a straight flight path from A to B stay clear of the surface? What is the signed distance of A, and why is it almost equal to its vertical clearance?
11. **Bank-angle limits.** (a) Why can an aircraft not hold level flight in a 90° bank? (b) At which bank angle is the load factor equal to 3? (c) Two aircraft fly level coordinated turns at the same bank angle, one at twice the speed of the other. Compare their turn radii and their load factors.
12. **Sine or cosine?** For $\mathbf{d}=(1,2,2)$ and $\mathbf{n}=(2,2,1)$ a student finds $\cos\alpha=\frac{8}{9}$ and reports $\alpha\approx 27.3^\circ$ as "the angle between the line and the plane". What went wrong? Give the correct angle.

13. **Corner reflector.** Three mirrors lie in the planes $x = 0$, $y = 0$ and $z = 0$. A ray with direction $\mathbf{d} = (d_1, d_2, d_3)$ is reflected successively by all three. Show that it leaves in the direction $-\mathbf{d}$, whatever the order of the reflections. Why does this make such a "corner" a good radar target?
14. **Three planes.** Two planes in general meet in a line. List all the possible intersections of *three* planes and say, in terms of the normals, when each occurs.

7.8.2 Computational Problems

Show all work and give exact answers unless the problem says otherwise. Use $g = 9.81\ \text{m/s}^2$ where needed.

1. The plane through $P_0(2, -1, 3)$ has normal $\mathbf{n} = (1, 2, -1)$. Find its equation, its normalised form, and its distance from the origin.
2. (a) Find the plane with x-intercept 2, y-intercept -1 and z-intercept 4. (b) Explain why "x-intercept 2, y-intercept -1, z-intercept 0" does not describe a plane in the intercept form.
3. Find the angle between the planes $2x - y + z = 4$ and $x + y - 2z = 1$.
4. Classify each pair as parallel, coincident, perpendicular or intersecting: (a) $x + 2y - z = 3$ and $2x + 4y - 2z = 6$; (b) $x + 2y - z = 3$ and $2x + 4y - 2z = 7$. In (b), find the distance between the planes.
5. Find the line in which $x + y + z = 3$ and $2x - y + z = 4$ meet.
6. Find the angle between $\mathbf{r}(t) = (1, 0, 2) + t(1, -2, 2)$ and the plane $2x + y - 2z = 3$, and the point in which they meet.
7. Decide whether $\mathbf{r}(t) = (0, 1, 2) + t(1, 2, -1)$ is parallel, perpendicular or neither to the plane $2x - y + z = 5$, and find the common point if there is exactly one.
8. Find the plane (a) through $P_1(1, 0, 0)$, $P_2(0, 2, 0)$, $P_3(0, 0, 3)$; (b) through $P_0(2, -1, 0)$ parallel to $\mathbf{u} = (1, 1, 0)$ and $\mathbf{v} = (0, 1, 1)$; (c) through $P_1(0, 0, 1)$ and $P_2(1, 1, 0)$ parallel to $\mathbf{v} = (1, 0, 1)$.
9. Find the distance from $P(3, -2, 1)$ to the plane $2x - y + 2z - 5 = 0$.
10. Find the plane through $P_0(1, 2, 3)$ perpendicular to both $x - y + z = 0$ and $2x + y - z = 3$.
11. Find the plane containing the line $\mathbf{r}(t) = (1, 1, 0) + t(1, 0, 1)$ and the point $P(0, 0, 2)$.
12. Find the distance between $3x - 4y + 12z - 26 = 0$ and $6x - 8y + 24z + 26 = 0$.
13. Find the foot of the perpendicular and the mirror image of $P(2, 1, -1)$ in the plane $x + 2y - 2z = 3$.
14. **Banked turn.** An aircraft flies a level coordinated turn at $v = 80$ m/s with bank angle 45°. Find (a) the turn radius, (b) the load factor, (c) the factor by which the stall speed increases, given that it grows like $\sqrt{n_L}$, and (d) the largest bank angle if the structure allows a load factor of 3.8.
15. **Glide path.** An aircraft at $A(-6000, 0, 330)$ m has the velocity $\mathbf{v} = (65, 0, -3.4)$ m/s. When and where does it touch the runway plane $z = 0$ if nothing changes, and what is its glide angle?
16. **Obstacles.** For the approach surface $x + 50z = 0$ test the tops of three obstacles $T_1(-1800, 20, 38)$, $T_2(-900, -35, 17.2)$ and $T_3(-3000, 0, 55)$ (metres). Which pierce the surface? Give the signed distances.
17. **Ramp.** A crate of mass 20 kg rests on the ramp $5x - 12z = 0$ (upward normal $(-5, 0, 12)$). Find the inclination of the ramp, the normal force, the force along the ramp, and the smallest coefficient of static friction that keeps the crate at rest.
18. **Radar facet.** A radar wave of direction $\mathbf{d} = (1, 0, 0)$ meets a flat facet with unit normal $\widehat{\mathbf{n}} = \frac{1}{3}(2, 1, 2)$. Find the reflected direction $\mathbf{r}$, check that $|\mathbf{r}| = 1$, and find the angle between $\mathbf{r}$ and the direction $-\mathbf{d}$ back to the transmitter.
19. **Flatness.** On a measuring table, the points $P_1(0, 0, 0)$, $P_2(300, 0, 0.6)$ and $P_3(0, 200, 0.2)$ (millimetres) fix a datum plane. Find its equation. Another point is $P_4(300, 200, 0.9)$. How far is P_4 from the datum plane? Is the panel within a tolerance of 0.15 mm?

8

Curves and Surfaces of the Second Order

Learning Objectives

After studying this chapter you should be able to:

- recognize and write the standard equations of the circle, ellipse, parabola and hyperbola, and find their key features (center, vertices, foci, asymptotes, directrix, eccentricity);
- classify a second-degree equation in x, y by completing the square, by the discriminant $B^2 - 4AC$, and (when there is an xy-term) by rotating the axes;
- explain why these curves are the *conic sections*, and what the focus–directrix property says about them;
- recognize the standard equations of the sphere, cylinder, cone, ellipsoid, hyperboloids and paraboloids, and sketch them from their cross-sections (traces);
- find the center and radius of a sphere, and classify a quadric surface, by completing the square in three variables;
- apply conics and quadrics to reflectors, orbits, projectiles, positioning, and engineering structures.

Equations of degree one in x, y give straight lines. The next simplest equations, of degree two, give a small family of beautifully symmetric curves (circles, ellipses, parabolas, hyperbolas) and, in space, a family of surfaces (spheres, ellipsoids, paraboloids, hyperboloids, cones, cylinders). They describe the orbits of planets, the shape of mirrors and antennas, the path of a thrown ball, and the towers of power stations. This chapter learns to recognize each shape from its equation, and each equation from its shape.

Intuition. **Why second order?** A line is the best straight-line approximation of a curve near a point; the next best approximation, which also captures how the curve *bends*, is a quadratic one (the second-order Taylor polynomial). Quadratic curves and surfaces are therefore the simplest shapes that can describe curvature, and they have a complete and beautiful classification with just a handful of cases. They also appear directly in nature, because many laws (inverse-square gravity, uniform acceleration, reflection) lead to them.

8.1 Second Order Curves

The general equation of a second order (quadratic) curve in the plane is

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0,$$

where A, B, C, D, E, F are constants and at least one of A, B, C is nonzero. Which curve it describes (circle, ellipse, parabola or hyperbola) depends on the coefficients, most directly on the sign of the **discriminant** $B^2 - 4AC$ (Section 8.1.5).

Intuition. **Why are they called conic sections?** Slice a double cone with a plane. A plane perpendicular to the axis gives a *circle*; tilt it a little and you get an *ellipse*; tilt it until it is parallel to a side of the cone and you get a *parabola*; tilt it further, so that it cuts both halves of the cone, and you get a *hyperbola* (two branches). Apollonius of Perga studied them this way more than two thousand years ago, and every curve of this section arises in this single way (Figure 8.1). A plane through the vertex gives the degenerate cases: a point, one line, or two crossing lines.

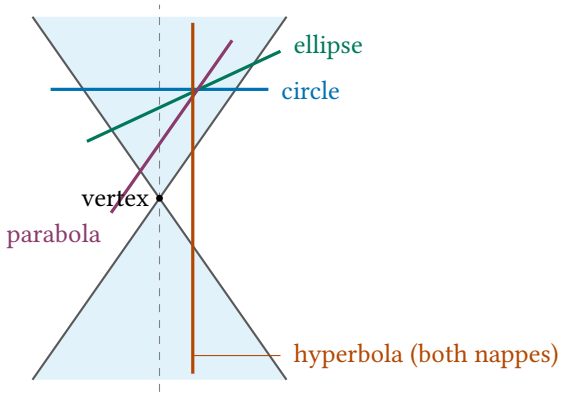


Figure 8.1: A double cone seen from the side, cut by planes seen edge-on. Horizontal plane: circle. Tilted plane through one nappe: ellipse. Plane parallel to a side line (generator) of the cone, here the right-hand one: parabola. Steeper plane through both nappes: the two branches of a hyperbola.

8.1.1 The Circle

Definition: Circle

A **circle** is the set of points at a fixed distance r (the **radius**) from a fixed point (h, k) (the **center**) (Figure 8.2a):

$$(x-h)^2 + (y-k)^2 = r^2 \qquad \Longleftrightarrow \qquad x^2 + y^2 + Dx + Ey + F = 0.$$

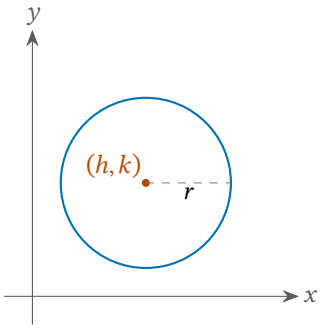
Completing the square in the second form gives $\left(x + \frac{D}{2}\right)^2 + \left(y + \frac{E}{2}\right)^2 = \frac{D^2+E^2}{4} - F$, so the center is $\left(-\frac{D}{2}, -\frac{E}{2}\right)$ and $r^2 = \frac{D^2+E^2}{4} - F$.

Common mistake. **The right-hand side must be positive.** After completing the square, $r^2 = \frac{D^2+E^2}{4} - F$ must be > 0. If it is 0 the "circle" is a single point; if it is negative the equation has *no* real points (for example $x^2 + y^2 + 1 = 0$).

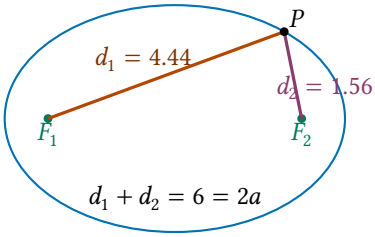
Definition: Ellipse

An **ellipse** is the set of points whose distances to two fixed points (the **foci**) sum to a constant, equal to the length $2a$ of the **major axis** (Figure 8.2b). Its key parameters, with center (h, k), are

- a: length of the **semi-major axis**;
- b: length of the **semi-minor axis** ($0 < b < a$);
- c: distance from center to each focus, $c^2 = a^2 - b^2$;
- **eccentricity** $e = \frac{c}{a}$, with $0 < e < 1$: e near 0 is nearly circular, e near 1 is very elongated.



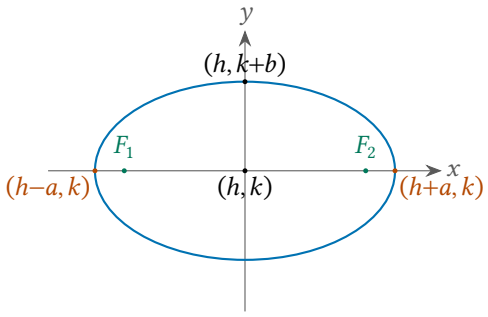
(a) A circle of radius r



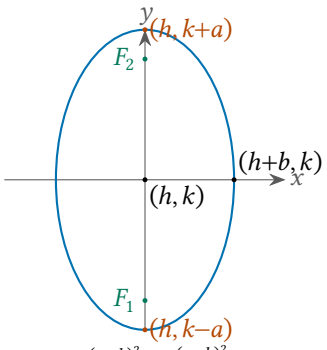
(b) The gardener's ellipse: $d_1 + d_2 = 2a$

Figure 8.2: (a) The circle $(x-h)^2 + (y-k)^2 = r^2$. (b) The ellipse with $a = 3$, $b = 2$, $c = \sqrt{5} \approx 2.24$: wherever the point P is on the curve, its distances to the two foci add up to the same number $2a = 6$.

Intuition. The gardener's ellipse. Push two pins into the ground (the foci), tie a loop of string of fixed length around them, and keep the string taut with a pencil: the pencil traces an ellipse (Figure 8.2b), because the two parts of the string always add up to the same length $2a$. Moving the pins closer together makes the ellipse rounder; when they coincide you draw a circle. An ellipse is also a circle *squashed* in one direction: $x = a\cos t$, $y = b\sin t$ is the unit circle stretched by a horizontally (Figure 8.3a) and b vertically (its area is πab) (Figure 8.3b).



(a) Horizontal: $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$, $a > b$



(b) Vertical: $\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$, $a > b$

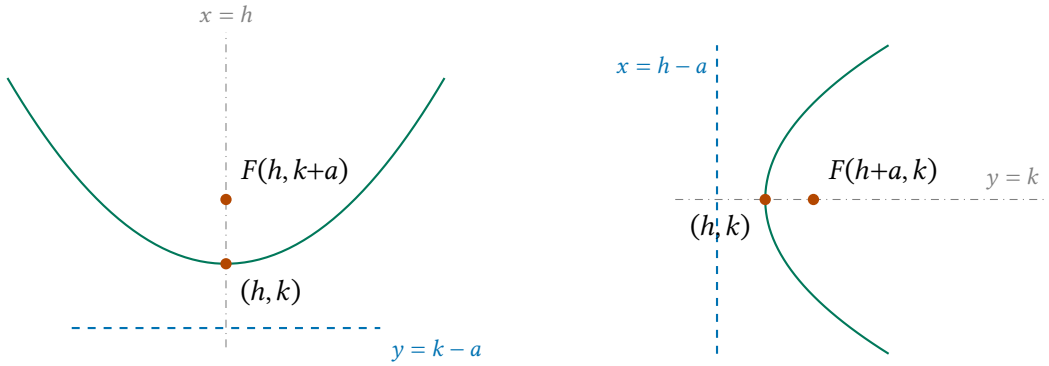
Figure 8.3: In both cases a measures the long direction, and the foci sit on that same axis, a distance $c = \sqrt{a^2 - b^2}$ from the center. The larger denominator belongs to a^2, whichever variable it sits under.

Common mistake. Which denominator is a^2? In an ellipse a is the *larger* semi-axis, so a^2 is the larger denominator, regardless of whether it stands under x^2 or under y^2. The foci lie on the major axis: for $\frac{x^2}{4} + \frac{y^2}{9} = 1$ the larger denominator 9 is under y, so the ellipse is *vertical* and the foci are $(0, \pm\sqrt{5})$. Also $c^2 = a^2 - b^2$ (a minus sign), so $c < a$.

8.1.2 The Parabola

Definition: Parabola

A **parabola** is the set of points equidistant from a fixed point (the **focus**) and a fixed line (the **directrix**). The **vertex** (h, k) is halfway between them; $a \neq 0$ is the directed distance from vertex to focus, Figure 8.4.



(a) vertical: $(x-h)^2 = 4a(y-k)$ **(b)** horizontal: $(y-k)^2 = 4a(x-h)$

Figure 8.4: A parabola opens toward its focus and away from its directrix; it opens in the positive direction when $a > 0$ and in the negative direction when $a < 0$. In both pictures $a > 0$.

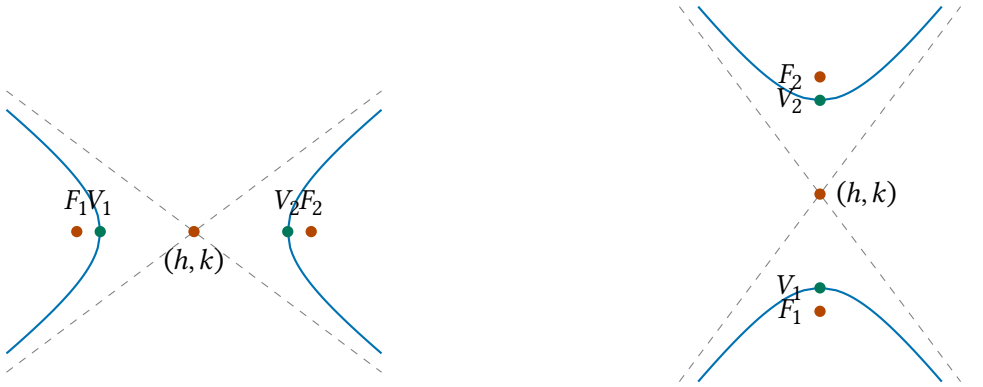
Why we need it. *The parabola is the curve of the thrown ball and of every reflector that must turn parallel rays into one point (or the reverse): satellite dishes, headlights, radio telescopes.*

Common mistake. The number $4a$ is not the focal distance. In $(x-h)^2 = 4a(y-k)$ the focus is at distance $|a|$ from the vertex, and $4a$ is the number in front of $(y-k)$. For $x^2 = 8y$ we have $4a = 8$, so $a = 2$: focus $(0, 2)$, directrix $y = -2$ (not $y = -8$). The sign of a gives the direction: $a < 0$ opens downward (vertical, Figure 8.4a) or to the left (horizontal, 8.4b).

8.1.3 The Hyperbola

Definition: Hyperbola

A **hyperbola** is the set of points whose distances to two foci differ, in absolute value, by a constant, equal to the length $2a$ of the **transverse axis** (Figure 8.5). With center (h, k): a is the center-to-vertex distance, b the semi-conjugate axis, and c the center-to-focus distance, related by $c^2 = a^2 + b^2$ (note the +, unlike the ellipse). The **eccentricity** is $e = \frac{c}{a}$, now with $e > 1$ always (as $e \to 1^+$ the branches become *narrower*, hugging the transverse axis; as e grows they open wider).



(a) horizontal: $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$ (b) vertical: $\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$

Figure 8.5: The dashed asymptotes are $y - k = \pm\frac{b}{a}(x-h)$ (horizontal case) or $y - k = \pm\frac{a}{b}(x-h)$ (vertical case); each branch hugs its asymptotes further from the center. In both pictures $a = 2$, $b = 1.5$, $c = 2.5$.

Intuition. **Asymptotes in one line.** Far from the center the constant 1 in $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is negligible compared with the other terms, so the curve looks like the solutions of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$, i.e. $\left(\frac{x}{a} - \frac{y}{b}\right)\left(\frac{x}{a} + \frac{y}{b}\right) = 0$, the two lines $y = \pm\frac{b}{a}x$. Replace the 1 by 0 and factor: that is the asymptote recipe for every hyperbola.

Common mistake. **Three traps for hyperbolas.** (i) $c^2 = a^2 + b^2$, with a plus sign (the foci are *outside* the vertices, $c > a$). (ii) a^2 is the denominator of the *positive* term, and it can be smaller than b^2: for $\frac{x^2}{4} - \frac{y^2}{9} = 1$, $a = 2$, $b = 3$. (iii) The asymptote slopes are $\pm\frac{b}{a}$ when the positive term is x^2, but $\pm\frac{a}{b}$ when it is y^2; use the replace-1-by-0 recipe instead of memorising.

8.1.4 One Family: Eccentricity and the Focus–Directrix Property

Fix a point F (the focus), a line ℓ (the directrix) not through F, and a number $e > 0$. The set of points P with

$$|PF| = e \cdot \operatorname{dist}(P, \ell)$$

is an **ellipse** if $0 < e < 1$, a **parabola** if $e = 1$, and one branch of a **hyperbola** if $e > 1$ (a circle is the limiting case $e = 0$). With the focus at the origin and the directrix perpendicular to the x-axis, the curve has the polar equation

$$r = \frac{p}{1 + e\cos\theta},$$

where p (the **semi-latus rectum**) is the distance from the focus to the curve, measured perpendicular to the axis; for an ellipse $p = a(1 - e^2)$.

Why we need it. *This single rule shows that the four curves are one family, with the eccentricity as a dial that turns a circle into an ellipse, then a parabola, then a hyperbola. In polar form it is exactly the equation of an orbit (Section 8.3).*

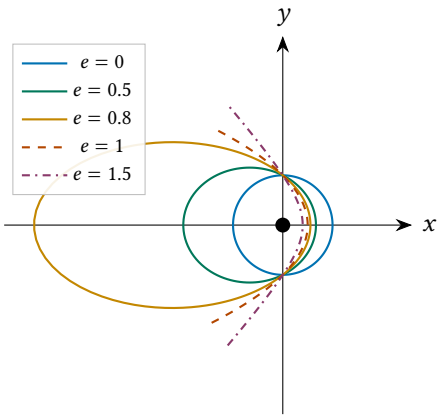


Figure 8.6: Conics with a common focus (the dot at the origin) and the same $p = 1$ in $r = \frac{p}{1+e\cos\theta}$: a circle ($e = 0$), two ellipses of increasing elongation ($e = 0.5, 0.8$), a parabola ($e = 1$) and one branch of a hyperbola ($e = 1.5$). The ellipse closes up on the left; as $e \to 1$ the far end runs off to infinity and becomes the parabola.

This continuous spectrum of curves is not just a geometric abstraction; eccentricity acts as the defining parameter that categorizes real celestial orbits and trajectories throughout our solar system, as summarized in Table 8.1.

Body	e	Orbit shape
Earth	0.017	ellipse, almost a circle
Mercury	0.206	ellipse
Pluto	0.249	clearly elongated ellipse
Halley's comet	0.967	extremely elongated ellipse
ideal escape trajectory	1	parabola (just barely escapes)
interstellar object 'Oumuamua	≈ 1.2	hyperbola (passes once and leaves)

Table 8.1: Eccentricities of some orbits. The smaller e, the rounder the ellipse.

8.1.5 Classifying a Second-Degree Equation

By completing the square, once the xy-term is absent ($B = 0$): group the x- and y-terms, complete each square, and read the type from the resulting standard form (Worked Problems 1–2).

By the discriminant, directly from $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$: compute $B^2 - 4AC$.

- $B^2 - 4AC < 0$: an **ellipse** (a circle if additionally $A = C$ and $B = 0$), or a single point, or no real curve;
- $B^2 - 4AC = 0$: a **parabola**, or two parallel (possibly coincident) lines, or no curve;
- $B^2 - 4AC > 0$: a **hyperbola**, or a pair of intersecting lines.

> **Common mistake. The discriminant cannot see degenerate cases.** $B^2 - 4AC < 0$ for both $x^2 + y^2 = 4$ (a circle) and $x^2 + y^2 = 0$ (a point) and $x^2 + y^2 = -4$ (nothing). Likewise $B^2 - 4AC > 0$ for $\frac{x^2}{4} - \frac{y^2}{9} = 1$ (hyperbola) and for $\frac{x^2}{4} - \frac{y^2}{9} = 0$ (two crossing lines). To decide whether the curve is genuine, complete the square (or check that the equation has real points). Also: completing the square separately in x and y works *only* when $B = 0$.

The xy-term: Rotating the Axes

If $B \neq 0$ the conic is *tilted* with respect to the coordinate axes (Figure 8.7). To straighten it, rotate the axes by an angle θ:

$$x = x' \cos\theta - y' \sin\theta, \qquad y = x' \sin\theta + y' \cos\theta.$$

The coefficient of $x'y'$ is $(C - A)\sin 2\theta + B\cos 2\theta$, which vanishes when

$$\cot 2\theta = \frac{A - C}{B} \qquad (\theta = 45^\circ \text{ if } A = C).$$

The new coefficients are $A' = A\cos^2\theta + B\sin\theta\cos\theta + C\sin^2\theta$ and $C' = A\sin^2\theta - B\sin\theta\cos\theta + C\cos^2\theta$.

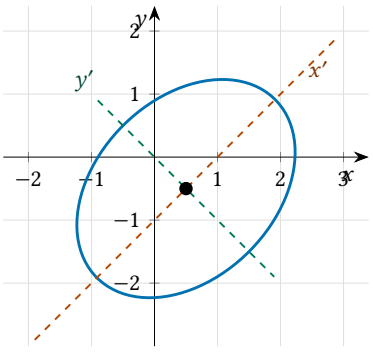


Figure 8.7: The conic $3x^2 - 2xy + 3y^2 - 4x + 4y - 6 = 0$ is an ellipse centered at $(\frac{1}{2}, -\frac{1}{2})$, tilted by 45°. In the rotated axes x', y' (dashed) it has the equation $\frac{x'^2}{4} + \frac{y'^2}{2} = 1$: semi-axes 2 along x' and $\sqrt{2}$ along y'.

Intuition. **A rotation does not change the shape, so it cannot change the type.** After the rotation we have $B' = 0$, and then $B'^2 - 4A'C' = -4A'C'$: an ellipse if A' and C' have the same sign, a hyperbola if opposite signs, a parabola if one of them is zero. The quantities $A + C$ and $B^2 - 4AC$ are *invariant* under rotation (they come out the same in both coordinate systems), and that is exactly why the discriminant test works in the original, tilted coordinates.

Example: The Hyperbola $xy = k$

The equation $xy = 4$ has $A = C = 0$, $B = 1$, so $B^2 - 4AC = 1 > 0$: a hyperbola. Since $A = C$, rotate by $\theta = 45°$: $xy = \frac{x'^2 - y'^2}{2} = 4$, i.e. $\frac{x'^2}{8} - \frac{y'^2}{8} = 1$, a hyperbola with $a = b = 2\sqrt{2}$ whose asymptotes are the original coordinate axes ($x = 0$ and $y = 0$). It describes every inverse proportion: pressure × volume = constant for a gas at fixed temperature (Boyle's law).

8.2 Second Order Surfaces

The general equation of a second order (**quadric**) surface in space is

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fzx + Gx + Hy + Iz + J = 0,$$

where at least one of $A, \dots, F$ is nonzero. By a translation and a rotation of the axes (exactly as for curves) every quadric reduces to one of the *standard forms* below.

Intuition. **Understand a surface by its slices.** A *trace* (cross-section) is the curve in which the surface meets a plane. For a quadric the traces in planes parallel to the coordinate planes ($x = k$, $y = k$ or $z = k$) are always conics. The strategy for every surface in this section is the same: (1) find the traces in the coordinate planes, (2) find the traces in the parallel planes $z = k$ (and $x = k$, $y = k$) and see how they grow, shrink or change type with k, (3) assemble the pictures, like the layers of an onion or the contour lines of a map. Each figure below shows the surface and, next to it, the traces from which it is built.

Common mistake. **A missing variable makes a cylinder.** In the plane $x^2 + y^2 = 1$ is a circle, but in *space* the same equation has no condition on z: it describes the whole infinite cylinder above and below that circle. Always ask which variables an equation contains: the missing one is the free direction of the surface.

8.2.1 The Sphere

Definition: Sphere

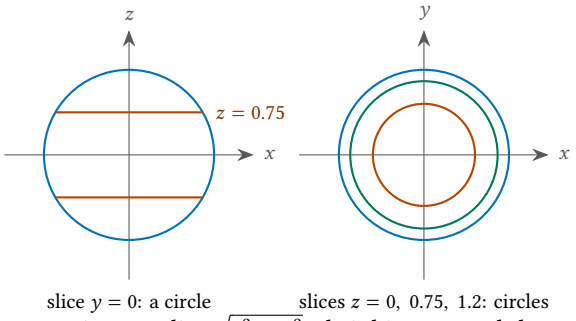
A **sphere** is the set of points $P(x, y, z)$ at a fixed distance r from a fixed center $C(a, b, c)$:

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2.$$

Expanding gives $x^2 + y^2 + z^2 + Gx + Hy + Iz + J = 0$; conversely such an equation is a sphere with center $\left(-\frac{G}{2}, -\frac{H}{2}, -\frac{I}{2}\right)$ and $r^2 = \frac{G^2+H^2+I^2}{4} - J$, provided $r^2 > 0$ (Figure 8.8).



(a) the sphere $x^2 + y^2 + z^2 = r^2$



slice $y = 0$: a circle slices $z = 0,\ 0.75,\ 1.2$: circles

(b) cross-sections: radius $\sqrt{r^2 - z^2}$, shrinking toward the poles

Figure 8.8: A sphere of radius $r = 1.5$. Every slice is a circle; slices at height z have radius $\sqrt{r^2 - z^2}$, from r at the equator down to a single point at the poles $z = \pm r$.

8.2.2 The Cylinder

Definition: Cylindrical Surface

A **cylindrical surface** consists of all lines (**elements**) parallel to a fixed direction that pass through a fixed plane curve (the **directrix**). Equivalently: take a curve in one coordinate plane and let the third variable be free. Its equation has *one variable missing*:

$$Ax^2 + By^2 + Gx + Hy + J = 0 \qquad \text{(the } z\text{-direction is free).}$$

The four types, according to the directrix, are the **circular** ($x^2 + y^2 = r^2$), **elliptic** ($\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$), **hyperbolic** ($\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$) and **parabolic** ($x^2 + 2ay = 0$) cylinders, Figure 8.9.



circular $x^2 + y^2 = r^2$ | elliptic $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ | hyperbolic $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ | parabolic $x^2 + 2ay = 0$

Figure 8.9: The four cylinders. Each is a plane curve (in the xy-plane) extended indefinitely along the missing variable, here the vertical z-direction.

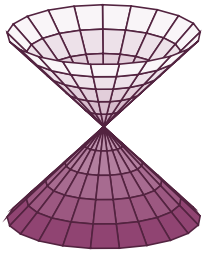
8.2.3 The Cone

Definition: Cone

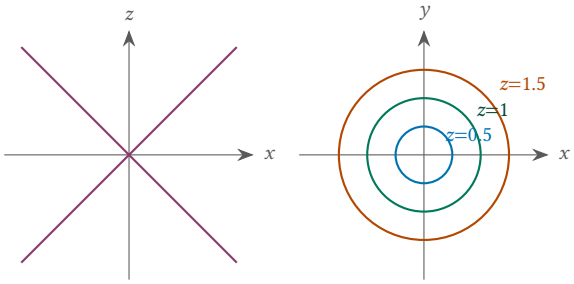
The **(elliptic) cone** with vertex at the origin and axis the z-axis is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2} \qquad \text{(circular cone if } a = b\text{).}$$

Horizontal cross-sections are ellipses (circles) that grow linearly with $|z|$ and shrink to a single point (the vertex) at $z = 0$; the sections by planes through the axis are pairs of straight lines, Figure 8.10.



(a) the cone $x^2 + y^2 = z^2$



slice $y = 0$: two lines $z = \pm x$ slices $z = k$: circles of radius $|k|$

(b) cross-sections

Figure 8.10: The circular cone $x^2 + y^2 = z^2$ (both nappes). A slice at height $z = k$ is a circle of radius $|k|$, so the radius grows in direct proportion to the distance from the vertex; a slice through the axis is a pair of crossing lines.

Intuition. Why $x^2/a^2 + y^2/b^2 = z^2/c^2$ is a cone: scale invariance. Every term of the equation has degree two, so if (x, y, z) satisfies it, so does (tx, ty, tz) for every number t: the whole line through the origin and that point lies on the surface. A surface made of lines through one point is a cone, and conversely *any* homogeneous second degree equation $Ax^2+By^2+Cz^2+Dxy+Eyz+Fzx = 0$ describes a cone (with vertex at the origin).

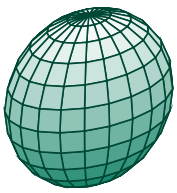
8.2.4 The Ellipsoid

Definition: Ellipsoid

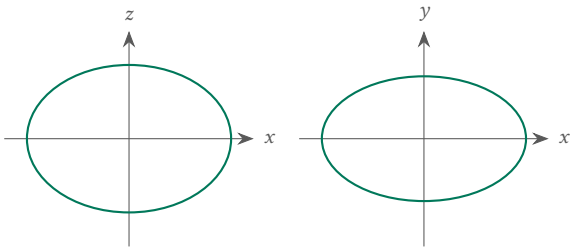
The **ellipsoid** with semi-axes $a, b, c > 0$ is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

Its intercepts are $(\pm a, 0, 0)$, $(0, \pm b, 0)$, $(0, 0, \pm c)$; parallel coordinate cross-sections are ellipses. If two semi-axes are equal, it is a **spheroid**; if all three, a sphere (Figure 8.11).



(a) $a = 1.8,\ b = 1.1,\ c = 1.3$



xz-trace: $\frac{x^2}{a^2} + \frac{z^2}{c^2} = 1$ xy-trace: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(b) two of the three principal traces

Figure 8.11: An ellipsoid with three different semi-axes ($a > c > b$). Each coordinate plane cuts it in an ellipse whose semi-axes are two of a, b, c.

Intuition. Spheroids: an ellipse spun about an axis. Rotating an ellipse about one of its axes gives a spheroid, with $a = b$: if $c > a$ it is *prolate* (elongated like a rugby ball or an egg), if $c < a$ it is *oblate* (flattened like a planet or a lentil). The Earth is an oblate spheroid whose equatorial radius is about 21 km larger than its polar radius, because of its rotation.

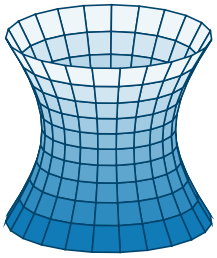
8.2.5 Hyperboloid of One Sheet

Definition: Hyperboloid of One Sheet

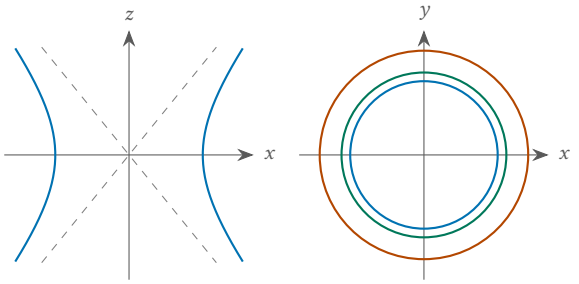
The **hyperboloid of one sheet** (along the z-axis, Figure 8.12) is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1.$$

Horizontal cross-sections $z = k$ are ellipses $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 + \frac{k^2}{c^2}$ growing with $|z|$ from a minimum at $z = 0$ (the **waist**); vertical cross-sections are hyperbolas or intersecting lines. It is connected .



(a) one sheet



xz-trace: $\frac{x^2}{a^2} - \frac{z^2}{c^2} = 1$ slices z=0, 0.8, 1.6: growing circles

(b) cross-sections

Figure 8.12: Hyperboloid of one sheet with $a = b = 1.3$, $c = 1.6$. The waist $z = 0$ is the smallest circle; the slices widen as $|z|$ grows, following the asymptotic cone (dashed lines in the xz-trace).

8.2.6 Hyperboloid of Two Sheets

Definition: Hyperboloid of Two Sheets

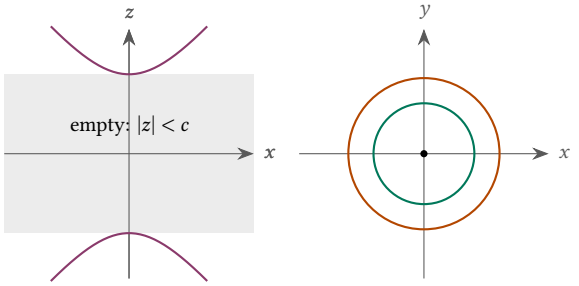
The **hyperboloid of two sheets** with axis the z-axis (Figure 8.13) is

$$\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

The equation forces $z^2 \geq c^2$, so the surface consists of two separate bowl-shaped sheets, one in $z \geq c$ and one in $z \leq -c$, with *no points at all for* $|z| < c$. Horizontal cross-sections are ellipses for $|z| > c$ and single points at $z = \pm c$; vertical cross-sections are hyperbolas.



(a) two sheets



xz-trace: $\frac{z^2}{c^2} - \frac{x^2}{a^2} = 1$ slices $z = \pm 1.4,\ \pm 1.8,\ \pm 2.2$

(b) cross-sections

Figure 8.13: Hyperboloid of two sheets with $a = b = 1.1$, $c = 1.4$. The sheets touch the axis at $z = \pm c$ and open away from each other; the slab $|z| < c$ contains no points of the surface.

Intuition. Count the minus signs. Write the equation with 1 on the right and all squares on the left, divided by their semi-axis squares. Then the number of *minus* signs decides the surface:

minus signs	surface
0	ellipsoid ($x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$)
1	hyperboloid of **one** sheet (the variable with the minus is the axis)
2	hyperboloid of **two** sheets (the variable with the plus is the axis)
3	no real points ($-x^2 - y^2 - z^2 = 1$)

The reason: a minus sign means that a variable can grow without bound at the expense of nothing. With one minus the other two terms add up to "1 plus something", which is always possible (the surface is connected, with circles that grow); with two minuses the lone positive term must reach at least 1 before the equation can be satisfied (hence the gap and the two separate sheets).

8.2.7 Elliptic and Hyperbolic Paraboloids

Definition: Paraboloids

Paraboloids with axis the z-axis are classified into two types:

- **Elliptic Paraboloid** (Figure 8.14):

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c} \quad \text{(circular if } a = b\text{)}.$$

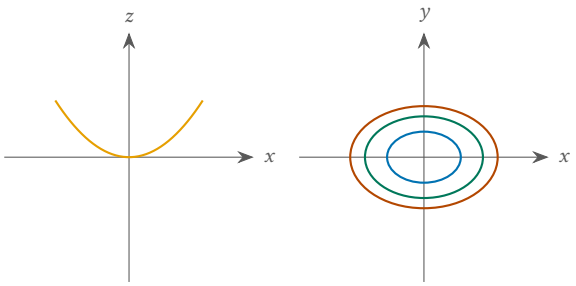
  For $c > 0$, horizontal slices $z = k > 0$ are growing ellipses shrinking to the **vertex** at $z = 0$ (empty for $z < 0$); vertical slices are parabolas.
- **Hyperbolic Paraboloid / Saddle** (Figure 8.15):

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{z}{c}.$$

  Horizontal slices $z = k$ are hyperbolas (opening along x if $k/c > 0$, along y if $k/c < 0$) or crossing lines $y = \pm\frac{b}{a}x$ at $z = 0$; vertical slices are orthogonal parabolas. The origin is a **saddle point** (minimum in x, maximum in y).



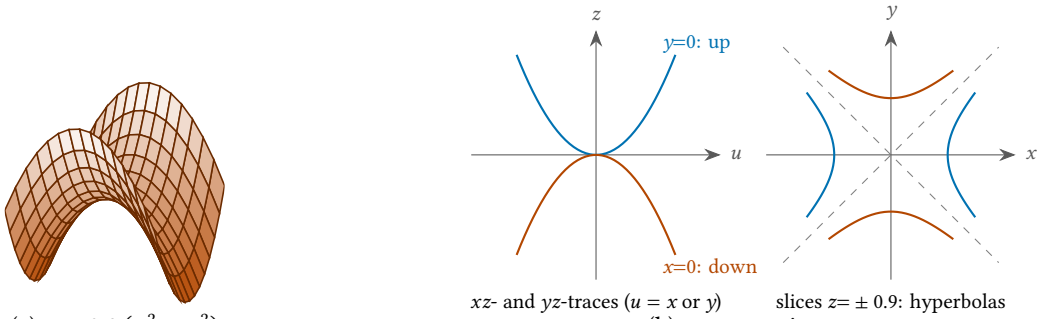
(a) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = z$



xz-trace: a parabola slices z=0.25, 0.64, 1: ellipses

(b) cross-sections

Figure 8.14: Elliptic paraboloid with $a = 1.3$, $b = 0.9$, $c = 1$. Vertical slices are parabolas; horizontal slices are ellipses whose semi-axes grow like $\sqrt{z}$, so the bowl gets wider more and more slowly as it gets deeper.



(a) $z = 0.9\,(x^2 - y^2)$ (b) cross-sections

Figure 8.15: The saddle $z = 0.9(x^2 - y^2)$. Left: the parabolas in the planes $y = 0$ (opening up, blue) and $x = 0$ (opening down, red) meet at the saddle point. Right: horizontal slices are hyperbolas, blue for $z > 0$ and red for $z < 0$, separated by the pair of lines $y = \pm x$ at $z = 0$.

8.2.8 Summary of the Quadric Surfaces

Surface	Standard equation	Slices $z = k$	Slices $x = k$ or $y = k$
Ellipsoid	$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$	ellipses, a point at $z = \pm c$	ellipses
Hyperboloid, one sheet	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$	ellipses for every k	hyperbolas (or two lines)
Hyperboloid, two sheets	$\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	ellipses for $\lvert k\rvert > c$, points at $\pm c$, none inside	hyperbolas
Cone	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$	ellipses, a point at $k = 0$	hyperbolas (two lines through the axis)
Elliptic paraboloid	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}$	ellipses for $k/c > 0$, a point at 0	parabolas
Hyperbolic paraboloid	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{z}{c}$	hyperbolas (two lines at $k = 0$)	parabolas, opening in opposite senses

Table 8.2: The six central/paraboloid quadrics (the sphere is the ellipsoid with $a = b = c$; the cylinders have one variable missing). All slices parallel to a coordinate plane are conics.

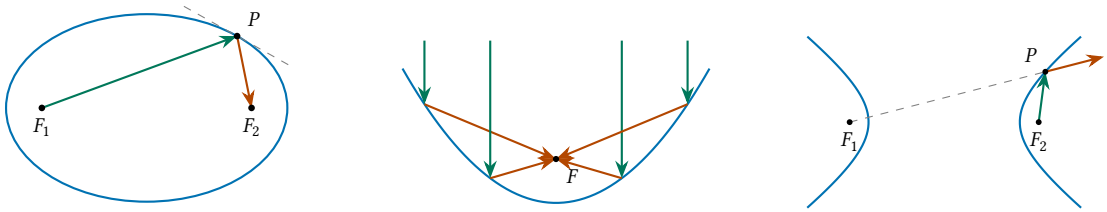
8.3 Conics and Quadrics in Physics and Engineering

Conics and quadrics are everywhere because they are the shapes that nature and design choose when something is *focused*, *falls*, *orbits* or must be both strong and simple to build. Parabolas collect parallel rays at one point and describe uniformly accelerated motion; ellipses describe bound orbits and join two foci optically; hyperbolas describe unbound orbits and positions found from differences of distances; and the quadric surfaces appear as dishes, cooling towers, saddle roofs, the surface of a spinning liquid and the shape of the Earth itself.

8.3.1 Reflectors, Antennas and Telescopes

Each conic has a *reflection property*: the tangent line at a point P makes equal angles with the two "rays" that the focal definition draws to P (Figure 8.16).

- **Parabola:** a ray parallel to the axis is reflected through the focus (and, reversed, a source at the focus sends a parallel beam). Use: satellite dishes, radio telescopes, solar cookers, car headlights, flashlights.
- **Ellipse:** a ray from one focus is reflected through the other focus. Use: lithotripsy (focusing shock waves on a kidney stone), whispering galleries, some lamp reflectors.
- **Hyperbola:** a ray aimed at one focus is reflected as if it came from the other. Use: the secondary mirror of Cassegrain telescopes.



(a) ellipse: $F_1 \to P \to F_2$ (b) parabola: parallel $\to F$ (c) hyperbola: as if from F_1

Figure 8.16: The reflection properties. Green: incoming ray, red: reflected ray. (a) Ellipse $a = 3$, $b = 2$: a ray from F_1 passes through F_2 (the dashed tangent at P bisects the exterior angle). (b) Parabola $y = x^2/4$ with focus $(0, 1)$: every vertical ray is sent through F. (c) Hyperbola $a = 2$, $b = 1.5$: a ray leaving F_2 is reflected along the line from F_1 through P (dashed).

Example: Where Should the Receiver of a Dish Be?

A satellite dish is a *paraboloid of revolution*, $z = \dfrac{x^2 + y^2}{4f}$, where f is the focal length (the receiver must sit at the focus $(0, 0, f)$). A dish of diameter $D = 3$ m has depth $d = 0.3$ m at its center. At the rim $r = D/2 = 1.5$ m the height is d, so

$$d = \frac{(D/2)^2}{4f} \quad \Longrightarrow \quad f = \frac{D^2}{16\,d} = \frac{9}{16 \cdot 0.3} = 1.875 \text{ m}.$$

The receiver is 1.875 m above the vertex, as drawn in every satellite-TV installation.

Common mistake. **A dish is a paraboloid, not a sphere.** A spherical mirror does *not* bring parallel rays to one point (this defect is called spherical aberration; the rays farther from the axis miss the focus). Only the parabola has the exact focusing property, which is why large antennas and telescope mirrors are ground to a paraboloid.

8.3.2 Orbits: Kepler and Newton

Kepler found (1609) that the planets move on ellipses with the Sun at one focus; Newton showed (1687) that an inverse-square force allows *exactly* the conic orbits $r = \frac{p}{1+e\cos\theta}$ of Figure 8.6. Which conic occurs is decided by the total energy: $E < 0$ gives an ellipse (a bound orbit), $E = 0$ a parabola (the escape-speed borderline), $E > 0$ a hyperbola (a fly-by that never returns).

For an elliptical orbit with semi-major axis a and eccentricity e, the nearest and farthest distances from the Sun (at the focus) are

$$r_{\min} = a(1 - e) \text{ (perihelion)}, \qquad r_{\max} = a(1 + e) \text{ (aphelion)}, \qquad b = a\sqrt{1 - e^2}.$$

Example: Earth and Halley's Comet

Earth: $a = 149.6$ million km, $e = 0.0167$. Then $r_{\min} = 147.1$ and $r_{\max} = 152.1$ million km, and $b = a\sqrt{1-e^2} = 149.58$ million km: the orbit is so close to a circle ($b/a = 0.99986$) that on a page it would be indistinguishable from one, but the Earth is still about 5 million km closer to the Sun in January than in July.

Halley's comet: $a = 17.8$ AU, $e = 0.967$, so $r_{\min} = 17.8 \cdot 0.033 \approx 0.59$ AU (inside the orbit of Venus) and $r_{\max} = 17.8 \cdot 1.967 \approx 35.0$ AU (beyond Neptune): an extremely elongated ellipse, which is why the comet spends most of its ≈ 76-year period far away and rushes past the Sun.

Common mistake. **The Sun is at a focus, not at the center.** The center of the ellipse is empty. The Sun–planet distance therefore changes between $a(1-e)$ and $a(1+e)$, and the planet moves faster near perihelion (Kepler's second law). The seasons are *not* caused by this change in distance (the tilt of the Earth's axis causes them): the distance is smallest in January.

8.3.3 Projectiles and Cables: Parabolas in Mechanics

A body moving under constant acceleration follows a parabola. Eliminating t from $x = v_0 \cos\alpha\, t$, $y = v_0 \sin\alpha\, t - \frac{1}{2}gt^2$ gives

$$y = x\tan\alpha - \frac{g}{2v_0^2\cos^2\alpha}\,x^2,$$

a parabola with a vertical axis opening downward (compare Chapter ??).

Example: A Thrown Ball and a Suspension Bridge

Ball: $v_0 = 20$ m/s, $\alpha = 45°$: $y = x - 0.0245\,x^2$. Completing the square gives the vertex at $x = \frac{1}{2\cdot 0.0245} \approx 20.4$ m, with height $y \approx 10.2$ m, and the range is the second root of $y = 0$, $x = \frac{1}{0.0245} \approx 40.8$ m, matching Chapter ??.

Bridge: the main cable of a suspension bridge carries a deck of uniform weight per horizontal meter, so it hangs in a parabola with its lowest point at midspan. For a span $L = 400$ m and sag $s = 50$ m, measured from the lowest point, $y = \frac{4s}{L^2}x^2 = 0.00125\,x^2$; the cable rises 50 m at $x = \pm 200$ m, and 12.5 m at $x = \pm 100$ m (the length of the shortest hangers above the lowest point).

Common mistake. **A freely hanging chain is not a parabola.** A chain or cable hanging under its own weight takes the shape of the *catenary* $y = a\cosh\frac{x}{a}$, which agrees with a parabola only near the lowest point. It is the *uniformly loaded* cable of a suspension bridge (load proportional to horizontal length) that is a parabola.

8.3.4 Hyperbolas: Positioning and Inverse Proportion

Positioning by time differences. If two stations F_1, F_2 send synchronised signals, a receiver that measures the *difference* of the arrival times knows the difference of its distances to the stations, $|PF_1 - PF_2| = 2a$, so it lies on a hyperbola with the stations as foci. A second pair of stations gives a second hyperbola, and the receiver sits at their intersection (the principle of the LORAN navigation system and of locating the source of a sound with microphones).

Example: A Position Line

Two stations are 300 km apart; place them at the foci $(\pm 150, 0)$. Radio signals travel 300 km per millisecond, and a receiver measures that the signals arrive $\frac{1}{3}$ ms apart, i.e. the distances differ by 100 km: $2a = 100$, $a = 50$, $c = 150$, $b^2 = c^2 - a^2 = 22\,500 - 2\,500 = 20\,000$, and the position line is

$$\frac{x^2}{2500} - \frac{y^2}{20\,000} = 1.$$

(Check: the point $(100, 245)$ is 350.0 km from F_1 and 250.0 km from F_2.) The receiver is on the branch nearer to the station that received the signal first.

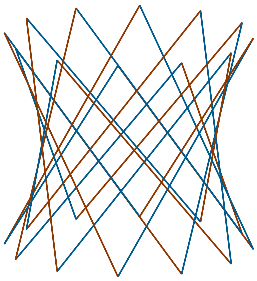
Inverse proportion. The graph of $xy = k$ is a (rectangular) hyperbola with the coordinate axes as asymptotes. At constant temperature a gas obeys Boyle's law $PV = k$: if $P = 2$ atm at $V = 3$ L then $k = 6$, and compressing the gas to $P = 4$ atm gives $V = \frac{6}{4} = 1.5$ L; the pressure–volume graph is one branch of this hyperbola.

8.3.5 Quadric Surfaces in Structures, Fluids and the Sky

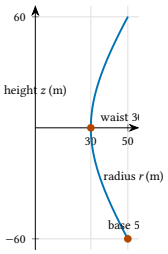
Cooling towers and ruled surfaces. The hyperboloid of one sheet is a *ruled surface*: it contains straight lines. Indeed, for $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ the lines

$$x = a(\cos\varphi - t\sin\varphi), \quad y = b(\sin\varphi + t\cos\varphi), \quad z = ct \qquad (t \in \mathbb{R})$$

lie on the surface for every φ, since $(\cos\varphi - t\sin\varphi)^2 + (\sin\varphi + t\cos\varphi)^2 = 1 + t^2$. Reversing the sign of t in the first two coordinates gives a second family (Figure 8.17). A concrete tower can therefore be built from *straight* steel beams or formwork, while its shape has a narrow waist that is strong against wind and aids the draught.



(a) two families of straight lines



(b) the profile of the tower

Figure 8.17: (a) The hyperboloid of one sheet $\frac{x^2+y^2}{1.3^2} - \frac{z^2}{1.6^2} = 1$ drawn only with its straight lines (blue and red families). (b) Profile $\frac{r^2}{900} - \frac{z^2}{2025} = 1$ of a cooling tower with waist radius 30 m and $c = 45$ m: at 60 m below the waist the radius is $30\sqrt{1 + 60^2/45^2} = 50$ m.

Example: Radius of a Cooling Tower

With waist radius $a = b = 30$ m and $c = 45$ m the tower is $\dfrac{x^2 + y^2}{900} - \dfrac{z^2}{2025} = 1$. At height z the cross-section is the circle $r = 30\sqrt{1 + z^2/2025}$; for example $r = 36.1$ m at $z = \pm 30$ m and $r = 50$ m at $z = \pm 60$ m. The hyperbolic paraboloid $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{z}{c}$ is also ruled (it contains the lines $\frac{x}{a} - \frac{y}{b} = \lambda$, $\frac{x}{a} + \frac{y}{b} = \frac{z}{c\lambda}$), which is why thin concrete "saddle" roofs can be cast with straight formwork.

A spinning liquid is a paraboloid. In a container rotating with angular velocity ω, the surface of the liquid is tangent to the combined force of gravity and the centrifugal force, so $\frac{dz}{dr} = \frac{\omega^2 r}{g}$ and

$$z = \frac{\omega^2}{2g} r^2 = \frac{x^2 + y^2}{4f}, \qquad f = \frac{g}{2\omega^2},$$

a circular (elliptic) paraboloid. For $\omega = 4$ rad/s: $z = 0.815\, r^2$, focal length $f = \frac{9.81}{32} \approx 0.31$ m. A rotating pool of mercury is a perfect parabolic mirror, used in "liquid-mirror" telescopes.

Ellipsoids, cones and spheres. Because of its rotation the Earth is an *oblate spheroid*: $\frac{x^2+y^2}{a^2} + \frac{z^2}{c^2} = 1$ with equatorial radius $a = 6378.137$ km and polar radius $c = 6356.752$ km, about 21.4 km shorter (flattening $\frac{a-c}{a} = \frac{1}{298.26}$); this is the surface on which GPS heights are defined. A projectile flying faster than sound ($M > 1$) carries a *cone* of shock waves with half-angle μ, $\sin\mu = \frac{1}{M}$: $\mu = 30^\circ$ at $M = 2$ and $\mu \approx 41.8^\circ$ at $M = 1.5$. In satellite positioning (*trilateration*) each measured distance puts the receiver on a *sphere* around the satellite.

Example: Intersecting Spheres

Two spheres $x^2 + y^2 + z^2 = 25$ and $(x-6)^2 + y^2 + z^2 = 25$ intersect where both hold. Subtracting the equations eliminates the squares of y, z: $x^2 - (x-6)^2 = 0$, so $12x - 36 = 0$, the plane $x = 3$. Substituting, $y^2 + z^2 = 25 - 9 = 16$: the intersection is a *circle* of radius 4 in the plane $x = 3$, centered at $(3, 0, 0)$. A third sphere would cut this circle in at most two points, and a fourth one selects between them; real GPS receivers also use the fourth signal to determine their clock error.

8.4 Facts and Curiosities

1. From Apollonius to Newton. Apollonius (about 200 BCE) named the ellipse, parabola and hyperbola and studied them as sections of a cone. For almost 1800 years they were pure geometry until Kepler (1609) found that Mars moves on an ellipse and Newton (1687) proved that an inverse-square attraction produces exactly the conic orbits.

2. A visitor on a hyperbola. The interstellar object *1I/'Oumuamua*, discovered in 2017, moved on a path with eccentricity about 1.2: a hyperbola. With $e > 1$ the orbit is unbound; it came from outside the solar system and left it again.

3. All parabolas look alike. Every ellipse has its own shape (its eccentricity), and so does every hyperbola; but *any two parabolas are similar*: one can be magnified to coincide with the other. The reason is that $e = 1$ is a single value. The circle has the same property for $e = 0$.

4. Five points determine a conic. A general conic has six coefficients, but only their ratios matter, which leaves five free parameters. So five points in general position lie on exactly one conic.

5. Potato chips. Stackable potato chips are modelled by a hyperbolic paraboloid: its saddle shape is stiff in both directions (hence it does not break easily) and identical chips nest on top of each other.

6. Straight lines on a curved surface. Only a few surfaces are curved yet contain straight lines (are "ruled"): the cylinder, the cone, the hyperboloid of one sheet and the hyperbolic paraboloid (and plane-type degenerate cases). The ellipsoid, the two-sheeted hyperboloid and the elliptic paraboloid contain *no* straight line at all.

8.5 Worked Problems

Problem 1: Equation of an Ellipse From Its Foci and Vertices

An ellipse has vertices $(\pm 5, 0)$ and foci $(\pm 3, 0)$. Find its equation.

Solution. Centered at the origin, horizontal (foci on x-axis): $a = 5$, $c = 3 \implies b^2 = a^2 - c^2 = 16 \implies \frac{x^2}{25} + \frac{y^2}{16} = 1$, with $e = \frac{c}{a} = 0.6$.

Problem 2: Center and Radius of a Circle

Find the center and radius of $x^2 + y^2 - 6x + 4y - 12 = 0$.

Solution. $(x^2 - 6x + 9) + (y^2 + 4y + 4) = 12 + 9 + 4 = 25$, so $(x-3)^2 + (y+2)^2 = 25$: center $(3, -2)$, radius 5.

Problem 3: Equation of a Parabola From Its Vertex and Focus

A parabola has vertex $(2, 1)$ and focus $(2, 4)$. Find its equation.

Solution. The focus lies directly above the vertex, so the parabola is vertical with $a = 4 - 1 = 3$ (focus distance, $a > 0$ since it opens upward):

$$(x-2)^2 = 12(y-1).$$

Problem 4: Equation of a Hyperbola From Its Vertices and Foci

A hyperbola has vertices $(0, \pm 2)$ and foci $(0, \pm\sqrt{13})$. Find its equation.

Solution. Vertical (vertices on the y-axis): $a = 2$, $c = \sqrt{13}$, so $b^2 = c^2 - a^2 = 13 - 4 = 9$:

$$\frac{y^2}{4} - \frac{x^2}{9} = 1,$$

with asymptotes $\frac{y^2}{4} - \frac{x^2}{9} = 0$, i.e. $y = \pm\frac{2}{3}x$.

Problem 5: Classify a Curve by Completing the Square

Identify the curve $4x^2 + 9y^2 - 16x + 18y - 11 = 0$, and find its center and key features.

Solution. Group and complete the square (remembering to multiply the added numbers by the leading coefficients):

$$4(x^2 - 4x) + 9(y^2 + 2y) = 11 \implies 4(x^2 - 4x + 4) + 9(y^2 + 2y + 1) = 11 + 16 + 9 = 36,$$

$$4(x-2)^2 + 9(y+1)^2 = 36 \implies \frac{(x-2)^2}{9} + \frac{(y+1)^2}{4} = 1.$$

This is an **ellipse**, center $(2, -1)$, $a^2 = 9 \Rightarrow a = 3$ (horizontal, larger denominator under x), $b = 2$, $c = \sqrt{9-4} = \sqrt{5}$: foci $\left(2 \pm \sqrt{5}, -1\right)$, eccentricity $e = \frac{\sqrt{5}}{3} \approx 0.75$.

Problem 6: Classify a Hyperbola by Completing the Square

Identify the curve $9x^2 - 4y^2 - 36x - 8y - 4 = 0$.

Solution.

$$9(x^2 - 4x) - 4(y^2 + 2y) = 4 \implies 9(x-2)^2 - 4(y+1)^2 = 4 + 36 - 4 = 36,$$

$$\frac{(x-2)^2}{4} - \frac{(y+1)^2}{9} = 1.$$

A **hyperbola**, center $(2, -1)$, horizontal transverse axis, $a = 2$, $b = 3$, $c = \sqrt{4+9} = \sqrt{13}$, asymptotes $y + 1 = \pm\frac{3}{2}(x-2)$.

Problem 7: A Tilted Conic: the Discriminant, the Center and a Rotation

Classify $3x^2 - 2xy + 3y^2 - 4x + 4y - 6 = 0$ and find its center and axes.

Solution. $A = 3$, $B = -2$, $C = 3$: $B^2 - 4AC = 4 - 36 = -32 < 0$, so the curve is an ellipse or a degenerate case. The partial derivatives $6x - 2y - 4 = 0$ and $-2x + 6y + 4 = 0$ give the center $\left(\frac{1}{2}, -\frac{1}{2}\right)$, where the left-hand side equals -8; translating the origin there leaves $3u^2 - 2uv + 3v^2 = 8$. The form $3u^2 - 2uv + 3v^2$ is positive, so the equation is satisfiable and the conic is a genuine ellipse. Since $A = C$, rotate by $45°$: $3u^2 - 2uv + 3v^2 = 2u'^2 + 4v'^2$, so

$$\frac{u'^2}{4} + \frac{v'^2}{2} = 1 \ :$$

semi-axes $a = 2$ (along the direction $(1, 1)$) and $b = \sqrt{2}$ (along $(-1, 1)$), eccentricity $e = \frac{\sqrt{4-2}}{2} = \frac{\sqrt{2}}{2} \approx 0.71$ (Figure 8.7).

Problem 8: Parabola From Focus and Directrix

Find the equation of the parabola with focus $(1, 3)$ and directrix $x = -1$.

Solution. The vertex is halfway between the focus and the directrix: x-coordinate $\frac{1+(-1)}{2} = 0$, same y as the focus, so vertex $(0, 3)$. The directrix is vertical, so the parabola is horizontal, with $a =$ (vertex-to-focus distance) $= 1 > 0$ (it opens toward the focus, to the right):

$$(y-3)^2 = 4 \cdot 1 \cdot (x - 0) = 4x.$$

(Check with the definition: $\sqrt{(x-1)^2 + (y-3)^2} = x + 1 \iff (y-3)^2 = 4x$.)

Problem 9: Sphere From Its Equation

Find the center and radius of $x^2 + y^2 + z^2 + 2x - 4y + 8z + 5 = 0$.

Solution.

$$(x+1)^2 + (y-2)^2 + (z+4)^2 = -5 + 1 + 4 + 16 = 16.$$

Center $(-1, 2, -4)$, radius 4.

Problem 10: Classify a Quadric Surface

Identify the surface $4x^2 + y^2 - 4z^2 - 16 = 0$, and describe its horizontal cross-sections.

Solution. Divide by 16:

$$\frac{x^2}{4} + \frac{y^2}{16} - \frac{z^2}{4} = 1.$$

One minus sign: a **hyperboloid of one sheet**, axis along z (the variable with the minus sign), $a = 2$ (under x), $b = 4$ (under y), $c = 2$. At $z = 0$ the cross-section is the ellipse $\frac{x^2}{4} + \frac{y^2}{16} = 1$, growing steadily in size for all $z \neq 0$: $\frac{x^2}{4} + \frac{y^2}{16} = 1 + \frac{z^2}{4}$.

Problem 11: Rotating an Axis-Aligned-Looking Ellipse

Identify $5x^2 - 6xy + 5y^2 = 32$ and give its semi-axes.

Solution. $B^2 - 4AC = 36 - 100 = -64 < 0$: an ellipse (the right side is positive, so it is real). Since $A = C$ rotate by 45°: $x = \frac{x' - y'}{\sqrt{2}}$, $y = \frac{x' + y'}{\sqrt{2}}$, so $5(x^2 + y^2) = 5(x'^2 + y'^2)$ and $-6xy = -3(x'^2 - y'^2)$, giving $2x'^2 + 8y'^2 = 32$:

$$\frac{x'^2}{16} + \frac{y'^2}{4} = 1.$$

Semi-axes 4 (along the line $y = x$) and 2 (along $y = -x$). Check the invariants: $A' + C' = 2 + 8 = 10 = 5 + 5$ and $-4A'C' = -64 = B^2 - 4AC$.

Problem 12: Center and Radius by Completing the Square

Find the center and radius of $x^2 + y^2 + z^2 - 4x + 6y - 2z - 2 = 0$.

Solution. Group and complete the square in each variable:

$$(x^2 - 4x + 4) + (y^2 + 6y + 9) + (z^2 - 2z + 1) = 2 + 4 + 9 + 1 \implies (x - 2)^2 + (y + 3)^2 + (z - 1)^2 = 16.$$

Center $(2, -3, 1)$, radius 4.

Problem 13: Classify a Quadric by Completing the Square

Identify $x^2 + y^2 - 2x + 4y - z + 5 = 0$.

Solution. z appears only to the first power, so look for a paraboloid: $(x^2 - 2x + 1) + (y^2 + 4y + 4) = z - 5 + 1 + 4 = z$, i.e.

$$(x - 1)^2 + (y + 2)^2 = z.$$

An **elliptic paraboloid** (here a paraboloid of revolution, with $a = b$), vertex $(1, -2, 0)$, opening upward; the cross-section $z = k$ is a circle of radius $\sqrt{k}$ for $k > 0$, a point for $k = 0$, and empty for $k < 0$.

Problem 14: A Satellite Dish

A dish with $\frac{x^2 + y^2}{4f} = z$ has diameter 2 m and the focus is 0.8 m above the vertex. How deep is the dish?

Solution. $f = 0.8$ m and the rim radius is $r = 1$ m, so the depth is $d = \dfrac{r^2}{4f} = \dfrac{1}{3.2} = 0.3125$ m. (Equivalently $f = \frac{D^2}{16d}$ gives $d = \frac{4}{12.8}$.)

Problem 15: A Comet's Orbit

A comet has perihelion distance 0.6 AU and eccentricity 0.9. Find its semi-major axis, aphelion distance, semi-minor axis and polar equation.

Solution. $r_{\min} = a(1-e) = 0.6$ gives $a = \frac{0.6}{0.1} = 6$ AU. Then $r_{\max} = a(1+e) = 11.4$ AU, $b = a\sqrt{1-e^2} = 6\sqrt{0.19} \approx 2.62$ AU, and $p = a(1-e^2) = 6 \cdot 0.19 = 1.14$ AU, so

$$r = \frac{1.14}{1 + 0.9\cos\theta} \quad (\text{check: } r(0) = \tfrac{1.14}{1.9} = 0.6,\ r(\pi) = \tfrac{1.14}{0.1} = 11.4).$$

Problem 16: Where Is the Receiver?

Stations at $(\pm 150, 0)$ km receive a signal from a transmitter, and the distances to the two stations differ by 100 km. If the transmitter has $x = 100$ km, find its y-coordinate.

Solution. The transmitter lies on $\frac{x^2}{2500} - \frac{y^2}{20\,000} = 1$ ($a = 50$, $c = 150$, $b^2 = 20\,000$). At $x = 100$: $\frac{y^2}{20\,000} = \frac{10\,000}{2500} - 1 = 3$, so $y^2 = 60\,000$ and $y = \pm 100\sqrt{6} \approx \pm 245$ km (two mirror-image positions; a second pair of stations decides which).

Problem 17: Intersection of Two Spheres

Find the intersection of $x^2 + y^2 + z^2 = 25$ and $x^2 + y^2 + (z-6)^2 = 25$.

Solution. Subtracting, $z^2 - (z-6)^2 = 0$, so $12z - 36 = 0$ and $z = 3$. Then $x^2 + y^2 = 25 - 9 = 16$: a circle of radius 4 in the plane $z = 3$, centered at $(0, 0, 3)$.

8.6 Conceptual & Computational Exercises

8.6.1 Conceptual Questions

Answer by *thinking*, with little or no calculation; give a reason or a counterexample. Answers are in the answer key.

1. **True or false?**
   (a) The set of points whose distances to two fixed points differ by a constant is an ellipse.
   (b) A circle is an ellipse with eccentricity 0.
   (c) In space, $x^2 + y^2 = 1$ is a circle.
   (d) The equation $z = x^2 + y^2$ describes a cone.
   (e) If $B^2 - 4AC = 0$ the curve is always a parabola.
   (f) A hyperboloid of two sheets contains points with $|z| < c$.
   (g) The tangent line to an ellipse at P is perpendicular to the segment from the center to P.
2. Why must $c < a$ for an ellipse but $c > a$ for a hyperbola?
3. What happens to an ellipse if its two foci move together until they coincide? What if one focus moves off to infinity?

4. In $4x^2 + y^2 = 4$ and in $4x^2 - y^2 = 4$, which variable carries the long (respectively the transverse) axis? How do you see it?
5. Describe how the shape of the conic $r = \dfrac{p}{1+e\cos\theta}$ changes as e increases from 0 to 2.
6. Why does the discriminant $B^2 - 4AC$ not change when the axes are rotated, and why does this make it useful?
7. Why do the equations of paraboloids contain z to the *first* power, while those of ellipsoids and hyperboloids contain z^2?
8. Count the minus signs in $\dfrac{x^2}{a^2} \pm \dfrac{y^2}{b^2} \pm \dfrac{z^2}{c^2} = 1$: what surface do you obtain for 0, 1, 2 and 3 minus signs?
9. What shape is the cross-section of $z = x^2 - y^2$ by a plane $x = k$? By a plane $z = k$ $(k \neq 0)$? By $z = 0$?
10. Why is a satellite dish made in the shape of a paraboloid and not of a sphere?
11. A planet moves on an ellipse with the Sun at a focus. Where is it fastest, and why do the "distance from the Sun" and "distance from the center of the ellipse" differ?
12. How can the hyperboloid $\dfrac{x^2+y^2}{a^2} - \dfrac{z^2}{c^2} = 1$, a surface that is curved in every direction, be built from straight beams?
13. Is a circle with equation $x^2 + y^2 + Dx + Ey + F = 0$ always a circle? What can go wrong?

8.6.2 Computational Problems

Solve the following problems. Show all work and give exact answers unless otherwise specified.

1. An ellipse has vertices $(\pm 6, 0)$ and foci $(\pm\sqrt{11}, 0)$. Find its equation.
2. A hyperbola has vertices $(0, \pm 3)$ and foci $(0, \pm 5)$. Find its equation and its asymptotes.
3. A parabola has focus $(2, -1)$ and directrix $y = 3$. Find its equation and the direction it opens.
4. Classify the curve $x^2 + 4y^2 + 6x - 16y + 9 = 0$ by completing the square, and find its center.
5. Classify the curve $x^2 - y^2 - 2x - 4y - 4 = 0$ by completing the square.
6. Use the discriminant to classify $5x^2 + 4xy + 2y^2 - 6x + 8y - 3 = 0$, and decide whether it is a genuine (nonempty) curve by locating its center.
7. Find the center and radius of the sphere $x^2 + y^2 + z^2 - 6x + 2y - 4z + 5 = 0$.
8. Classify the quadric surface $x^2 + y^2 - z^2 + 4 = 0$ (hint: rewrite with z^2 isolated). What does a cross-section at constant z look like, and for which z is it nonempty?
9. A cylinder has directrix $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$ in the xy-plane and axis along the z-axis. Write its equation, and name the type of cylinder.
10. Identify $3x^2 + 2xy + 3y^2 = 16$ by rotating the axes, and give its semi-axes and the directions in which they point.
11. A dish of diameter 2.4 m is 0.36 m deep. Find the focal length and the position of the receiver.
12. A planet moves on an ellipse with $a = 2.5$ AU and $e = 0.6$ with the Sun at a focus. Find the perihelion and aphelion distances, the semi-minor axis, and the polar equation $r = \dfrac{p}{1+e\cos\theta}$.
13. Two stations are at $(\pm 100, 0)$ km. A signal's distances to the stations differ by 60 km. Find the equation of the position line.
14. A cooling tower has waist radius 25 m and $c = 30$ m. Find its equation and its radius at height 40 m above the waist.
15. Find the intersection of the spheres $x^2 + y^2 + z^2 = 36$ and $x^2 + y^2 + (z - 8)^2 = 36$.
16. Classify $x^2 - y^2 + 2x + 4y - z = 0$ and find its saddle point.
17. Classify $x^2 - y^2 - z^2 - 2x + 4y - 4 = 0$, give its center and the axis of symmetry, and say for which x the surface has points.

18. A liquid rotates with $\omega = 6$ rad/s, so its surface is a paraboloid $z = \frac{\omega^2}{2g} r^2$. Find its focal length f and the height of the surface at radius $r = 0.15$ m above its lowest point (the vertex on the axis).
19. A suspension bridge has span 300 m and sag 40 m. Write the equation of the (parabolic) cable measured from its lowest point, and find the height above the lowest point at 75 m from the center.
20. A jet of water leaves a nozzle with path $y = x - \frac{x^2}{50}$ (x, y in meters). Find the maximum height and the range.